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DEPARTMENT OF DEFENSE STANDARD PRACTICE

ELECTRIC PLANT INSTALLATION STANDARD METHODS FOR SURFACE SHIPS AND SUBMARINES (CABLEWAYS)

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FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. This standard disseminates up-to-date information detailing requirements for standard installation methods employed for submarine and surface ship electrical distribution systems.

3. These criteria apply to work on a specific ship or ships only when invoked by the Ship Specifications or similar contractual documents.

4. These criteria are for application to new construction, conversion, and alteration of existing ships.

5. Considering the magnitude of this standard, along with the changing requirements imposed on the Electric Plant, it is inevitable that changes will be required to update these criteria. Therefore, as comments arise, they should be forwarded to Naval Sea Systems Command (NAVSEA) 05Z3 to keep this standard as current as possible through subsequent revisions. Revisions will be accomplished by the issuance of additional or revised figures to be inserted in the basic standard parts. Superseded pages may be retained for reference if so desired.

6. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1. SCOPE

1.1 Scope. This standard covers standard practices for electric plant installation of cableways on surface ships and submarines.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-915	-	Cable, Electrical, for Shipboard Use, General Specification for
MIL-I-3064	-	Insulation, Electrical, Plastic-Sealer
MIL-T-16366	-	Terminals, Electrical Lug and Conductor Splices, Crimp-Style
MIL-PRF-23236	-	Coating Systems for Ship Structures
MIL-S-24149	-	Studs, Welding and Arc Shields (Ferrules), General Specification for
MIL-J-24445	-	Joint, Bimetallic Bonded, Aluminum to Steel
MIL-DTL-24640	-	Cables, Light-Weight, Electric, for Shipboard Use, General Specification for
MIL-DTL-24640/15	-	Cable, Electric, 600 Volts, Light-Weight, Multi-Pair, AWG 22, Type 2XSAW (Including Variation Type 2XSAOW)
MIL-DTL-24643	-	Cables and Cords, Electric, Low Smoke, For Shipboard Use, General Specification for
MIL-DTL-24643/24	-	Cable, Electrical, Types LSTCJX, LSTCKX and LSTCTX
MIL-DTL-24643/25	-	Cable, Electrical, Type LSPI
MIL-DTL-24643/26	-	Cable, Electrical, 600 Volts, Types LSDPS, LSFPS, LSTPS and LS7PS
MIL-DTL-32180	-	Cable Assembly, Aircraft Electrical Service

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-299	-	Cable Comparison Handbook Data Pertaining to Electrical Shipboard Cable

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(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DESIGN DATA SHEETS

DDS 305-1 - Designation and Marking of Electric System

(Copies of this document are available from Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160, or by email at CommandStandards@navy.mil with the subject line "DDS request".)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9AA0-AB-GOS-010/ GSO Section 075	-	General Specifications for Ships of the United States Navy, Threaded Fasteners, Section 075
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
S9407-AB-HBK-010	-	Handbook of Shipboard Electromagnetic Shielding Practices

(Copies of these documents are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Bottom loading. Cable secured to the underside of a cableway tier.

3.2 Cableway tier. Cableway tier is one row of cable installed in a cableway.

3.3 Double banking. Double banking is two rows of cable installed in a single tier of a cableway.

3.4 Multiple banking. Multiple banking is three or more rows of control and/or instrumentation cables on a single tier or in a cable tray.

4. GENERAL REQUIREMENTS

4.1 Location of cable runs. Main fore-and-aft cableways shall be located port and starboard, high and low in ship; lower cableways shall be through machinery spaces and corresponding platform decks, upper cableways under the main deck. This is to provide a quadrangular pattern to allow maximum athwartship and vertical separation of cables to loads requiring two sources of power for systems requiring alternate reliability circuits. For example, a ship service power feeder in the lower starboard cableway should have its complementary emergency or alternate feeder in the upper port cableway.

4.1.1 Vertical cable runs. Vertical cable runs shall be organized on the basis of the fore-and-aft or athwartship separation.

4.1.2 Emergency diesel generator seawater booster pumps. Cables supplying emergency diesel generator seawater booster pumps shall be run in the same compartment with, and as close as practicable to, but shall not be attached to the water piping connecting the pump to the diesel engine.

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4.1.3 Interior communications and weapons control. The main runs of interior communications and weapons control system cables shall follow cable runs of lighting and power installations. Interior communication and weapons control cables for circuits having several interrelated main and interrelated auxiliary circuits shall be run so that main circuits including normal power supply will be in one group and auxiliary circuits including alternate power supply will be in another group.

4.1.4 Control from more than one location. Where equipment is controlled from more than one location, the control cabling from each location to the equipment shall be installed in separate wireways.

4.1.5 Weather exposure. Cables shall not be exposed to the weather except where they penetrate a deck or bulkhead to supply equipment. Cables exposed to the weather shall be kept to a minimum. Where possible, cables exposed to the weather shall be protected by installation within conduit, masts, staffs, macks, and yardarms.

4.1.6 Circuit trunking. Location and arrangement of cable runs shall provide circuit trunking avoiding physical interferences with piping and ventilation ducts. Spaces considered harmful to cable or subject to explosion hazards shall be avoided. Cableways shall be arranged so that ventilation will be adequate for maintaining the minimum practicable temperature of cables within the cableway. Cableways shall be located so the cables will not be disturbed by disassembly or removal of machinery.

4.1.7 Armored cable. Armored cable has been identified as detrimental to the ability of low smoke cable to self extinguish. As a result, armored cable is no longer permitted aboard Naval ships except where necessary to provide physical protection to certain cables. These applications are limited aboard ship. Armored cable shall not be installed in weather exposed areas unless run in conduit. The following armored cables are permitted aboard ships:

a. MIL-DTL-24640/15, 2XSAWA

b. MIL-DTL-24643/24, LSTCKX, LSTCTX, and LSTCJX

c. MIL-DTL-24643/25, LSPI

d. MIL-DTL-24643/26, LSDPS, LSTPS, LSFPS, and LS7PS

4.1.8 Dead-ended cable. Cables installed through error or rendered useless as a result of modification shall be removed.

4.1.9 Gun mounts. Cables to gun mounts and directors shall be routed from the deck below through the center column. Watertight integrity shall be maintained. Cable shall be located and installed to avoid the possibility of chafing. Nonflexing cables shall terminate in submersible or watertight changeover connection boxes at the point of connection to the flexible cables. Length of flexible cables between changeover connection boxes and the slack cable enclosure shall be kept to a minimum. The slack cable enclosure shall be entered through stuffing tubes. In compartments containing hydraulic systems, the cable installation shall not impair the air-tightness or watertightness of decks and bulkheads forming the boundaries of the compartment.

4.1.10 Rotating missile launchers. Cable connections to rotating missile launchers shall comply with the Government Furnished Information (GFI) for the specific ordnance system being connected.

4.1.11 Ballistic structures. Cables shall not be supported or attached to ballistic structures unless approved by NAVSEA. Preferences for cable routing to avoid ballistic structure is as follows:

a. First preference shall be given to routing cables on the inboard or after bulkheads in the forward half of the ship, and on the inboard or forward bulkheads in the after half of the ship.

b. Second preference shall be given to routing cables on channels, or in cable racks on angles, attached to overhead deck beams.

4.1.12 Bulkheads and overheads. Cables shall not be routed on the outboard or forward bulkheads in the forward half of the ship, and on the outboard or after bulkheads in the after half of the ship, when the plating is 50 pounds per square foot (lb/ft^2) or heavier. This routing shall also be avoided for lighter plating unless no other route is feasible. The installation of cables on the overhead and on bulkheads shall comply with NAVSEA standard methods specified herein. Cables for vital systems such as interior communication and weapons control systems shall not be secured to the overhead, or to shell planking, to shell plating or to ballistic bulkheads.

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4.1.13 Sonar system cables. Sonar system cables and other low frequency system cables shall be installed in accordance with the requirements of NAVSEA S9407-AB-HBK-010.

4.2 Spare cable space allowance. In the organization of principal cableways, spare cable space of approximately 20 percent of that to be occupied by the final cable installation (as known at time of delivery of the ship) shall be reserved on cableways and in cable penetration areas for future cable installations. The additional cable space may consist of unused hangers or combination of unused hangers and space available on used hangers, assuming that for future addition of cable, double banking will be allowed. During the design phase, the contractor shall provide cableway space in excess of the spare 20 percent in order to accommodate cables added as a result of developments occurring during the construction period. Through horizontal cable runs in aircraft carriers' hanger spaces will not be permitted. Through vertical runs such as those from the second deck to the gallery or flight deck levels shall be grouped to reduce the number of protective casings required and shall be protected from fire.

5. DETAILED REQUIREMENTS

(See figures and requirements listed in this section.)

5.1 Cable slack. Cable slack requirements are as follows:

a. Cables shall be installed so that sufficient slack exists to allow for deflection of bulkheads under shock conditions.

b. Sag between hangers shall be uniform for each row of cables so that clearance between rows will be the same throughout the cable run.

c. Where cables spread out to enter bulkhead stuffing tubes or multiple cable penetrators, bends shall have liberal sweep to provide as much flexibility as practicable.

d. Cables having only a minimum spread where they pass through bulkhead stuffing tubes or multiple cable penetrators shall have enough slack to give them the same flexibility as other cables in the group.

e. Cables from switchboards or other electric equipment shall enter cableways in a curve of sufficient radius to prevent transmission of stresses to the equipment during severe cableway deflection.

f. Cables crossing expansion joints, such as those under the flight deck and gallery deck of aircraft carriers, shall have slack allowance at such points at least equal to the maximum movement of the expansion joints.

5.2 Cable bend radius. Cable bend radii shall be in accordance with MIL-HDBK-299. The following general requirements are also applicable:

a. Bends at terminal entrances, where stuffing tubes are required, shall be made through angle stuffing tubes.

b. Straight stuffing tubes may be used in place of angle stuffing tubes if the bend in the cable is not less than the minimum bend radius.

c. Measurement point for minimum radius of bend is that surface of the cable jacket which is on the innermost portion of the cable bend.

5.3 Protection of cable.

5.3.1 Battle damage. Protection afforded by the ship structure shall be used to the greatest extent practicable. Cables shall not be run on the exterior of deck houses or similar structures above the main deck (including the island structure of aircraft carriers) except where necessary because of the location of the equipment served.

5.3.2 Mechanical damage. Cables subject to mechanical damage because of their proximity to areas frequented by personnel shall be protected by metal casings. Cableways in areas where their misuse as steps or handholds would cause damage shall be protected. Protective plates shall be installed over the cableways in all passages where cables might be stepped on. At hatch openings and in trunks where objects are raised or lowered, cables shall be protected by steel casings.

5.3.2.1 Auxiliary ship installation. Auxiliary (cargo) ship cable installations shall be routed outside cargo spaces wherever practicable. Where routing through cargo spaces is unavoidable, cables shall be protected from mechanical damage, including damage due to shifting of cargo.

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5.3.2.2 Riser boxes. Riser boxes, or multiple cable penetrators with mechanical protection, shall be installed for three or more cables passing through a deck in a group. Where fewer than three cables pass through a deck, kickpipes shall be installed. Kickpipes installed in the vicinity of hydraulic equipment, where the possibility of oil leakage exists, shall be welded in place. Details of kickpipe installations and the installation of top bracing, where required, shall conform to this standard.

5.3.2.3 Oiler weather deck. Cableways installed on the weather deck of oilers shall be protected by a substantial open bottom steel enclosure. Cableways installed alongside or under weather deck catwalks shall be completely enclosed in a steel enclosure, the bottom section of which shall consist of a removable perforated steel plate. Enclosures shall be constructed so as to permit periodic inspection and maintenance of cables and hangers.

5.3.3 Excessive heat. Installation of cables in locations subject to excessive heat shall be avoided. Where required, heat insulating barriers shall be installed. Cable shall not be installed adjacent to machinery, piping, or other surfaces having an exposed surface temperature greater than 70 °C. Cables shall not be run over boilers in the upper portions of firerooms or in other locations where they may be surrounded by hot air. Cables shall not be run in locations where they will be exposed to hot stack gases.

5.3.4 Excessive moisture. Cables shall not be installed in locations where they may be subjected to excessive moisture. Where cable routing near firemains, water, steam, oil, or other piping is unavoidable, dripproof shields shall be provided for protection. Where cables must be installed in spaces subject to flooding, they shall be installed as high as practicable within the space.

5.3.5 Cables installed through submerged spaces and voids. Cables that would normally be submerged, except for cables in the sonar dome, shall be enclosed in a single pipe in a manner similar to that for cables passing through tanks. Cable runs in voids and other dead air spaces shall be avoided. If it is not practicable to avoid cable runs in such spaces, cables shall be installed only in those spaces which are not provided with a means for flooding, and then only when temperature calculations show that the spaces have sufficient radiation area to maintain an ambient temperature at or below the temperature in which the cable was selected to operate. Cables shall be supported clear of decks and bulkheads to avoid condensate which might form on such surfaces.

5.3.6 Cable installations in hazardous locations. Cable runs shall be avoided through hazardous spaces such as the following (except such cable runs that terminate in or cannot be routed around these spaces):

a. Compartments where hazardous materials are stowed, such as explosive signaling apparatus, ready service ammunition, chemical materials, film, paint, compressed oxygen, chlorine, and combustible gases.

b. Diesel, JP-5, and fuel tanks and stowage compartments (stowage in separate tanks or containers).

c. Gasoline hazard areas.

d. Hangar spaces.

e. LOX (liquid oxygen) handling and transfer spaces.

f. Magazines.

g. Missile magazines.

h. Warhead, depth charge, mine charge, aerial bomb, and eased explosives magazines.

Cables within hazardous spaces shall be protected against mechanical damage. This protection shall be in the form of non-tight metal guards (expanded or solid) of sufficient strength to provide the required protection. Clearance shall be provided between the protective guards and the cables to provide ample ventilation. Protective guards shall be provided with drainage holes where required. Cables protected by the ship structure or by permanently installed equipment are considered adequately protected. Only cables having conductor insulation of silicone rubber (except for interior communication and weapons control cables totally within the compartment) shall be used. Through cables shall be of unbroken length within compartments. Cables terminating at lighting fixtures or other equipment shall be of broken lengths. Separate cables to each lighting fixture are not required; a single cable may be run between fixtures where more than one fixture is installed in a space.

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5.3.7 Channel rubber. Channel rubber shall be used with banding straps in cableways to protect the outer jacket of cables from being cut and coaxial cables from being deformed by the banding strap. Cables such as coaxial, category, profibus, and other data and communication cables should be run in the middle of the cableway for increased mechanical protection. Channel rubber is also required in the following instances:

a. For each banding strap of a vertical cableway.

b. For each banding strap at a cableway bend, including breakout bends.

c. Before and after each cable penetration of a deck or bulkhead.

Where hose clamp retention devices are used, channel rubber shall be required under the following conditions:

a. When coaxial, category, profibus, and other sensitive data and communication cable would otherwise be in contact with the hose clamp.

b. On all vertical wireways.

c. On all sound isolated equipment supplied by armored cable.

d. On a single hanger to properly secure cables of varying diameters.

e. On any cable hanger where the banding material (hose clamp) could damage the cable, i.e., a case-by-case basis.

5.4 Installation of propulsion system cables. Propulsion system cables shall be installed in cableways separate from those used for other electrical system cables. Direct current or alternating current propulsion system cables shall be arranged and installed in such manner as to minimize heating of adjacent structural material by induced currents. Alternating current propulsion cables installed on insulating hangers shall be grounded at approximately the midpoint of the cable run. Propulsion system cables shall not be installed in bilges. When cables are located near dripping water, or oil, they shall be protected by drip covers. These protective covers shall be spaced away from the cables so as not to restrict ventilation. Cables entering propulsion system equipment from above or from the sides shall enter through stuffing tubes or other NAVSEA-approved watertight enclosure penetration device and shall be braced and secured to prevent dislodgment under vibration, shock, and magnetic stresses. The ends of propulsion system cables 9,000 circular mils and larger shall be sealed with solderless waterseal type lug terminals in accordance with MIL-T-16366.

5.5 Cable handling and stowage. All cables except flexing service and silicone rubber insulated type of MIL-DTL-915, MIL-DTL-24643, and MIL-DTL-24640 shall be given the following special handling at temperatures below 35 °F:

a. If the compartment in which the cable is to be installed cannot be heated, the cable shall first be stored in an ambient temperature of at least 50 °F but not above 120 °F until it is warm enough to be completely installed before it cools to 35 °F.

b. If cable must be installed when its temperature is 35 °F or lower, extra care will be required. The radius of bends shall be not shorter than the minimum values in MIL-HDBK-299. Before bending to the final radius, that portion of the cable comprising the bend shall be warmed thoroughly by a portable warm air blower.

c. Cable shall be stored in a dry place, which is not subject to accidental flooding, is protected from the weather, and is subjected to a minimum variation of temperature.

d. Portable cables for aircraft servicing in accordance with MIL-DTL-32180 shall be stowed in bins. In hangar bay areas, they shall be stowed on cable racks.

e. Cable that has been in storage for prolonged period may be installed provided a visual inspection shows that it has sustained no mechanical damage that would impair the watertight integrity of its outer sheath.

f. Cable armor showing signs of corrosion shall be cleaned and a coat of zinc-chromate primer applied to all corroded areas.

5.6 Cable tags. All permanently installed cables shall be tagged to each point of connection, and on both sides of decks and bulkheads except as follows:

a. Where through cable runs within a compartment are direct (such as a vertical run between decks), a single tag will suffice.

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b. For cables with both points of connection within a compartment and which can be readily traced, a single cable tag will suffice.

c. Where compartments are subdivided by internal bulkheads or where machinery or installed equipment makes tracking of cable runs difficult, additional tags shall be provided.

d. For multiple cable penetrations of decks and bulkheads (main cableways), individual cable tags can be omitted, and in lieu thereof, an identification plate shall be installed adjacent to the cableway penetration area showing each cable designation in the order of location in the penetration area.

Cable tags shall be in accordance with DDS 305-1. Capital letters shall be used on cable tags; height of all letters and numbers shall be not less than 3/16 inch, and letters and numbers shall be embossed to at least 1/64 inch above the surface.

5.7 Cable hangers and supports. Cable hangers and supports shall meet the following requirements:

a. MIL-S-901 shock.

b. MIL-STD-167-1 vibration.

c. Minimum weight.

d. Minimum maintenance on cable supporting systems.

e. Facilitate cabling procedures.

f. Convenience of servicing system.

g. Fireproof.

h. Material used is non-toxic under all conditions.

i. Free of sharp protrusions, corners, items, and so forth, dangerous to personnel.

j. Shall be designed to permit repair with local market items. It is not intended that extensive back-up stocks be supplied for item by item replacement.

k. Compatible with and non-damaging to cables.

l. Self-locking nuts or hex nuts with spring lockwashers shall be used in the installation of cable hangers and supports.

5.7.1 Insulation spacer block. The use of insulation spacer blocks is mandatory for propulsion system cables in order to prevent bunching.

5.7.2 Shock design. All cable supporting arrangements shall be tested for resistance to shock and vibration. Deformation to supporting members and retaining members is satisfactory providing cables are prevented from becoming whips or missiles and retainers, racks and supports do not become missiles. Shock and vibration tests shall be conducted at an approved laboratory. No mechanical damage shall occur to cable specimens, and insulation resistance shall comply with cable specifications before and after the testing. These tests shall be conducted by each manufacturer for each type hanger supplied. Retesting shall be required when design, manufacturing process, or material is changed. Testing of a hanger with lesser tiers or lesser dimensions and spacing shall not be extended or exempt from testing for other hangers. The cable supports when tested shall be loaded with maximum design cable load. Material is not satisfactory unless it has successfully passed these tests. Shock tests shall be in accordance with MIL-S-901, Class I. Cable supporting devices shall be arranged for testing as specified in 5.7.2.1 and 5.7.2.2.

5.7.2.1 Multiple cable test racks (15 inches free length of cable support). If multiple cable racks are being tested for ability of cable support member only to withstand shock and vibration as fully loaded, a minimum of three supporting stations with 32-inch, 21-inch, or 16-inch spacing shall be provided. The maximum number of the largest cables that can be accommodated on these racks shall be applied. Cable overhang at each end shall be 16 inches or one-half the distance between supports. If tiers of multiple cable racks and supporting devices for attachments to ships' structure are to be tested for capability to withstand shock and vibration, a duplicate of the assembly proposed for shipboard installation with a minimum of three supporting stations with 32-inch, 21-inch, or 16-inch spacing shall be provided. The maximum number of the largest cables that can be accommodated on these racks shall be applied. Cable overhang at each end shall be 16 inches.

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5.7.2.2 Single cable supports. All single cable supports shall be tested. These shall be arranged using a minimum of three supporting stations spaced 32 inches, 21 inches, or 16 inches apart. Cable overhang shall be 16 inches. Maximum size cables shall be mounted with these single cable supports. Two, three, or four cables shall be arranged for tests in a manner similar to that for single cables and supported using one of the methods shown in figures 4C1-4C12. A single tier of cables shall be arranged as described for multiple cable racks.

5.7.3 Installation.

5.7.3.1 Spacing. Spacing of steel and aluminum cable supports shall be 32 inches except as follows:

a. The spacing of hangers shall be 21 inches to reduce cable sag in cable racks when headroom clearance under the cable racks before installation of cables is 6 feet 7 inches or less (6 feet 4 inches minimum).

b. Hangers for overhead multiple tier aluminum cableways, when welded to aluminum decks, shall be spaced 16 inches and the number of tiers shall be limited to three. If aluminum hangers or supports are being utilized from steel or structurally fire protected aluminum decks, a steel hanger or support or both shall be installed in lieu of an aluminum hanger or support every 5 feet 4 inches as shown in figure 4B37.

c. The spacing of hanger supports for single cables or single tier of cables shall be 32 inches but may be adjusted between 32 inches and 16 inches as necessary to suit particular ship conditions.

5.7.3.2 Cable racks. A maximum of five single tiers of cables may be supported from an overhead in one cable rack. Aluminum tiers shall be limited to three. A maximum of two tiers of cables may be supported from bulkheads in one cable rack. Eight tiers of double-banked cables may be supported in main cableways in machinery spaces and boiler rooms, when approved by NAVSEA. Backing plates are not required for cable support except that when bulkheads are 1/16-inch thick or less and cable run is over 4-inch width, backing washers shall be used. Multiple tiers of cable shall not exceed two on bulkheads except in congested locations, in which case 3 tiers may be used on request and subject to NAVSEA approval. Cable support angles shall be secured to ship structure by 180-degree weld except that in the following cases, the weld shall be 360 degrees:

a. Where the number of tiers of 15-inch cable rows in wireway exceeds two.

b. When welding to ballistic structure.

c. When the welds are in wet spaces such as weather decks or shower rooms.

5.7.3.2.1 Support components. Stainless corrosion resistant steel cable standoffs and clamps shall be used for cableway installations outboard or in corrosion environment locations. Cable support components shall be of the following materials:

Nonmagnetic vessels

Inboard	Outboard
Hangers – aluminum	Hangers – aluminum
Supports – aluminum	Supports – aluminum
Ferrules – CRES, aluminum, or nylon	Ferrules – CRES or nylon
Bolts and nuts – CRES or aluminum	Bolts and nuts – CRES
Banding – aluminum	Banding – CRES

Other vessels

Inboard	Outboard
Hangers – aluminum or steel (see 5.7.3.2.2)	Hangers – aluminum or steel (see 5.7.3.2.2)
Supports – aluminum or steel	Supports – aluminum or steel
Bolts and nuts – steel – zinc plated	Bolts and nuts – CRES
Banding – steel – zinc plated or CRES	Banding – CRES
	Ferrules – CRES or nylon

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Appropriate method details specified as submarine only may be utilized for surface ship applications, particularly for weather deck, wet/humid areas, and hull applications. Aluminum cable supports may be attached to steel deck and bulkhead where suitable design and interface is approved. Cable hangers shall be bracketed away from deck and bulkhead surfaces in food handling and food storage spaces to permit spraying for insect control.

5.7.3.3 Double banking of cable. Not more than one row of cables shall be installed on a cable hanger. Where space is limited, and if specific approval has been obtained, double banking of cables on hangers will be permitted provided the clearance between the top of the upper layer of cables and the hanger above shall be at least ⅜ inch. Multiple banking of control and instrumentation cables is allowed provided sufficient order is maintained in the installation to prevent intertwining of the cables and to allow removal of any cables in the run without affecting other cables in the run.

5.7.3.4 Cable retention. Devices for cable retention meeting the requirements of 5.7 and 5.7.2 are required on horizontal cable supports in multiple racks. Retention of cables on supports can be accomplished by the use of retainers such as contour straps, soft iron flat bars bent over the cables, semi-contour straps or angle iron retainers. Non-toxic strips or channel material shall be used with semi-contour straps, bars, and angle retainers to reduce cable damage, distortion, and chafing. On cable runs in which the cable is vertical, cable retention is required at every hanger.

a. Cable retention spacing shall be as specified in 5.7.3.1.

b. Cable straps shall be omitted on horizontal cable runs except as follows:

(1) Where the hanger has no side brackets.

(2) At those locations where the cable runs change direction or pass through beams or bulkheads.

(3) Where four consecutive hangers would not require straps. In this case, a minimum of one strap shall be installed on every fourth hanger.

c. Marine snaplock type cable clamps may be used (22-inch maximum diameter).

5.7.3.5 Channel rubber. Channel rubber shall be installed on all vertical wireways, before and after each breakout hanger and where wireway changes direction.

5.7.3.6 Plastic sealer. After the cables are properly secured, plastic sealer electrical insulation, MIL-I-3064, Type HF, shall be used to seal the space around the cable as follows:

a. In cable clamps and bushings entering the top or side, with no drip loop, of an electrical enclosure.

b. In bushings, collars or nipples used for passing cables through light-tight and fume-tight bulkheads.

5.7.3.7 Welding. Welding shall be in accordance with NAVSEA S9074-AR-GIB-010/278.

5.7.3.8 Threaded fastener. Threaded fasteners shall be in accordance with S9AA0-AB-GOS-010, GSO Section 075. Unless otherwise specified, threaded fasteners of normally stocked lengths shall be used. Male threads on threaded fasteners, after being installed and tightened, shall protrude at least one thread beyond the top of the nut or plastic locking element. Excessive protrusion shall be avoided, particularly where necessary clearances, accessibility, and safety are important. Where practicable, the number of threads protruding shall not exceed five. Washers shall not be used for the sole purpose of lessening thread protrusion.

5.7.3.9 Collar studs. Collar studs shall be in accordance with MIL-S-24149.

5.7.3.10 Bimetallic joints. Bimetallic joints in accordance with MIL-J-24445 may be used to secure aluminum or steel cable hangers to dissimilar decks or bulkheads in lieu of other methods identified for dissimilar materials. Sources for bimetallic bonding material are E.I. Dupont Denemours Co., Inc., Revere Copper, and Brass Co., North Technical Industries, Inc., or equal.

5.7.4 Dead-ended cable. Cables installed through error or rendered useless as a result of modifications shall be removed. Vacated and unused penetrations shall be sealed by methods which satisfy the tightness requirements of the structure penetrated.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This section specifies the requirements for submarine and surface ship cableways and protection methods to be employed both on surface ships and submarines. Standard methods identified for electric plant installation are intended for new construction, conversion, and alteration of existing ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

6.3 Designation of electric plant installation standard methods figures. MIL-STD-2003-4 contains drawings that depict standard methods that are applicable for general electric plant installation on both surface ships and submarines. Each drawing has been assigned a figure number. The methods shown on the figures are grouped together providing similar functions. These groups are:

MIL-STD-2003-4 (Cableways)

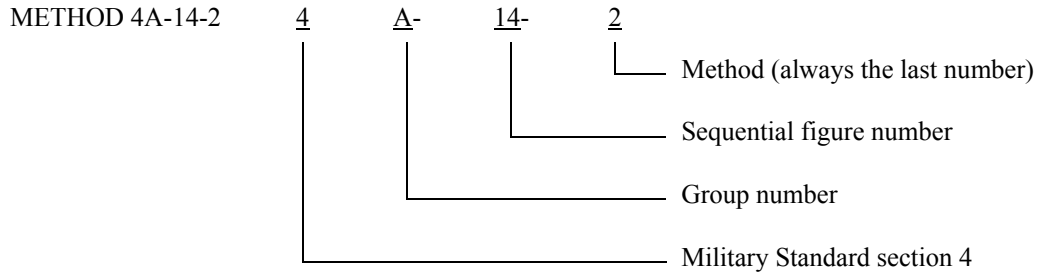
Group A. Cableways (Submarines)

B. Cableways (Surface Ships)

C. Cableways (General)

D. Cable Protection

The methods shown on the figures are identified by the following alphanumeric designation system:



Thus, Method 4A-14-2 identifies Method 2, figure 14 in Group A of MIL-STD-2003-4.

6.4 Subject term (key word) listing.

Double banking

Hangers

Tier

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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GROUP 4A - CABLEWAYS (SUBMARINES)

A.1 SCOPE

A.1.1 Scope. This appendix describes standard methods for the installation of cableways on submarines.

A.2. APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-6855	-	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-W-12133	-	Washer, Spring Tension, General Specification for
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-S-24149	-	Studs, Welding and Arc Shields (Ferrules), General Specification for
MIL-S-24149/1	-	Stud, Welding, and Arc Shields (Ferrules); Type I, CLASS 1, 2, 3, and Type II, Class 1, 4, 5, 5A, 6, Carbon Steel, for Direct Energy, Arc Welding
MIL-S-24149/2	-	Studs, Welding, and Arc Shields (Ferrules); Type III, Class 1, 2, 3, and Type IV, Class 1, 2, 3, 4, 5, 6, Aluminum Alloy, for Direct Energy Arc Welding
MIL-S-24149/3		Studs, Welding, & Arc Shields (Ferrules); Type V, Class 1, 4, 5, 5A, Corrosion-Resistant Steel, for Direct Energy Arc Welding
MIL-P-24691/3	-	Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

A.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil.)

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A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM25027	-	Nut, Self-Locking, 250 °F, 450 °F, and 800 °F

(Copies of this document are available from Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3901 or online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM A240/A240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

(Copies of this document are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-QQ-S-763	-	Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings
SAE-AS23190	-	Straps, Clamps, and Mounting Hardware, Plastic and Metal for Cable Harness Tying and Support

(Copies of these documents are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.)

A.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3 REQUIRED EQUIPMENT AND MATERIALS

A.3.1 Required equipment and materials. The equipment and materials in the tables and referenced in the text of this section shall be used for the installation of cableways on submarines. For fiber optic cable, refer to MIL-STD-2042-4 for additional information and guidance.

A.4 NOTES AND PROCEDURES

A.4.1 Dimensions. For figures and tables in this section, all dimensions are in inches unless otherwise noted.

A.4.2 Figures. Table 4AI provides information for the figures in this group. The details and methods shown on figures 4A1 and 4A2 specifically support the installation of battery cables on submarines, however, they may be used for general installations where applicable.

TABLE 4AI. Figures for cable installations on submarines.

Figure number	Cable installation	Page
4A1	Main cableway for battery cables (submarines)	14
4A2	Main cableway for battery cables (submarines)	17
4A3	Addition of cable to existing battery cableway (submarines)	19
4A4	Cableway adjacent to bulkhead penetration (submarines)	20
4A5	Spacer blocks for submarine cableways	21

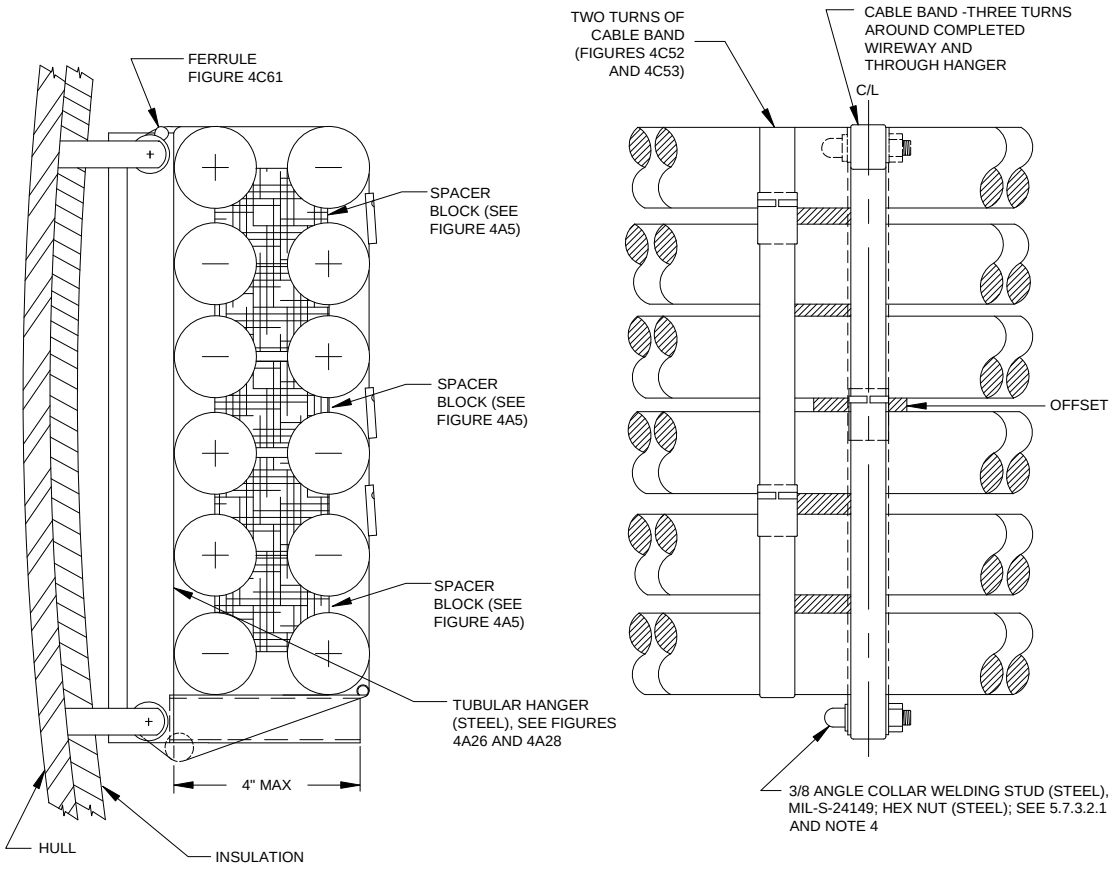
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TABLE 4AI. Figures for cable installations on submarines - Continued.

Figure number	Cable installation	Page
4A6	Cableway for auxiliary cables (submarines)	22
4A7	Cableway for auxiliary cables (submarines)	24
4A8	Cableway for auxiliary cables (submarines only)	27
4A9	Single row cableway (submarines)	29
4A10	Cableways through coamings (submarines)	31
4A11	Cableway for double bank of main propulsion cables (submarines)	33
4A12	Cableway for double bank of main propulsion cables (submarines)	34
4A13	External wireways installation (submarines)	36
4A14	External wireways installation (submarines)	41
4A15	External wireways installation (submarines)	46
4A16	External wireways installation standard methods (submarines)	51
4A17	External wireways installation standard methods (submarines)	56
4A18	External wireways installation standard methods (submarines)	59
4A19	External wireways installation standard methods (submarines)	62
4A20	External wireways installation standard methods (submarines)	65
4A21	External wireways installation standard methods (submarines)	68
4A22	External wireways installation methods (submarines) – cable rack assembly	71
4A23	External wireways installation methods (submarines) – section A-A for figure 4A22	74
4A24	External wireways installation methods (submarines)	77
4A25	External wireways installation methods (submarines) – pad and section views	80
4A26	Tubular hanger details “L” shape for submarines	83
4A27	Tubular hanger details “L” shape for submarines	84
4A28	Tubular hanger details “U” shape for submarines	85

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METHOD 4A-1-1
BANK OF 12 CABLES ARRANGED
IN DOUBLE VERTICAL TIERS WITH SPACERS



NOTES:

1. Hangers with 16" center spacing is the maximum size that can withstand HI shocks. Where the number of cables exceeds the maximum size of support shown, additional wireways shall be provided.
2. Methods shown on this sheet may be used for general installation where applicable.
3. See note 4 below for table of dimensions and stock numbers (NSN) for angle collar studs.
4. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750

FIGURE 4A1. Main cableway for battery cables (submarines).

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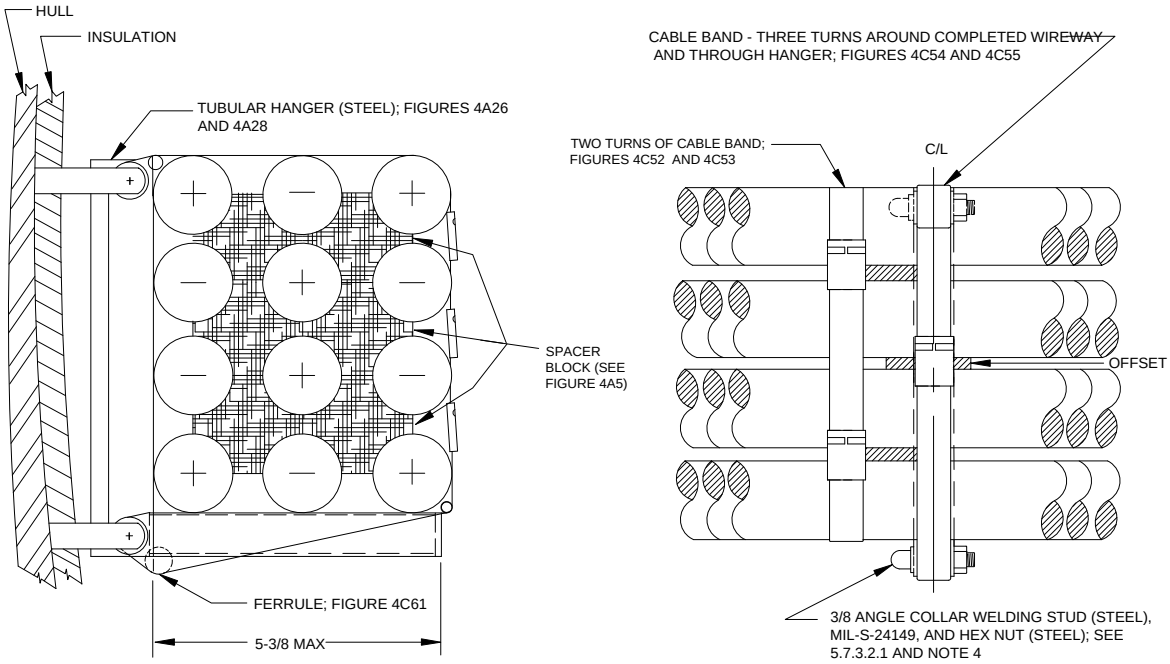
NOTES (continued):

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, Class 8	MIL-S-24149/2 aluminum type IV, class 4
2.250			5307-01-294-1097
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A1. Main cableway for battery cables (submarines) - Continued.

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METHOD 4A-2-1
BANK OF 12 CABLES ARRANGED IN TRIPLE VERTICAL TIERS WITH SPACER BLOCKS



NOTES:

1. Methods shown on this sheet may be used for general installation where applicable.
2. Hangers with 16" center spacing is the maximum size that can withstand HI shocks. Where the number of cables exceed the maximum size of support shown, additional wireways shall be provided.
3. See note 4 below for table of dimensions and stock numbers (NSN) for angle collar studs.
4. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147

FIGURE 4A2. Main cableway for battery cables (submarines).

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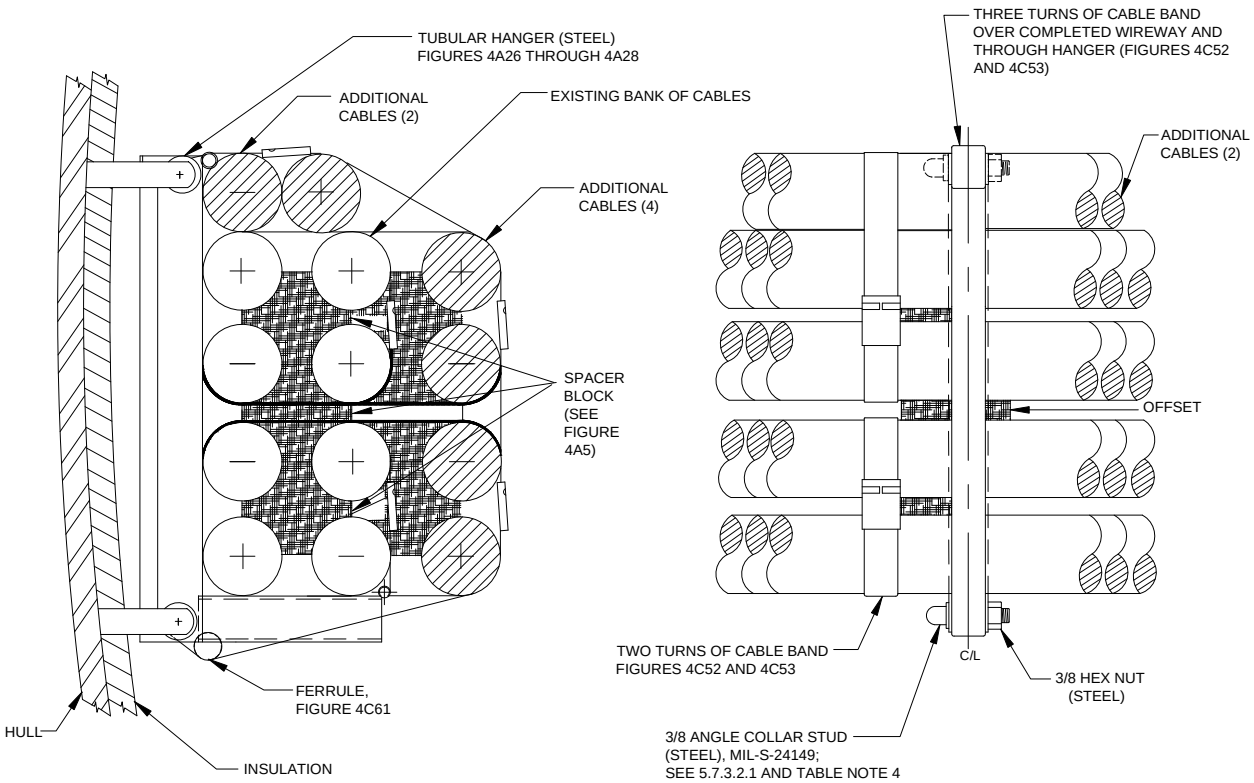
NOTES (continued):

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
1.875			5307-01-294-0268
2.250			5307-01-294-1097
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A2. Main cableway for battery cables (submarines) - Continued.

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METHOD 4A-3-1
ADDITION OF CABLES TO EXISTING BATTERY CABLEWAY



NOTES:

1. Hangers with 16" center-to-center spacing is the maximum size that can withstand HI shock. Where the number of cables exceeds the maximum size of support, additional wireways shall be provided.
2. Procedures shown on this sheet may be used for general installation.
3. See note 4 below for table of dimensions and stock numbers (NSN) for angle collar studs.

FIGURE 4A3. Addition of cable to existing battery cableway (submarines).

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NOTES (continued):

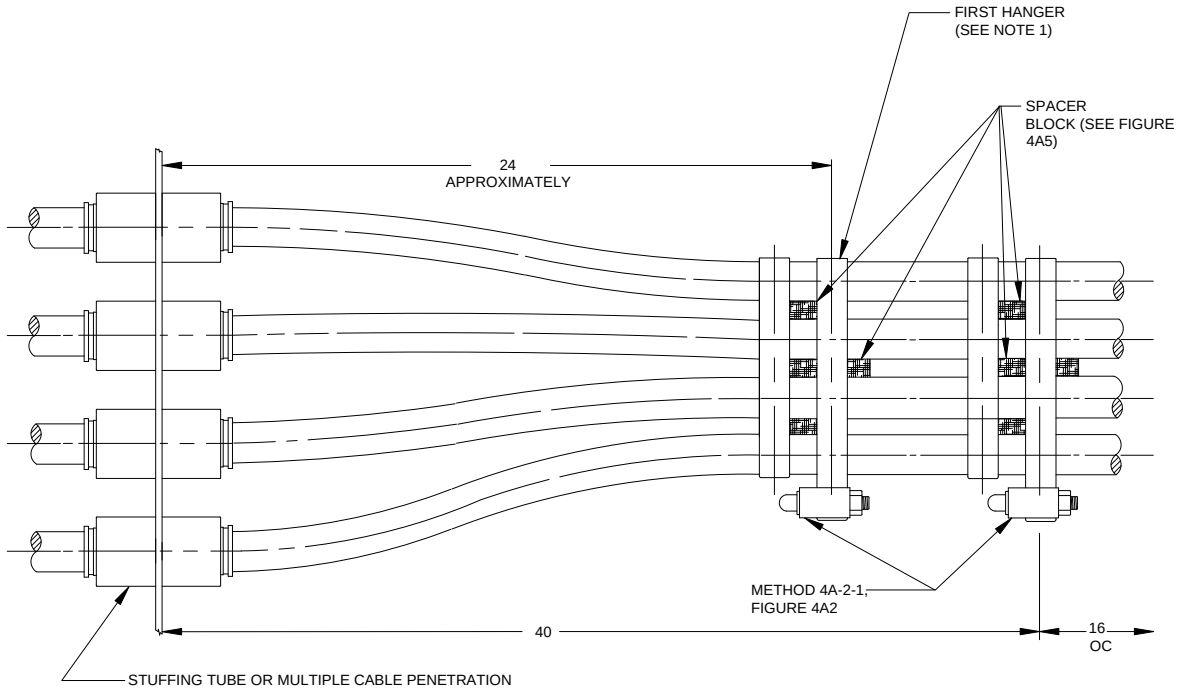
4. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A3. Addition of cable to existing battery cableway (submarines) - Continued.

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METHOD 4A-4-1
CABLEWAY ADJACENT TO BULKHEAD PENETRATION
(ALL DIMENSIONS ARE IN INCHES)



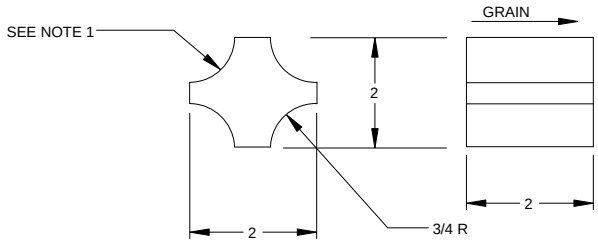
NOTE:

1. The first hanger next to the bulkhead shall be located to suit cable bends at penetration but shall not exceed 32".

FIGURE 4A4. Cableway adjacent to bulkhead penetration (submarines).

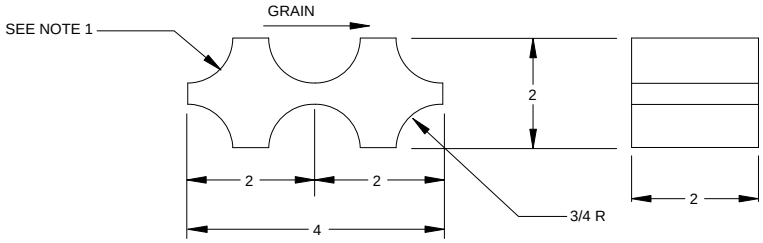
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METHOD 4A-5-1
SPACER BLOCK
FOR FOUR CABLES



ALL DIMENSIONS ARE IN INCHES

METHOD 4A-5-2
SPACER BLOCK
FOR SIX CABLES



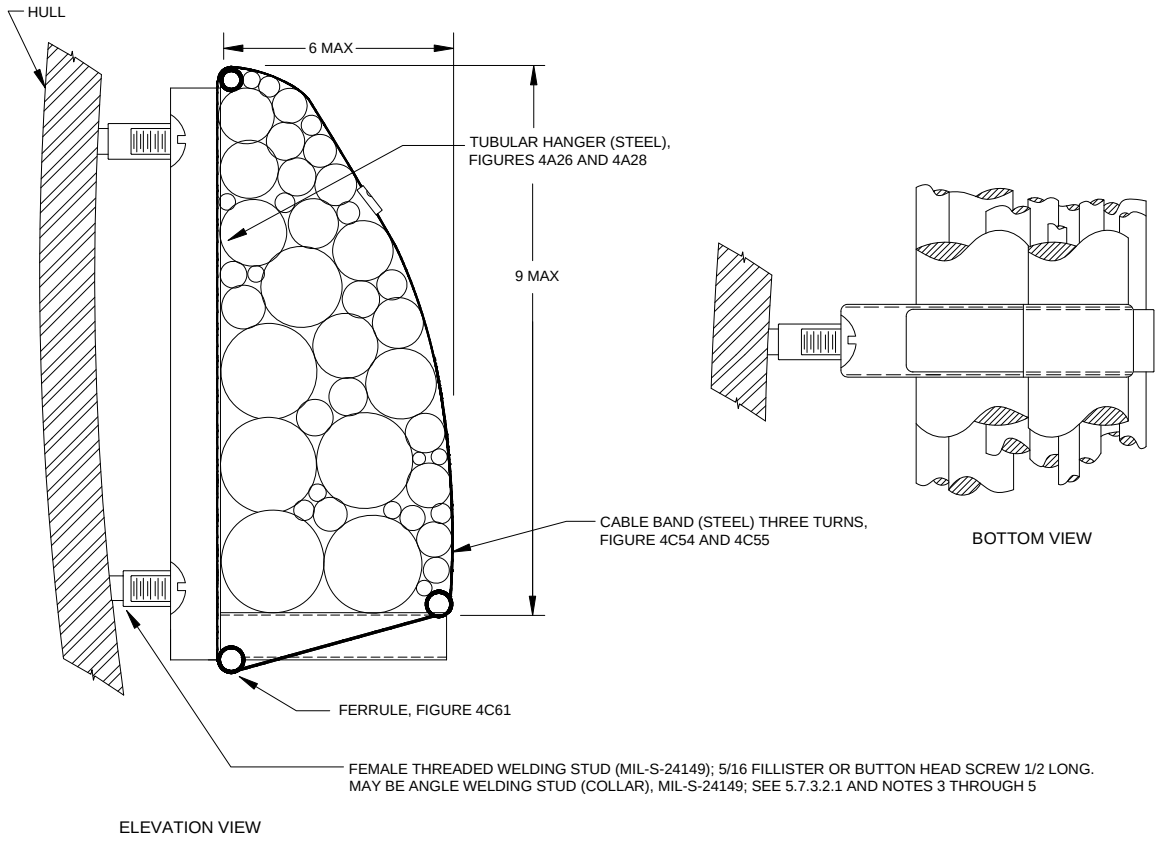
NOTE:

1. Material - material may be maple wood finished with two coats of marine varnish or melamine in accordance with ASTM D704.

FIGURE 4A5. Spacer blocks for submarine cableways.

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METHOD 4A-6-1
CABLE BANKS SUPPORTED BY WELDING STUDS ON HULL WITHOUT INSULATION



NOTES:

1. Hangers with 16" center-to-center spacing is the maximum size that can withstand HI shock. Where the number of cables exceeds the maximum size of support, additional wireways shall be provided.
2. See note 6 below for table of dimensions and stock numbers (NSN) for angle collar studs.
3. For the installation of auxiliary cables (figures 4A6, 4A7, and 4A8), the cables shall be arranged so that the larger cables are in the center of the hanger if possible.
4. National stock numbers for 5/16-17 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

FIGURE 4A6. Cableway for auxiliary cables (submarines).

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NOTES (continued):

5. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

6. National stock numbers for 3/8-16 angled threaded collar studs are as follows:

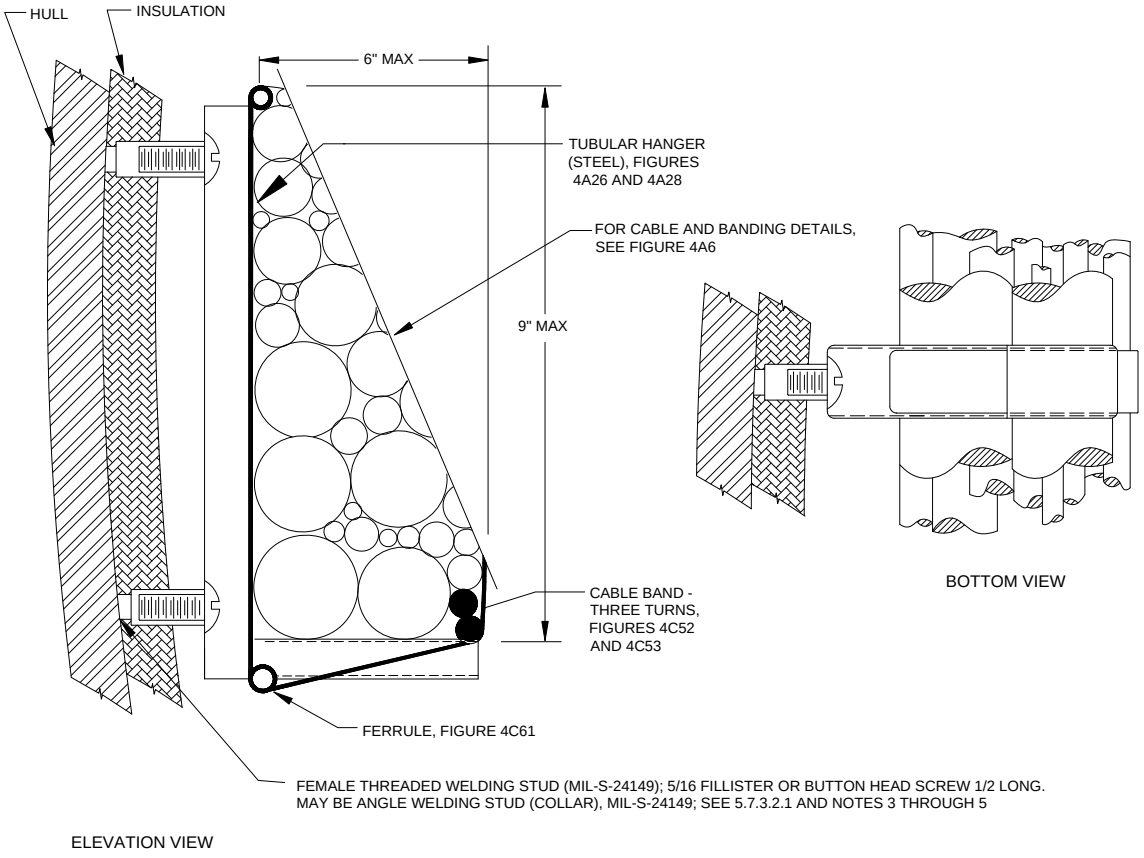
Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A6. Cableway for auxiliary cables (submarines) - Continued.

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METHOD 4A-7-1

CABLE BANKS SUPPORTED BY WELDING STUDS ON HULL WITH INSULATION



NOTES:

1. Hangers with 16" center-to-center spacing is the maximum size that can withstand HI shock. Where the number of cables exceeds the maximum size of support, additional wireways shall be provided.
2. See note 6 below for table of dimensions and stock numbers (NSN) for angle collar studs.
3. For the installation of auxiliary cables (figures 4A6, 4A7, and 4A8), the cables shall be arranged so that the larger cables are in the center of the hanger if possible.

FIGURE 4A7. Cableway for auxiliary cables (submarines).

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NOTES (continued):

4. National stock numbers for 5/16-17 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

5. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, Class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

FIGURE 4A7. Cableway for auxiliary cables (submarines) - Continued.

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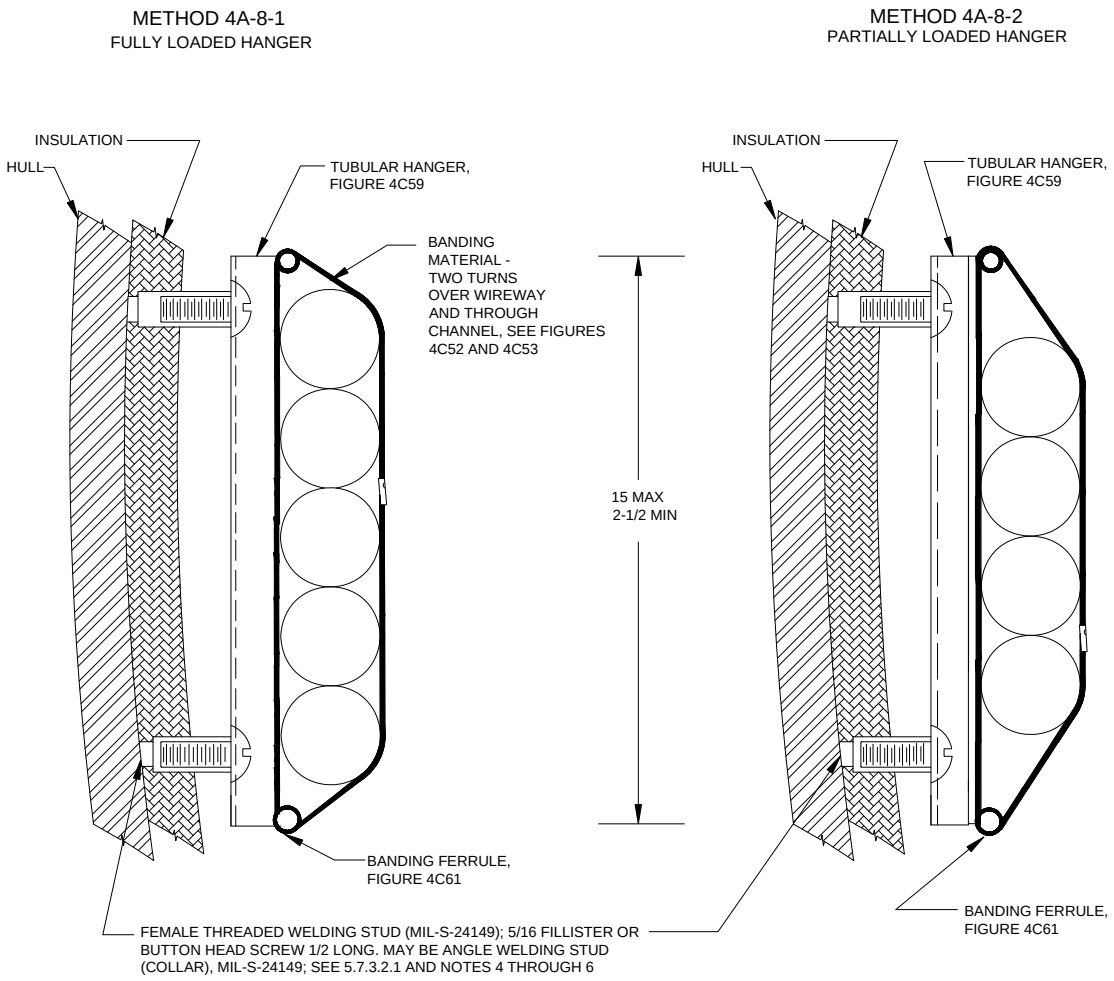
NOTES (continued):

6. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A7. Cableway for auxiliary cables (submarines) - Continued.

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NOTES:

1. Methods shown on this figure may be used for general installation where applicable.
2. Cables shall be arranged so that the larger cables are in the center of the hanger if possible.
3. See notes 5 and 6 below for table of sizes and stock numbers (NSN) for threaded welding studs.
4. For the installation of auxiliary cables (figures 4A6, 4A7, and 4A8), the cables shall be arranged so that the larger cables are in the center of the hanger if possible.
5. National stock numbers for 5/16-17 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

FIGURE 4A8. Cableway for auxiliary cables (submarines only).

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NOTES (continued):

6. National stock numbers for 5/16-18 internal thread welding studs are as follows:

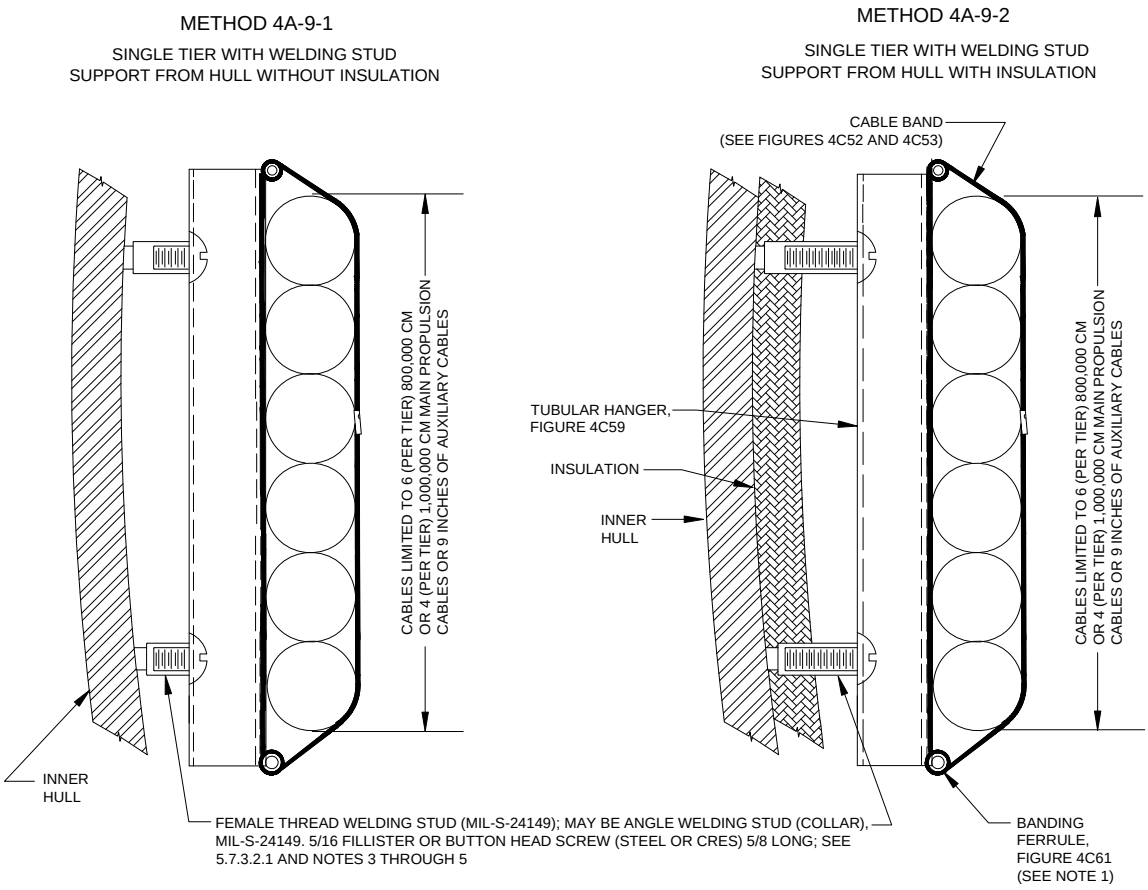
Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

7. National stock numbers for 3/8-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A8. Cableway for auxiliary cables (submarines only) - Continued.

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NOTES:

1. Ferrules (figure 4C61) are required for 7" through 9" hanger lengths where cable diameters are over 2".
2. See notes 3 and 4 below for table of sizes and stock numbers (NSN) for threaded welding studs.
3. National stock numbers for $\frac{5}{16}$-17 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

FIGURE 4A9. Single row cableway (submarines).

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NOTES (continued):

4. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

5. National stock numbers for 3/8-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A9. Single row cableway (submarines) - Continued.

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METHOD 4A-10-1
CABLEWAY THROUGH COAMING AND HATCHWAYS
FOR VERTICAL CABLE RUNS ONLY

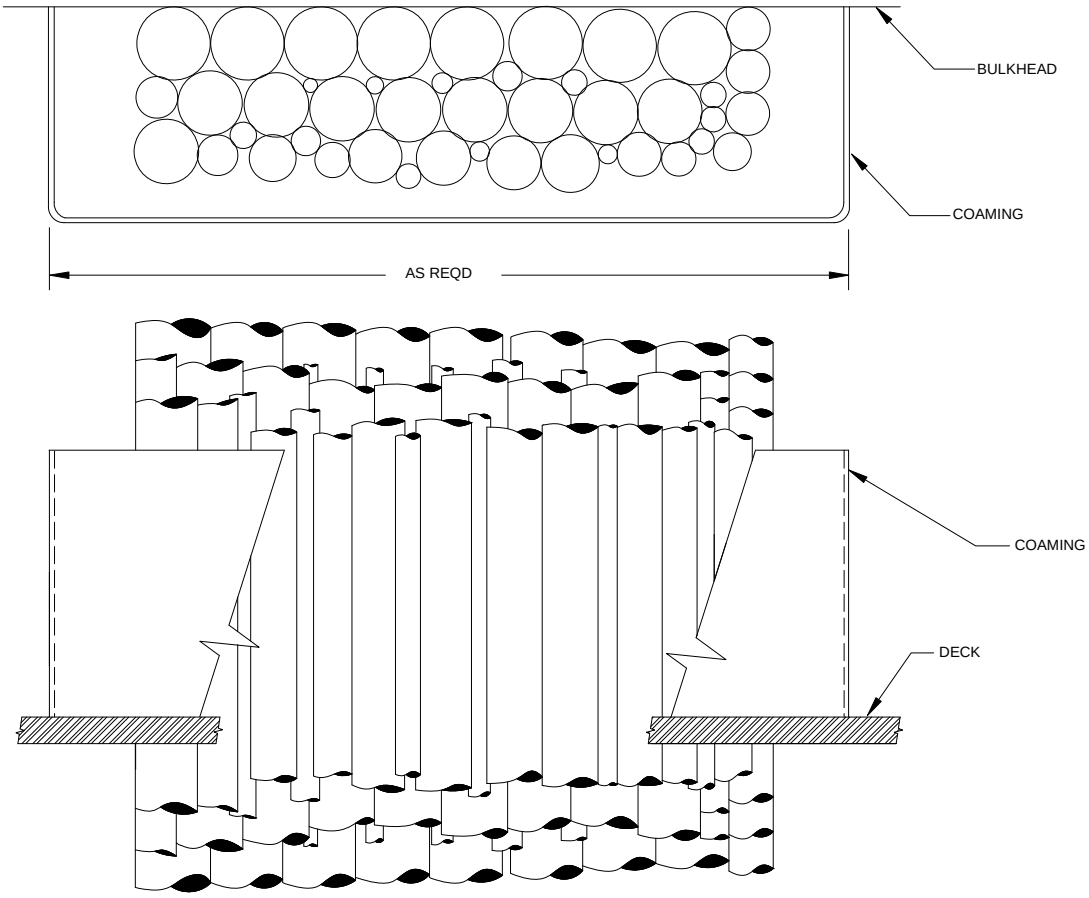
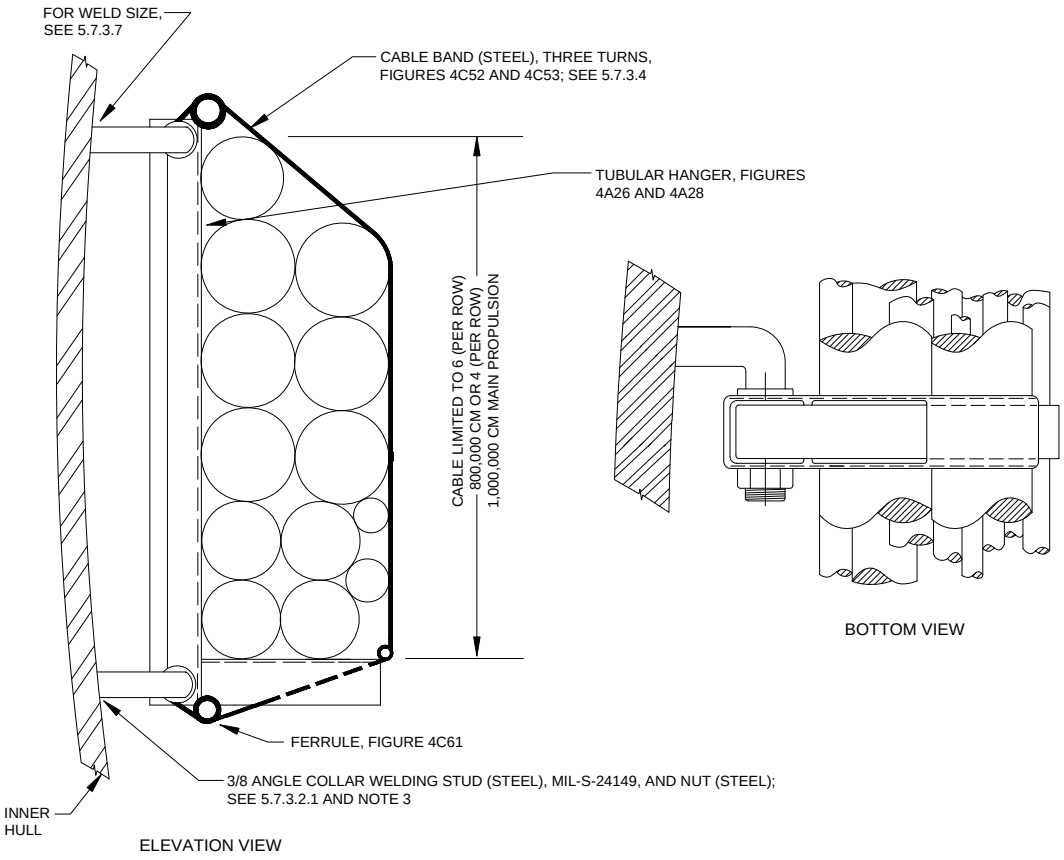


FIGURE 4A10. Cableways through coamings (submarines).

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METHOD 4A-11-1

CABLE BANKS SUPPORTED BY WELDING STUDS
ON INNER HULL WITHOUT INSULATION



NOTES:

1. Hangers with 16" center-to-center spacing is the maximum size that can withstand HI shock. Where number of cables exceeds the maximum size of support shown, additional wireways shall be provided.
2. See note 3 below for table of sizes and stock numbers (NSN) for angle collar studs.

FIGURE 4A11. Cableway for double bank of main propulsion cables (submarines).

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NOTES (continued):

3. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

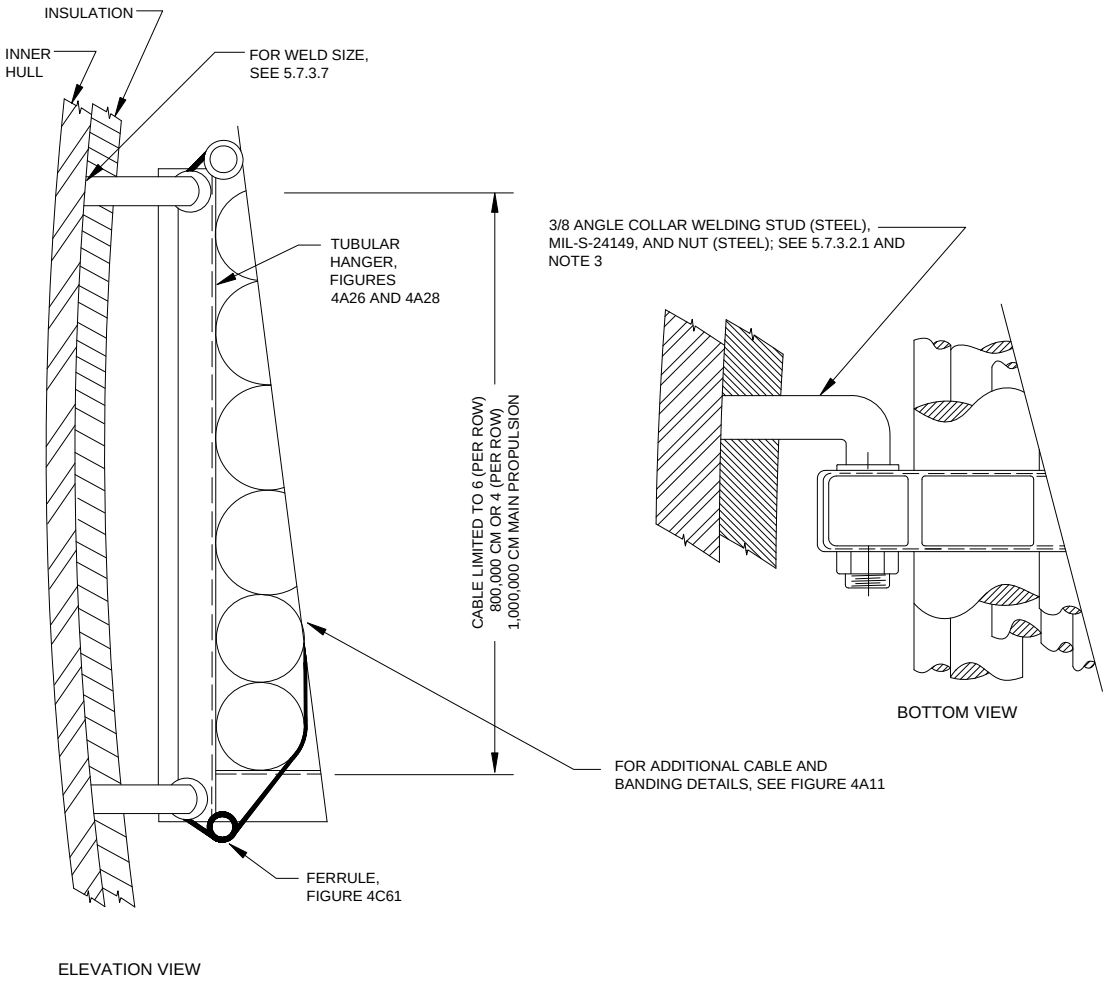
Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
1.875			5307-01-294-0268
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A11. Cableway for double bank of main propulsion cables (submarines) - Continued.

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METHOD 4A-12-1

CABLE BANKS SUPPORTED BY WELDING STUDS ON INNER HULL WITH INSULATION



NOTES:

1. Hangers with 16" center-to-center spacing is the maximum size that can withstand HI shock. Where number of cables exceeds the maximum size of support shown, additional wireways shall be provided.
2. See note 3 below for table of sizes and stock numbers (NSN) for angle collar studs.

FIGURE 4A12. Cableway for double bank of main propulsion cables (submarines).

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NOTES (continued):

3. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

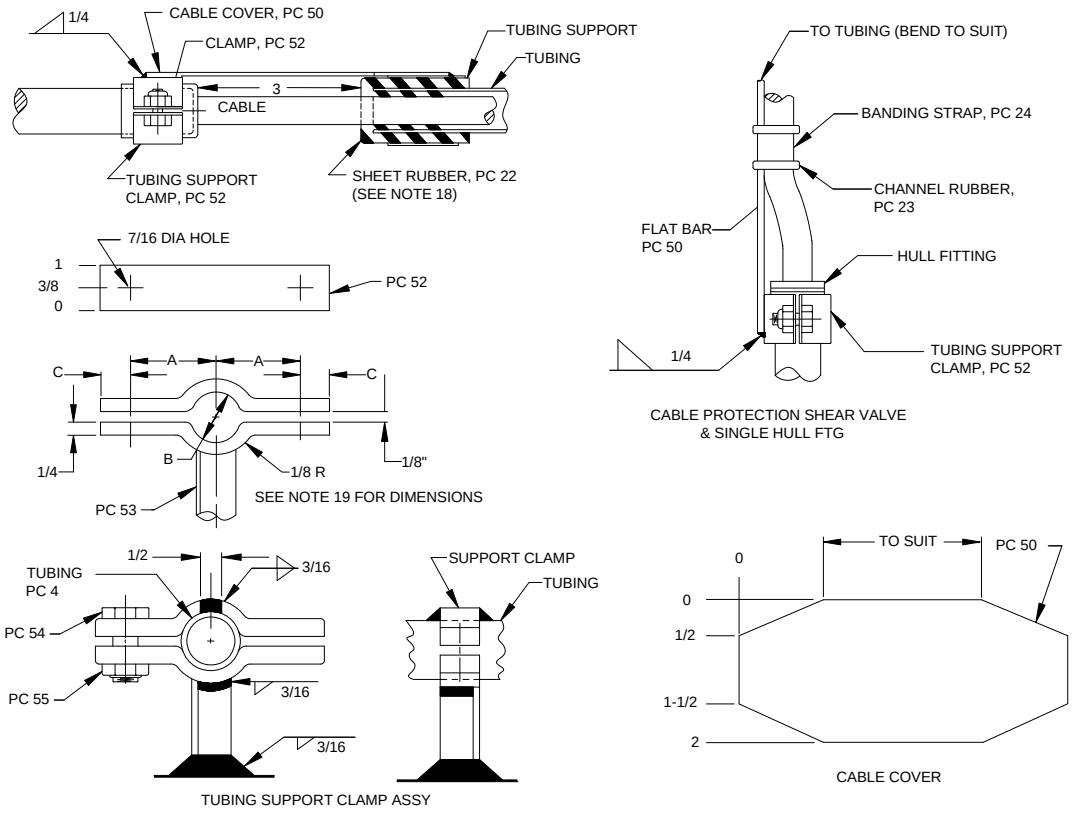
Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4A12. Cableway for double bank of main propulsion cables (submarines) - Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS



NOTES:

1. Cables shall be completely protected for their entire length by the methods shown.
2. Cable pans (PCs 1 through 3), angle bar (PCs 20 and 21), and tubing (PC 4) shall be mounted from the hull or frame at each frame space or three-foot maximum spacing and no more than six feet from the ends. Cables in pans shall be banded through each slot provided and banding shall be spaced not more than 16" apart. Cable angle bars shall be banded every 12". Cables shall not sag. Covers shall be double banded every 16". All welding shall be on frame lines only. Banding of cables on angle bar wireways and cable pans shall be one turn minimum.
3. Channel rubber (PC 23) shall be used in all cases between cables and banding straps in a manner to prevent any possible contact between the banding straps and the cables.
4. All welding shall comply with NAVSEA S9074-AR-GIB-010/278.
5. Enough cable slack shall be allowed for one complete turn around the hull fitting cable rack assembly. Where cable is shortened due to initial molding error, the cable shall be installed in rack as shown in section "A-A" of figure 4A23.
6. Cable pan covers shall be installed at all times except during installation of cables in the pans. Cable pans shall be clean of all foreign matter prior to installation of cables.

FIGURE 4A13. External wireways installation (submarines).

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NOTES (continued):

7. Radius of tubing bends shall be a minimum of ten (10) times the diameter of the cable installed where practicable.
8. Tubing (PC 4) shall not be used except in special installations where specified.
9. All sharp edges shall be rounded, and all burrs shall be removed.
10. Tubing support clamps shall be assembled without rubber lining. Weld as shown for tubing support clamp assembly on figure 4A13.
11. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
12. All cable pan and angle bar runs shall be continuous and shall be broken only as indicated on ships installation plan.
13. The cable pan, angle bars, mounting brackets, and tubing shall be painted in accordance with the material requirements of the submarine painting schedule.
14. All cable slack shall be taken up on the cable rack around the hull fitting. There shall be no slack in the cable pans. See section "A-A", figure 4A23. Radii of the cable bends shall be ten (10) times the diameter of the cable where possible.
15. Banding straps shall be installed so as not to deform the cable sheath but shall be tightened sufficiently to preclude loosening of cable under environmental conditions.
16. Mounting brackets (PC 46), Type A, shall be staggered from each other.
17. Mounting brackets (PC 45), Type A, shall be staggered from each other.
18. Rubber shall be inserted into tubing (2" minimum) and folded back 2" over tubing. Tubing shall not be flared. Remove all sharp edges. The rubber shall be held in place with tubing support clamp. A piece of 1/4" by 2" flat-bar shall be welded to the top half of the clamp. The thickness of the rubber and size of the clamp depends on the size of tubing and cables in it. There shall be no mounting leg on clamps.
19. Dimensions for tubing support clamp assemblies are as follows:

Type	Tubing (O.D.)	Dimensions		
		A	B	C
A	0.840	1 3/16	23/32	7/16
B	1.050	1 7/32	1 1/16	11/32
C	1.315	1 9/32	1 3/16	11/32
D	1.660	1 17/32	1 9/16	11/32
E	1.900	1 31/32	1 13/16	11/32
F	2.375	2 1/8	2 13/32	1/2
G	2.875	2 5/16	2 7/8	1/2
H	3.500	2 5/8	3 1/2	1/2
J	4.500	3 1/8	4 1/2	1/2
K	5.563	3 5/8	5 1/2	1/2

20. See List of Material below.

FIGURE 4A13. External wireways installation (submarines) - Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
1		Pan, 12⅛ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
2		Pan, 8⅞ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
3		Pan, 6⅞ Wide, 15 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
4		Tubing, see notes 8 and 19	Cu-Ni	MIL-T-16420
5		Cover, 7⅞ Wide, 16 GA, use with PC 1	CRES, CL FS-304	ASTM A240/A240M
6		Cover, 5⅝ Wide, 16 GA, use with PCs 2 and 3	CRES, CL FS-304	ASTM A240/A240M
7		Pan, Offset, Left-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
8		Pan, Offset, Right-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
9		Pan, Offset, Left-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
10		Pan, Offset, Right-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
11		Cover, Offset, Left-Hand, 13⅝ x 12, 16 Gauge, use with PC 9 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
12		Cover, Offset, Right-Hand, 13⅝ x 12, 16 Gauge, use with PC 10 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
13		Cover, Offset, Left-Hand, 13⅝ x 12, 16 Gauge, use with PC 11 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
14		Cover, Offset, Right-Hand, 13⅝ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
15		Dropout, 3 x 6⅛, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
16		Dropout, 3 x 4⅝, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
17		Dropout, 3 x 5½, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
18		Dropout, 3 x 3¾, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
19		Channel Rubber, see figure 4A20, notes 1-4 for bonding information	Rubber	MIL-PRF-6855
20		Angle Bar, 1½ x 1½ x 3⁄16	CRES – CL FS-304	SAE-AMS-QQ-S-763
21		Angle Bar, 2 x 2 x 3⁄16	CRES – CL FS-304	SAE-AMS-QQ-S-763
22		Rubber, Synthetic, ⅛ Thick	Class 2, Grade 60	MIL-PRF-6855
23		Channel Rubber, ¾ Wide x ½ Channel x 5⁄16 High	Class 2, Grade 60	MIL-PRF-6855
24		Banding Strip, ½ Wide x 0.020 Thick	CRES – CL FS-304	
25		Buckle Banding, Use With PC 24	CRES	ASTM A240/A240M
26		Tubing, 0.250 x 0.049 Wall	CRES	MIL-P-24691/3
27		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A13. External wireways installation (submarines) - Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
28		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
29		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
30		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
31		Elbow – 90° Inside (8⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
32		Elbow – 90° Inside (6⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

Dimensions for PCs 1, 2, and 3.

Pan					Cover			
PC no.	A	B	C	Max. number of SS cables	PC no.	D	E	Remarks
1	6⅛	3	2⅝	50	5	6 7/16	½	U/W PC 1
2	4⅝	2⅛	1⅝	30	6	4 15/16	½	U/W PC 2 & 3
3	4⅝	1⅛	¾	16				

Dimensions for PCs 7 through 14.

Offset						Offset - cover				
PC no.		A	B	C	D	PC no.		E	F	G
LH	RH					LH	RH			
7	8	6⅛	3	12	13⅜	11	12	6 7/16	12 5/16	13⅜
9	10	4⅝	2⅛	12	14⅛	13	14	4-15/16	12-5/16	14⅛

NOTE:
1. All dimensions are in inches.

Dimensions for PCs 15 through 18.

Dropout			
PC no.	Width	Use with	Remarks
15	6⅛	PC 1	
16	4⅝	PC 2	
17	5½	PC 7 & 8	2 Tk Ftgs
18	3¾	PC 7 & 8	3 Tk Ftgs

FIGURE 4A13. External wireways installation (submarines) - Continued.

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Dimensions for PCs 27 through 41.

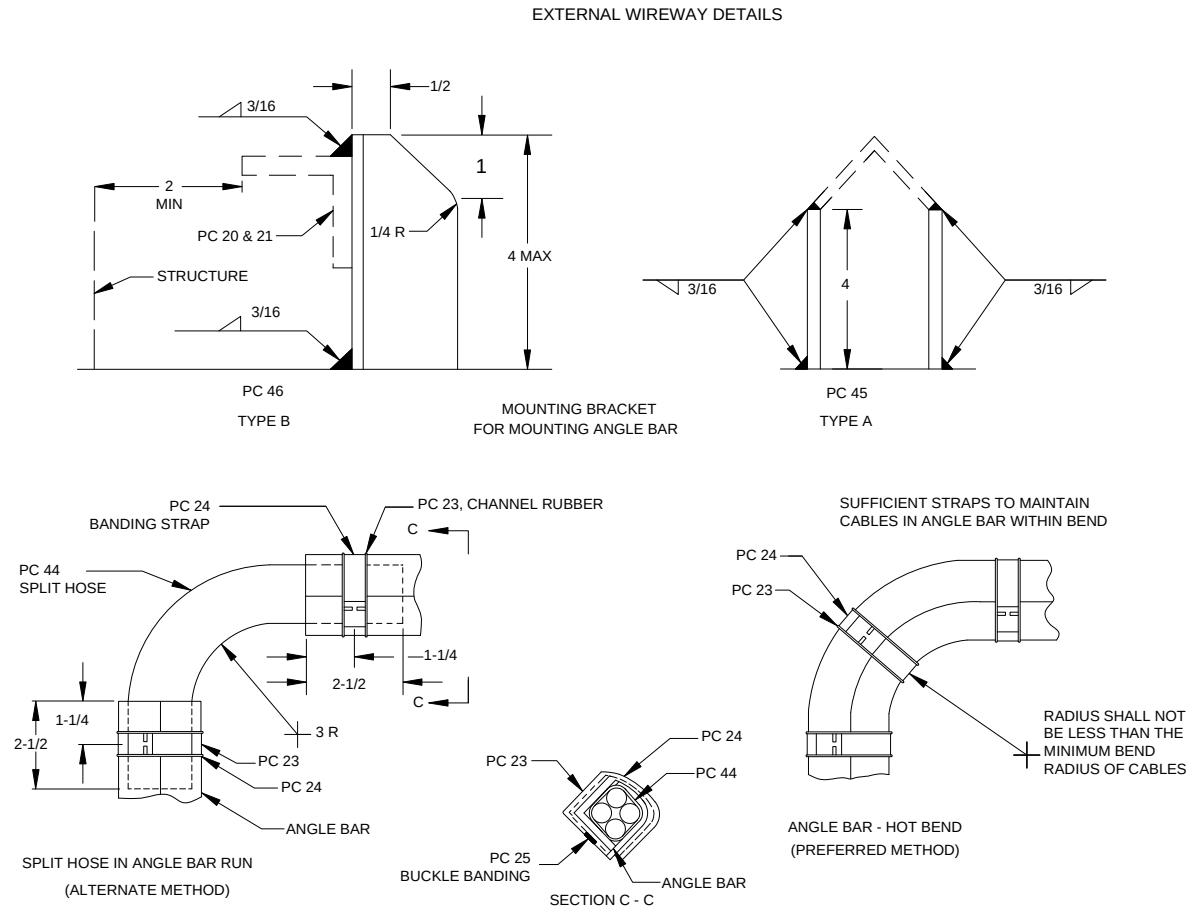
Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC no.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

FIGURE 4A13. External wireways installation (submarines) - Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.



NOTES:

1. Cables shall be completely protected for their entire length by the methods shown.
2. Cable pans (PCs 1 through 3), angle bar (PCs 20 and 21), and tubing (PC 4) shall be mounted from the hull or frame at each frame space or three-foot maximum spacing and no more than six feet from the ends. Cables in pans shall be banded through each slot provided and banding shall be spaced not more than 16" apart. Cable angle bars shall be banded every 12". Cables shall not sag. Covers shall be double banded every 16". All welding shall be on frame lines only. Banding of cables on angle bar wireways and cable pans shall be one turn minimum.
3. Channel rubber (PC 23) shall be used in all cases between cables and banding straps in a manner to prevent any possible contact between the banding straps and the cables.
4. All welding shall comply with NAVSEA S9074-AR-GIB-010/278.
5. Enough cable slack shall be allowed for one complete turn around the hull fitting cable rack assembly. Where cable is shortened due to initial molding error, the cable shall be installed in rack as shown in section “A-A” of figure 4A23.
6. Cable pan covers shall be installed at all times except during installation of cables in the pans. Cable pans shall be clean of all foreign matter prior to installation of cables.

FIGURE 4A14. External wireways installation (submarines).

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NOTES (continued):

7. Radius of tubing bends shall be a minimum of ten (10) times the diameter of the cable installed where practicable.
8. Tubing (PC 4) shall not be used except in special installations where specified.
9. All sharp edges shall be rounded, and all burrs shall be removed.
10. Tubing support clamps shall be assembled without rubber lining. Weld as shown for tubing support clamp assembly on figure 4A13.
11. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
12. All cable pan and angle bar runs shall be continuous and shall be broken only as indicated on ships installation plan.
13. The cable pan, angle bars, mounting brackets, and tubing shall be painted in accordance with the material requirements of the submarine painting schedule.
14. All cable slack shall be taken up on the cable rack around the hull fitting. There shall be no slack in the cable pans. See section "A-A", figure 4A23. Radii of the cable bends shall be ten (10) times the diameter of the cable where possible.
15. Banding straps shall be installed so as not to deform the cable sheath but shall be tightened sufficiently to preclude loosening of cable under environmental conditions.
16. Mounting brackets (PC 46), Type A, shall be staggered from each other.
17. Mounting brackets (PC 45), Type A, shall be staggered from each other.
18. Rubber shall be inserted into tubing (2" minimum) and folded back 2" over tubing. Tubing shall not be flared. Remove all sharp edges. The rubber shall be held in place with tubing support clamp. A piece of ¼" by 2" flat-bar shall be welded to the top half of the clamp. The thickness of the rubber and size of the clamp depends on the size of tubing and cables in it. There shall be no mounting leg on clamps.
19. Dimensions for tubing support clamp assemblies are as follows:

Type	Tubing (O.D.)	Dimensions		
		A	B	C
A	0.840	1 3/16	23/32	7/16
B	1.050	1 7/32	1 1/16	11/32
C	1.315	1 9/32	1 3/16	11/32
D	1.660	1 17/32	1 9/16	11/32
E	1.900	1 31/32	1 13/16	11/32
F	2.375	2 1/8	2 13/32	½
G	2.875	2 5/16	2 7/8	½
H	3.500	2 5/8	3 ½	½
J	4.500	3 1/8	4 ½	½
K	5.563	3 5/8	5 ½	½

20. See List of Material below.

FIGURE 4A14. External wireways installation (submarines) - Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
1		Pan, 12⅛ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
2		Pan, 8⅞ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
3		Pan, 6⅞ Wide, 15 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
4		Tubing, see notes 8 and 19	Cu-Ni	MIL-T-16420
5		Cover, 7 7/16 Wide, 16 GA, use with PC 1	CRES, CL FS-304	ASTM A240/A240M
6		Cover, 5 5/16 Wide, 16 GA, use with PCs 2 and 3	CRES, CL FS-304	ASTM A240/A240M
7		Pan, Offset, Left-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
8		Pan, Offset, Right-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
9		Pan, Offset, Left-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
10		Pan, Offset, Right-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
11		Cover, Offset, Left-Hand, 13 5/16 x 12, 16 Gauge, use with PC 9 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
12		Cover, Offset, Right-Hand, 13 5/16 x 12, 16 Gauge, use with PC 10 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
13		Cover, Offset, Left-Hand, 13 5/16 x 12, 16 Gauge, use with PC 11 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
14		Cover, Offset, Right-Hand, 13 5/16 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
15		Dropout, 3 x 6⅛, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
16		Dropout, 3 x 4⅝, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
17		Dropout, 3 x 5½, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
18		Dropout, 3 x 3¾, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
19		Channel Rubber, see figure 4A20, notes 1-4 for bonding information	Rubber	MIL-PRF-6855
20		Angle Bar, 1½ x 1½ x 3/16	CRES – CL FS-304	SAE-AMS-QQ-S-763
21		Angle Bar, 2 x 2 x 3/16	CRES – CL FS-304	SAE-AMS-QQ-S-763
22		Rubber, Synthetic, ⅛ Thick	Class 2, Grade 60	MIL-PRF-6855
23		Channel Rubber, ¾ Wide x ½ Channel x 5/16 High	Class 2, Grade 60	MIL-PRF-6855
24		Banding Strip, ½ Wide x 0.020 Thick	CRES – CL FS-304	
25		Buckle Banding, Use With PC 24	CRES	ASTM A240/A240M
26		Tubing, 0.250 x 0.049 Wall	CRES	MIL-P-24691/3
27		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A14. External wireways installation (submarines) - Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
28		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
29		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
30		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
31		Elbow – 90° Inside (8⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
32		Elbow – 90° Inside (6⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

Dimensions for PCs 1, 2, and 3.

Pan					Cover			
PC no.	A	B	C	Max. number of SS cables	PC no.	D	E	Remarks
1	6⅛	3	2⅝	50	5	6 7/16	½	U/W PC 1
2	4⅝	2⅛	1⅝	30	6	4 15/16	½	U/W PC 2 & 3
3	4⅝	1⅛	¾	16				

Dimensions for PCs 7 through 14.

Offset						Offset - cover				
PC no.		A	B	C	D	PC no.		E	F	G
LH	RH					LH	RH			
7	8	6⅛	3	12	13⅜	11	12	6 7/16	12 5/16	13⅜
9	10	4⅝	2⅛	12	14⅛	13	14	4-15/16	12-5/16	14⅛

NOTE:
1. All dimensions are in inches.

Dimensions for PCs 15 through 18.

Dropout			
PC no.	Width	Use with	Remarks
15	6⅛	PC 1	
16	4⅝	PC 2	
17	5½	PC 7 & 8	2 Tk Ftgs
18	3¾	PC 7 & 8	3 Tk Ftgs

FIGURE 4A14. External wireways installation (submarines) - Continued.

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Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC No.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

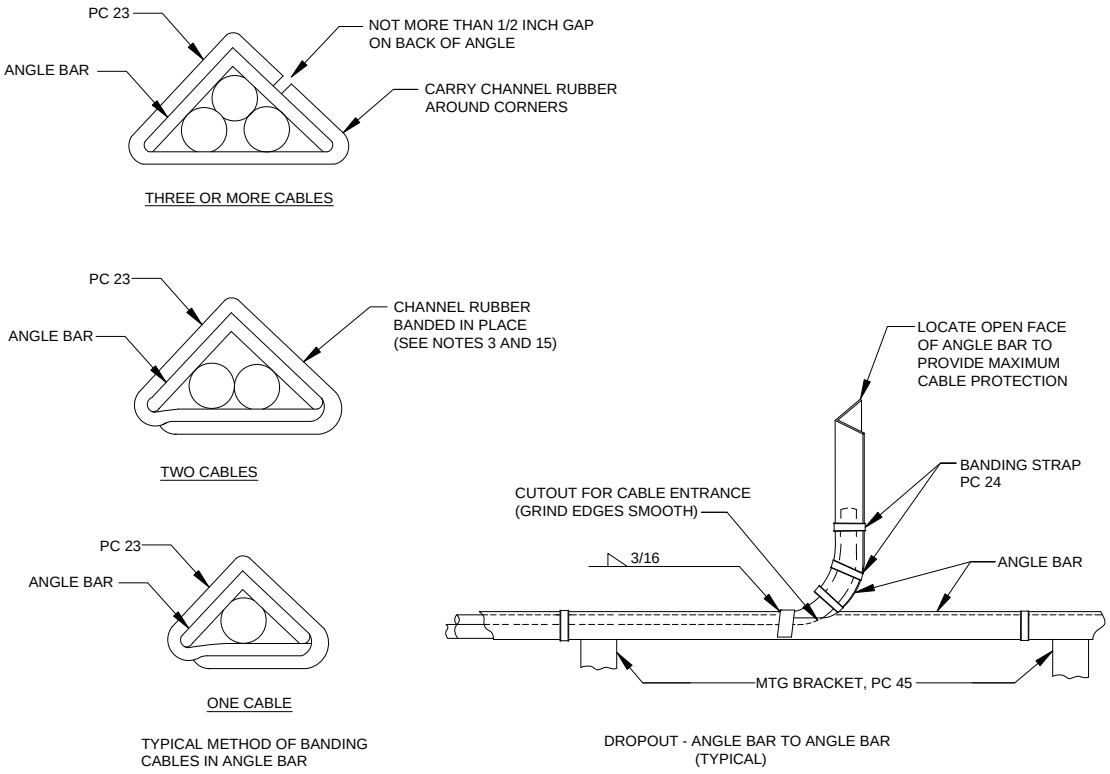
NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

FIGURE 4A14. External wireways installation (submarines) - Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS



NOTES:

1. Cables shall be completely protected for their entire length by the methods shown.
2. Cable pans (PCs 1 through 3), angle bar (PCs 20 and 21), and tubing (PC 4) shall be mounted from the hull or frame at each frame space or three-foot maximum spacing and no more than six feet from the ends. Cables in pans shall be banded through each slot provided and banding shall be spaced not more than 16" apart. Cable angle bars shall be banded every 12". Cables shall not sag. Covers shall be double banded every 16". All welding shall be on frame lines only. Banding of cables on angle bar wireways and cable pans shall be one turn minimum.
3. Channel rubber (PC 23) shall be used in all cases between cables and banding straps in a manner to prevent any possible contact between the banding straps and the cables.
4. All welding shall comply with NAVSEA S9074-AR-GIB-010/278.
5. Enough cable slack shall be allowed for one complete turn around the hull fitting cable rack assembly. Where cable is shortened due to initial molding error, the cable shall be installed in rack as shown in section “A-A” of figure 4A23.
6. Cable pan covers shall be installed at all times except during installation of cables in the pans. Cable pans shall be clean of all foreign matter prior to installation of cables.

FIGURE 4A15. External wireways installation (submarines).

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NOTES (continued):

7. Radius of tubing bends shall be a minimum of ten (10) times the diameter of the cable installed where practicable.
8. Tubing (PC 4) shall not be used except in special installations where specified.
9. All sharp edges shall be rounded, and all burrs shall be removed.
10. Tubing support clamps shall be assembled without rubber lining. Weld as shown for tubing support clamp assembly on figure 4A13.
11. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
12. All cable pan and angle bar runs shall be continuous and shall be broken only as indicated on ships installation plan.
13. The cable pan, angle bars, mounting brackets, and tubing shall be painted in accordance with the material requirements of the submarine painting schedule.
14. All cable slack shall be taken up on the cable rack around the hull fitting. There shall be no slack in the cable pans. See section "A-A", figure 4A23. Radii of the cable bends shall be ten (10) times the diameter of the cable where possible.
15. Banding straps shall be installed so as not to deform the cable sheath but shall be tightened sufficiently to preclude loosening of cable under environmental conditions.
16. Mounting brackets (PC 46), Type A, shall be staggered from each other.
17. Mounting brackets (PC 45), Type A, shall be staggered from each other.
18. Rubber shall be inserted into tubing (2" minimum) and folded back 2" over tubing. Tubing shall not be flared. Remove all sharp edges. The rubber shall be held in place with tubing support clamp. A piece of 1/4" by 2" flat-bar shall be welded to the top half of the clamp. The thickness of the rubber and size of the clamp depends on the size of tubing and cables in it. There shall be no mounting leg on clamps.
19. Dimensions for tubing support clamp assemblies are as follows:

Type	Tubing (O.D.)	Dimensions		
		A	B	C
A	0.840	1 3/16	23/32	7/16
B	1.050	1 7/32	1 1/16	11/32
C	1.315	1 9/32	1 3/16	11/32
D	1.660	1 17/32	1 9/16	11/32
E	1.900	1 31/32	1 13/16	11/32
F	2.375	2 1/8	2 13/32	1/2
G	2.875	2 5/16	2 7/8	1/2
H	3.500	2 5/8	3 1/2	1/2
J	4.500	3 1/8	4 1/2	1/2
K	5.563	3 5/8	5 1/2	1/2

20. See List of Material below.

FIGURE 4A15. External wireways installation (submarines) - Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
1		Pan, 12⅛ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
2		Pan, 8⅞ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
3		Pan, 6⅞ Wide, 15 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
4		Tubing, see notes 8 and 19	Cu-Ni	MIL-T-16420
5		Cover, 7⅞ Wide, 16 GA, use with PC 1	CRES, CL FS-304	ASTM A240/A240M
6		Cover, 5⅝ Wide, 16 GA, use with PCs 2 and 3	CRES, CL FS-304	ASTM A240/A240M
7		Pan, Offset, Left-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
8		Pan, Offset, Right-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
9		Pan, Offset, Left-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
10		Pan, Offset, Right-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
11		Cover, Offset, Left-Hand, 13⅝ x 12, 16 Gauge, use with PC 9 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
12		Cover, Offset, Right-Hand, 13⅝ x 12, 16 Gauge, use with PC 10 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
13		Cover, Offset, Left-Hand, 13⅝ x 12, 16 Gauge, use with PC 11 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
14		Cover, Offset, Right-Hand, 13⅝ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
15		Dropout, 3 x 6⅛, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
16		Dropout, 3 x 4⅝, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
17		Dropout, 3 x 5½, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
18		Dropout, 3 x 3¾, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
19		Channel Rubber, see figure 4A20, notes 1-4 for bonding information	Rubber	MIL-PRF-6855
20		Angle Bar, 1½ x 1½ x 3⁄16	CRES – CL FS-304	SAE-AMS-QQ-S-763
21		Angle Bar, 2 x 2 x 3⁄16	CRES – CL FS-304	SAE-AMS-QQ-S-763
22		Rubber, Synthetic, ⅛ Thick	Class 2, Grade 60	MIL-PRF-6855
23		Channel Rubber, ¾ Wide x ½ Channel x 5⁄16 High	Class 2, Grade 60	MIL-PRF-6855
24		Banding Strip, ½ Wide x 0.020 Thick	CRES – CL FS-304	
25		Buckle Banding, Use With PC 24	CRES	ASTM A240/A240M
26		Tubing, 0.250 x 0.049 Wall	CRES	MIL-P-24691/3
27		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A15. External wireways installation (submarines) - Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
28		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
29		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
30		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
31		Elbow – 90° Inside (8⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
32		Elbow – 90° Inside (6⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

Dimensions for PCs 1, 2, and 3.

Pan					Cover			
PC no.	A	B	C	Max. number of SS cables	PC no.	D	E	Remarks
1	6⅛	3	2⅝	50	5	6 7/16	½	U/W PC 1
2	4⅝	2⅛	1⅝	30	6	4 15/16	½	U/W PC 2 & 3
3	4⅝	1⅛	¾	16				

Dimensions for PCs 7 through 14.

Offset						Offset - cover				
PC no.		A	B	C	D	PC no.		E	F	G
LH	RH					LH	RH			
7	8	6⅛	3	12	13⅜	11	12	6 7/16	12 5/16	13⅜
9	10	4⅝	2⅛	12	14⅛	13	14	4-15/16	12-5/16	14⅛

NOTE:
1. All dimensions are in inches.

Dimensions for PCs 15 through 18.

Dropout			
PC no.	Width	Use with	Remarks
15	6⅛	PC 1	
16	4⅝	PC 2	
17	5½	PC 7 & 8	2 Tk Ftgs
18	3¾	PC 7 & 8	3 Tk Ftgs

FIGURE 4A15. External wireways installation (submarines) - Continued.

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Dimensions for PCs 27 through 41.

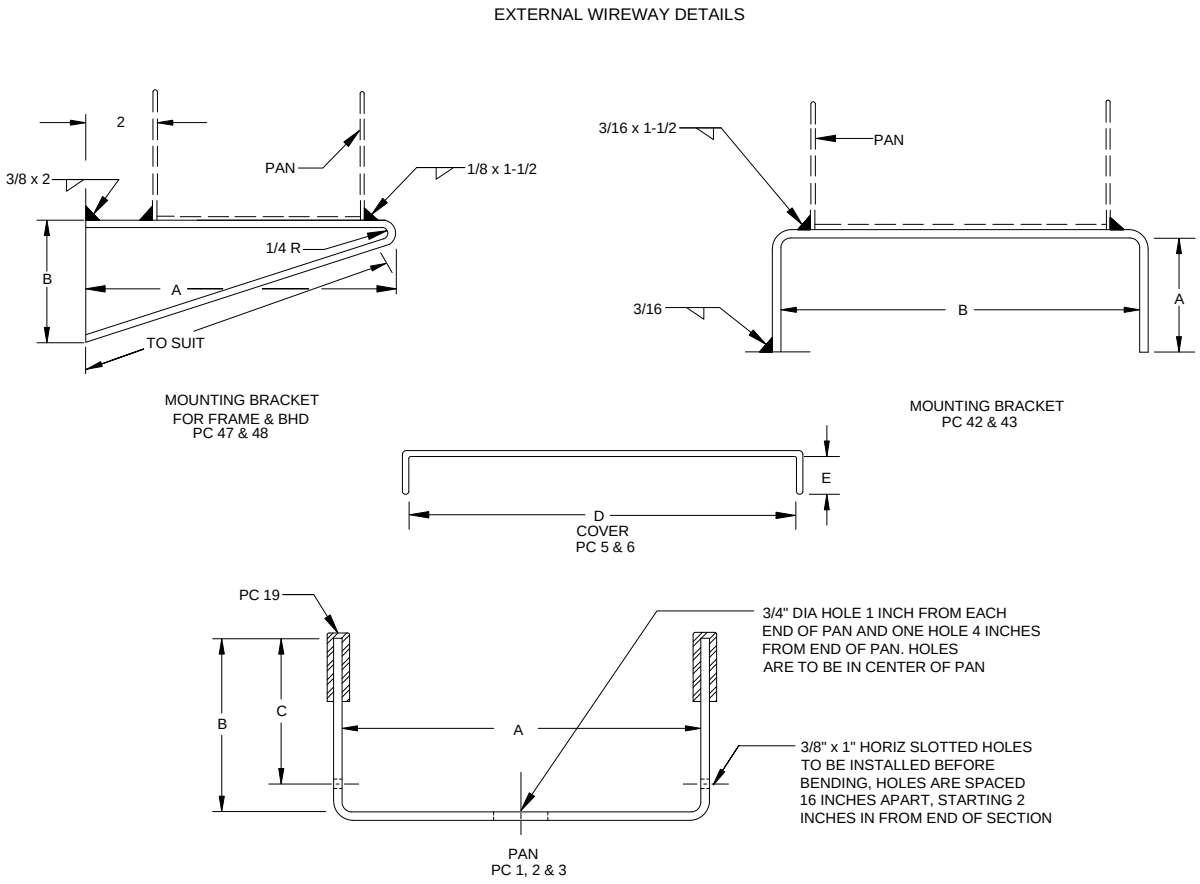
Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC no.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

FIGURE 4A15. External wireways installation (submarines) - Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.



NOTES:

1. Cables shall be completely protected for their entire length by the methods shown.
2. Cable pans (PCs 1 through 3), angle bar (PCs 20 and 21), and tubing (PC 4) shall be mounted from the hull or frame at each frame space or three-foot maximum spacing and no more than six feet from the ends. Cables in pans shall be banded through each slot provided and banding shall be spaced not more than 16" apart. Cable angle bars shall be banded every 12". Cables shall not sag. Covers shall be double banded every 16". All welding shall be on frame lines only. Banding of cables on angle bar wireways and cable pans shall be one turn minimum.
3. Channel rubber (PC 23) shall be used in all cases between cables and banding straps in a manner to prevent any possible contact between the banding straps and the cables.
4. All welding shall comply with NAVSEA S9074-AR-GIB-010/278.
5. Enough cable slack shall be allowed for one complete turn around the hull fitting cable rack assembly. Where cable is shortened due to initial molding error, the cable shall be installed in rack as shown in section “A-A” of figure 4A23.
6. Cable pan covers shall be installed at all times except during installation of cables in the pans. Cable pans shall be clean of all foreign matter prior to installation of cables.

FIGURE 4A16. External wireways installation standard methods (submarines).

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NOTES (continued):

7. Radius of tubing bends shall be a minimum of ten (10) times the diameter of the cable installed where practicable.
8. Tubing (PC 4) shall not be used except in special installations where specified.
9. All sharp edges shall be rounded, and all burrs shall be removed.
10. Tubing support clamps shall be assembled without rubber lining. Weld as shown for tubing support clamp assembly on figure 4A13.
11. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
12. All cable pan and angle bar runs shall be continuous and shall be broken only as indicated on ships installation plan.
13. The cable pan, angle bars, mounting brackets, and tubing shall be painted in accordance with the material requirements of the submarine painting schedule.
14. All cable slack shall be taken up on the cable rack around the hull fitting. There shall be no slack in the cable pans. See section "A-A", figure 4A23. Radii of the cable bends shall be ten (10) times the diameter of the cable where possible.
15. Banding straps shall be installed so as not to deform the cable sheath but shall be tightened sufficiently to preclude loosening of cable under environmental conditions.
16. Mounting brackets (PC 46), Type A, shall be staggered from each other.
17. Mounting brackets (PC 45), Type A, shall be staggered from each other.
18. Rubber shall be inserted into tubing (2" minimum) and folded back 2" over tubing. Tubing shall not be flared. Remove all sharp edges. The rubber shall be held in place with tubing support clamp. A piece of ¼ by 2" flat-bar shall be welded to the top half of the clamp. The thickness of the rubber and size of the clamp depends on the size of tubing and cables in it. There shall be no mounting leg on clamps.
19. Dimensions for tubing support clamp assemblies are as follows:

Type	Tubing (O.D.)	Dimensions		
		A	B	C
A	0.840	1 3/16	23/32	7/16
B	1.050	1 7/32	1 1/16	11/32
C	1.315	1 9/32	1 3/16	11/32
D	1.660	1 17/32	1 9/16	11/32
E	1.900	1 31/32	1 13/16	11/32
F	2.375	2 1/8	2 13/32	½
G	2.875	2 5/16	2 7/8	½
H	3.500	2 5/8	3 ½	½
J	4.500	3 1/8	4 ½	½
K	5.563	3 5/8	5 ½	½

20. See List of Material below.

FIGURE 4A16. External wireways installation standard methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
1		Pan, 12⅛ Wide, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
2		Pan, 8⅞ Wide, 16 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
3		Pan, 6⅞ Wide, 15 Gauge, 8 Sect (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
4		Tubing, see notes 8 and 19	Cu-Ni	MIL-T-16420
5		Cover, 7⅞ Wide, 16 GA, use with PC 1	CRES, CL FS-304	ASTM A240/A240M
6		Cover, 5⅝ Wide, 16 GA, use with PCs 2 and 3	CRES, CL FS-304	ASTM A240/A240M
7		Pan, Offset, Left-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
8		Pan, Offset, Right-Hand, 18 x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
9		Pan, Offset, Left-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
10		Pan, Offset, Right-Hand, 16¼ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
11		Cover, Offset, Left-Hand, 13⅝ x 12, 16 Gauge, use with PC 9 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
12		Cover, Offset, Right-Hand, 13⅝ x 12, 16 Gauge, use with PC 10 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
13		Cover, Offset, Left-Hand, 13⅝ x 12, 16 Gauge, use with PC 11 (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
14		Cover, Offset, Right-Hand, 13⅝ x 12, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
15		Dropout, 3 x 6⅛, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
16		Dropout, 3 x 4⅝, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
17		Dropout, 3 x 5½, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
18		Dropout, 3 x 3¾, 16 Gauge (dimensions below)	CRES, CL FS-304	ASTM A240/A240M
19		Channel Rubber, see figure 4A20, notes 1-4 for bonding information	Rubber	MIL-PRF-6855
20		Angle Bar, 1½ x 1½ x 3⁄16	CRES – CL FS-304	SAE-AMS-QQ-S-763
21		Angle Bar, 2 x 2 x 3⁄16	CRES – CL FS-304	SAE-AMS-QQ-S-763
22		Rubber, Synthetic, ⅛ Thick	Class 2, Grade 60	MIL-PRF-6855
23		Channel Rubber, ¾ Wide x ½ Channel x 5⁄16 High	Class 2, Grade 60	MIL-PRF-6855
24		Banding Strip, ½ Wide x 0.020 Thick	CRES – CL FS-304	
25		Buckle Banding, Use With PC 24	CRES	ASTM A240/A240M
26		Tubing, 0.250 x 0.049 Wall	CRES	MIL-P-24691/3
27		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
28		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A16. External wireways installation standard methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
29		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
30		Elbow, 90° Outside, 12⅛ Wide x 18 Long, 16 Gauge (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
31		Elbow – 90° Inside (8⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
32		Elbow – 90° Inside (6⅞ Wide x 18 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

Dimensions for PCs 1, 2, and 3.

Pan					Cover			
PC no.	A	B	C	Max. number of SS cables	PC no.	D	E	Remarks
1	6⅛	3	2⅝	50	5	6 7/16	½	U/W PC 1
2	4⅝	2⅛	1⅝	30	6	4 15/16	½	U/W PC 2 & 3
3	4⅝	1⅛	¾	16				

Dimensions for PCs 7 through 14.

Offset						Offset - cover				
PC no.		A	B	C	D	PC no.		E	F	G
LH	RH					LH	RH			
7	8	6⅛	3	12	13⅜	11	12	6 7/16	12 5/16	13⅜
9	10	4⅝	2⅛	12	14⅛	13	14	4-15/16	12-5/16	14⅛

NOTE:
1. All dimensions are in inches.

Dimensions for PCs 15 through 18.

Dropout			
PC no.	Width	Use with	Remarks
15	6⅛	PC 1	
16	4⅝	PC 2	
17	5½	PC 7 & 8	2 Tk Ftgs
18	3¾	PC 7 & 8	3 Tk Ftgs

FIGURE 4A16. External wireways installation standard methods (submarines) – Continued.

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Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC no.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

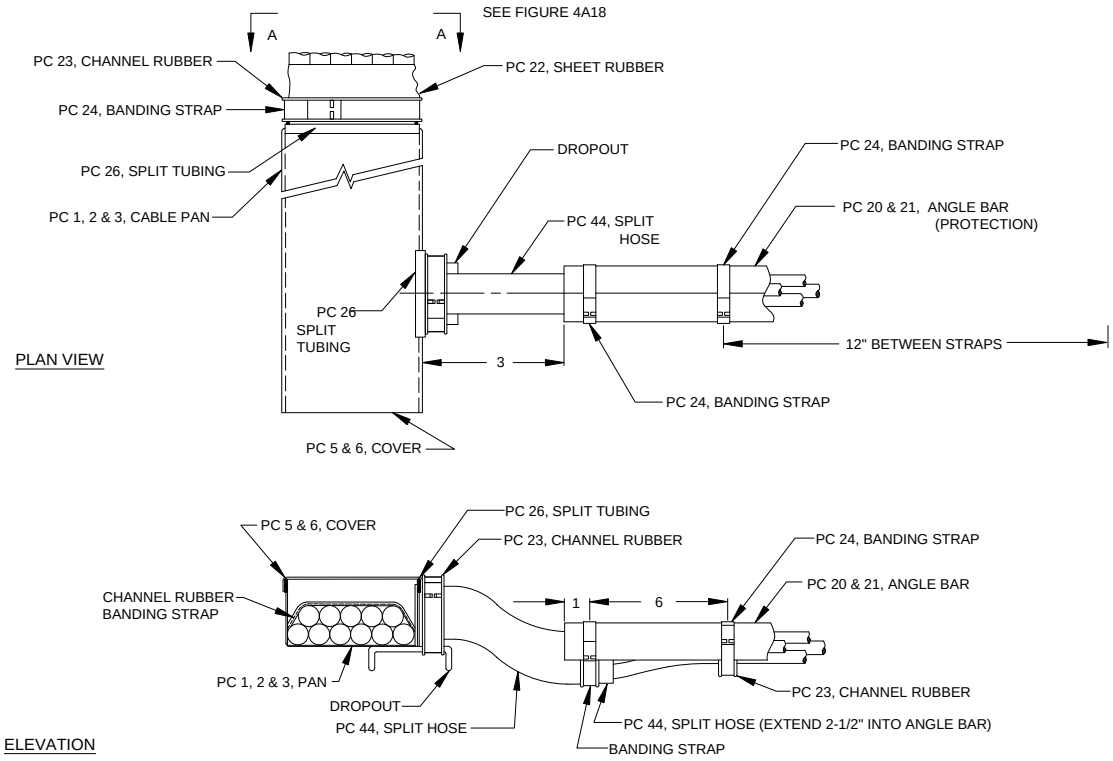
NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

FIGURE 4A16. External wireways installation standard methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with $\frac{3}{16}$ lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside ($7\frac{7}{16}$ Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside ($5\frac{15}{16}$ Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A17. External wireways installation standard methods (submarines).

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
38		Cover, 90° Elbow Inside (7⁷⁄₁₆ Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5⁷⁄₁₆ Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 14¾ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 13¼ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1¼ ID x ³⁄₁₆" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), ¼ x 1½ Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1¼ x 1¼ x ³⁄₁₆ Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x ³⁄₁₆	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, ¼ x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, ¼ x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1¼ x 1¼ x ⅛	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, ⅜-16 x 1¼ Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, ⅜ –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, ¼ x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, ¼ x 1	STEEL, TY A	MIL-S-22698
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

FIGURE 4A17. External wireways installation standard methods (submarines) – Continued.

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Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC no.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

Dimensions for PCs 42, 43, 47, and 48.

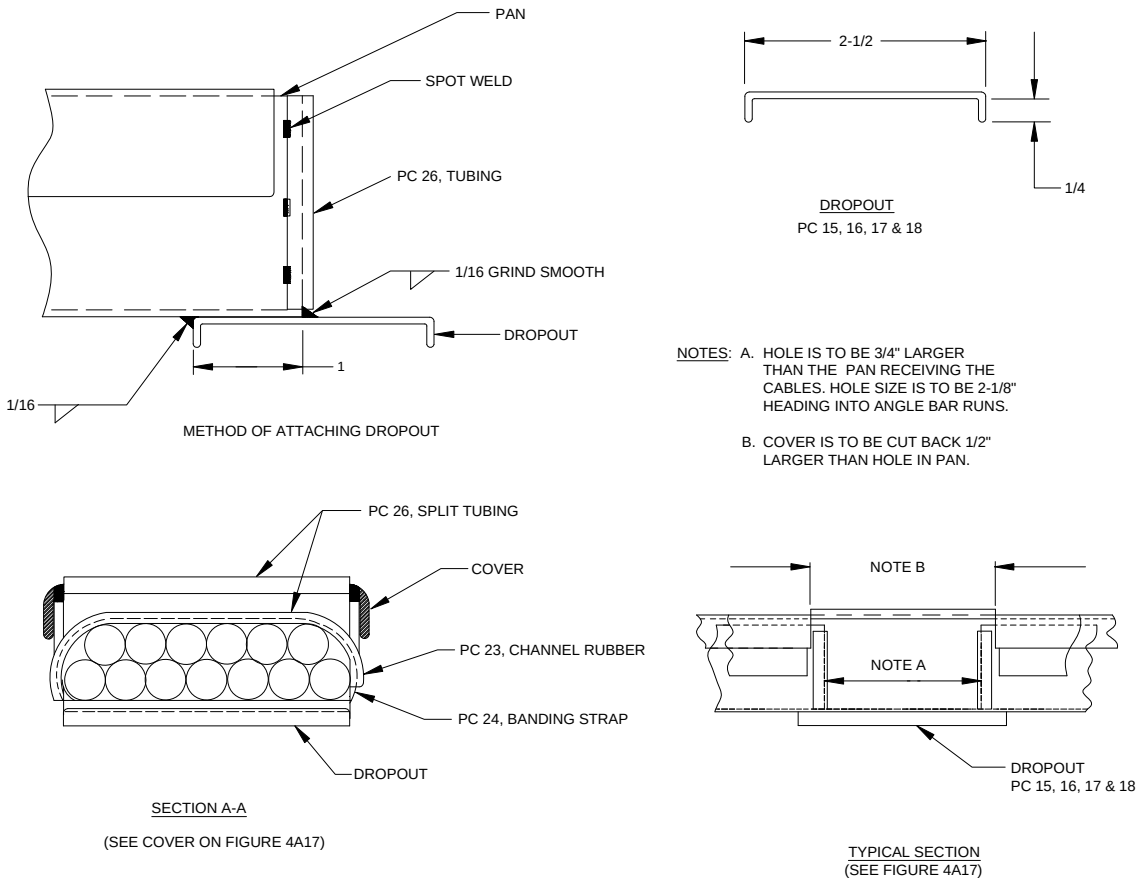
PC no	Pan ID	A	B	Remarks
42	6 1/8	4" max	6 3/4	U/W PC 1
43	4 5/8	4" max	5 1/4	U/W PC 2 & 3
47	6 1/8	9 1/4	6	U/W PC 1
48	4 5/8	7 3/4	6	U/W PC 2 & 3

FIGURE 4A17. External wireways installation standard methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS
SEE A.3.2.7 AND TABLE 4A-II



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A18. External wireways installation standard methods (submarines).

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, 1/4 x 1 1/2 Flat Bar x 14 3/4 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, 1/4 x 1 1/2 Flat Bar x 13 1/4 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1 1/4 ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), 1/4 x 1 1/2 Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1 1/4 x 1 1/4 x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, 1/4 x 1 1/2 Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, 1/4 x 1 1/2 Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, 1/4 x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, 1/4 x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1 1/4 x 1 1/4 x 1/8	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, 3/8-16 x 1 1/4 Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, 3/8 –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, 1/4 x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, 1/4 x 1	STEEL, TY A	MIL-S-22698

FIGURE 4A18. External wireways installation standard methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	A	B	C	F	G	K	PC no.	D	E	H	J	Type
27	6⅛	3				2⅝	36	6 7/16	11 1/16			90° Outside
28	4⅝	2⅛				1⅝	37	4 15/16	10 3/16			90° Outside
29	4⅝	1⅛				¾						
30	6⅛	3	11 1/16			2⅝	38	6 7/16				90° Inside
31	4⅝	2⅛	10 3/16			1⅝	39	4 15/16				90° Inside
32	4⅝	1⅛	9 3/16			¾						
33	6⅛	3		6	14 3/16	2⅝	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4⅝	2⅛		6	12 11/16	1⅝	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4⅝	1⅛		6	12 11/16	¾						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

Dimensions for PCs 42, 43, 47, and 48.

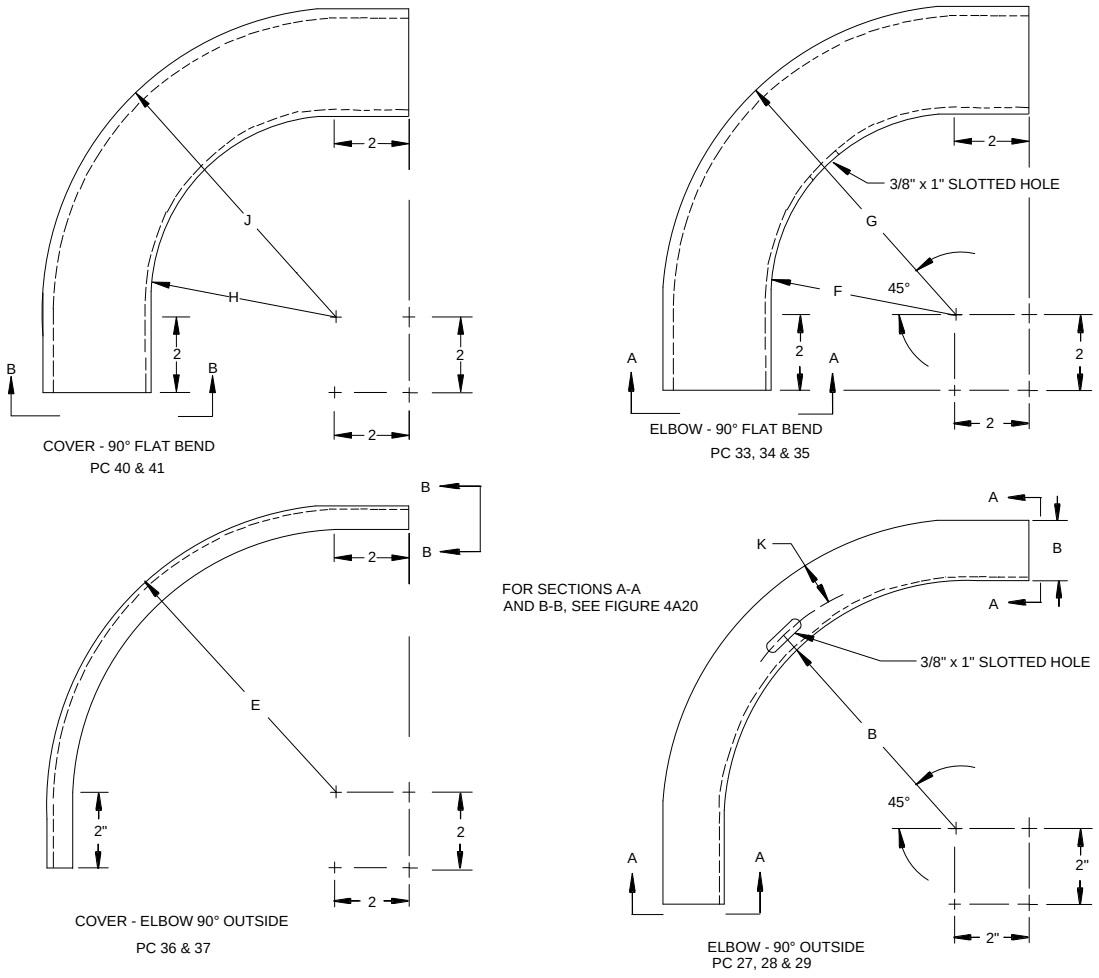
PC no.	Pan ID	A	B	Remarks
42	6⅛	4" max	6¾	U/W PC 1
43	4⅝	4" max	5¼	U/W PC 2 & 3
47	6⅛	9¼	6	U/W PC 1
48	4⅝	7¾	6	U/W PC 2 & 3

FIGURE 4A18. External wireways installation standard methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with $\frac{3}{16}$ lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A19. External wireways installation standard methods (submarines).

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, 1/4 x 1 1/2 Flat Bar x 14 3/4 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, 1/4 x 1 1/2 Flat Bar x 13 1/4 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1 1/4 ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), 1/4 x 1 1/2 Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1 1/4 x 1 1/4 x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, 1/4 x 1 1/2 Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, 1/4 x 1 1/2 Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, 1/4 x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, 1/4 x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1 1/4 x 1 1/4 x 1/8	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, 3/8-16 x 1 1/4 Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, 3/8 –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, 1/4 x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, 1/4 x 1	STEEL, TY A	MIL-S-22698

FIGURE 4A19. External wireways installation standard methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	A	B	C	F	G	K	PC no.	D	E	H	J	Type
27	6⅛	3				2⅝	36	6 7/16	11 1/16			90° Outside
28	4⅝	2⅛				1⅝	37	4 15/16	10 3/16			90° Outside
29	4⅝	1⅛				¾						
30	6⅛	3	11 1/16			2⅝	38	6 7/16				90° Inside
31	4⅝	2⅛	10 3/16			1⅝	39	4 15/16				90° Inside
32	4⅝	1⅛	9 3/16			¾						
33	6⅛	3		6	14 3/16	2⅝	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4⅝	2⅛		6	12 11/16	1⅝	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4⅝	1⅛		6	12 11/16	¾						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

Dimensions for PCs 42, 43, 47, and 48.

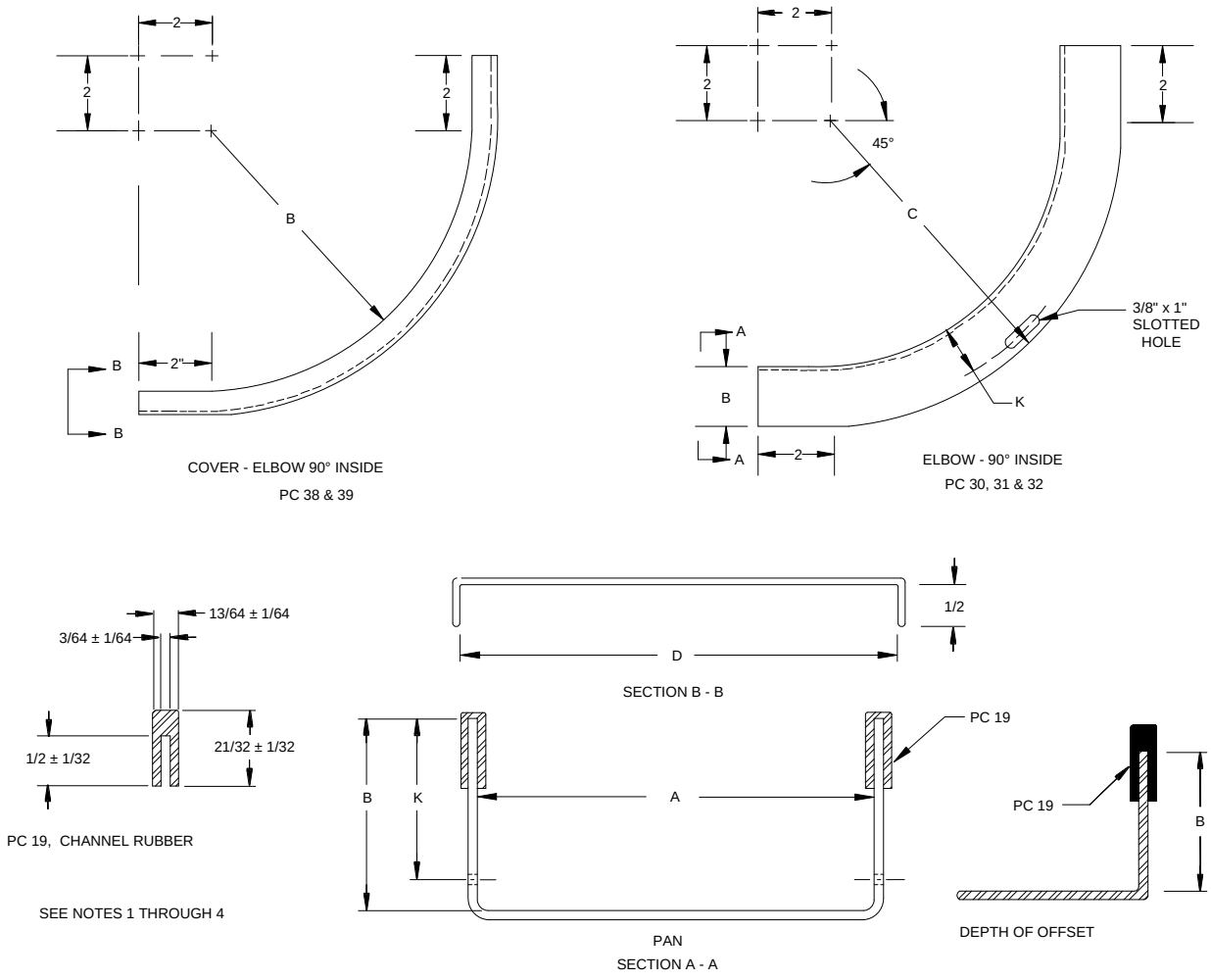
PC no.	Pan ID	A	B	Remarks
42	6⅛	4" max	6¾	U/W PC 1
43	4⅝	4" max	5¼	U/W PC 2 & 3
47	6⅛	9¼	6	U/W PC 1
48	4⅝	7¾	6	U/W PC 2 & 3

FIGURE 4A19. External wireways installation standard methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3⁄16 lettering.
2. See List of Material below.

FIGURE 4A20. External wireways installation standard methods (submarines).

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 14¾ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 13¼ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1¼ ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), ¼ x 1½ Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1¼ x 1¼ x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, ¼ x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, ¼ x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1¼ x 1¼ x ⅛	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, ⅜-16 x 1¼ Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, ⅜ –16	CRES	AIA/NAS NASM25027

FIGURE 4A20. External wireways installation standard methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, ¼ x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, ¼ x 1	STEEL, TY A	MIL-S-22698
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	A	B	C	F	G	K	PC no.	D	E	H	J	Type
27	6⅛	3				2⅝	36	6 7/16	11 1/16			90° Outside
28	4⅝	2⅛				1⅝	37	4 15/16	10 3/16			90° Outside
29	4⅝	1⅛				¾						
30	6⅛	3	11 1/16			2⅝	38	6 7/16				90° Inside
31	4⅝	2⅛	10 3/16			1⅝	39	4 15/16				90° Inside
32	4⅝	1⅛	9 3/16			¾						
33	6⅛	3		6	14 3/16	2⅝	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4⅝	2⅛		6	12 11/16	1⅝	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4⅝	1⅛		6	12 11/16	¾						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

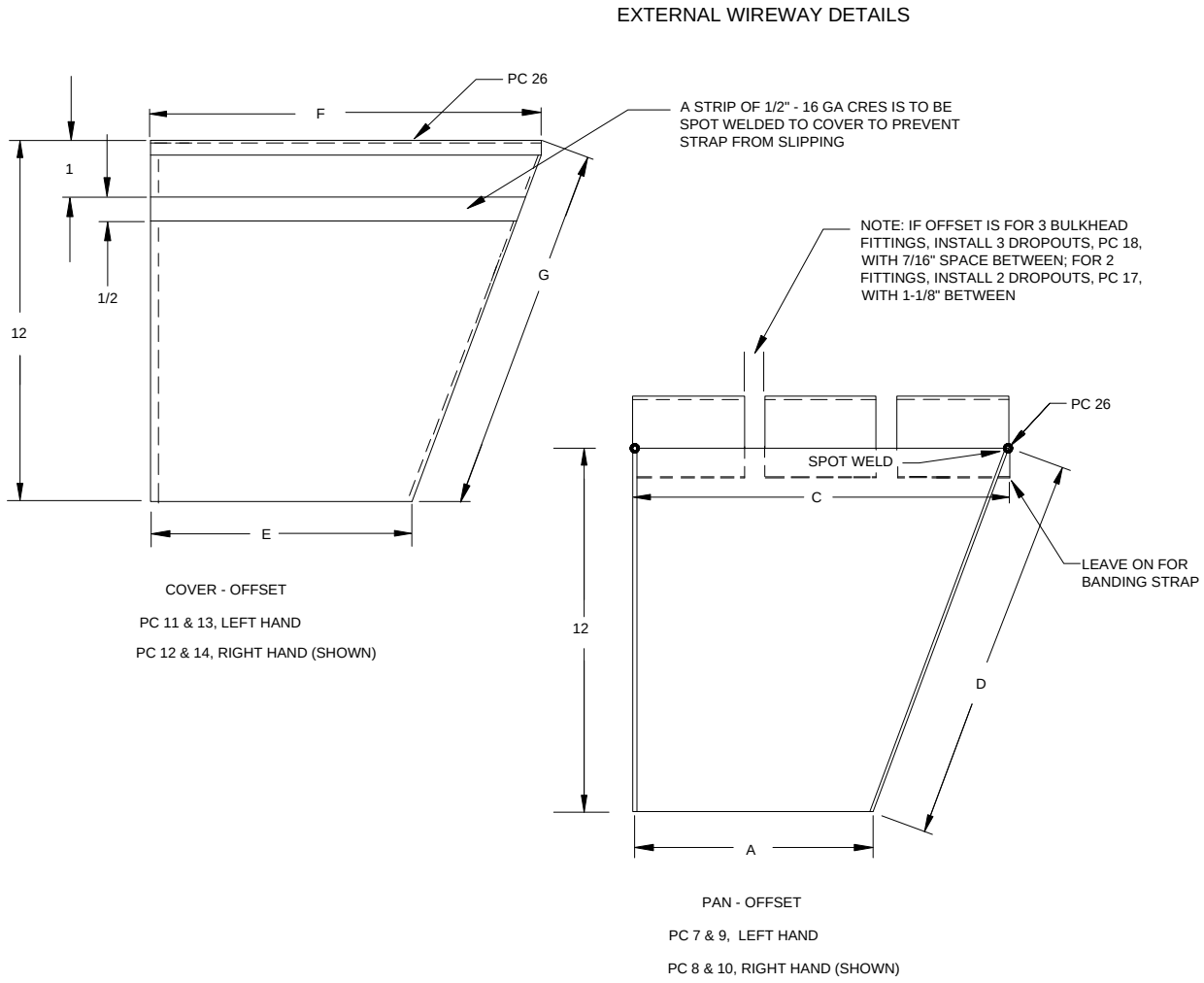
Dimensions for PCs 42, 43, 47, and 48.

PC no.	Pan ID	A	B	Remarks
42	6⅛	4" max	6¾	U/W PC 1
43	4⅝	4" max	5¼	U/W PC 2 & 3
47	6⅛	9¼	6	U/W PC 1
48	4⅝	7¾	6	U/W PC 2 & 3

FIGURE 4A20. External wireways installation standard methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3⁄16 lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A21. External wireways installation standard methods (submarines).

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 14¾ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 13¼ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1¼ ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), ¼ x 1½ Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1¼ x 1¼ x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, ¼ x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, ¼ x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1¼ x 1¼ x ⅛	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, ⅜-16 x 1¼ Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, ⅜ –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190

FIGURE 4A21. External wireways installation standard methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
58		Flat Bar, ¼ x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, ¼ x 1	STEEL, TY A	MIL-S-22698
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	A	B	C	F	G	K	PC no.	D	E	H	J	Type
27	6⅛	3				2⅝	36	6 7/16	11 1/16			90° Outside
28	4⅝	2⅛				1⅝	37	4 15/16	10 3/16			90° Outside
29	4⅝	1⅛				¾						
30	6⅛	3	11 1/16			2⅝	38	6 7/16				90° Inside
31	4⅝	2⅛	10 3/16			1⅝	39	4 15/16				90° Inside
32	4⅝	1⅛	9 3/16			¾						
33	6⅛	3		6	14 3/16	2⅝	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4⅝	2⅛		6	12 11/16	1⅝	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4⅝	1⅛		6	12 11/16	¾						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

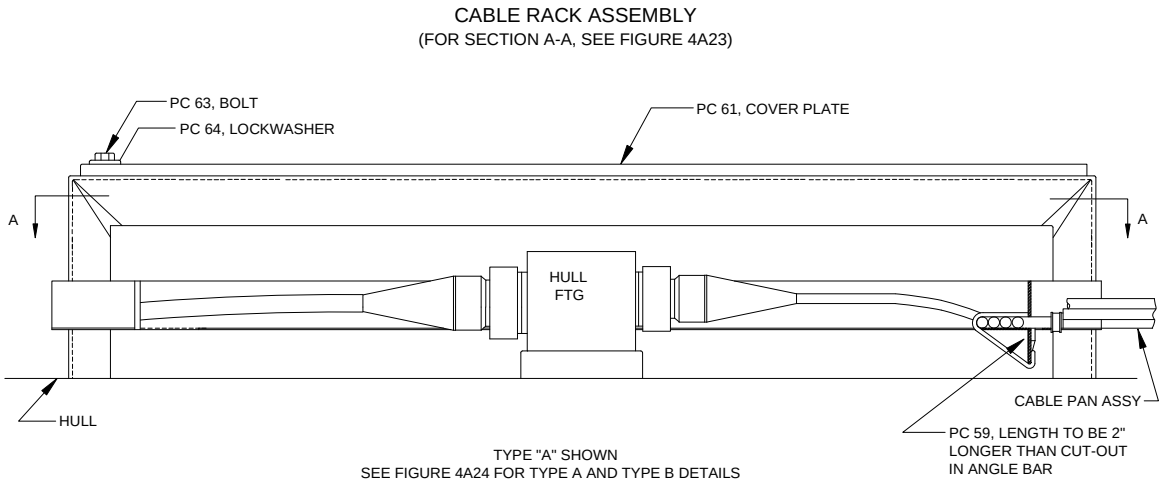
Dimensions for PCs 42, 43, 47, and 48.

PC no.	Pan ID	A	B	Remarks
42	6⅛	4" max	6¾	U/W PC 1
43	4⅝	4" max	5¼	U/W PC 2 & 3
47	6⅛	9¼	6	U/W PC 1
48	4⅝	7¾	6	U/W PC 2 & 3

FIGURE 4A21. External wireways installation standard methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3⁄16 lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7⁄16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15⁄16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7⁄16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7⁄16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A22. External wireways installation methods (submarines) – cable rack assembly.

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Check the source to verify that this is the current version before use.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
42		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 14¾ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 13¼ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1¼ ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), ¼ x 1½ Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1¼ x 1¼ x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, ¼ x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, ¼ x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1¼ x 1¼ x ⅛	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, ⅜-16 x 1¼ Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, ⅜ –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, ¼ x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, ¼ x 1	STEEL, TY A	MIL-S-22698
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

FIGURE 4A22. External wireways installation methods (submarines) – cable rack assembly – Continued.

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Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC no.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

Dimensions for PCs 42, 43, 47, and 48.

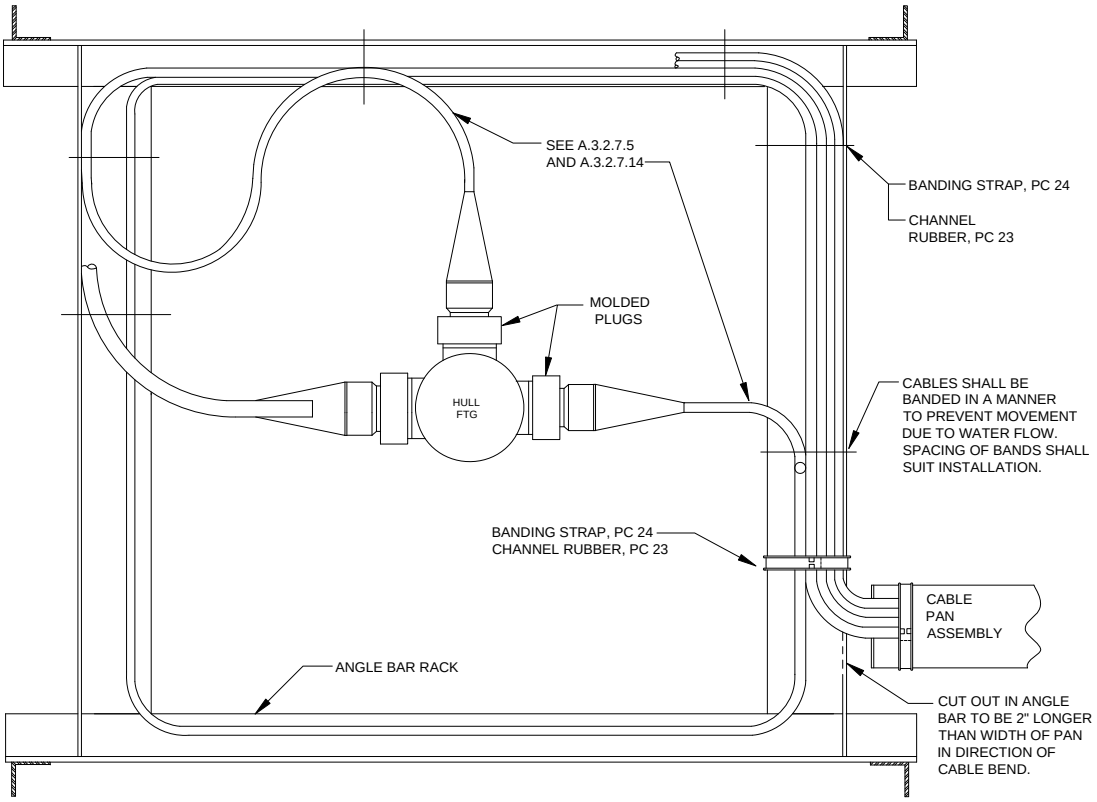
PC no.	Pan ID	A	B	Remarks
42	6 1/8	4" max	6 3/4	U/W PC 1
43	4 5/8	4" max	5 1/4	U/W PC 2 & 3
47	6 1/8	9 1/4	6	U/W PC 1
48	4 5/8	7 3/4	6	U/W PC 2 & 3

FIGURE 4A22. External wireways installation methods (submarines) – cable rack assembly – Continued.

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Check the source to verify that this is the current version before use.

NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

SECTION A-A FOR FIGURE 4A22



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate,. PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A23. External wireways installation methods (submarines) – section A-A for figure 4A22.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 14¾ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 13¼ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1¼ ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), ¼ x 1½ Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1¼ x 1¼ x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, ¼ x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, ¼ x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1¼ x 1¼ x ⅛	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, 3/8-16 x 1¼ Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, 3/8 –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, ¼ x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, ¼ x 1	STEEL, TY A	MIL-S-22698
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, 3/8-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, 3/8-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, 3/8	CRES	MIL-W-12133

FIGURE 4A23. External wireways installation methods (submarines) – section A-A for figure 4A22 – Continued.

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Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	**A**	**B**	**C**	**F**	**G**	**K**	**PC no.**	**D**	**E**	**H**	**J**	**Type**
27	6 1/8	3				2 5/8	36	6 7/16	11 1/16			90° Outside
28	4 5/8	2 1/8				1 5/8	37	4 15/16	10 3/16			90° Outside
29	4 5/8	1 1/8				3/4						
30	6 1/8	3	11 1/16			2 5/8	38	6 7/16				90° Inside
31	4 5/8	2 1/8	10 3/16			1 5/8	39	4 15/16				90° Inside
32	4 5/8	1 1/8	9 3/16			3/4						
33	6 1/8	3		6	14 3/16	2 5/8	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4 5/8	2 1/8		6	12 11/16	1 5/8	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4 5/8	1 1/8		6	12 11/16	3/4						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

Dimensions for PCs 42, 43, 47, and 48.

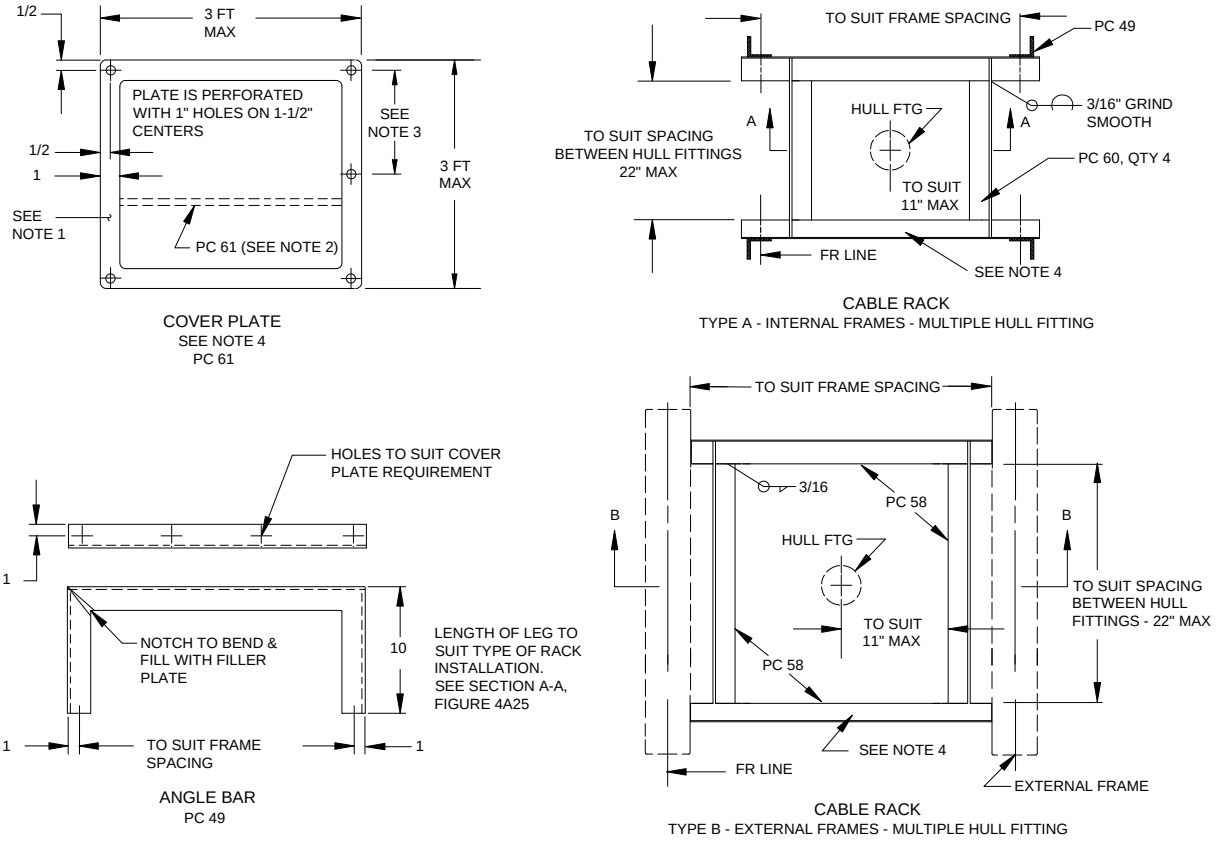
PC no.	Pan ID	A	B	Remarks
42	6 1/8	4" max	6 3/4	U/W PC 1
43	4 5/8	4" max	5 1/4	U/W PC 2 & 3
47	6 1/8	9 1/4	6	U/W PC 1
48	4 5/8	7 3/4	6	U/W PC 2 & 3

FIGURE 4A23. External wireways installation methods (submarines) – section A-A for figure 4A22 – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.

EXTERNAL WIREWAY DETAILS



NOTES:

1. If 1" margin is cut off to suit installation, a piece of flat bar (PC 59) shall be welded on the edge for mounting.
2. Flat bar (PC 59) shall be welded to the underside of plate if needed to prevent vibrations. Number and spacing to suit plate.
3. The number of mounting holes shall be determined by the size of the plate; the maximum distance between holes is 8".
4. Cable racks and cover plates shall be painted to the requirements of the ballast tanks as stated in submarine painting schedule.
5. For outboard identification tags, use Ty-Rap identification plate, PC 56, and “Ty-Rap ties,” PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3/16 lettering.
6. See List of Material below.

FIGURE 4A24. External wireways installation methods (submarines).

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 14¾ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, ¼ x 1½ Flat Bar x 13¼ Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1¼ ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), ¼ x 1½ Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1¼ x 1¼ x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, ¼ x 1½ Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, ¼ x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, ¼ x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1¼ x 1¼ x ⅛	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, ⅜-16 x 1¼ Long	CRES	SAE-AMS-QQ-S-763

FIGURE 4A24. External wireways installation methods (submarines) – Continued.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
55		Elastic Stop Nut, ⅜ –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 5 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 5 above	SAE-AS23190
58		Flat Bar, ¼ x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, ¼ x 1	STEEL, TY A	MIL-S-22698
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	A	B	C	F	G	K	PC no.	D	E	H	J	Type
27	6⅛	3				2⅝	36	6 7/16	11 1/16			90° Outside
28	4⅝	2⅛				1⅝	37	4 15/16	10 3/16			90° Outside
29	4⅝	1⅛				¾						
30	6⅛	3	11 1/16			2⅝	38	6 7/16				90° Inside
31	4⅝	2⅛	10 3/16			1⅝	39	4 15/16				90° Inside
32	4⅝	1⅛	9 3/16			¾						
33	6⅛	3		6	14 3/16	2⅝	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4⅝	2⅛		6	12 11/16	1⅝	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4⅝	1⅛		6	12 11/16	¾						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

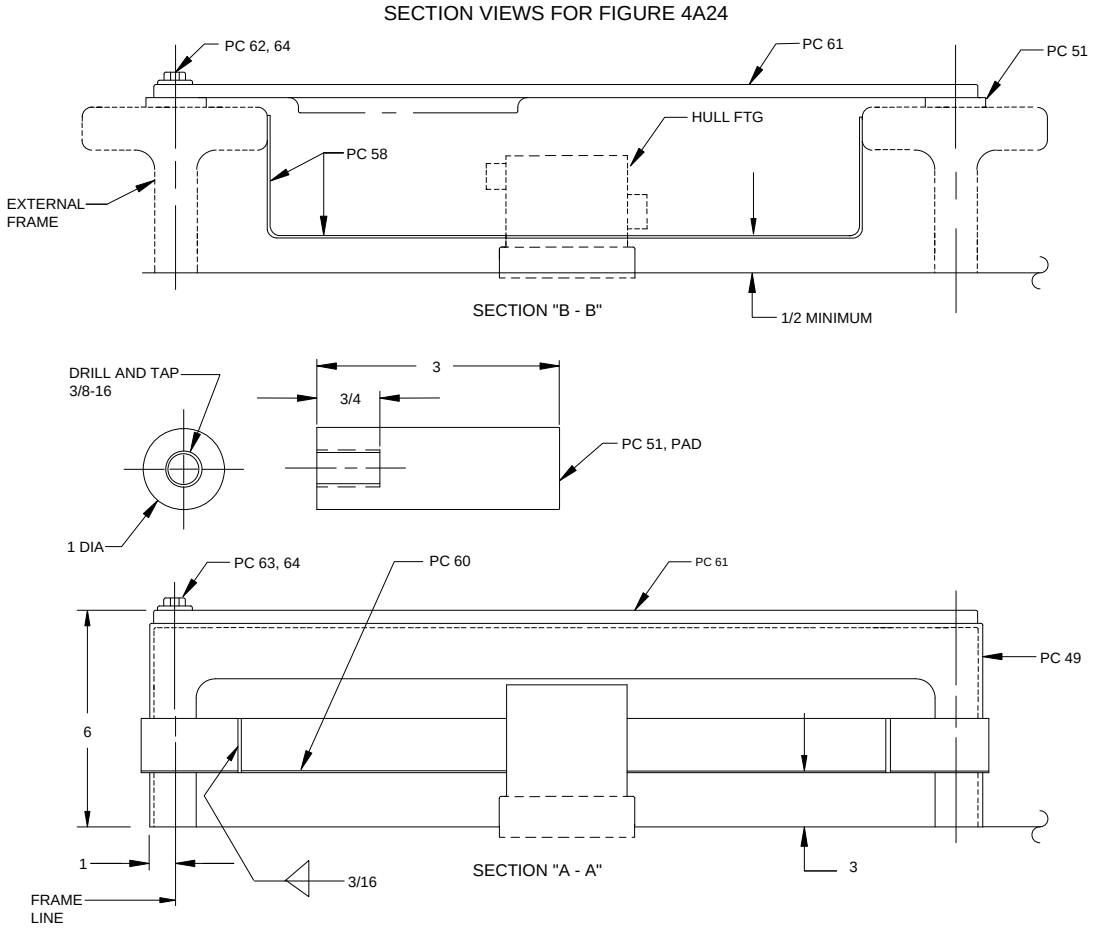
Dimensions for PCs 42, 43, 47, and 48.

PC no.	Pan ID	A	B	Remarks
42	6⅛	4" max	6¾	U/W PC 1
43	4⅝	4" max	5¼	U/W PC 2 & 3
47	6⅛	9¼	6	U/W PC 1
48	4⅝	7¾	6	U/W PC 2 & 3

FIGURE 4A24. External wireways installation methods (submarines) – Continued.

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NOTE: Figures 4A13 through 4A25 and their notes provide guidelines for the installation of all cabling external to the submarine hull. The figures illustrate the methods to be used for protecting and securing the cabling from an outboard appliance to an electrical hull fitting. These methods are not applicable to submarine antenna and mast assemblies. All piece numbers referred to in the following procedures are from the table below each figure. Quantities shall be as required for the installation.



NOTES:

1. For outboard identification tags, use Ty-Rap identification plate, PC 56, and "Ty-Rap ties," PC 57, or equivalent. Plate, PC 56 or equivalent, shall be indelibly marked with 3⁄16 lettering.
2. See List of Material below.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
33		Elbow – 90° Flat Bend (20 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
34		Elbow – 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M

FIGURE 4A25. External wireways installation methods (submarines) – pad and section views.

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PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
35		Elbow – 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
36		Cover, 90° Elbow Outside (7 7/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
37		Cover, 90° Elbow Outside (5 15/16 Wide x 22 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
38		Cover, 90° Elbow Inside (7 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
39		Cover, 90° Elbow Inside (5 7/16 Wide x 17 Long – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
40		Cover, 90° Flat Bend (17 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
41		Cover, 90° Flat Bend (16 Sq – 16 Gauge) (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
42		Mounting Bracket, Hull & Bulkhead, 1/4 x 1 1/2 Flat Bar x 14 3/4 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
43		Mounting Bracket, Hull & Bulkhead, 1/4 x 1 1/2 Flat Bar x 13 1/4 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
44		Hose, 1 1/4 ID x 3/16" Thick	Class 2, Grade 60	MIL-PRF-6855
45		Mounting Bracket (use with Angle Bar), 1/4 x 1 1/2 Flat Bar x 4 Long	CRES – CL FS-304	ASTM A240/A240M
46		Mounting Bracket (use with Angle Bar), 1 1/4 x 1 1/4 x 3/16 Angle Bar	CRES – CL FS-304	ASTM A240/A240M
47		Mounting Bracket, Frame & Bulkhead, 1/4 x 1 1/2 Flat Bar x 20 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
48		Mounting Bracket, Frame & Bulkhead, 1/4 x 1 1/2 Flat Bar x 18 Long (dimensions below)	CRES – CL FS-304	ASTM A240/A240M
49		Angle Bar, 2 x 2 x 3/16	STEEL, HT, TY A	MIL-S-22698
50		Flat Bar, 1/4 x 2	CRES	SAE-AMS-QQ-S-763
51		Pad, 1 Round Bar, 3 Long	GRADE AH-36	MIL-S-22698
52		Tubing Support Clamp Assembly, 1/4 x 1 Flat Bar	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
53		Angle Bar, 1 1/4 x 1 1/4 x 1/8	CRES, CL 304, Cond A	SAE-AMS-QQ-S-763
54		Bolt, Hex Head, 3/8-16 x 1 1/4 Long	CRES	SAE-AMS-QQ-S-763
55		Elastic Stop Nut, 3/8 –16	CRES	AIA/NAS NASM25027
56		Ty-Rap, Identification Plate, Part No 650-4045-5	see note 1 above	Mfg Code No 59730
57		Ty-Rap Ties, Type 25m	see note 1 above	SAE-AS23190
58		Flat Bar, 1/4 x 3	STEEL, TY A	MIL-S-22698
59		Flat Bar, 1/4 x 1	STEEL, TY A	MIL-S-22698

FIGURE 4A25. External wireways installation methods (submarines) – pad and section views – Continued.

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Check the source to verify that this is the current version before use.

PC no.	Qty	Item name (all dimensions are in inches)	Material requirement	Material specification or standard
60		Angle Bar, 2 x 3 x ¼	GRADE AH-36	MIL-S-22698
61		Cover Plate, 10.2#	COML STEEL	MIL-S-22698
62		Bolt, Hex Head, ⅜-16 x ¾ Long	CRES	SAE-AMS-QQ-S-763
63		Bolt, Hex Head, ⅜-16 x 1 Long	CRES	SAE-AMS-QQ-S-763
64		Lockwasher, ⅜	CRES	MIL-W-12133

Dimensions for PCs 27 through 41.

Elbow							Cover					
PC no.	A	B	C	F	G	K	PC no.	D	E	H	J	Type
27	6⅛	3				2⅝	36	6 7/16	11 1/16			90° Outside
28	4⅝	2⅛				1⅝	37	4 15/16	10 3/16			90° Outside
29	4⅝	1⅛				¾						
30	6⅛	3	11 1/16			2⅝	38	6 7/16				90° Inside
31	4⅝	2⅛	10 3/16			1⅝	39	4 15/16				90° Inside
32	4⅝	1⅛	9 3/16			¾						
33	6⅛	3		6	14 3/16	2⅝	40	6 7/16		7 29/32	14 11/32	90° Flat
34	4⅝	2⅛		6	12 11/16	1⅝	41	4 15/16		7 29/32	2 27/32	90° Flat
35	4⅝	1⅛		6	12 11/16	¾						

NOTES:
1. These 90° elbows are to be cut to obtain elbows of lesser degrees.
2. All dimensions are in inches.

Dimensions for PCs 42, 43, 47, and 48.

PC no.	Pan ID	A	B	Remarks
42	6⅛	4" max	6¾	U/W PC 1
43	4⅝	4" max	5¼	U/W PC 2 & 3
47	6⅛	9¼	6	U/W PC 1
48	4⅝	7¾	6	U/W PC 2 & 3

FIGURE 4A25. External wireways installation methods (submarines) – pad and section views – Continued.

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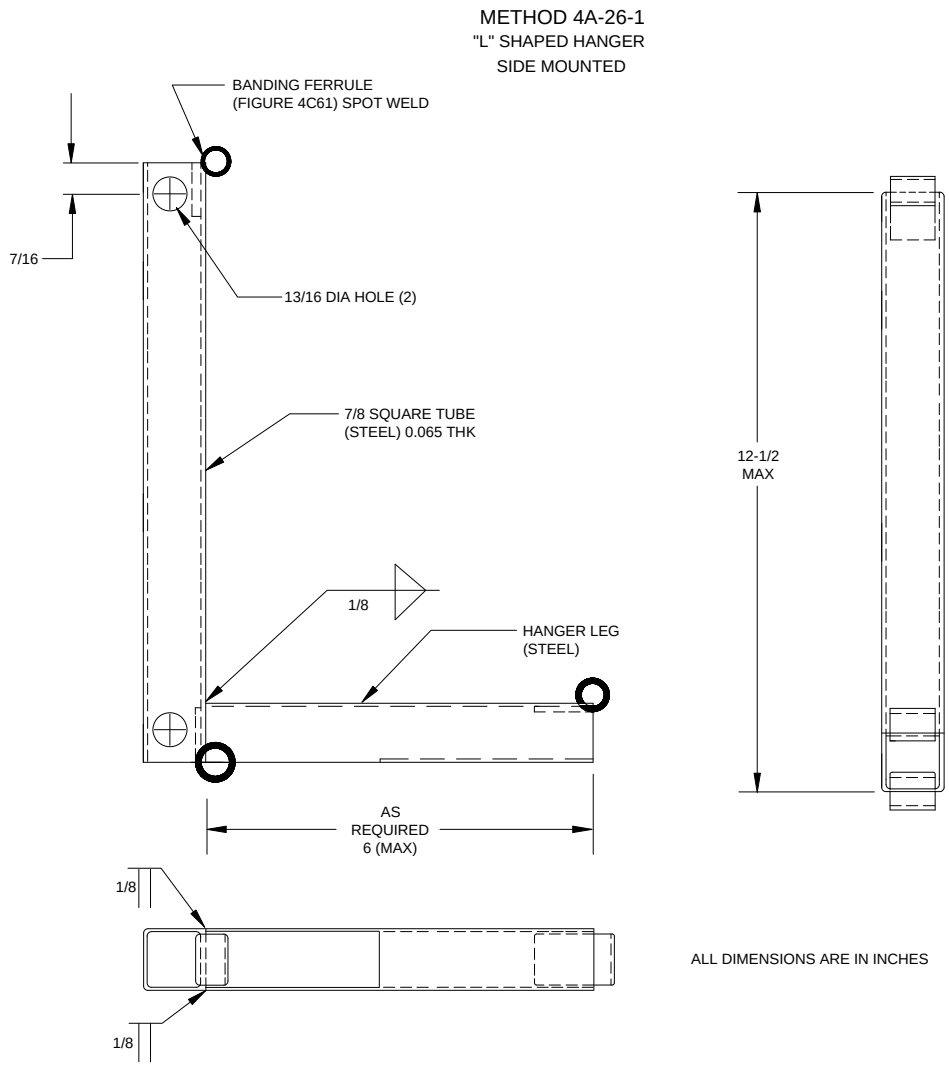


FIGURE 4A26. Tubular hanger details “L” shape for submarines.

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METHOD 4A-27-1
"L" SHAPED HANGER
BOTTOM MOUNTED

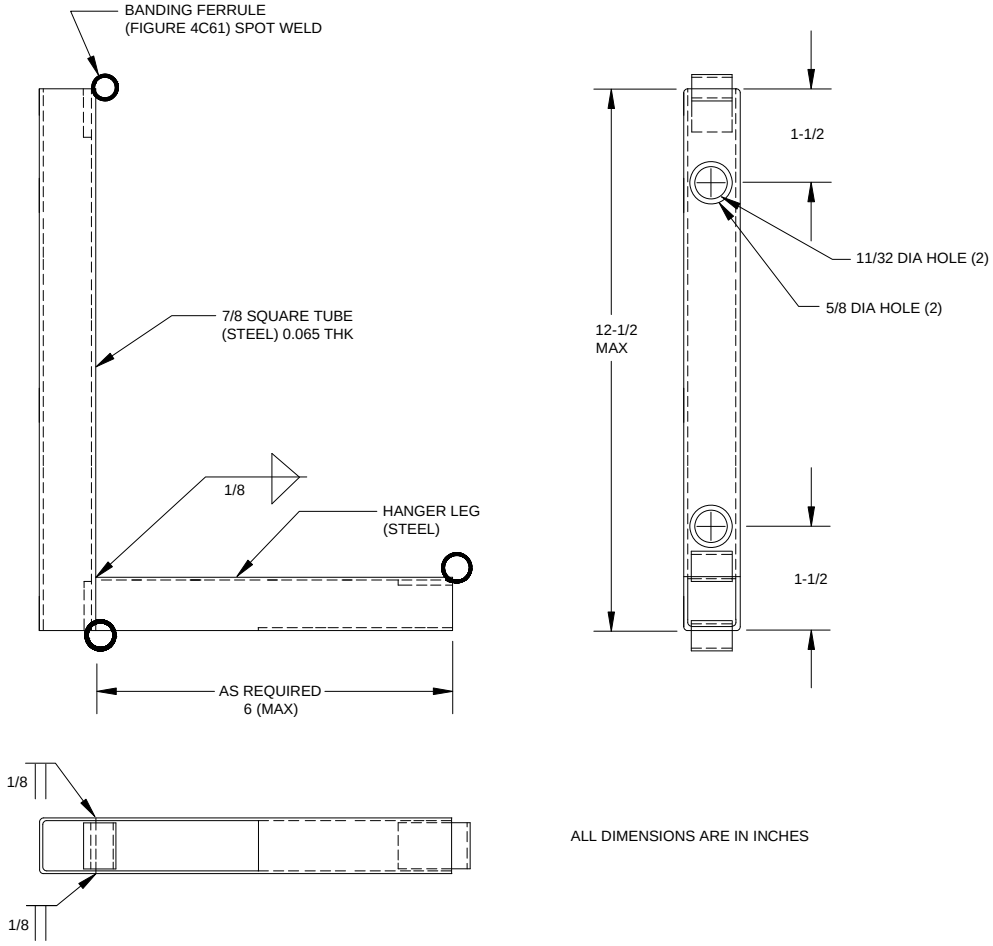


FIGURE 4A27. Tubular hanger details “L” shape for submarines.

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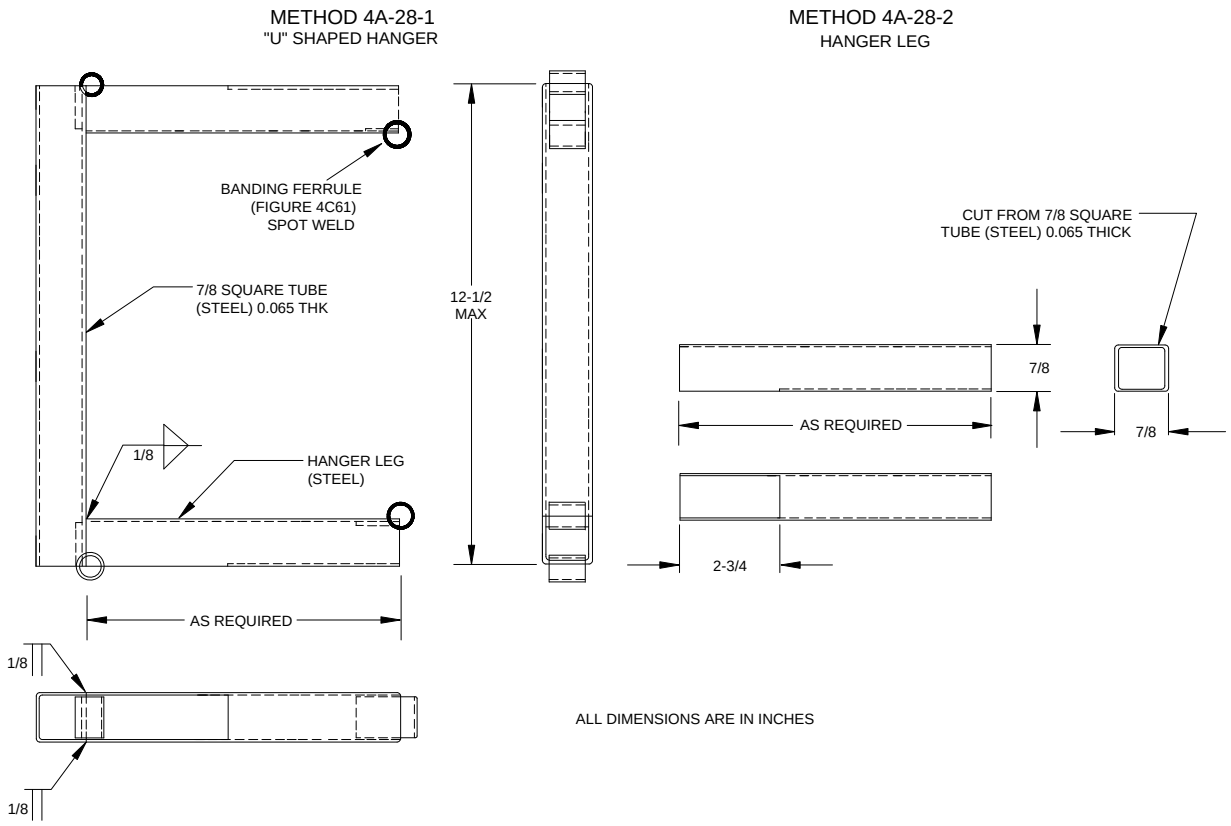


FIGURE 4A28. Tubular hanger details “U” shape for submarines.

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GROUP 4B – CABLEWAYS (SURFACE SHIPS)

B.1 SCOPE

B.1.1 Scope. This appendix establishes procedures for the installation of cableways on surface ships.

B.2. APPLICABLE DOCUMENTS

B.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

B.2.2 Government documents.

B.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-I-3064	-	Insulation, Electrical, Plastic-Sealer
MIL-DTL-5015		Connectors, Electrical, Circular Threaded, AN Type, General Specification for
MIL-S-24149	-	Studs, Welding and Arc Shields (Ferrules), General Specification for
MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for
MIL-S-24235/9	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Brass and Steel, for Decks and Bulkheads with Pipe Protection

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-299	-	Cable Comparison Handbook Data Pertaining to Electrical Shipboard Cable

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

B.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DESIGN DATA SHEETS

DDS 100-1	-	Reinforcement of Openings in Structure of Surface Ships other than in Protective Plating

(Copies of this document are available from Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160, or by email at CommandStandards@navy.mil with the subject line "DDS request".)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9086-CM-STM-010	-	NSTM Chapter 078, Volume 1 - Seals; Naval Ships Technical Manual
S9086-CM-STM-020	-	NSTM Chapter 078, Volume 2 - Gaskets and Packing

(Copies of these documents are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil.)

B.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A36/A36M	-	Standard Specification for Carbon Structural Steel
ASTM A501	-	Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing
ASTM B221	-	Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B241/B241M	-	Standard Specification for Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-DTL-23053/5	-	Insulation Sleeving, Electrical, Heat-shrinkable, Polyolefin, Flexible, Crosslinked
SAE-AMS-DTL-23053/15	-	Insulation Sleeving, Electrical, Heat-shrinkable, Polyolefin, Heavy-Wall, Coated, Flexible, Outer Wall Crosslinked
SAE-AS34001	-	Connector, Receptacle, Electric, Wall Mounting, Front Release, Crimp Contact, An Type
SAE-AS85049	-	Connector Accessories, Electrical General Specification For

(Copies of these documents are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.)

B.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

B.3 REQUIRED EQUIPMENT AND MATERIALS

B.3.1 Required equipment and materials. The equipment and materials referenced in the text of this section shall be used for the installation of cableways on surface ships. For fiber optic cable, refer to MIL-STD-2042-4 for additional information and guidance.

B.4 NOTES AND PROCEDURES

B.4.1 Dimensions. For figures and tables in this section, all dimensions are in inches unless otherwise noted.

B.4.2 Figures. Table 4BI provides information for the figures in this group.

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TABLE 4BI. Figures for cable installations on surface ships.

Figure number	Cable installation	Page
4B1	Cableways for A.C. propulsion cables (surface ships) – one group and one tier	94
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4B3	Cableways for A.C. propulsion cables (surface ships) – one group and two tiers	96
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4B5	Cableways for A.C. propulsion cables (surface ships) – one group and three tiers	98
4B6	Cableways for A.C. propulsion cables (surface ships) – two groups and three tiers	99
4B7	Cableways for A.C. propulsion cables (surface ships)	101
4B8	Cableways for A.C. propulsion cables (surface ships)	102
4B9	Cableways for A.C. propulsion cables (surface ships) – two groups and two tiers	103
4B10	Cableways for A.C. propulsion cables (surface ships) – two groups and three tiers	104
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4B12	A.C. propulsion cableway (surface ships) – three cables on transverse frames	106
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4B14	Cableways for A.C. propulsion cables – using stuffing tubes, fabricated	108
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4B23	Multiple cableways on steel decks and bulkheads (surface ships)	118
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4B27	Multiple cables on steel bulkheads (surface ships)	123
4B28	Multiple cables on steel bulkheads – cables supported on stiffener (surface ships)	124
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4B33	Multiple cableways supported from aluminum or steel decks (surface ships)	130
4B34	Multiple cableways on insulated steel or aluminum decks or bulkheads (surface ships)	131
4B35	Multiple cableway supports on aluminum decks and bulkheads (surface ships only)	132
4B36	Angle down comer for figure 4B35	133

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TABLE 4BI. Figures for cable installations on surface ships - Continued.

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4B38	Detail of fire protection installation	134
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4B40	Single cableway on aluminum structures – local runs (surface ships)	136
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TABLE 4BI. Figures for cable installations on surface ships - Continued.

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4B77	Cable support for degaussing cables (surface ships)	176
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4B83	Cableways on wooden ships (surface ships) (Guidance only)	181
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4B90	Cableways on wooden ships (surface ships) (Guidance only)	188
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4B99	Cableways of main cable runs on small vessels (surface ships)	197
4B100	Cableways of main cable runs on small vessels (surface ships) – section A-A	198

B.4.3 Details for cableways for A.C. propulsion cables on surface ships.

B.4.3.1 Heat dissipation. Where cables enter propulsion electrical equipment, care shall be exercised to avoid bunching or closely grouping the cables in any manner that may restrict dissipation of heat from the cables.

B.4.3.2 Phase arrangements. Phase arrangements as shown in table 4BII shall apply to horizontal, vertical or inclined wireway runs. Any arrangements of cables other than shown must be specifically approved by NAVSEA.

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NOTE: Where “Inverted T” type hangers are used, the downcomer shall not separate any of the phase groups.

TABLE 4BII. Cable arrangement and phase relationship.
(see figures 4B1 through 4B13)

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

B.4.3.3 Separation of propulsion cables. Propulsion cables in wireways shall be run separate from other cables.

B.4.3.4 End sealing. When non-waterblocked cable is approved for use, ends shall be water-sealed by terminating in water-seal solderless lugs in accordance with MIL-STD-2003-1, Group A, figure 1A9.

B.4.3.5 Use of stuffing tubes for propulsion cables. Where cables to propulsion equipment are required to enter through stuffing tubes, the cables and stuffing tubes shall be installed in accordance with MIL-STD-2003-3.

B.4.3.6 Phase marking of single conductor cables. In order to ensure that the proper arrangement of single conductor cables is maintained, the phase of single conductor cables shall be indicated at suitable intervals along the length of each cable and at the extremities.

B.4.3.7 Cable support mounting. Unless otherwise approved, no cable support shall be mounted in such a manner as to be directly connected to strength frames or beams, particularly where near the skin of the ship and at or below the waterline.

B.4.3.8 Temperature limit exposure. Avoid running propulsion cable wireways in locations where ambient temperatures exceed 50 °C during operation.

B.4.3.9 Hanger spacing. Hangers shall be spaced at intervals of not more than 16" on center.

B.4.3.10 Cable slack for shock and vibration. Cables shall have sufficient slack when entering equipment for shock excursion (MIL-S-901) and vibration (MIL-STD-167-1). Cable bend radii shall be in accordance with MIL-HDBK-299 and 5.2.

B.4.4 Detail cable arrangements for D.C. propulsion cables on surface ships.

B.4.4.1 Minimizing external magnetic fields. The methods shown indicate the recommended arrangements of direct current single-conductor propulsion cables in wireways to reduce, to a minimum, external magnetic fields due to the propulsion current. Each method illustrates the arrangement of cables for one circuit consisting of a positive and a negative leg, each leg containing two or more single conductor cables in parallel. Whenever practicable, each circuit should contain an even number of single conductor cables per leg in preference to an odd number per leg.

B.4.4.1.1 Arrangements. The detail arrangements apply to horizontal, vertical or inclined wireways. When several propulsion circuits are installed in one wireway, each circuit shall be arranged as indicated. The circuits may be located adjacent on the same hanger or hangers or separated on individual hangers.

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B.4.4.1.2 Parallel cables. The external magnetic fields due to direct current in double conductor cables are relatively small if the two conductors in each cable are connected to carry current of opposite polarity. Accordingly, when several double-conductor cables are used in parallel, the two conductors in each cable shall be of opposite polarity.

TABLE 4BIII. Maximum allowable current capacities for D.C. propulsion cables in wireways and racks – individual adjacent cables on each hanger without spacing.

Number of hangers (horizontal or vertical)	Percentage of current ratings specified in MIL-HDBK-299 (at 50 °C ambient) (see B.3.4.6)			
	Individual cables on each hanger adjacent without spacing			
	Cables in horizontal hangers next to bulkheads without insulation	Cables in horizontal hangers next to bulkheads with insulation	Cables in horizontal hangers under decks without insulation or in vertical hangers next to bulkheads without insulation	Cables in horizontal hangers under decks with insulation or in vertical hangers next to bulkheads with insulation
1	95	90	90	80
2	85	80	80	70
3	80	75	70	65

TABLE 4BIV. Maximum allowable current capacities for D.C. propulsion cables in wireways and racks – individual cables on each hanger spaced ½" apart.

Number of hangers (horizontal or vertical)	Percentage of current ratings specified in MIL-HDBK-299 (at 50 °C ambient) (see B.3.4.6)			
	Individual cables on each hanger spaced ½" apart			
	Cables in horizontal hangers next to bulkheads without insulation	Cables in horizontal hangers next to bulkheads with insulation	Cables in horizontal hangers under decks without insulation or in vertical hangers next to bulkheads without insulation	Cables in horizontal hangers under decks with insulation or in vertical hangers next to bulkheads with insulation
1	100	95	100	95
2	100	95	100	95
3	95	90	95	90

B.4.4.1.3 Connection to terminals. Cable connections to the terminals of electrical equipment shall be accomplished with a minimum amount of fanning of the individual conductors. Loops formed by conductors of opposite polarity should be reduced in area to a minimum and avoided wherever possible. When unavoidable, the plane of the loop formed by the conductors of opposite polarity in each circuit should lie vertical in preference to horizontal.

B.4.4.2 Polarity marking of single conductor cables. In order to ensure that the proper arrangement of single conductor cables is maintained, the polarity of single conductor cables shall be indicated at suitable intervals along the length of each cable and at the extremities.

B.4.4.3 Allowable current capacities. Tables 4BIII and 4BIV provide allowable current capacities for D.C. propulsion cables in wireways and racks. For propulsion cables in enclosed ducts, cables shall be rated at 50 percent of the values listed in the tables. The ratings in the tables are based on four or more cables on each hanger. For less than four cables per hanger, an additional 5 percent increase in the percentages specified in the tables is permissible.

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B.4.4.4 Separation in wireways. Propulsion cables in wireways shall run separate from other cables where practicable.

B.4.4.5 Retention in wireways. Propulsion cables shall be adequately clamped or strapped to avoid loosening or dislodgement due to vibration, shock, or magnetic stresses.

B.4.4.6 Temperature limit exposure. Avoid running propulsion cable wireways in location where ambient temperatures exceed 50 °C during operation, as in the vicinity of boilers, evaporators, steam-pipes, etc.

B.4.4.7 Dissipation of heat. Where cables enter propulsion electrical equipment, care shall be exercised to avoid bunching or closely grouping the cables in any manner that may restrict dissipation of heat from the cables.

B.4.4.8 End sealing. When non-waterblocked cable is approved for use, the ends of propulsion cables shall be adequately sealed by approved methods to avoid the entrance of water or moisture.

B.4.4.9 Other details. Details not specifically covered in D.C. propulsion methods shall be in accordance with the applicable standard methods.

B.4.4.10 Cable slack for shock and vibration. Cables shall have sufficient slack when entering equipment for shock excursion (MIL-S-901) and vibration (MIL-STD-167-1). Cable bend radii shall be in accordance with MIL-HDBK-299 and 5.2.

B.4.5 Hanger spacing, tiers and weight. Spacing, number of tiers, and cable weight shall not exceed the values specified in table 4BV. Downward extension to the lesser condition of fewer tiers for hanger assemblies (with cable load not exceeding the maximum per tier as specified) is acceptable.

B.4.5.1 Corrosion protection. Contact areas between adapter, downcomer and attaching bolts shall be painted with two coats of primer in accordance with MIL-PRF-23236.

B.4.5.2 Banding below cable support bar. Cables may be banded below cable support bars on overhead mounted cable hangers provided they are banded separately from the banding for cables over the support bar and the total weight does not exceed 450 pounds per cable hanger.

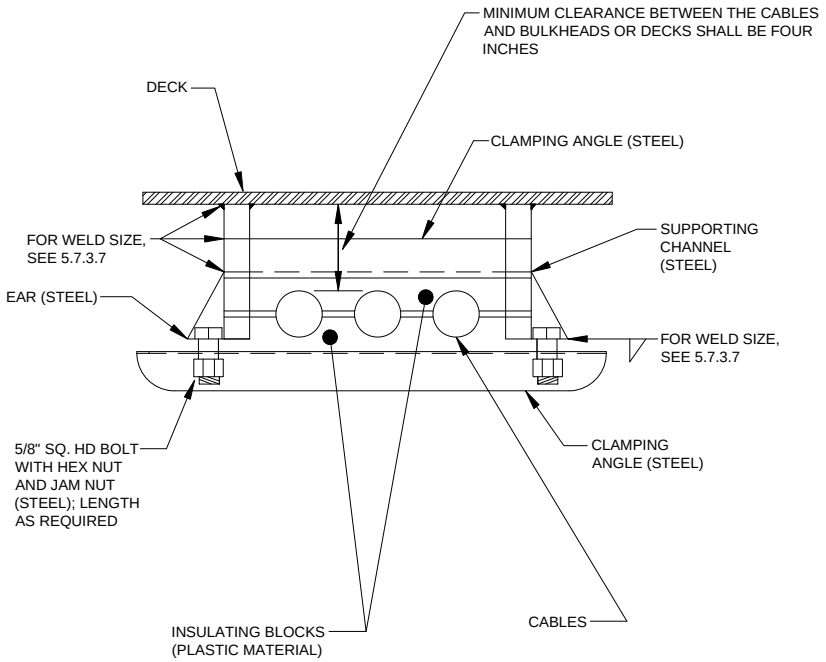
TABLE 4BV. Spacing and maximum number of tiers for horizontal and vertical cable runs. (see figures 4B26, 4B31, 4B46, and 4B53)

Cable run	Assembly		Center-to-center spacing (inches)	Maximum number of tiers	Maximum cable weight (pounds) supported by each tier
	Mounting type	Dimensions (inches)			
Horizontal	Deck	36 x 17	16	3	45.5
Horizontal	Deck	36 x 17	32	5	91
Vertical	Bulkhead	20 x 17	24	2	68
Vertical	Bulkhead	20 x 17	24	2	45.5

B.4.6 Mounting aluminum hangers on steel decks. For mounting aluminum hangers on steel decks, the steel (ASTM A501) adapter, detailed on figure 4B50, shall be used. Contact areas between the adapter, downcomer and attaching bolts shall be painted with two coats of primer in accordance with MIL-PRF-23236.

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METHOD 4B-1-1
ONE GROUP AND ONE TIER OF CABLES
FROM STEEL DECKS OR BULKHEADS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS AND PHASE RELATIONSHIP)



NOTE:

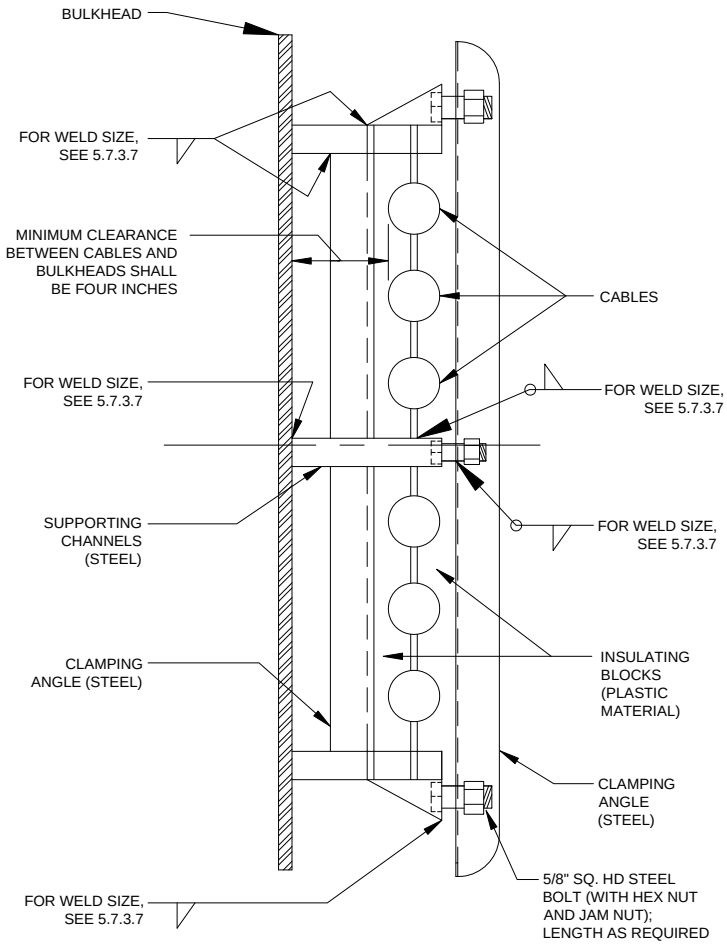
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B1. Cableways for A.C. propulsion cables (surface ships) – one group and one tier.

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METHOD 4B-2-1
TWO GROUPS AND ONE TIER OF CABLES
FROM STEEL DECKS OR BULKHEADS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS AND PHASE RELATIONSHIP)



NOTE:

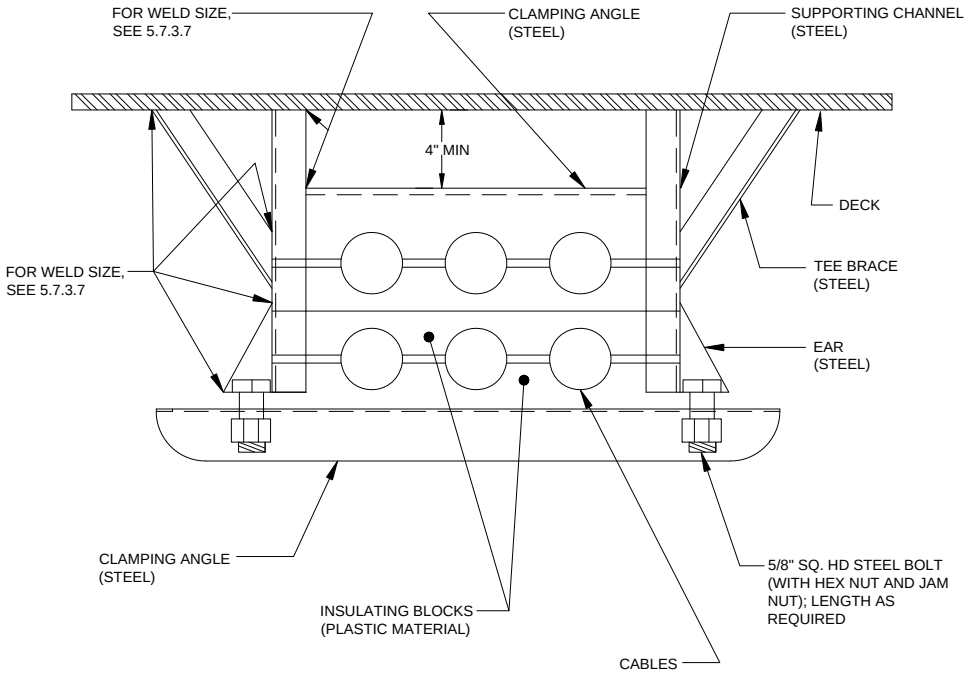
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B2. Cableways for A.C. propulsion cables (surface ships) – two groups and one tier.

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METHOD 4B-3-1
ONE GROUP AND TWO TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS AND PHASE RELATIONSHIP)



NOTE:

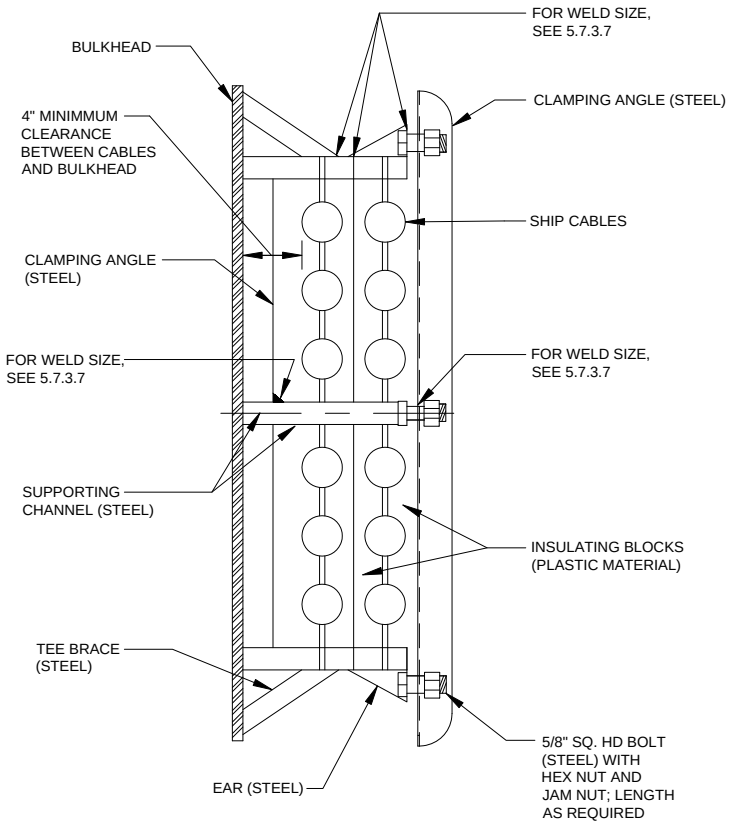
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B3. Cableways for A.C. propulsion cables (surface ships) – one group and two tiers.

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METHOD 4B-4-1
TWO GROUPS AND TWO TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS AND PHASE RELATIONSHIP)



NOTE:

1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B4. Cableways for A.C. propulsion cables (surface ships) – two groups and two tiers.

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METHOD 4B-5-1
ONE GROUP AND THREE TIERS OF CABLES
FROM STEEL DECKS OR BULKHEADS

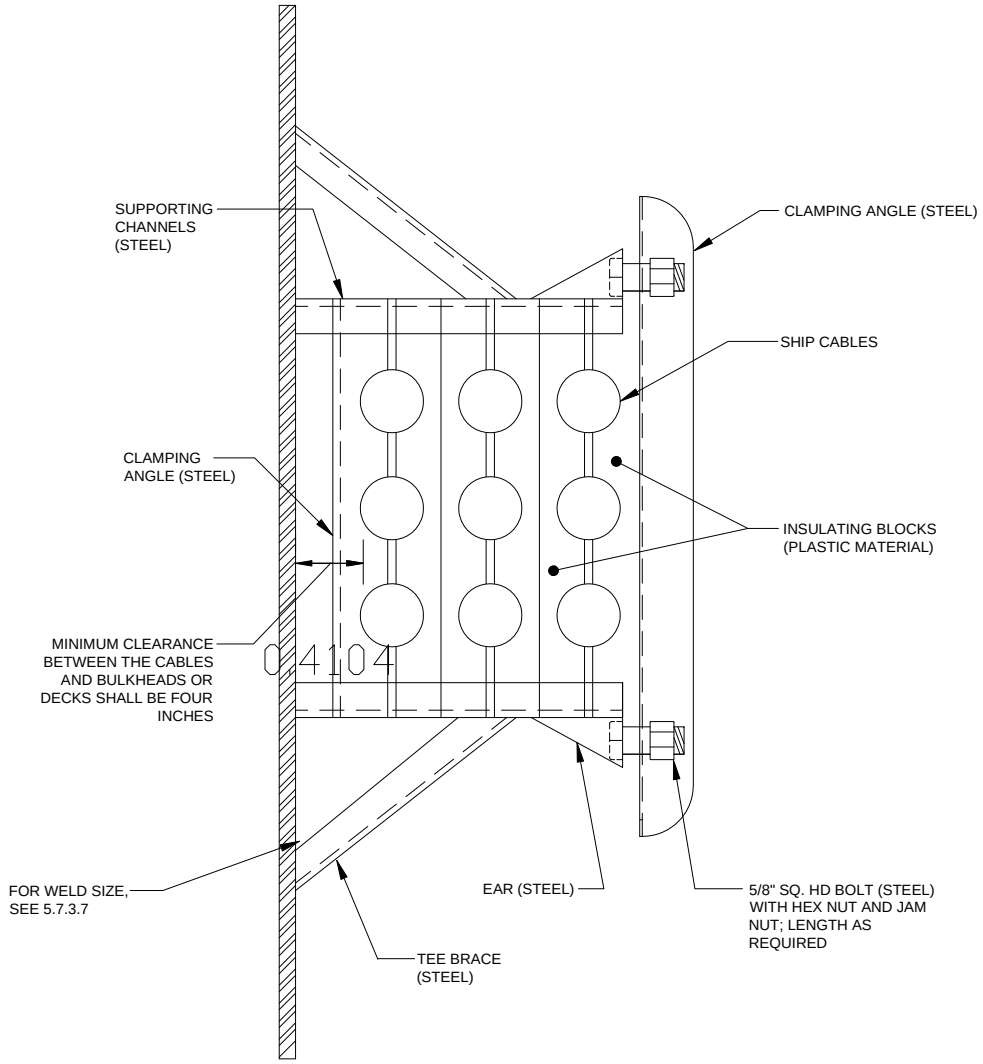


FIGURE 4B5. Cableways for A.C. propulsion cables (surface ships) – one group and three tiers.

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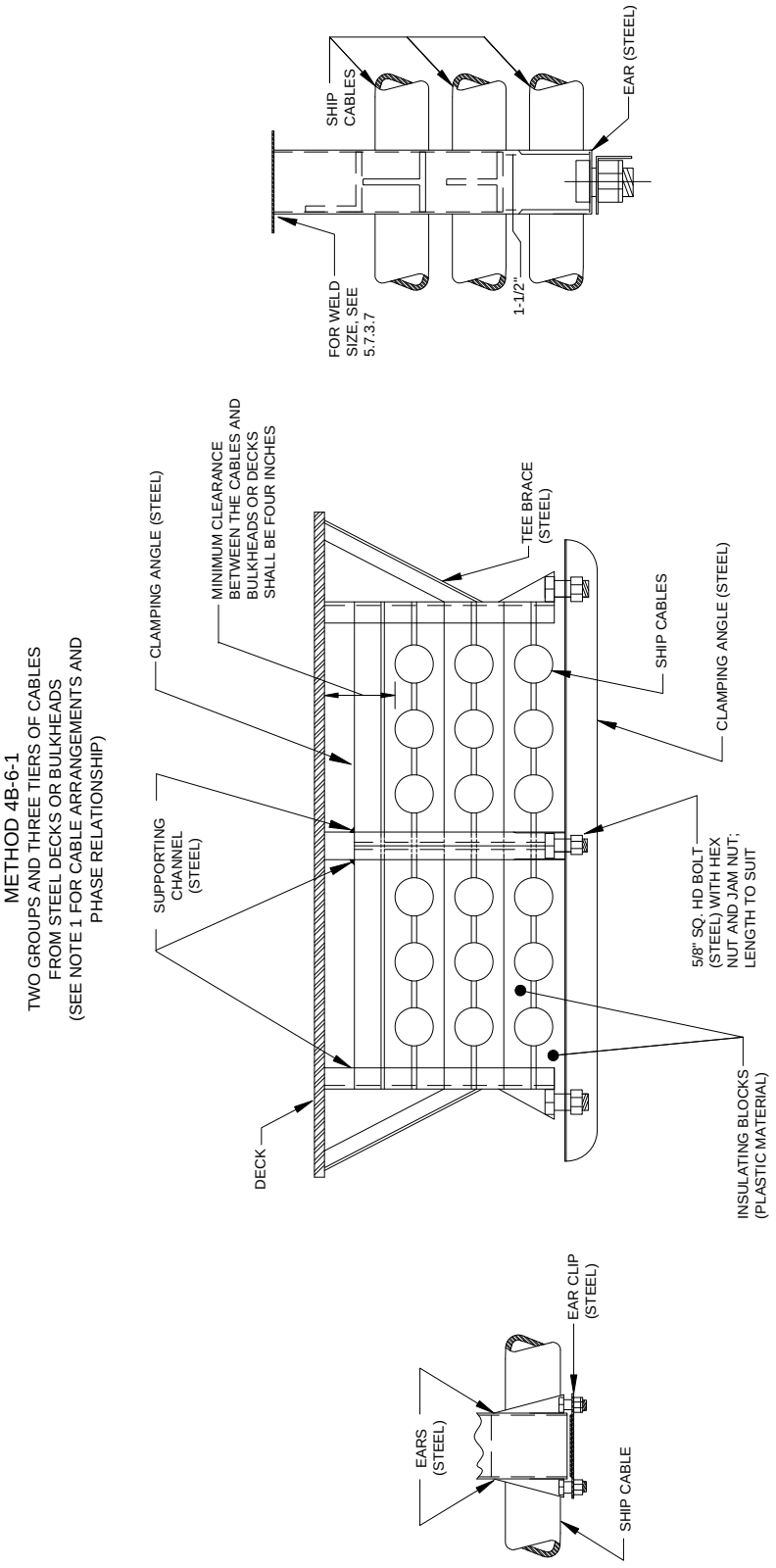


FIGURE 4B6. Cableways for A.C. propulsion cables (surface ships) – two groups and three tiers.

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NOTE:

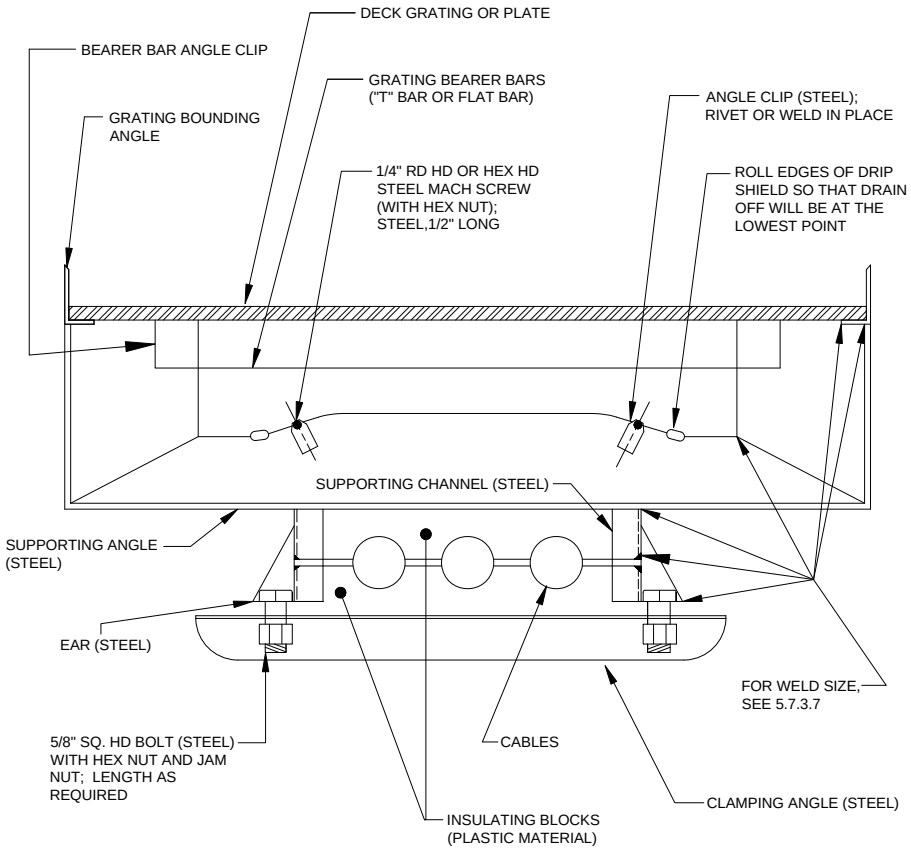
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B6. Cableways for A.C. propulsion cables (surface ships) – two groups and three tiers - Continued.

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METHOD 4B-7-1
ONE GROUP AND ONE TIER OF CABLES
UNDER DECK GRATINGS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS
AND PHASE RELATIONSHIP)



NOTE:

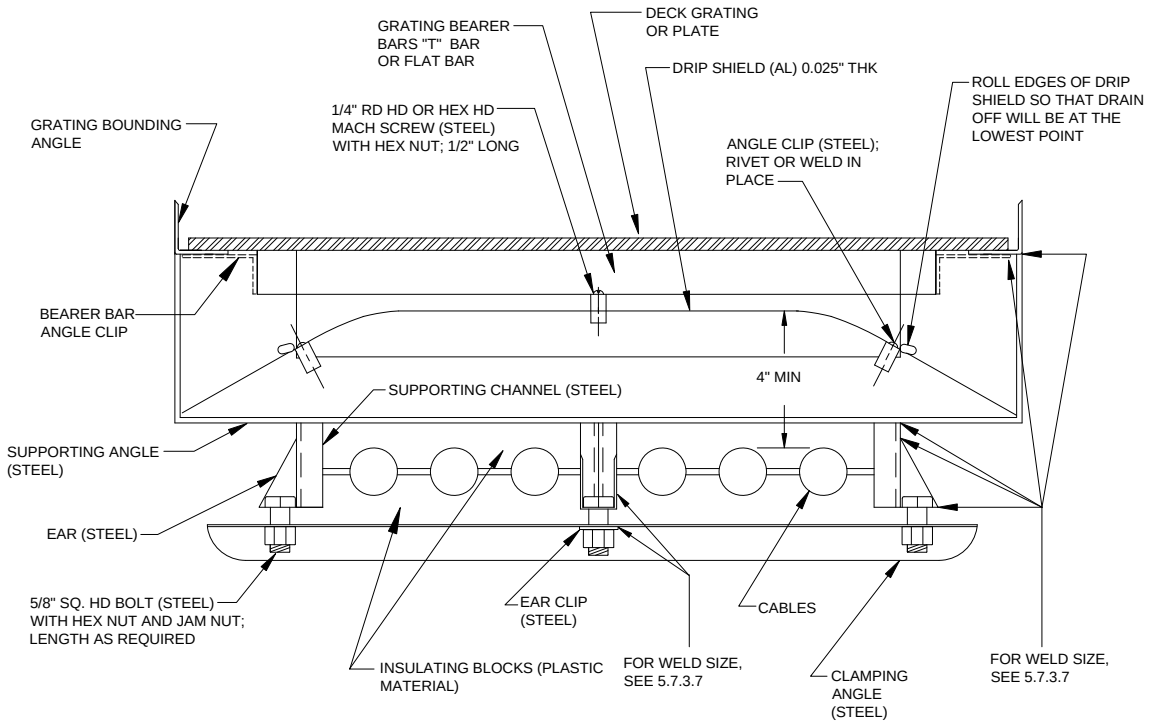
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B7. Cableways for A.C. propulsion cables (surface ships).

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METHOD 4B-8-1
TWO GROUPS AND ONE TIER OF CABLES
UNDER DECK GRATINGS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS
AND PHASE RELATIONSHIP)



NOTE:

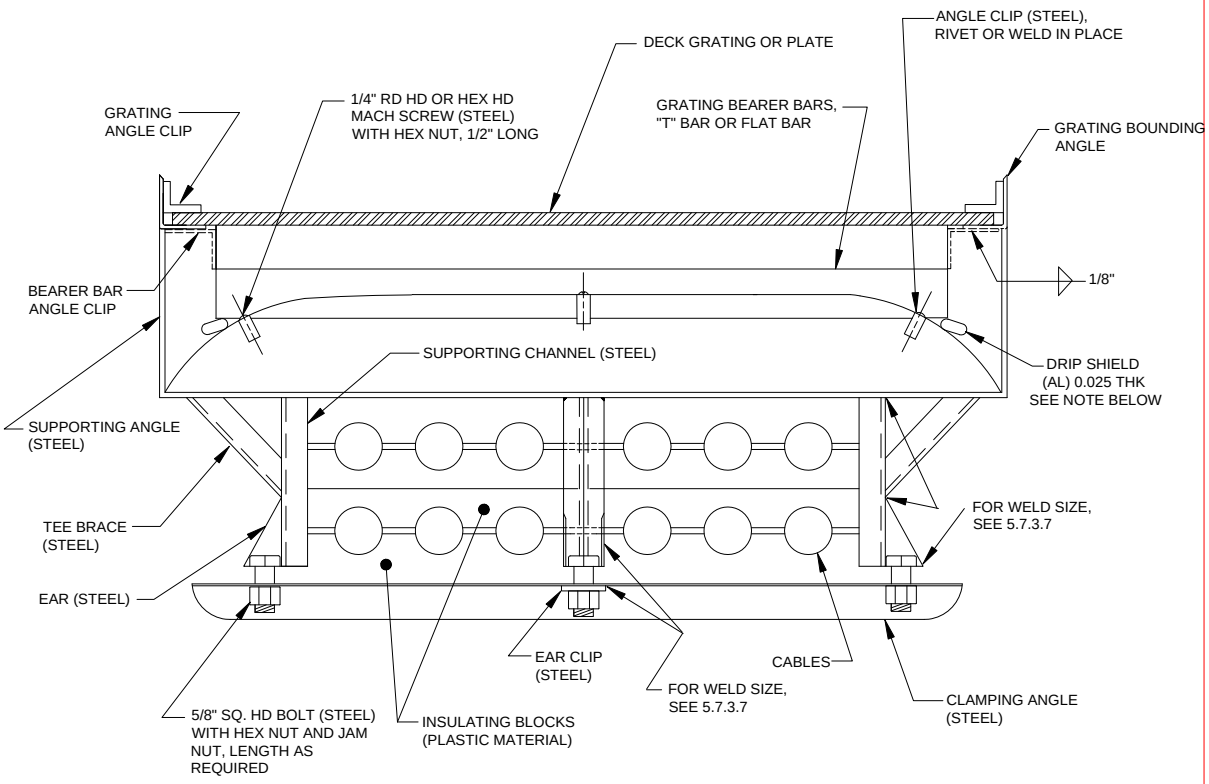
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B8. Cableways for A.C. propulsion cables (surface ships).

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METHOD 4B-9-1
TWO GROUPS AND TWO TIERS OF CABLES
UNDER DECK GRATINGS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS AND PHASE RELATIONSHIP)



NOTE: FOR DRIP SHIELD, ROLL EDGES OF DRIP SHIELD SO THAT THE DRAIN OFF WILL BE AT THE LOWEST POINT.

NOTE:

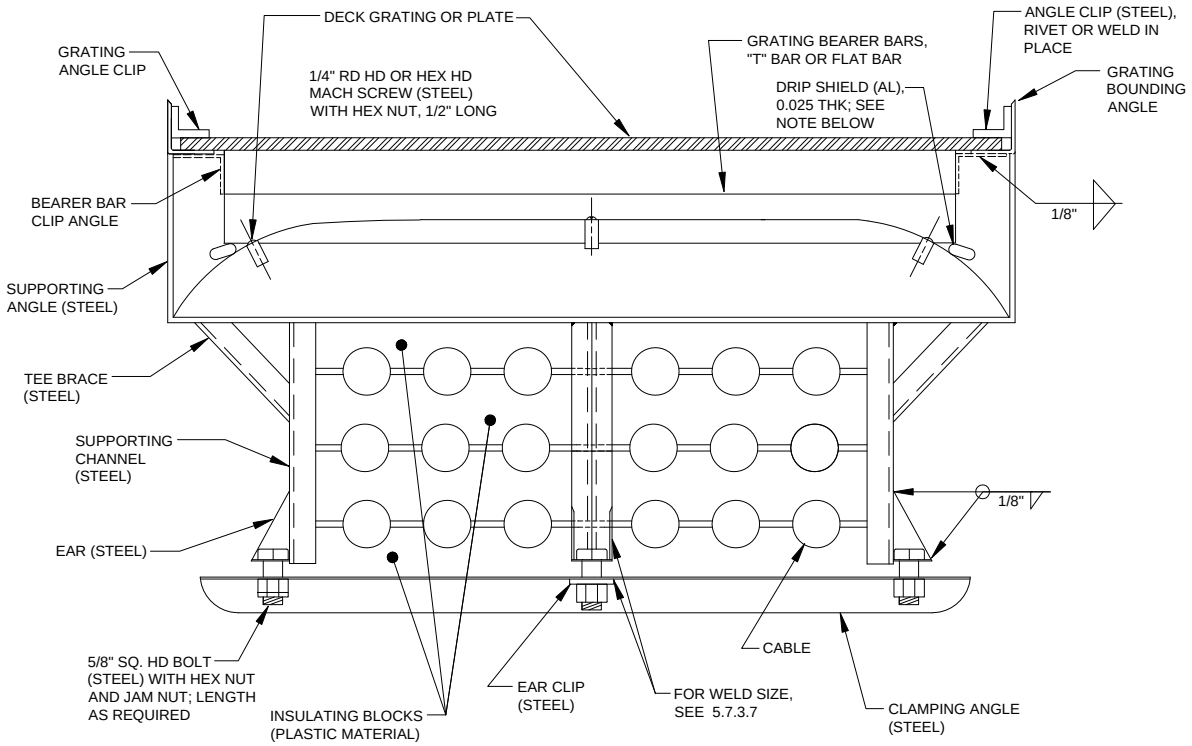
1. Cable arrangements and phase relationships are as follows:

One group – one tier			**Two groups – one tier**					
A	B	C	A	B	C	C	B	A
One group – two tiers			**Two groups – two tiers**					
A	B	C	A	B	C	C	B	A
A	B	C	A	B	C	C	B	A
One group – three tiers			**Two groups – three tiers**					
A	B	C	A	B	C	C	B	A
C	A	B	C	A	B	B	A	C
B	C	A	B	C	A	A	C	B

FIGURE 4B9. Cableways for A.C. propulsion cables (surface ships) – two groups and two tiers.

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METHOD 4B-10-1
TWO GROUPS AND THREE TIERS OF CABLES
UNDER DECK GRATINGS
(SEE NOTE 1 FOR CABLE ARRANGEMENTS AND PHASE RELATIONSHIP)



NOTE FOR DRIP SHIELD: ROLL EDGES OF DRIP SHIELD SO THAT DRAIN OFF WILL BE AT LOWEST POINT.

FIGURE 4B10. Cableways for A.C. propulsion cables (surface ships) – two groups and three tiers.

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METHOD 4B-11-1
CABLE RUNS ACROSS STEEL BEAMS
SHOWING DRIP SHIELD UNDER STEAM PIPE

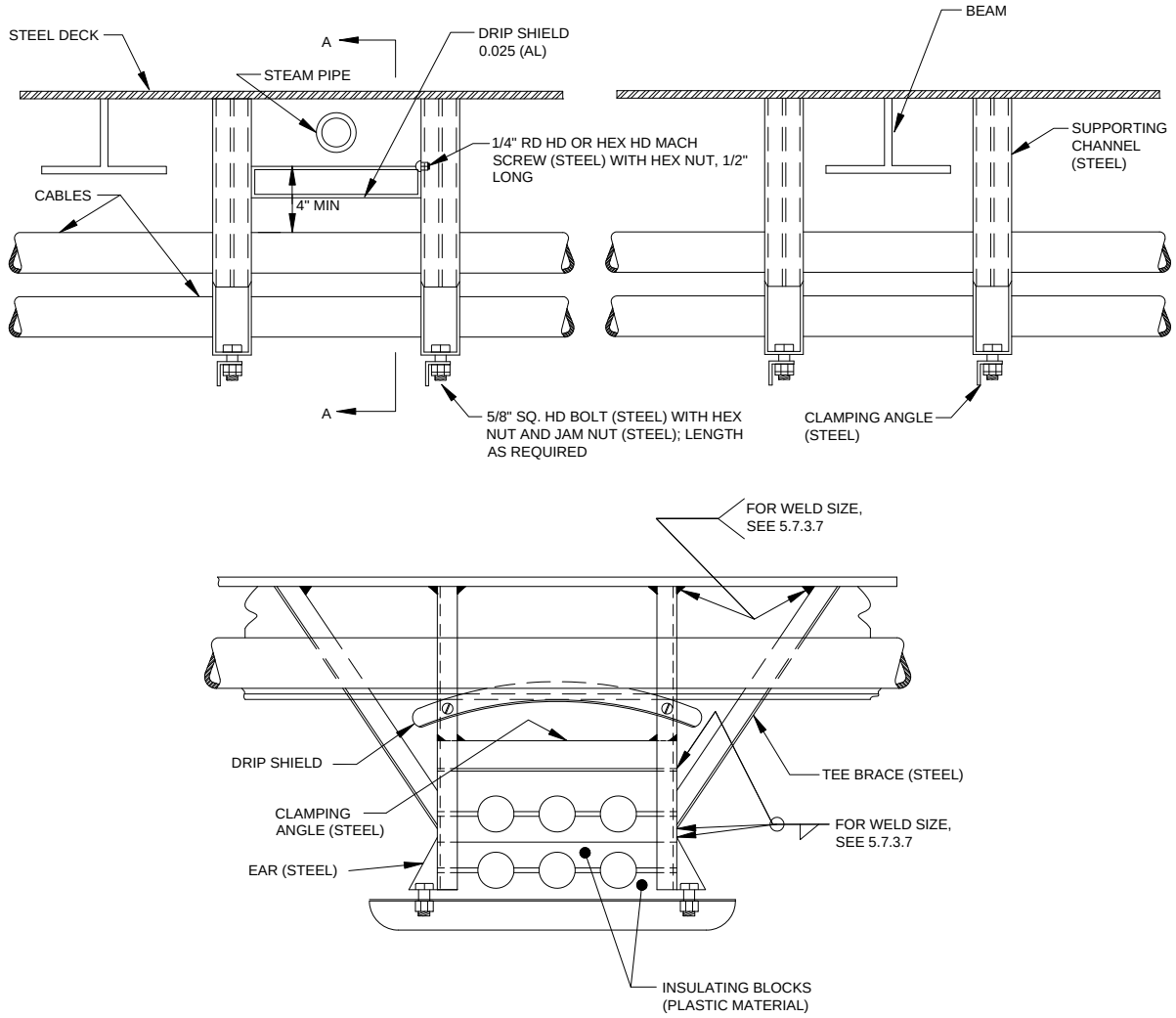
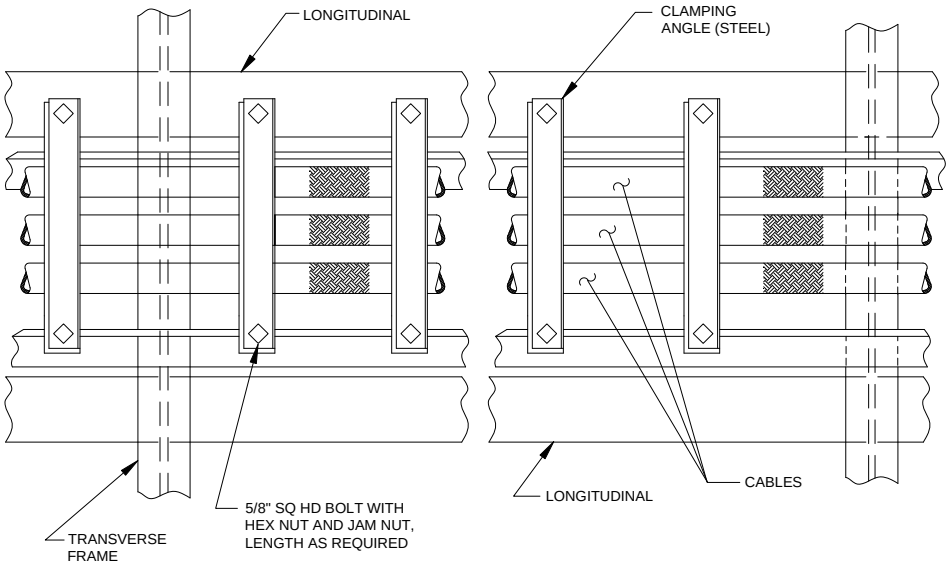


FIGURE 4B11. A.C. propulsion cableway across steel beams (surface ships).

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METHOD 4B-12-1
THREE CABLES ON TRANSVERSE FRAMES



END VIEW FOR METHOD 4B-12-1
THREE CABLES ON TRANSVERSE FRAMES

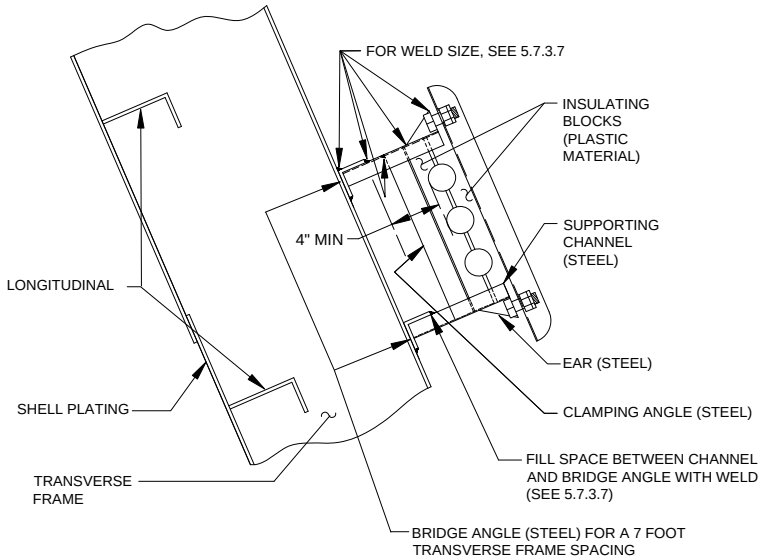


FIGURE 4B12. A.C. propulsion cableway (surface ships) – three cables on transverse frames.

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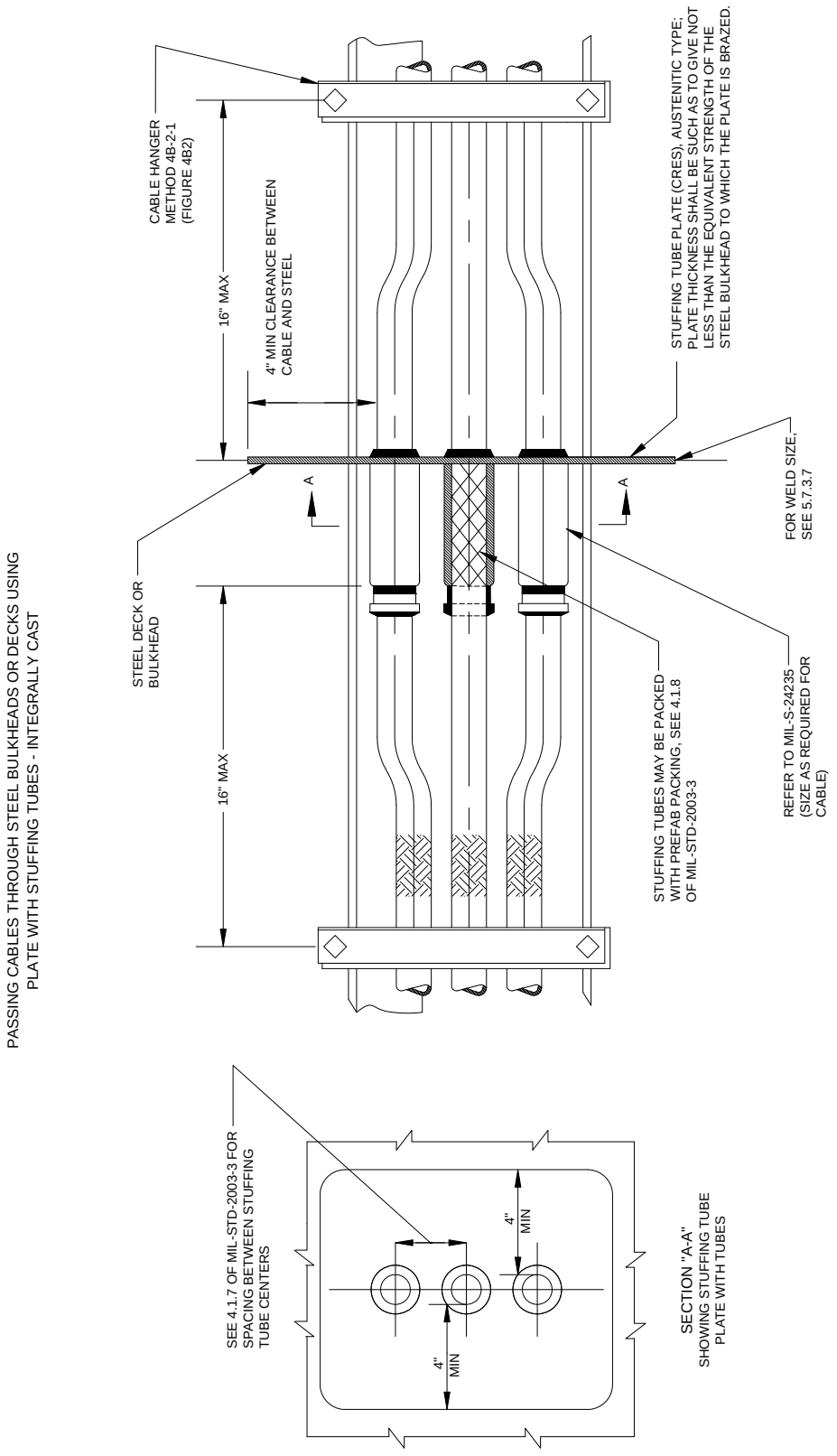


FIGURE 4B13. Cableways for A.C. propulsion cables – using stuffing tubes, integrally cast.

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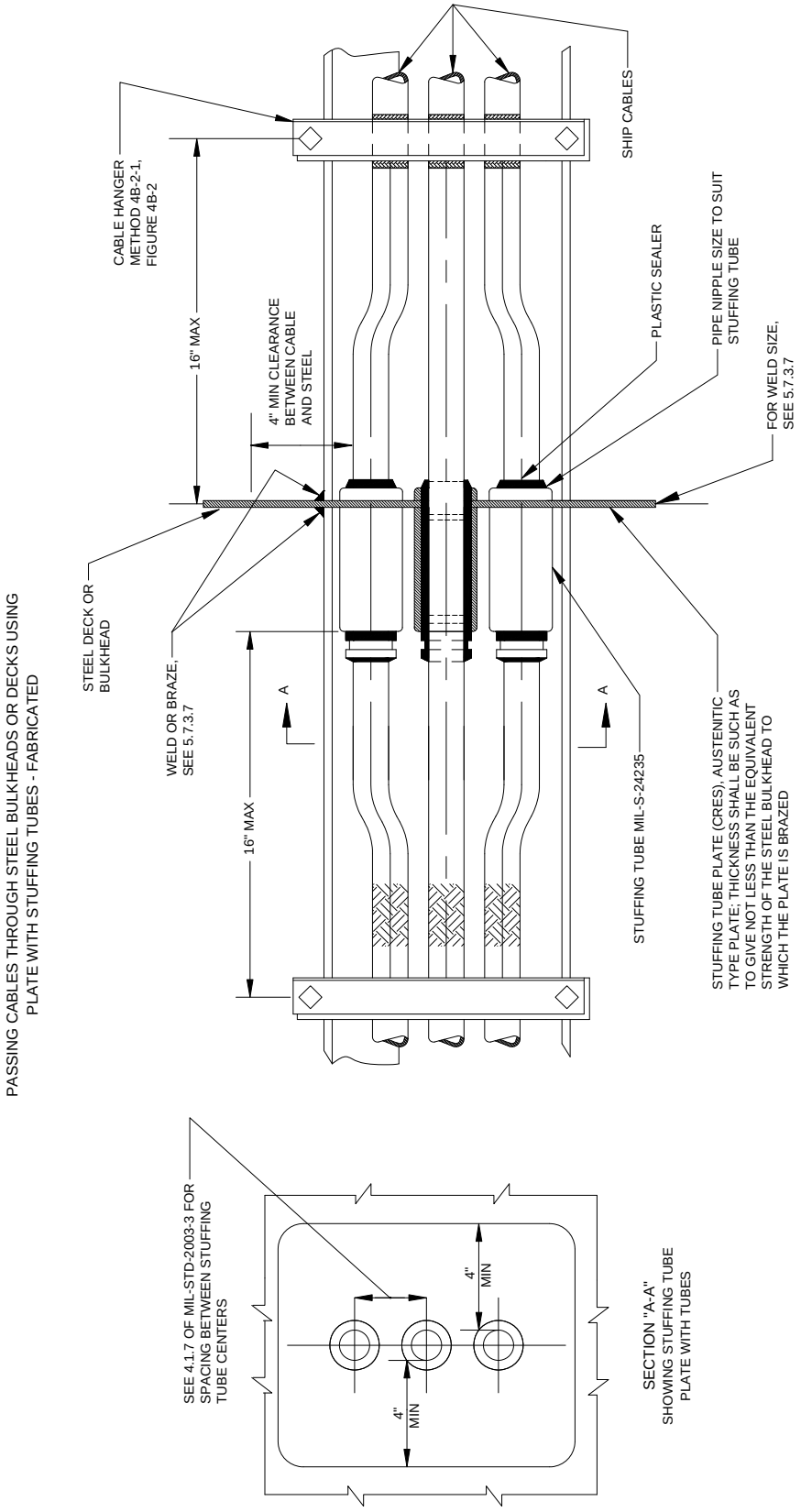
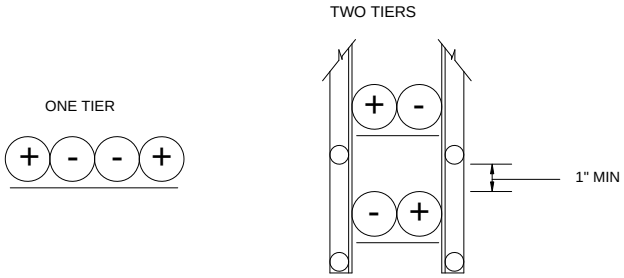


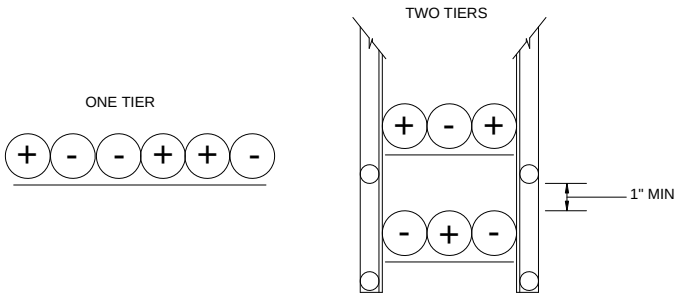
FIGURE 4B14. Cableways for A.C. propulsion cables – using stuffing tubes, fabricated.

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METHOD 4B-15-1
ARRANGEMENTS FOR TWO SINGLE CONDUCTOR CABLES PER LEG



METHOD 4B-15-2
ARRANGEMENTS FOR THREE SINGLE CONDUCTOR CABLES PER LEG



METHOD 4B-15-3
ARRANGEMENTS FOR FOUR SINGLE CONDUCTOR CABLES PER LEG

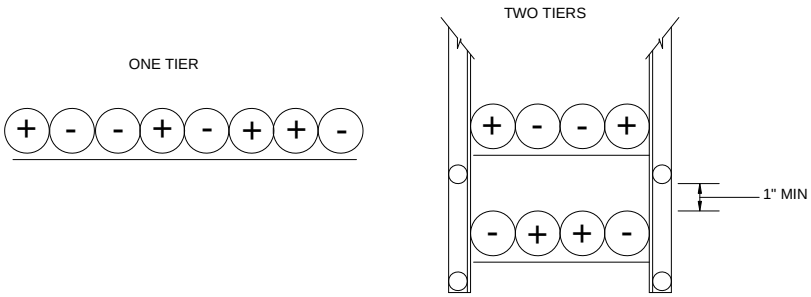
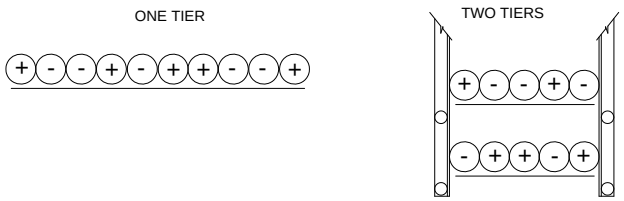


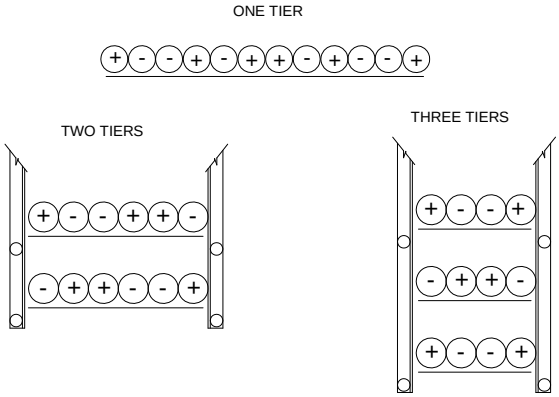
FIGURE 4B15. Cableways for D.C. propulsion cables (surface ships).

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Check the source to verify that this is the current version before use.

METHOD 4B-16-1
ARRANGEMENTS FOR FIVE SINGLE CONDUCTOR CABLES PER LEG



METHOD 4B-16-2
ARRANGEMENTS FOR SIX SINGLE CONDUCTOR CABLES PER LEG



METHOD 4B-16-3
ARRANGEMENTS FOR SEVEN SINGLE CONDUCTOR CABLES PER LEG

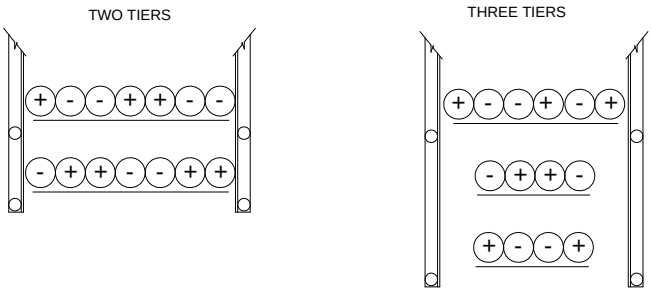


FIGURE 4B16. Cableways for D.C. propulsion cables (surface ships).

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Check the source to verify that this is the current version before use.

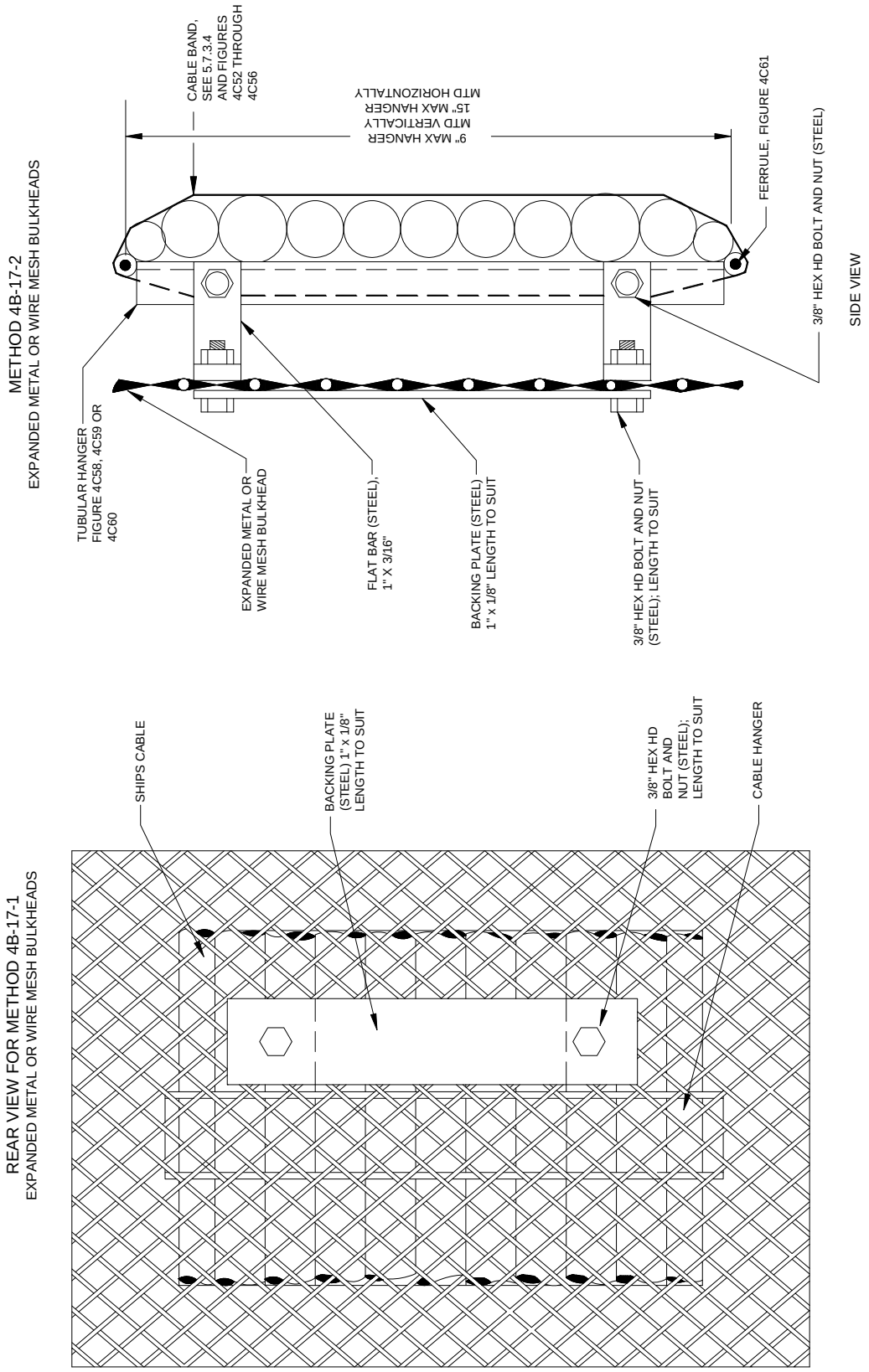


FIGURE 4B17. Single cableways on non-structural bulkhead (surface ships).

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METHOD 4B-18-1
WATERTIGHT DECK OR BULKHEAD WITH INSULATION AND SHEATHING
CABLES SECURED TO STUD PADS OR TAP PADS

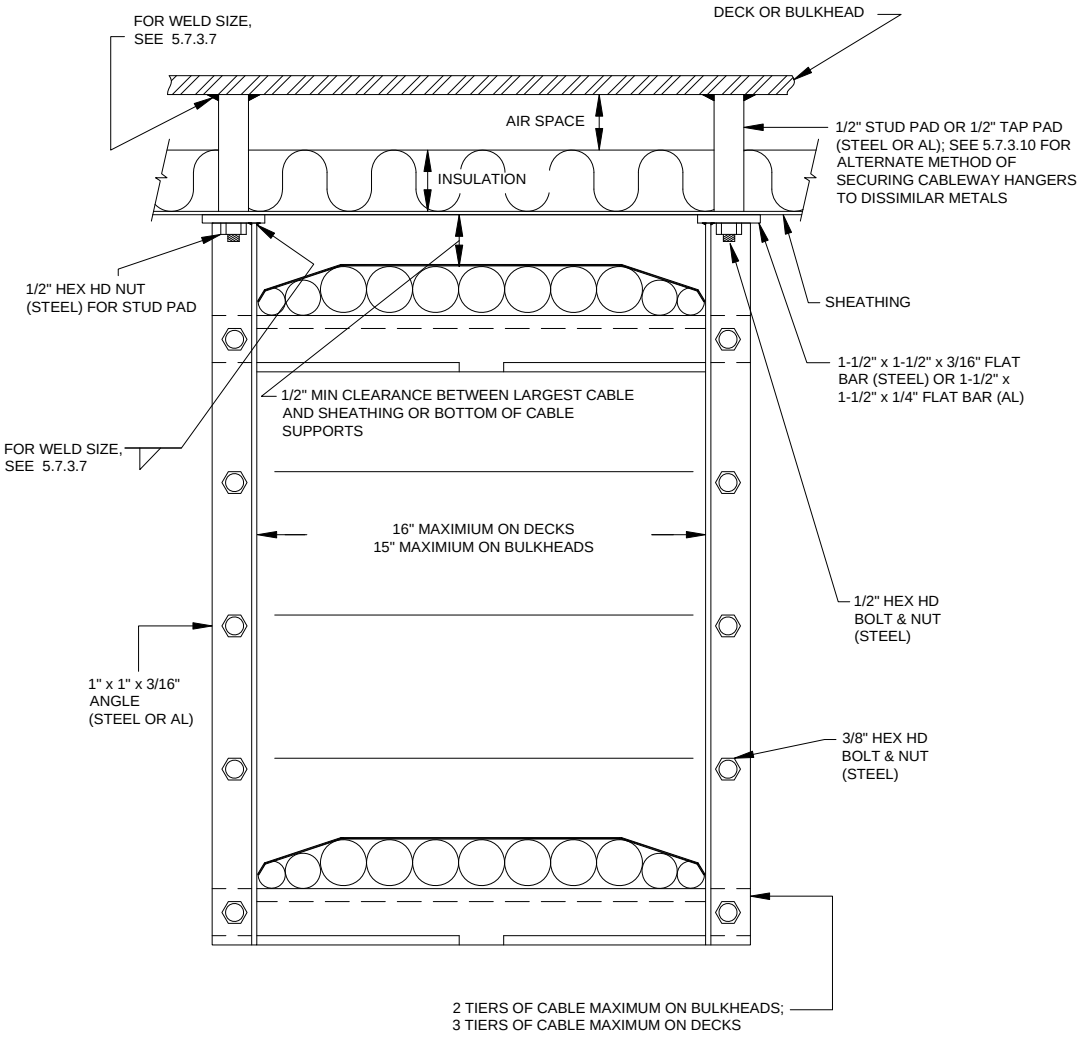


FIGURE 4B18. Cableways in acoustical locations (surface ships).

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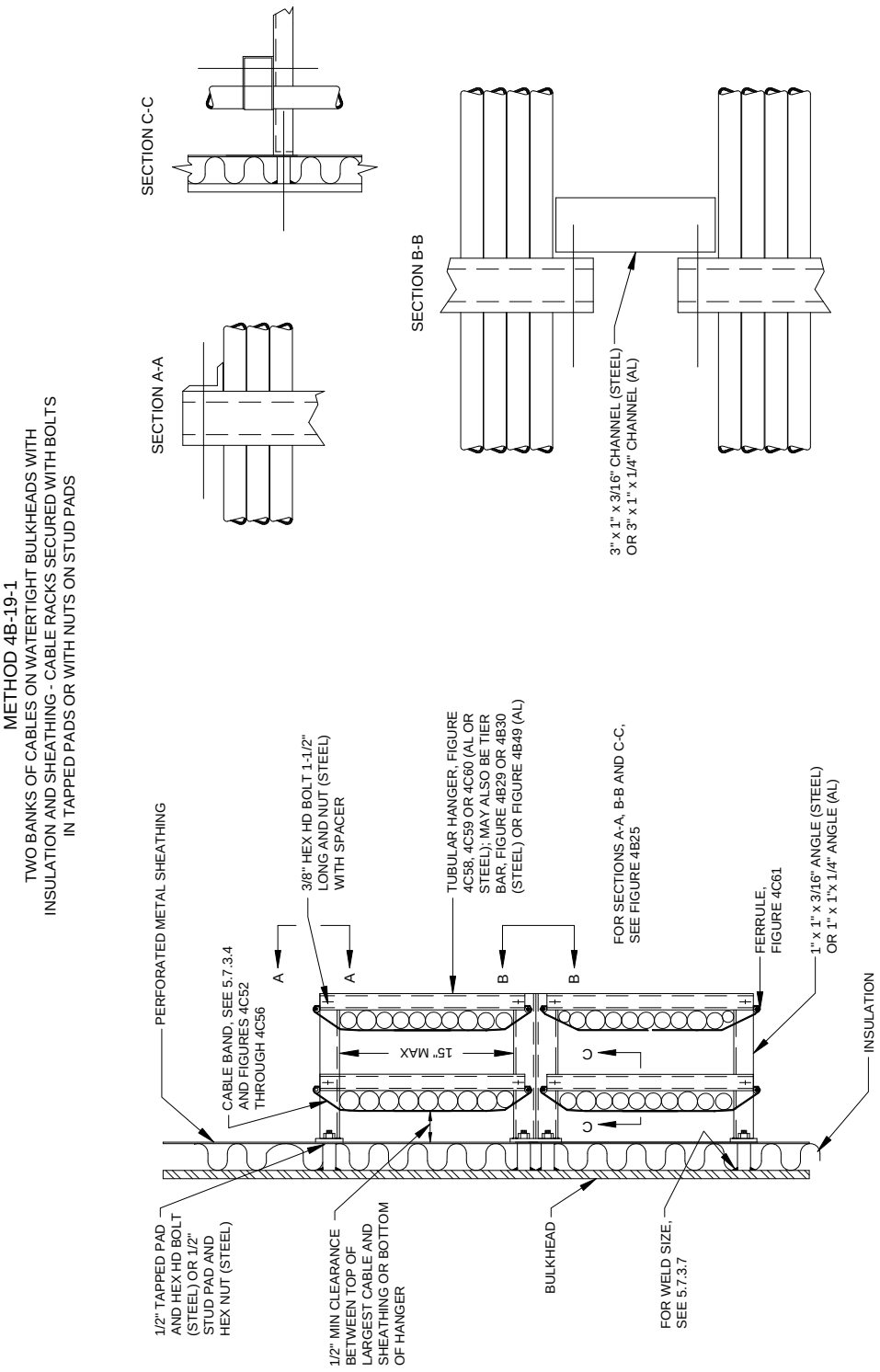
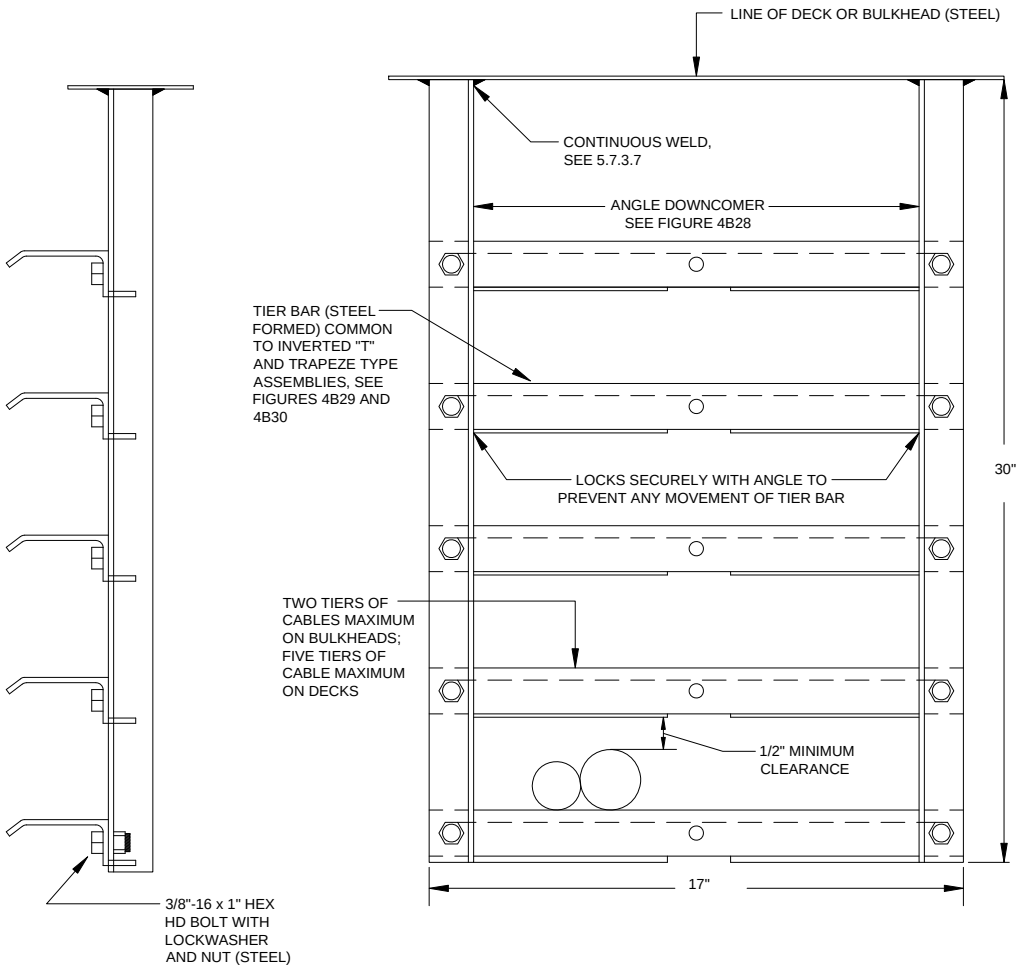


FIGURE 4B19. Cableways in acoustical locations (surface ships).

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METHOD 4B-20-1
TRAPEZE TYPE CABLE HANGER FOR MAIN WIREWAYS (STEEL)
SEE FIGURE 4B21 FOR ATTACHING STEEL HANGERS TO ALUMINUM DECKS OR BULKHEADS



NOTES:

1. Spacing, number of tiers, and cable weight shall not exceed the values specified in table 4BV. Downward extension to the lesser condition of fewer tiers for hanger assemblies (with cable load not exceeding the maximum per tier as specified) is acceptable.
2. Sharp corners shall be slightly rounded. All parts shall have smooth surfaces.
3. Contact areas between adapter, downcomer, and attaching bolts shall be painted with two coats of MIL-PRF-23236 primer.
4. See 5.7.3.10 for alternate method of securing cableway hangers to dissimilar metals.
5. Cables may be banded below cable support bars on overhead mounted cable hangers provided they are banded separately from the banding for cables over the support bar and the total weight does not exceed 450 pounds per cable hanger.

FIGURE 4B20. Multiple cableways on steel decks and bulkheads (surface ships).

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NOTES (continued):

6. Spacing and maximum number of tiers for horizontal and vertical cable runs are as follows:

Cable run	Assembly		Center-to-center spacing (inches)	Maximum number of tiers	Maximum cable weight (pounds) supported by each tier
	Mounting type	Dimensions (inches)			
Horizontal	Deck	36 x 17	16	3	45.5
Horizontal	Deck	36 x 17	32	5	91
Vertical	Bulkhead	20 x 17	24	2	68
Vertical	Bulkhead	20 x 17	24	2	45.5

FIGURE 4B20. Multiple cableways on steel decks and bulkheads (surface ships) - Continued.

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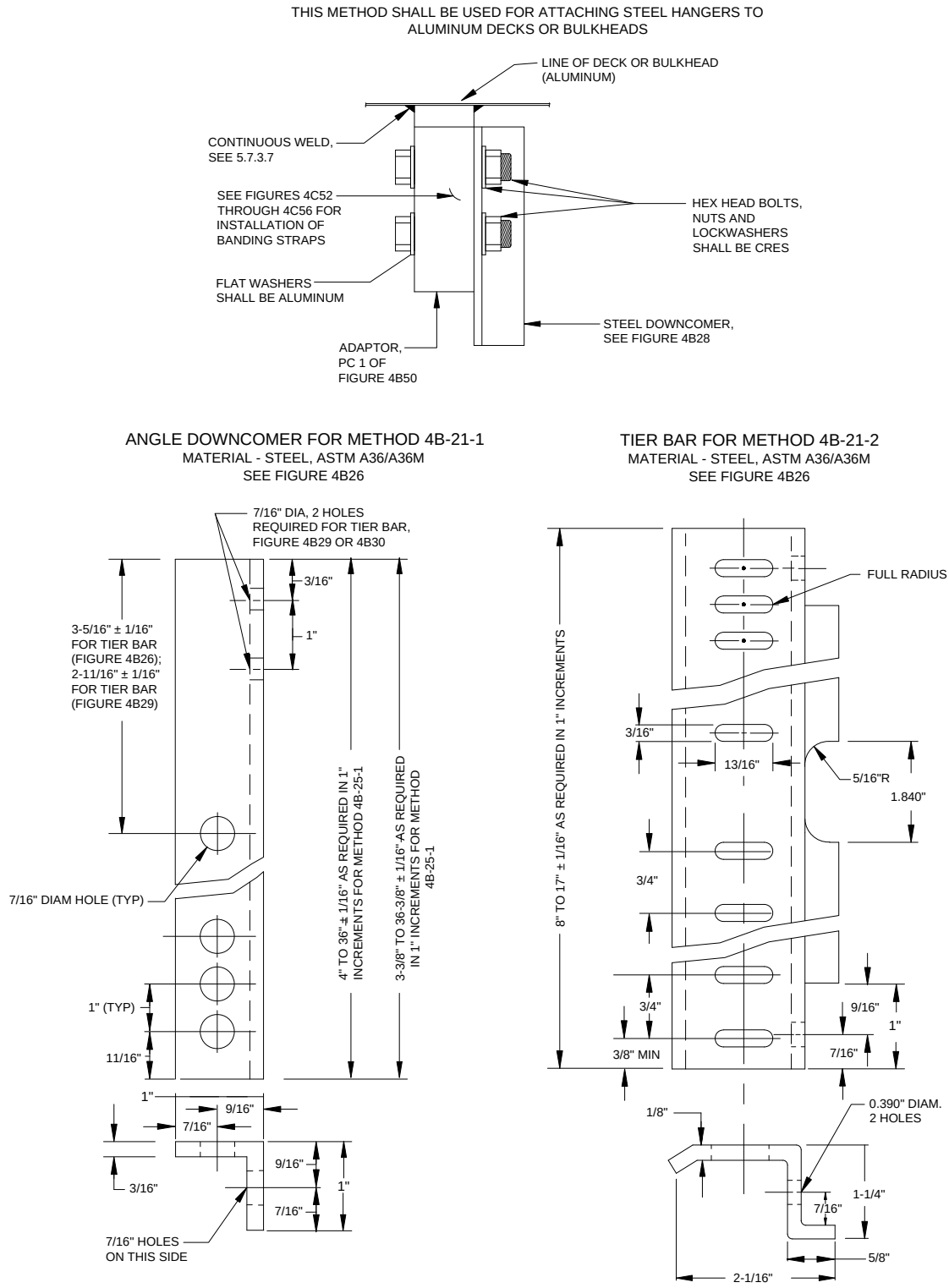


FIGURE 4B21. Multiple cableways on steel decks and bulkheads (surface ships).

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TIER BAR
ALTERNATE DESIGN
MATERIAL - STEEL, ASTM A36/A36M OR A1011/A1011M
SEE FIGURE 4B26

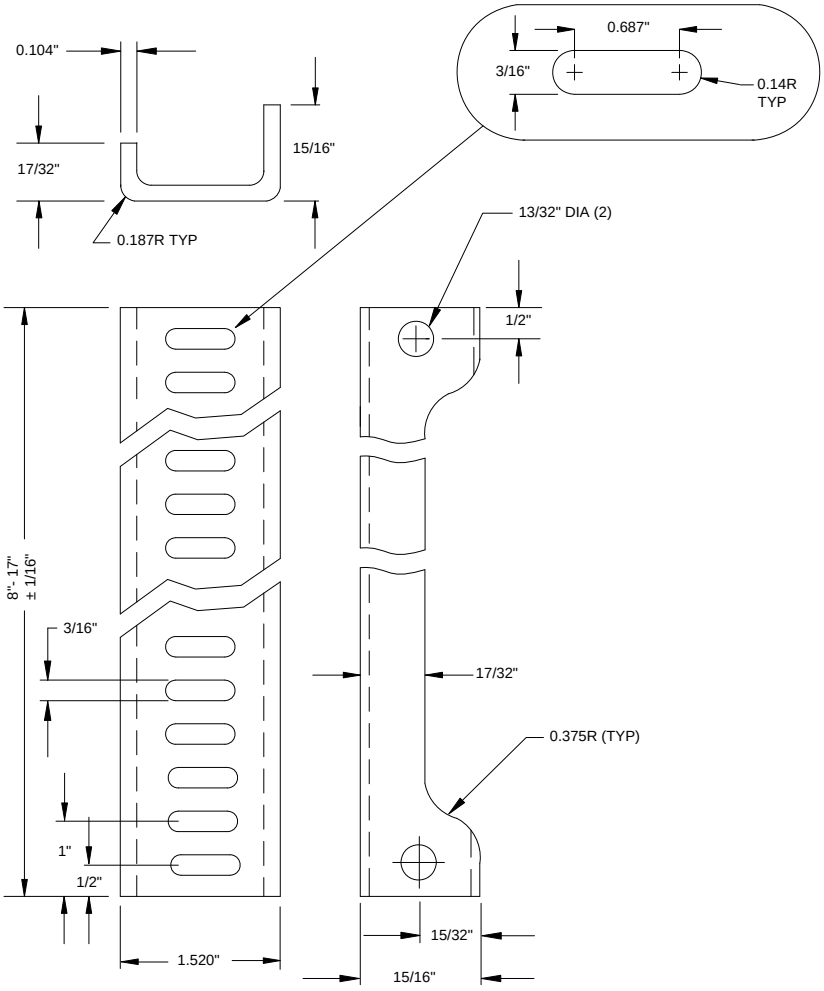
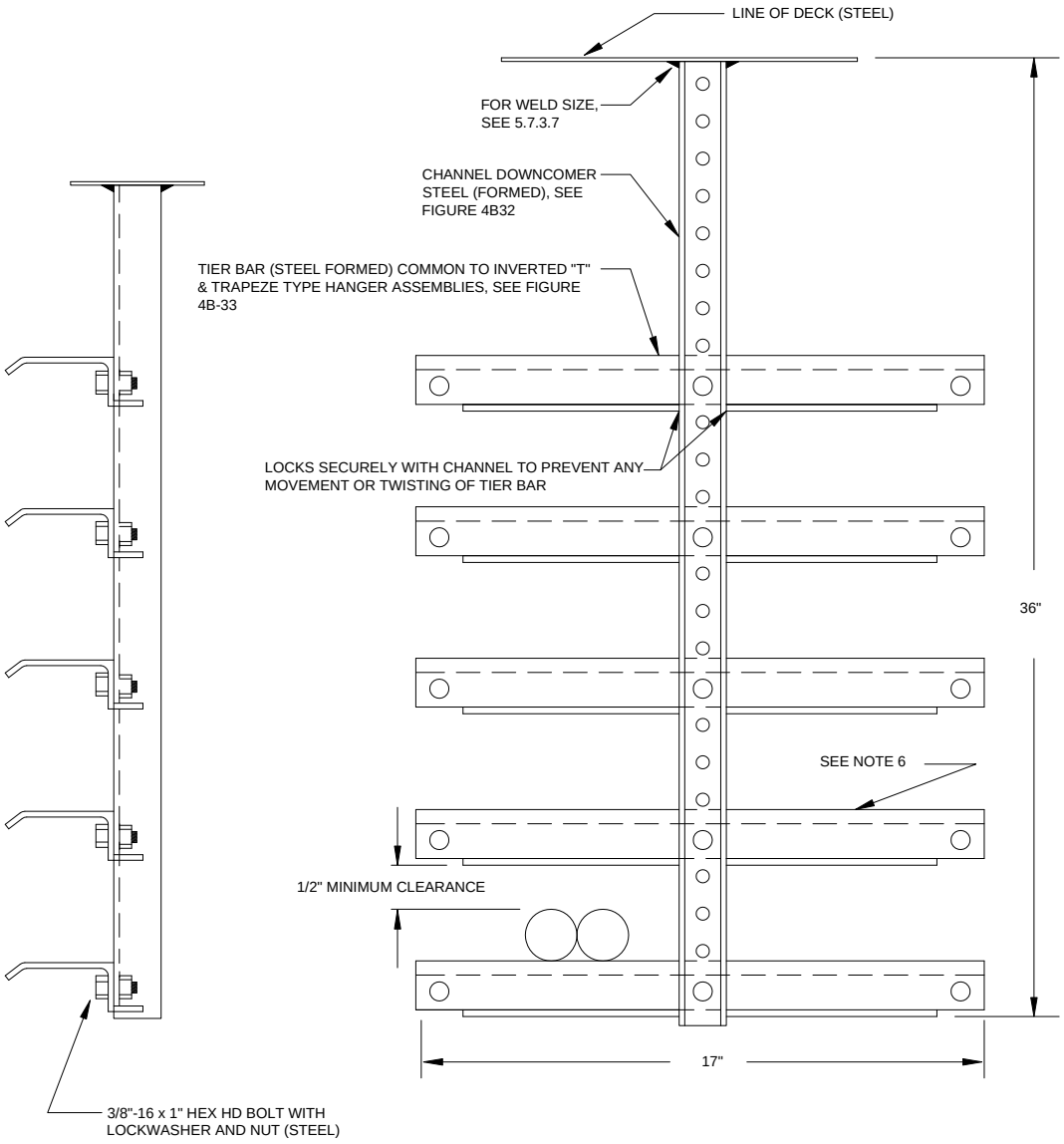


FIGURE 4B22. Tier bar (alternate design) for figure 4B26.

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METHOD 4B-23-1
INVERTED "T" TYPE CABLE HANGER FOR MAIN WIREWAYS (STEEL)



NOTES:

1. Spacing, number of tiers, and cable weight shall not exceed the values specified in table 4BV. Downward extension to the lesser condition of fewer tiers for hanger assemblies (with cable load not exceeding the maximum per tier as specified) is acceptable.
2. Sharp corners shall be slightly rounded. All parts shall have smooth surfaces.
3. Contact areas between adapter, downcomer, and attaching bolts shall be painted with two coats of MIL-PRF-23236 primer.
4. See 5.7.3.10 for alternate method of securing cableway hangers to dissimilar metals.
5. Cables may be banded below cable support bars on overhead mounted cable hangers provided they are banded separately from the banding for cables over the support bar and the total weight does not exceed 450 pounds per cable hanger.

FIGURE 4B23. Multiple cableways on steel decks and bulkheads (surface ships).

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Check the source to verify that this is the current version before use.

NOTES (continued):

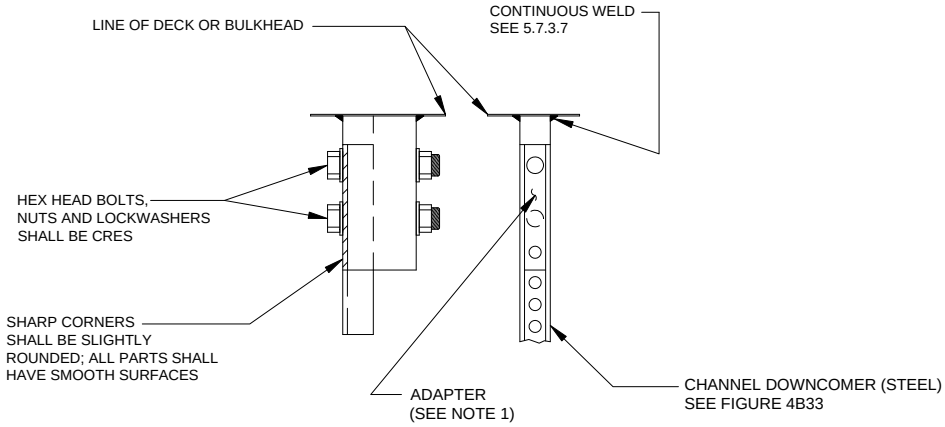
6. Spacing and maximum number of tiers for horizontal and vertical cable runs are as follows:

Cable run	Assembly		Center-to-center spacing (inches)	Maximum number of tiers	Maximum cable weight (pounds) supported by each tier
	Mounting type	Dimensions (inches)			
Horizontal	Deck	36 x 17	16	3	45.5
Horizontal	Deck	36 x 17	32	5	91
Vertical	Bulkhead	20 x 17	24	2	68
Vertical	Bulkhead	20 x 17	24	2	45.5

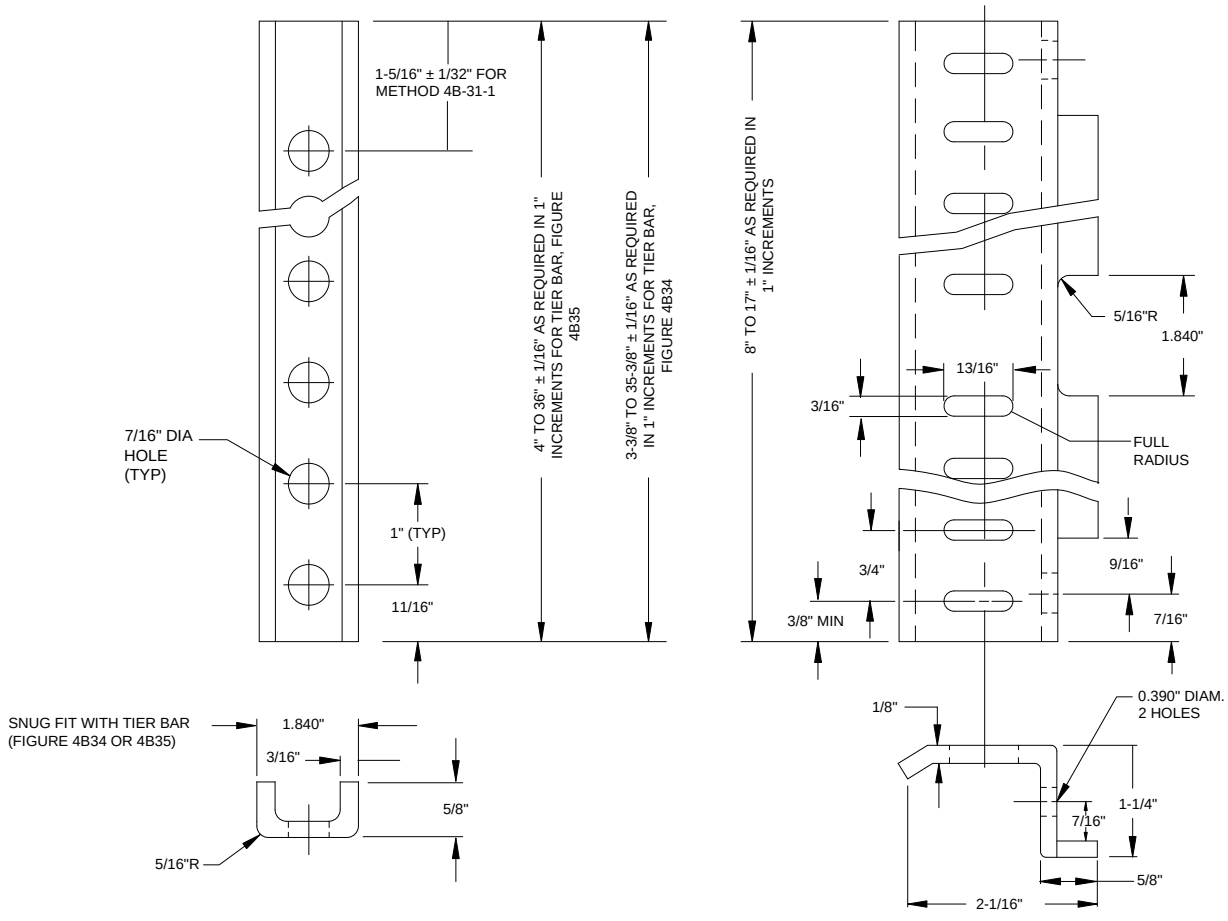
FIGURE 4B23. Multiple cableways on steel decks and bulkheads (surface ships) - Continued.

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METHOD 4B-24-1
THIS METHOD SHALL BE USED FOR ATTACHING STEEL HANGERS TO ALUMINUM DECKS



CHANNEL DOWNCOMER
STEEL, ASTM A36/A36M
SEE FIGURE 4B31



NOTE:

1. Contact areas between adapter, downcomer, and attaching bolts shall be painted with two coats of MIL-PRF-23236 primer.

FIGURE 4B24. Multiple cableways on steel decks (surface ships).

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TIER BAR
ALTERNATE DESIGN
STEEL, ASTM A36/A36M OR A1011/A1011M
SEE FIGURE 4B31

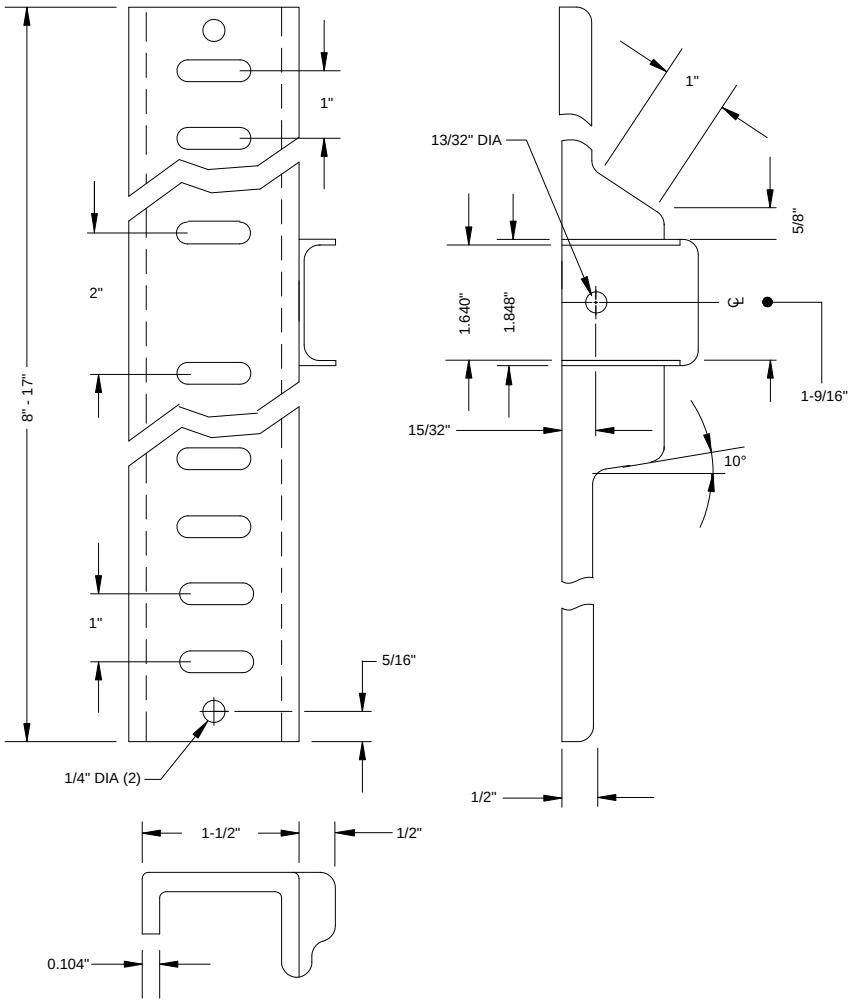


FIGURE 4B25. Tier bar (alternate design for figure 4B24.

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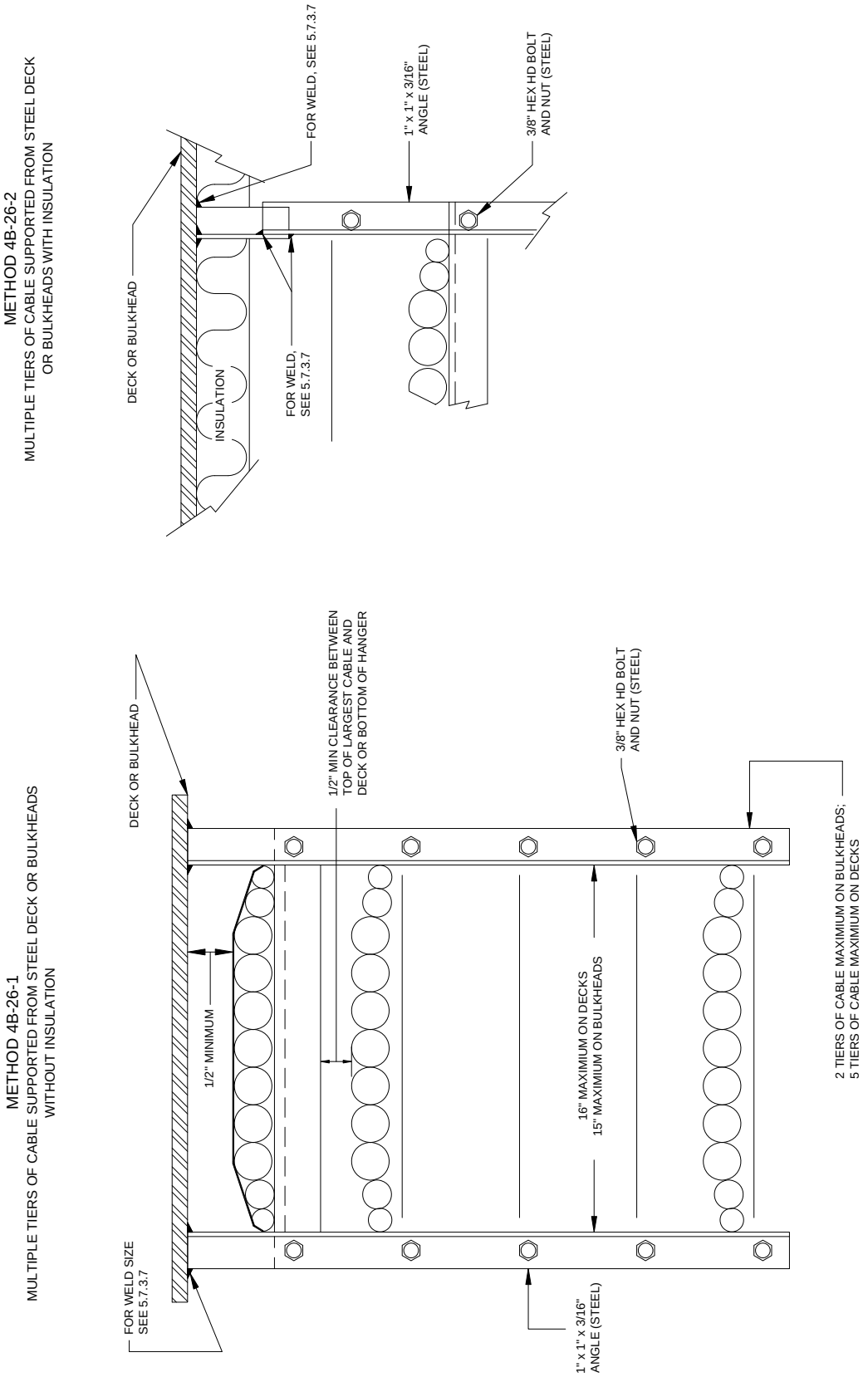


FIGURE 4B26. Multiple cables on steel decks and bulkheads (surface ships).

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Check the source to verify that this is the current version before use.

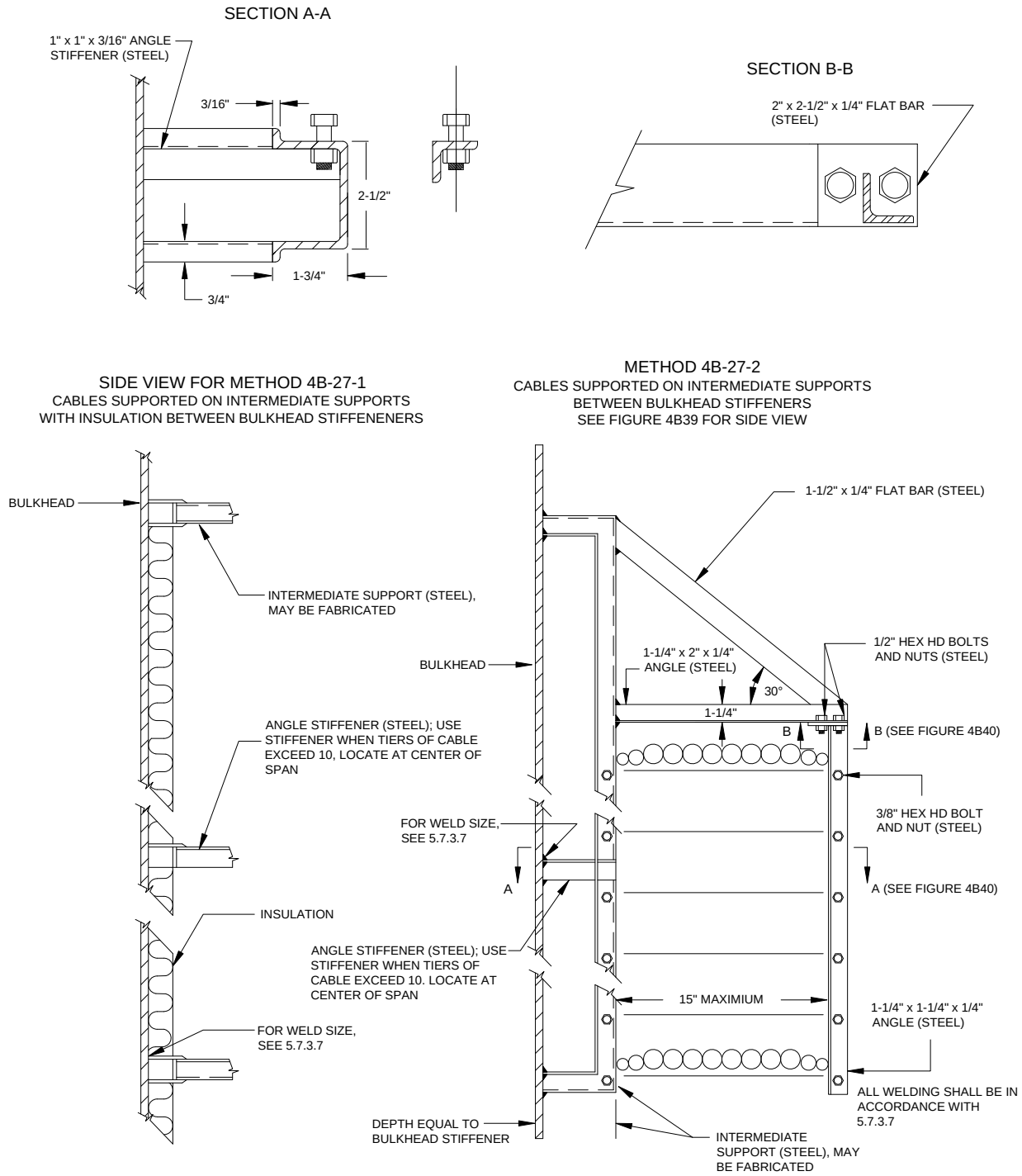


FIGURE 4B27. Multiple cables on steel bulkheads (surface ships).

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METHOD 4B-28-1
CABLES SUPPORTED ON STIFFENER

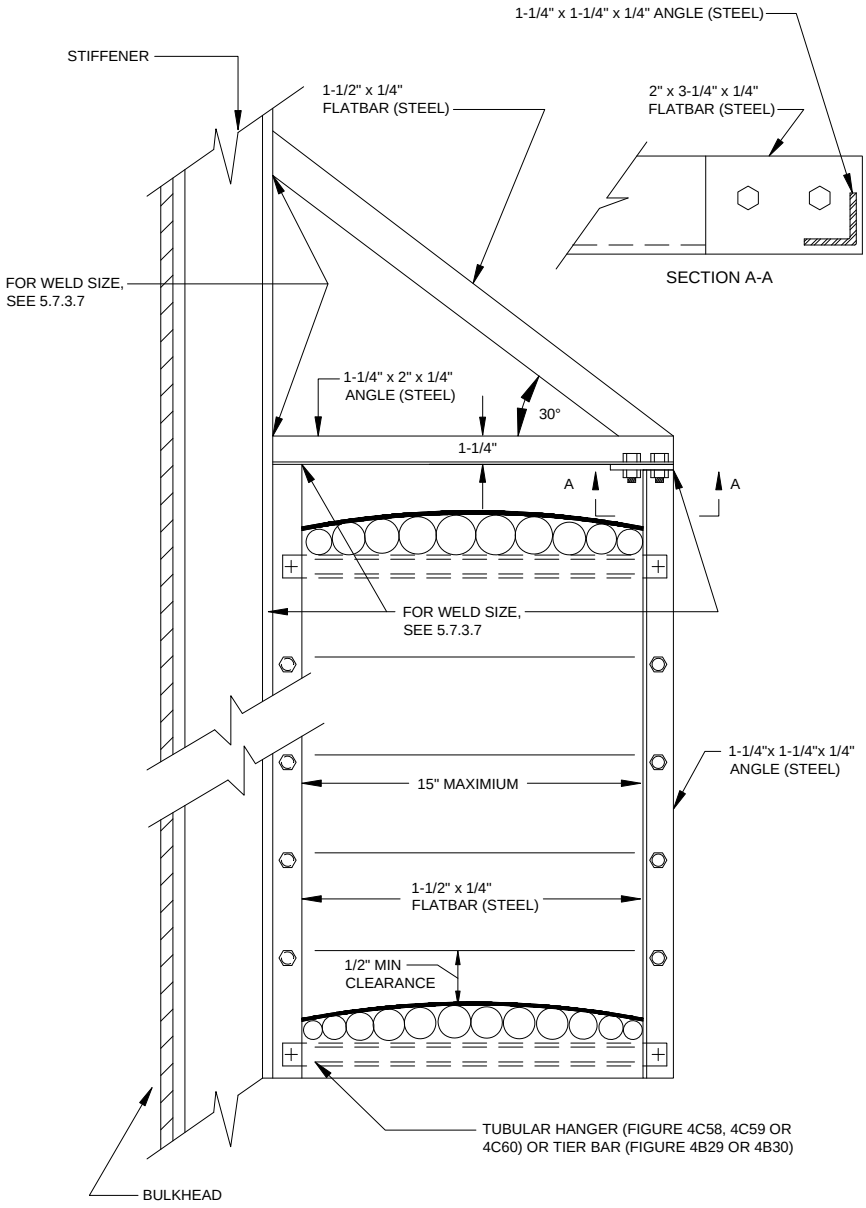


FIGURE 4B28. Multiple cables on steel bulkheads – cables supported on stiffener (surface ships).

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METHOD 4B-29-1
CABLES SUPPORTED ON BULKHEAD WITHOUT INSULATION
(SEE FIGURE 4B41 FOR ADDITIONAL DETAILS)

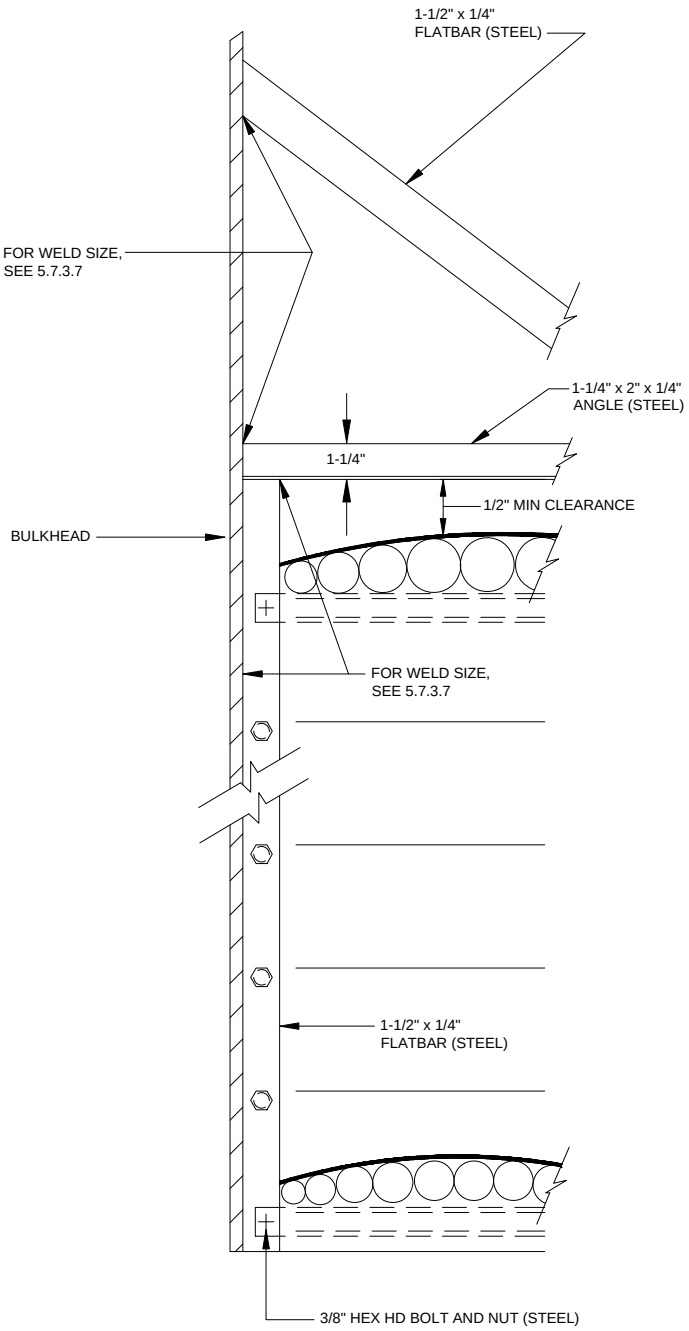


FIGURE 4B29. Multiple cables on steel bulkheads without insulation (surface ships).

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METHOD 4B-30-1
CABLES SUPPORTED ON BULKHEAD WITH INSULATION
(SEE FIGURE 4B41 FOR ADDITIONAL DETAILS)

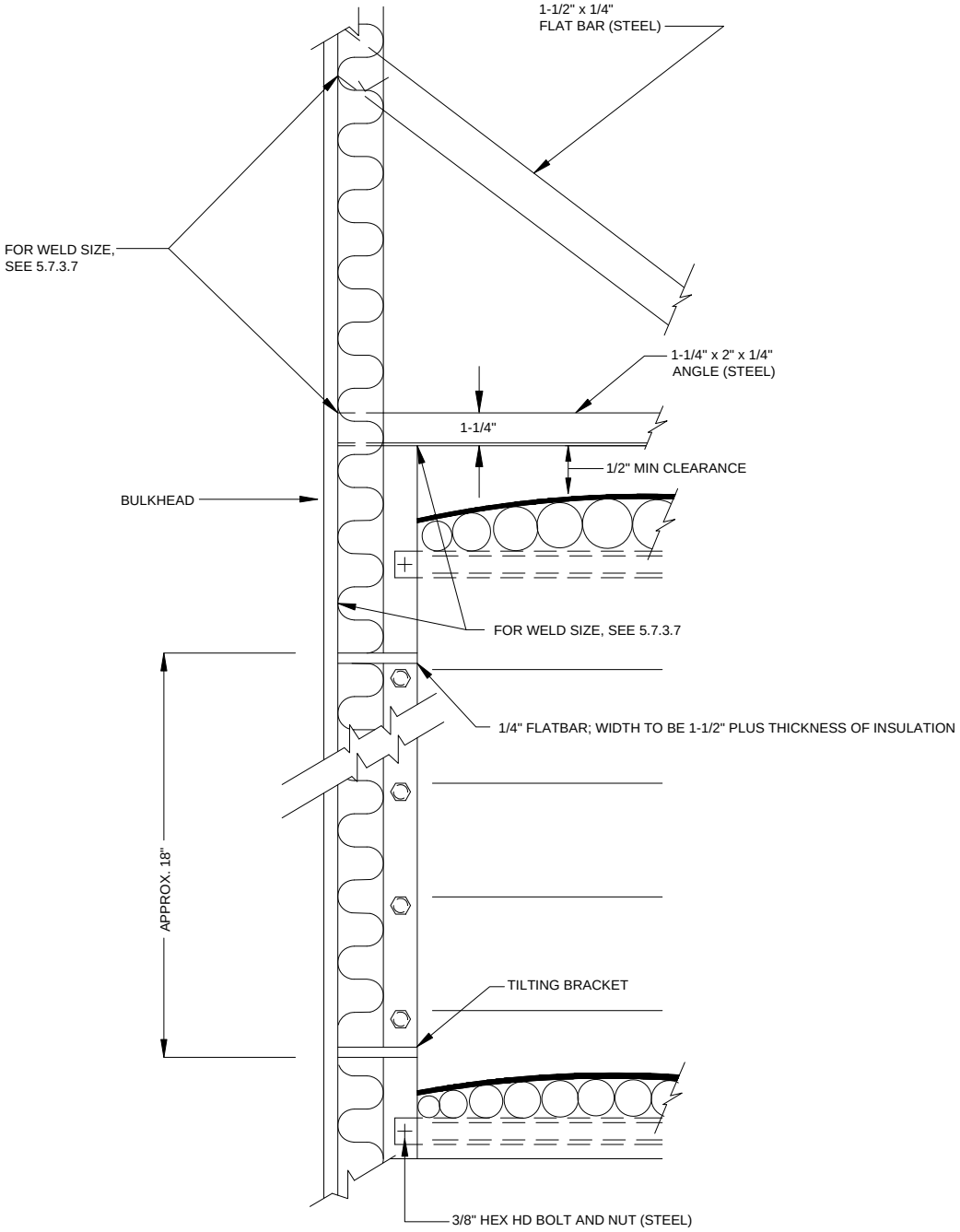


FIGURE 4B30. Multiple cables on steel bulkheads with insulation (surface ships).

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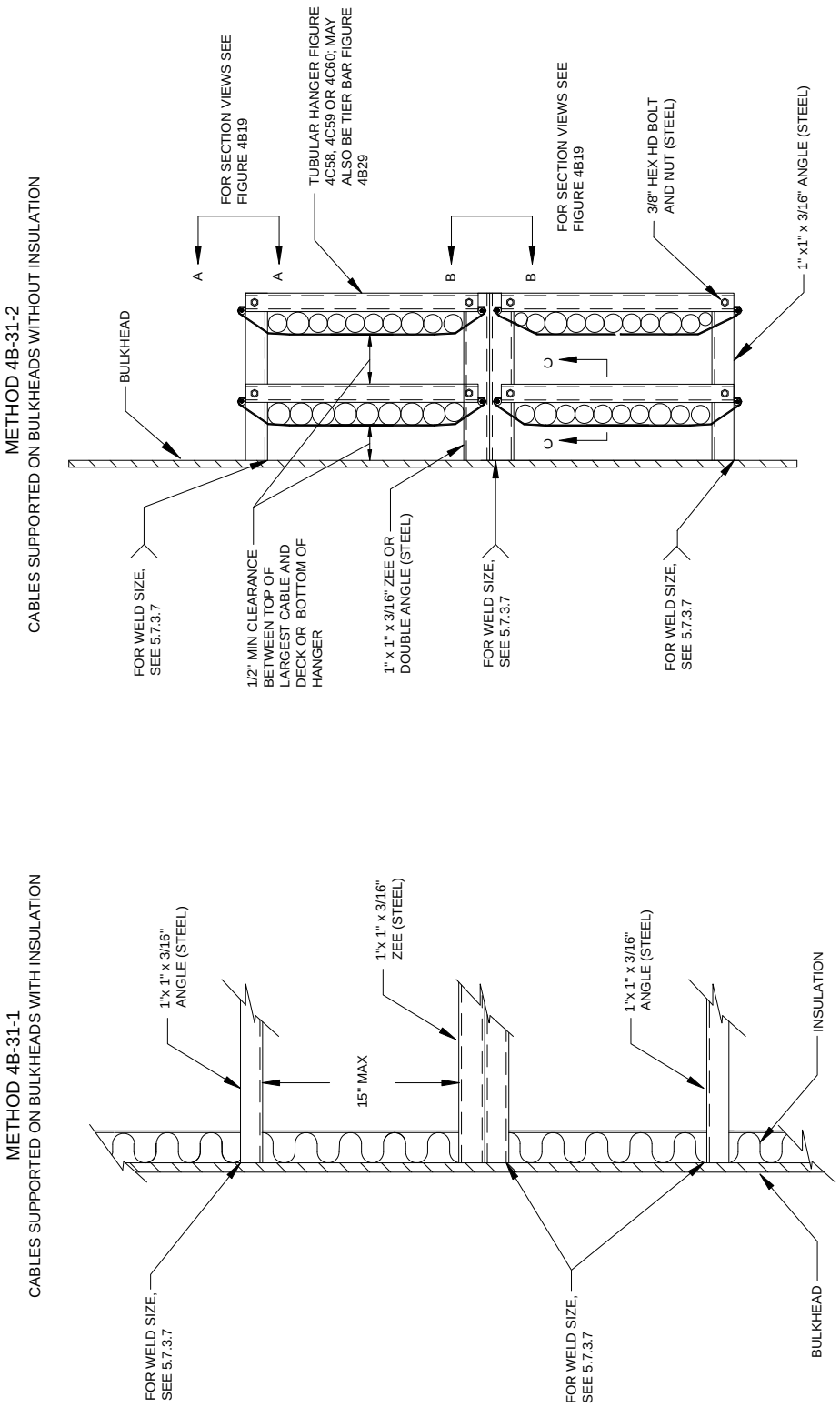
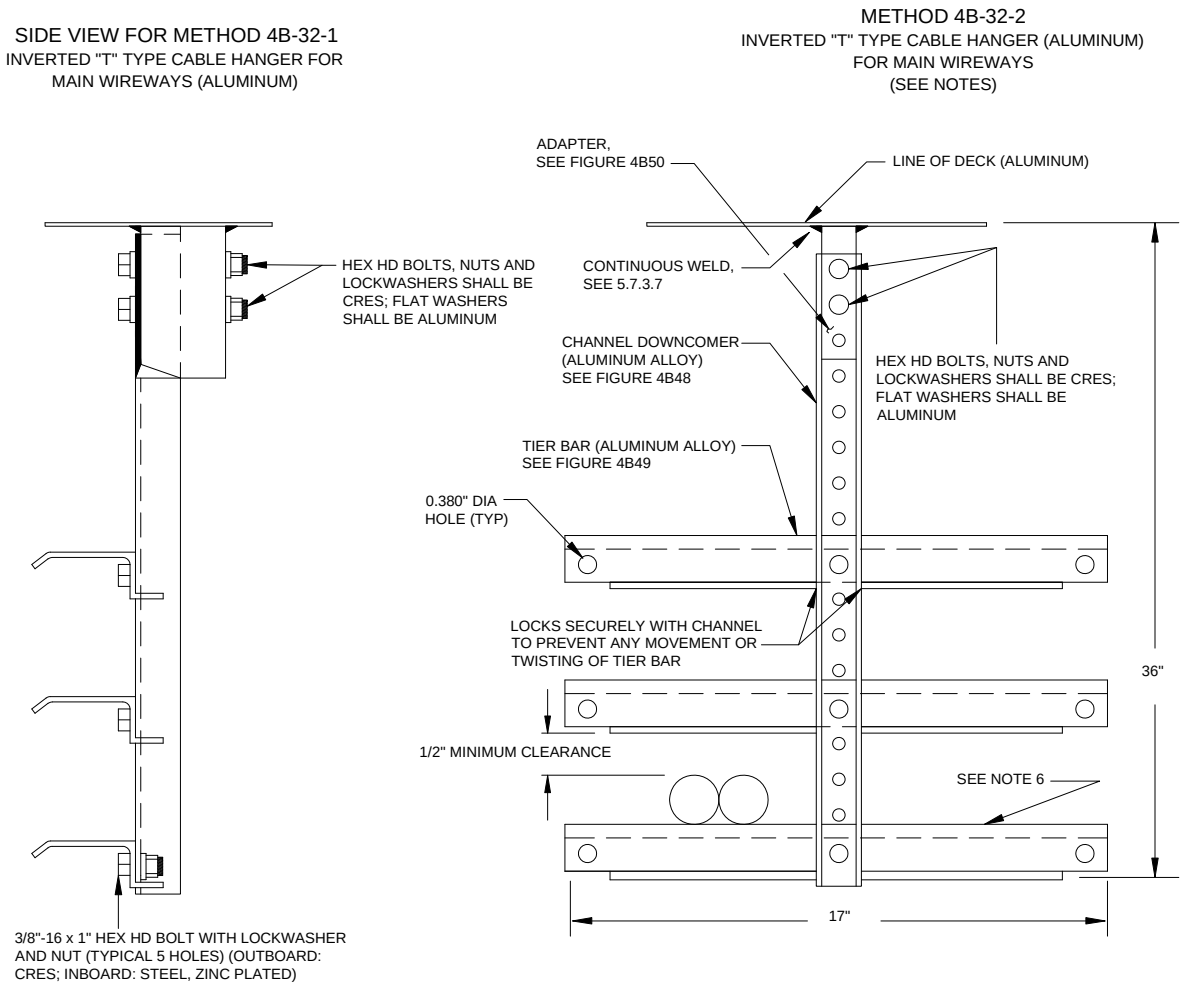


FIGURE 4B31. Multiple cables on steel bulkheads (surface ships).

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NOTES:

1. Spacing, number of tiers, and cable weight shall not exceed the values specified in table 4BV. Downward extension to the lesser condition of fewer tiers for hanger assemblies (with cable load not exceeding the maximum per tier as specified) is acceptable.
2. Sharp corners shall be slightly rounded. All parts shall have smooth surfaces.
3. Contact areas between adapter, downcomer, and attaching bolts shall be painted with two coats of primer in accordance with MIL-PRF-23236.
4. See 5.7.3.10 for alternate method of securing cableway hangers to dissimilar metals.
5. Banding of cabling is required on each inverted “T” cableway hangar.

FIGURE 4B32. Multiple cableways supported from aluminum or steel decks (surface ships).

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NOTES (continued):

6. Cables may be banded below cable support bars on overhead mounted cable hangers provided they are banded separately from the banding for cables over the support bar and the total weight does not exceed 450 pounds per cable hanger.
7. Spacing and maximum number of tiers for horizontal and vertical cable runs are as follows:

Cable run	Assembly		Center-to-center spacing (inches)	Maximum number of tiers	Maximum cable weight (pounds) supported by each tier
	Mounting type	Dimensions (inches)			
Horizontal	Deck	36 x 17	16	3	45.5
Horizontal	Deck	36 x 17	32	5	91
Vertical	Bulkhead	20 x 17	24	2	68
Vertical	Bulkhead	20 x 17	24	2	45.5

FIGURE 4B32. Multiple cableways supported from aluminum or steel decks (surface ships) - Continued.

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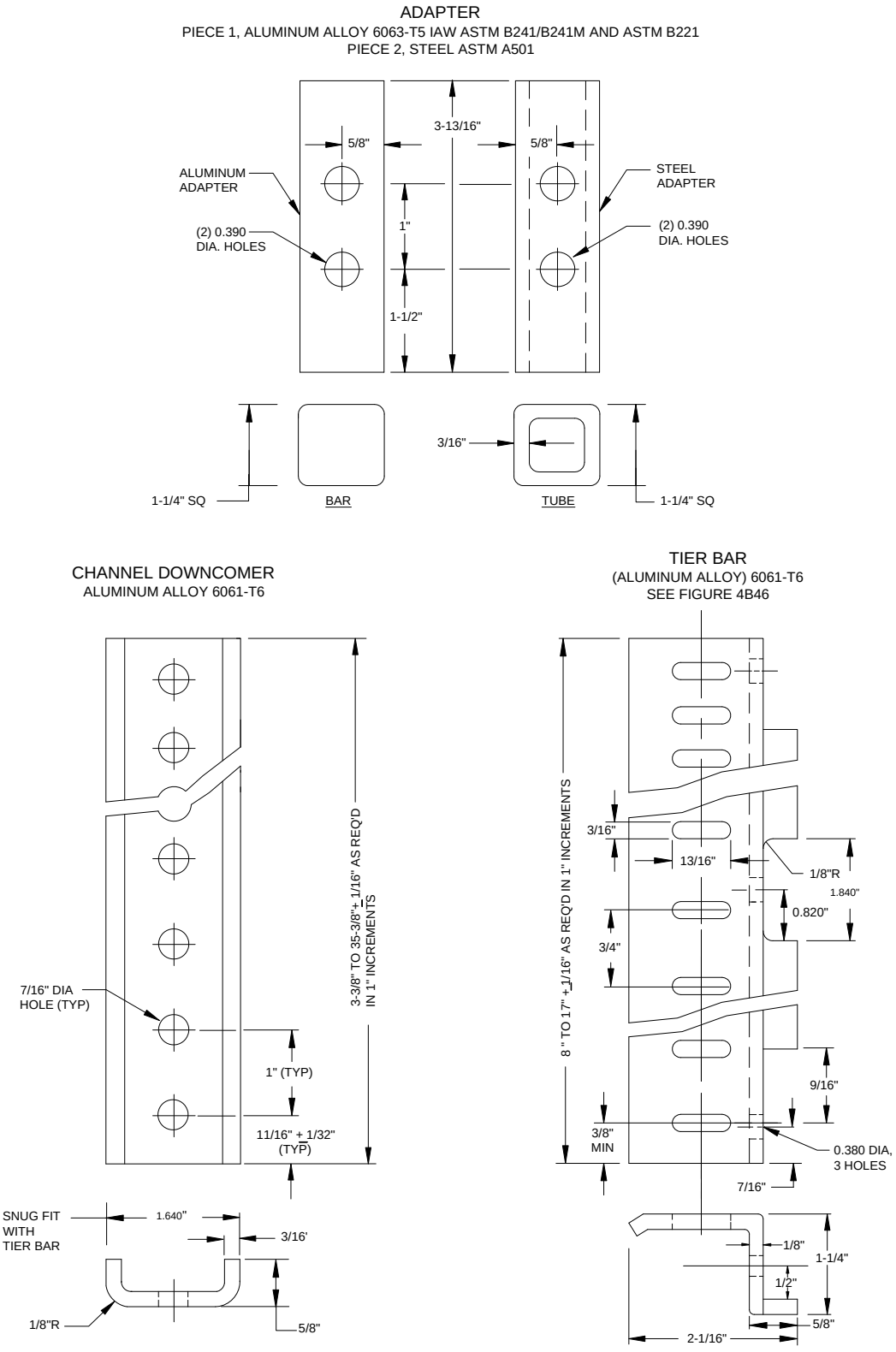


FIGURE 4B33. Multiple cableways supported from aluminum or steel decks (surface ships).

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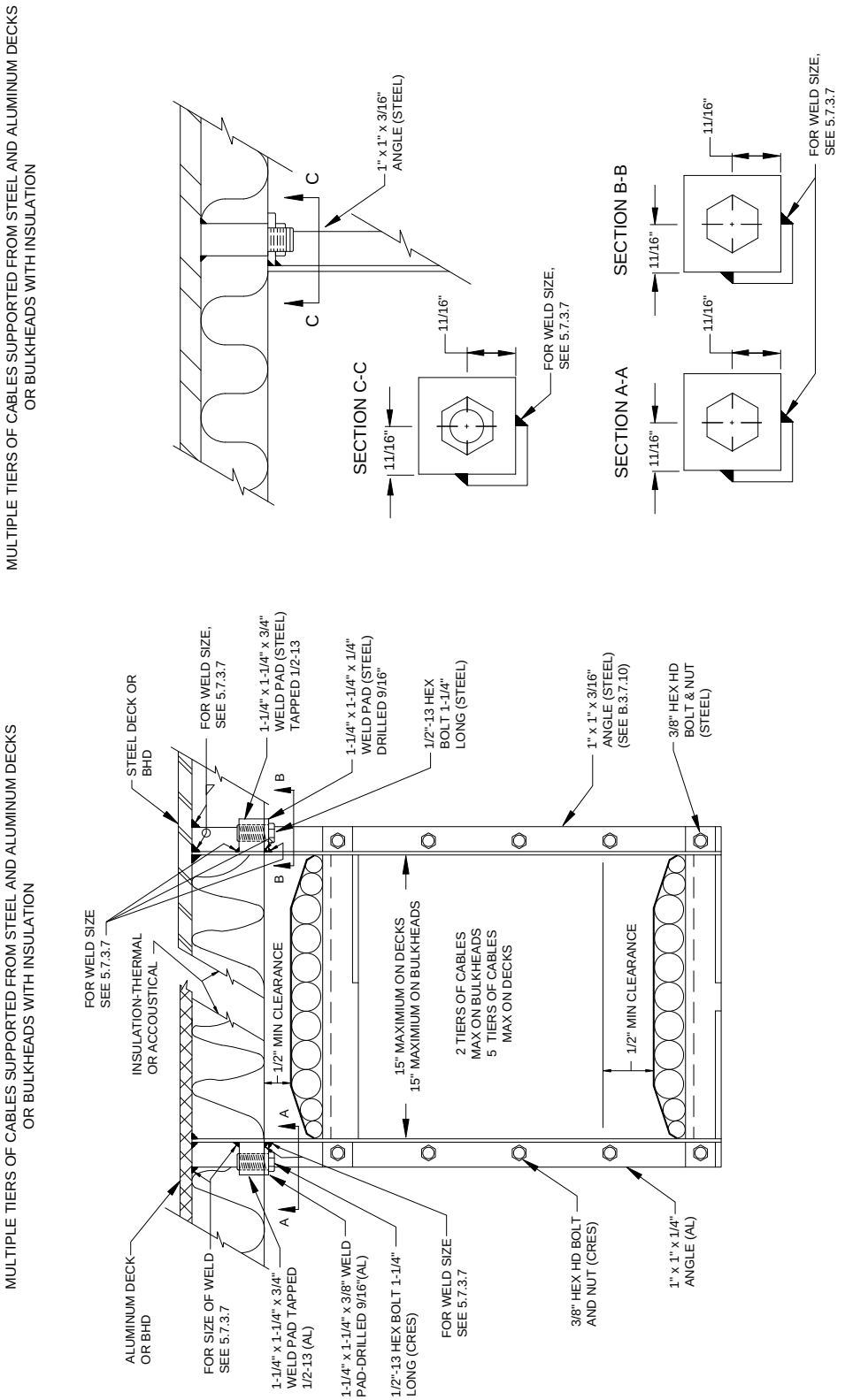


FIGURE 4B34. Multiple cableways on insulated steel or aluminum decks or bulkheads (surface ships).

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METHOD 4B-35-1
TRAPEZE TYPE CABLE HANGER
FOR MAIN WIREWAYS
(ALUMINUM)

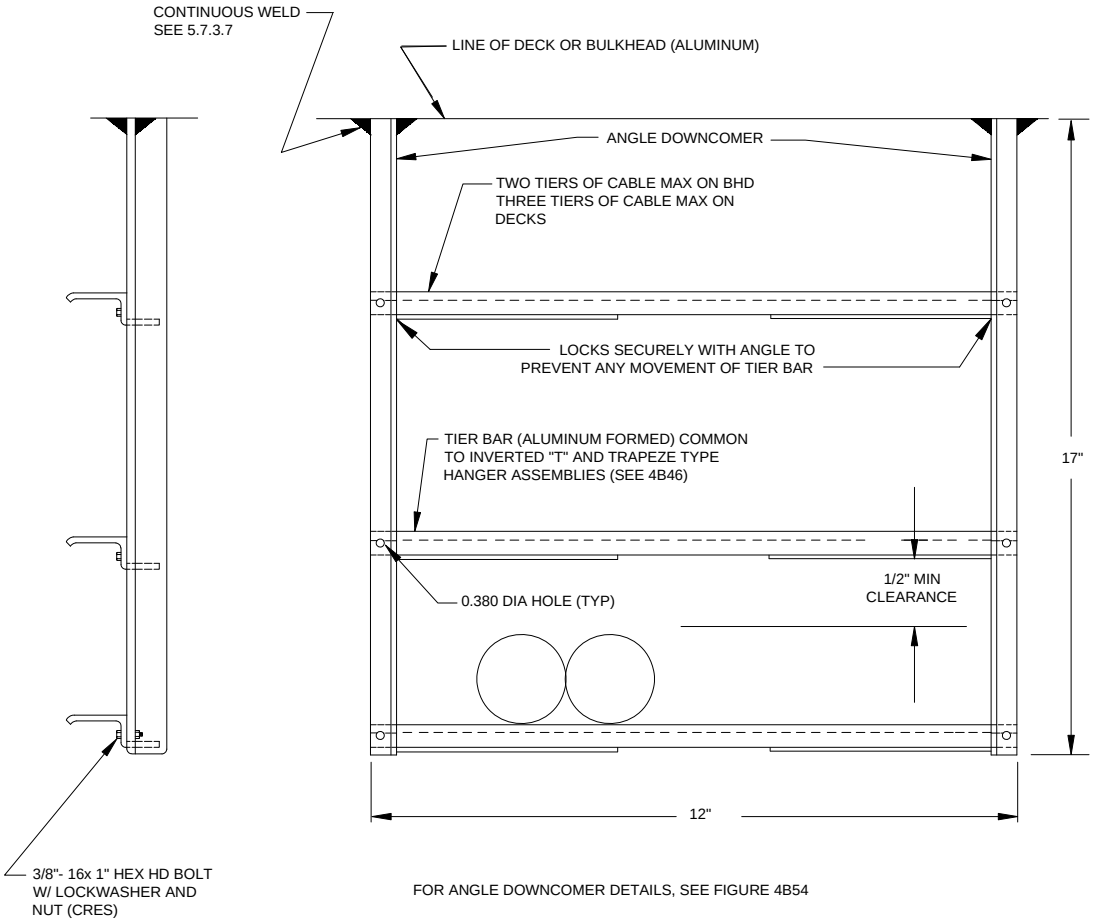


FIGURE 4B35. Multiple cableway supports on aluminum decks and bulkheads (surface ships only).

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ANGLE DOWNCOMER DETAILS FOR METHOD 4B-35
ALUMINUM ALLOY 6061-T6

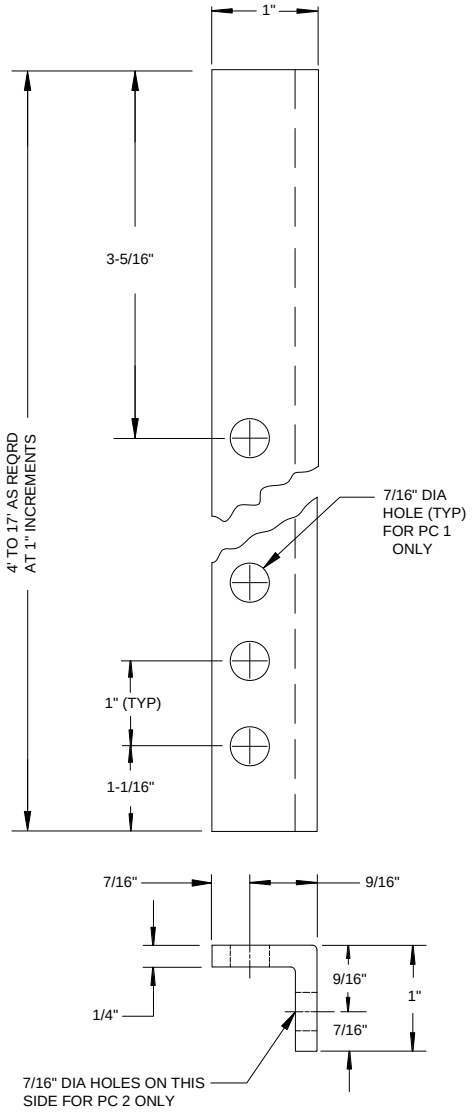
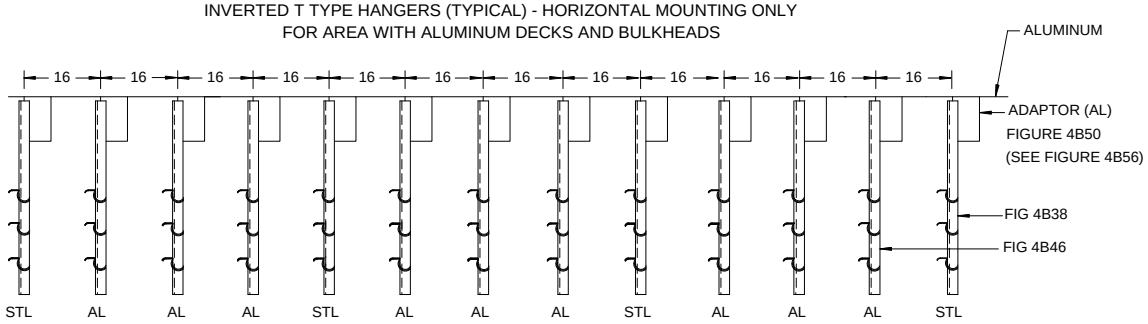


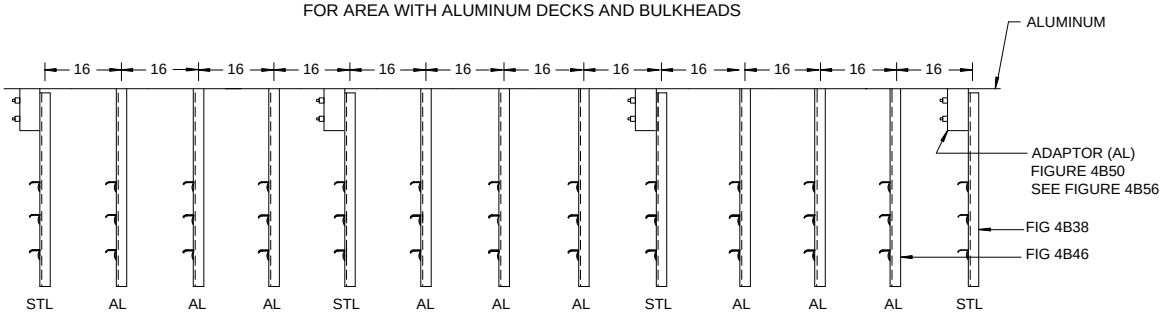
FIGURE 4B36. Angle down comer for figure 4B35.

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METHOD 4B-37-1
INVERTED T TYPE HANGERS (TYPICAL) - HORIZONTAL MOUNTING ONLY
FOR AREA WITH ALUMINUM DECKS AND BULKHEADS



METHOD 4B-37-2
TRAPEZE TYPE HANGERS (TYPICAL) - HORIZONTAL OR VERTICAL MOUNTING
FOR AREA WITH ALUMINUM DECKS AND BULKHEADS



SEE PARAGRAPH 5.7.3.10 FOR ALTERNATE METHOD OF SECURING CABLEWAY HANGERS TO DISSIMILAR METALS

FIGURE 4B37. Spacing of multiple cableway supports for aluminum decks and bulkheads (surface ships only).

METHOD 4B-38
(TYPICAL FOR ALUMINUM DECKS AND BULKHEADS)
SEE 5.7.3.10 FOR ALTERNATE METHODS OF SECURING CABLEWAY HANGERS TO DISSIMILAR METALS

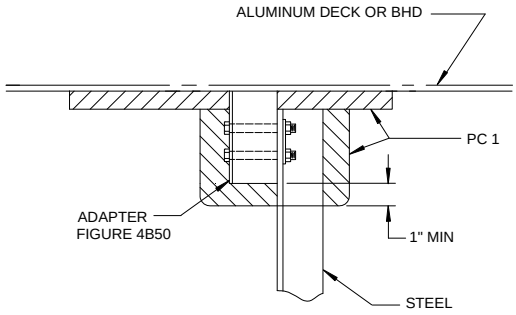
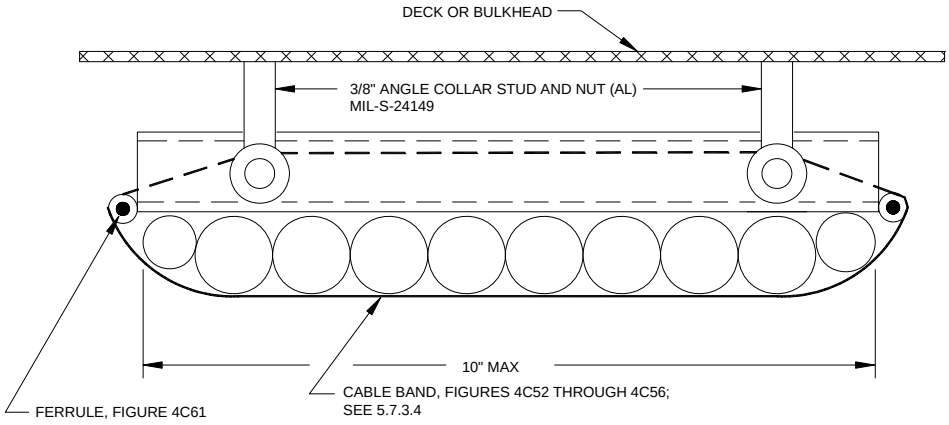


FIGURE 4B38. Detail of fire protection installation.

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METHOD 4B-39-1
SINGLE TIER SUPPORTED WITH ANGLE COLLAR STUDS
ON UNINSULATED DECKS OR BULKHEADS
CABLE - ONE INCH DIAMETER MAXIMUM
MAXIMUM CABLE WEIGHT: THREE POUNDS PER STUD



METHOD 4B-39-2
SINGLE TIER SUPPORTED WITH ANGLE COLLAR STUDS
ON INSULATED DECKS OR BULKHEADS
CABLE - ONE INCH DIAMETER MAXIMUM
MAXIMUM CABLE WEIGHT: THREE POUNDS PER STUD

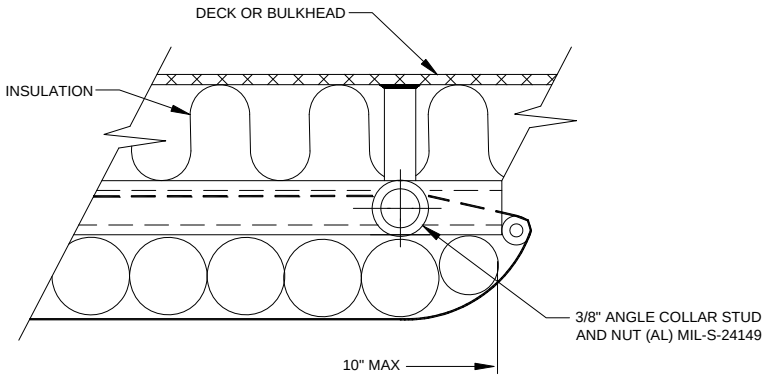
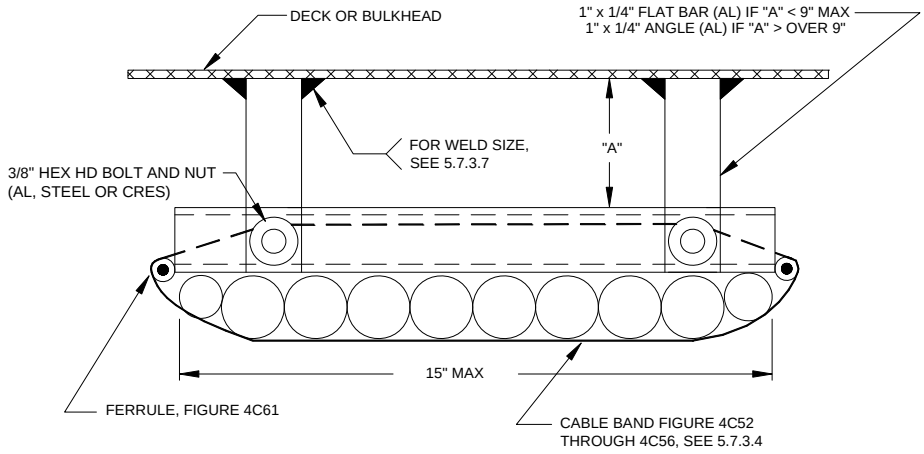


FIGURE 4B39. Single cableway on aluminum structures – local runs (surface ships).

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METHOD 4B-40-1
SINGLE TIER SUPPORTED WITH FLAT BAR OR ANGLE
ON UNINSULATED DECKS OR BULKHEADS
SEE FIGURE 4B41 FOR OPTIONAL METHOD



METHOD 4B-40-2
SINGLE TIER SUPPORTED WITH FLAT BAR OR ANGLE
ON INSULATED DECKS OR BULHEADS

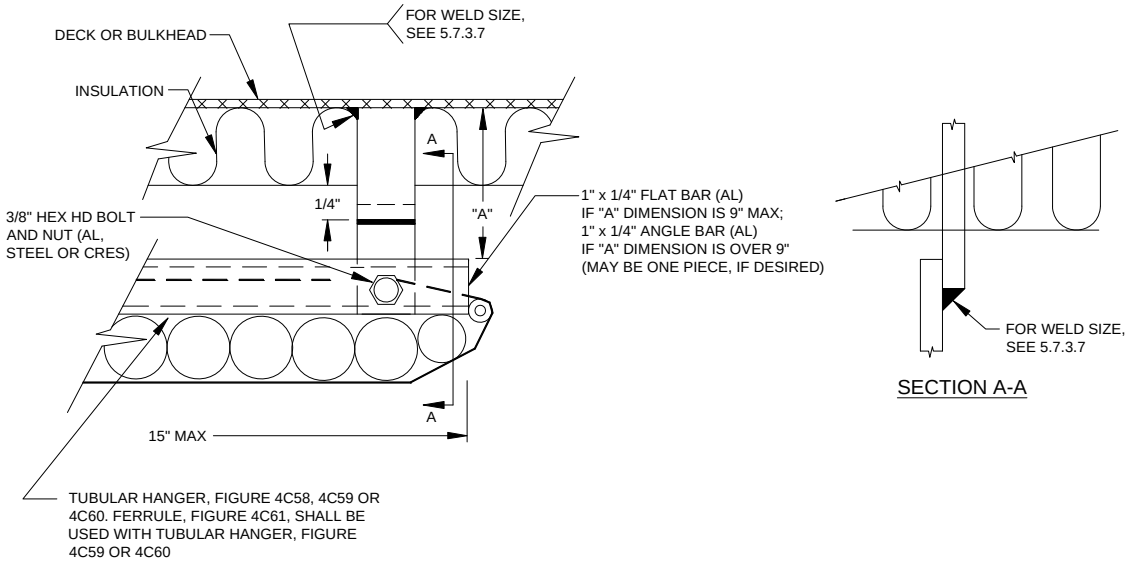
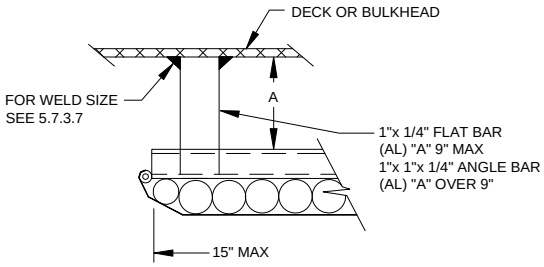


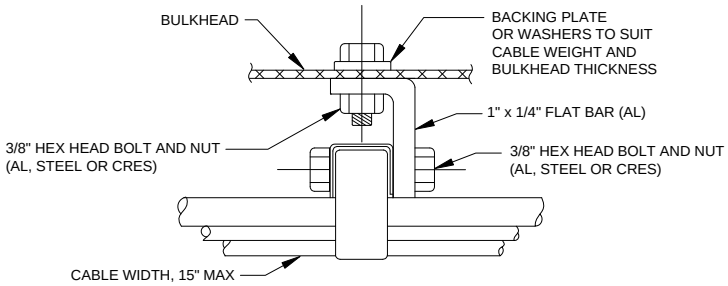
FIGURE 4B40. Single cableway on aluminum structures – local runs (surface ships).

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METHOD 4B-41-1
SINGLE TIER SECURED WITH FLAT BAR OR ANGLE AND WELD STUD
OPTION TO METHOD 4B-40-1



METHOD 4B-41-2
SINGLE TIER ON NON-WATERTIGHT BULKHEAD
SECURED BY BOLTING



METHOD 4B-41-3
SINGLE TIER SECURED WITH WELD STUDS
OPTION TO METHOD 4B-40-2

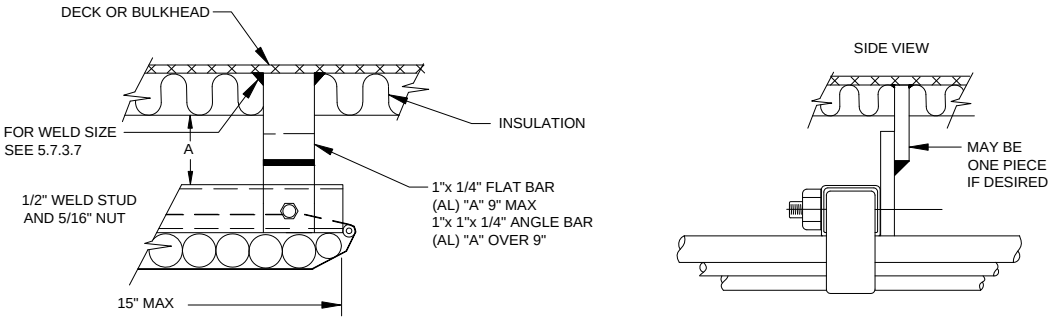
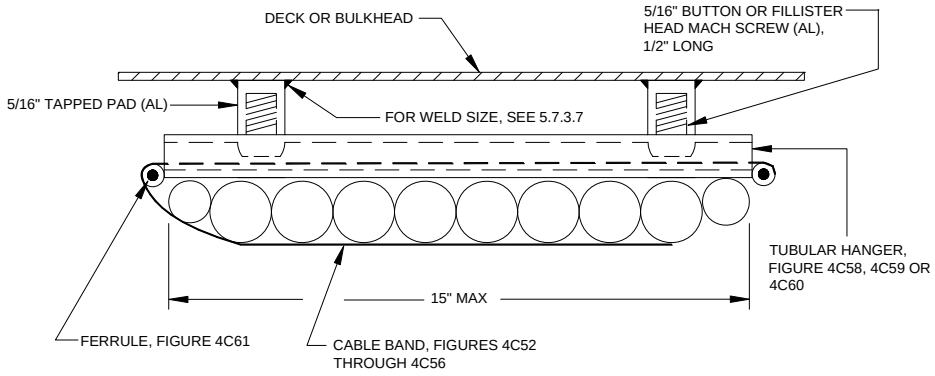


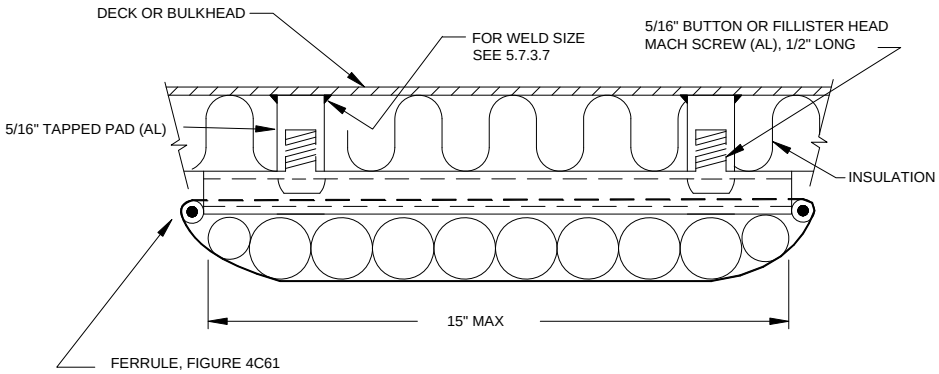
FIGURE 4B41. Single cableway on aluminum structures – local runs (surface ships).

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METHOD 4B-42-1
SINGLE TIER SECURED WITH TAPPED PADS
ON ALUMINUM DECKS AND BULKHEADS



METHOD 4B-42-2
SINGLE TIER SECURED WITH TAPPED PADS
ON INSULATED ALUMINUM DECKS AND BULKHEADS



METHOD 4B-42-3
SINGLE TIER SECURED WITH A SINGLE WELD STUD
ON ALUMINUM DECKS AND BULKHEADS
(MAXIMUM CABLE WEIGHT: THREE POUNDS PER SUPPORT)

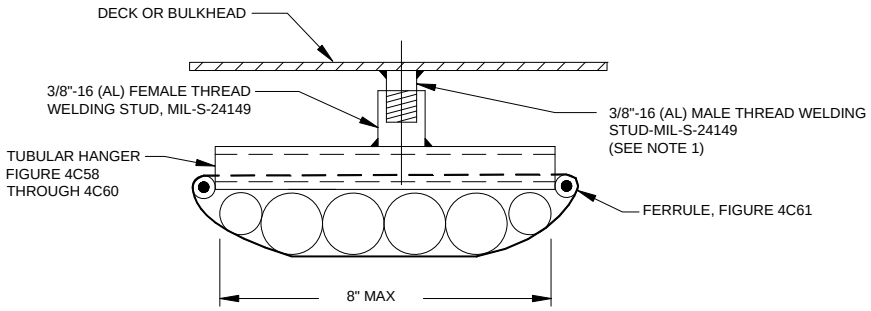


FIGURE 4B42. Single cableway on aluminum structures (surface ships).

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NOTE:

1. National stock numbers for ⅜-16 external thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 1	MIL-S-24149/3 CRES 316 type V, class 1	MIL-S-24149/2 aluminum type IV, class 1
0.750	5307-01-293-7981	5307-01-293-8797	5307-01-298-5394
0.875	5307-01-294-3148	5307-01-294-9538	5307-01-293-5428
1.000	5307-01-293-1818	5307-01-297-4107	5307-01-293-5427
1.250	5307-01-294-2149	5307-01-294-4356	5307-01-293-5428
1.500	5307-01-244-6967	5307-01-294-4357	5307-01-293-5429
1.750	5307-01-280-9436	5307-01-294-4358	5307-01-293-5430
2.000	5307-01-293-1819	5307-01-294-4359	5307-01-294-4370
2.500	5307-01-292-9609	5307-01-294-4360	5307-01-293-5431
3.000	5307-01-292-9610	5307-01-294-4361	5307-01-293-5432
3.500	5307-01-294-3150	5307-01-294-4362	5307-01-294-4371
4.000	5307-01-295-1098	5307-01-294-4363	5307-01-293-7980

FIGURE 4B42. Single cableway on aluminum structures (surface ships) - Continued.

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METHOD 4B-43-1
CABLES SUPPORTED BETWEEN SHIPS LONGITUDINALS AND
THROUGH SHIPS TRANSVERSE FRAMING
FOR SIDE VIEW, SEE FIGURE 4B44

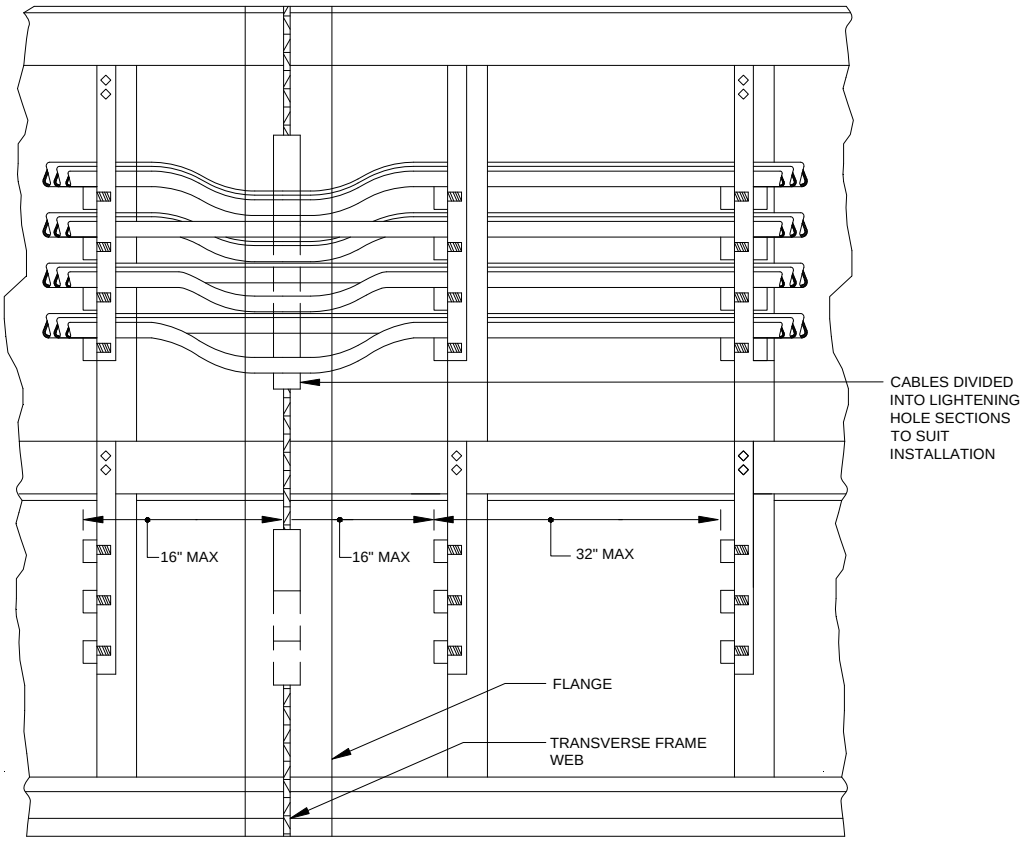
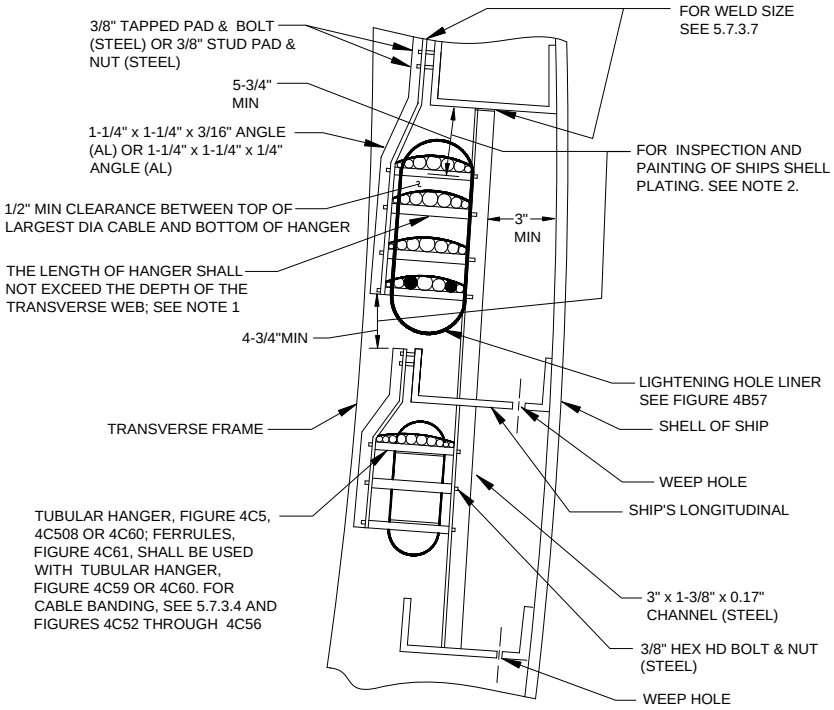


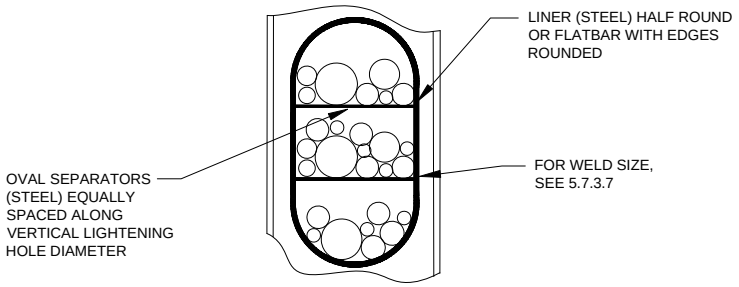
FIGURE 4B43. Multiple cableways in machinery spaces (surface ships).

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SIDE VIEW FOR METHOD 4B-44-1
CABLES SUPPORTED BETWEEN SHIPS LONGITUDINALS
THROUGH SHIPS TRANVERSE FRAMING - SIDE VIEW



LIGHTENING HOLE VIEW FOR METHOD 4B-44-2
VIEW SHOWING LIGHTENING HOLE LINER ASSEMBLY
WITH CABLE GROUPING



NOTES:

1. The number of hangers in groups between longitudinals shall be determined on a basis of the number of cables that can be suitably carried through lightening holes in the transverse webs.
2. For the inspection and painting of the ship's shell plating, clearance shall be provided between the top of the hanger and the ship's longitudinal above the hanger installation and between the hanger and the ship's longitudinal below the hanger installation.
3. Three cableway rows may be installed in congested locations on request and subject to specific approval by NAVSEA.

FIGURE 4B44. Multiple cableways in machinery spaces (surface ships).

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METHOD 4B-45-1

CABLES THROUGH CONDUIT; WATERTIGHT INSTALLATIONS IN NON-HAZARDOUS LOCATIONS

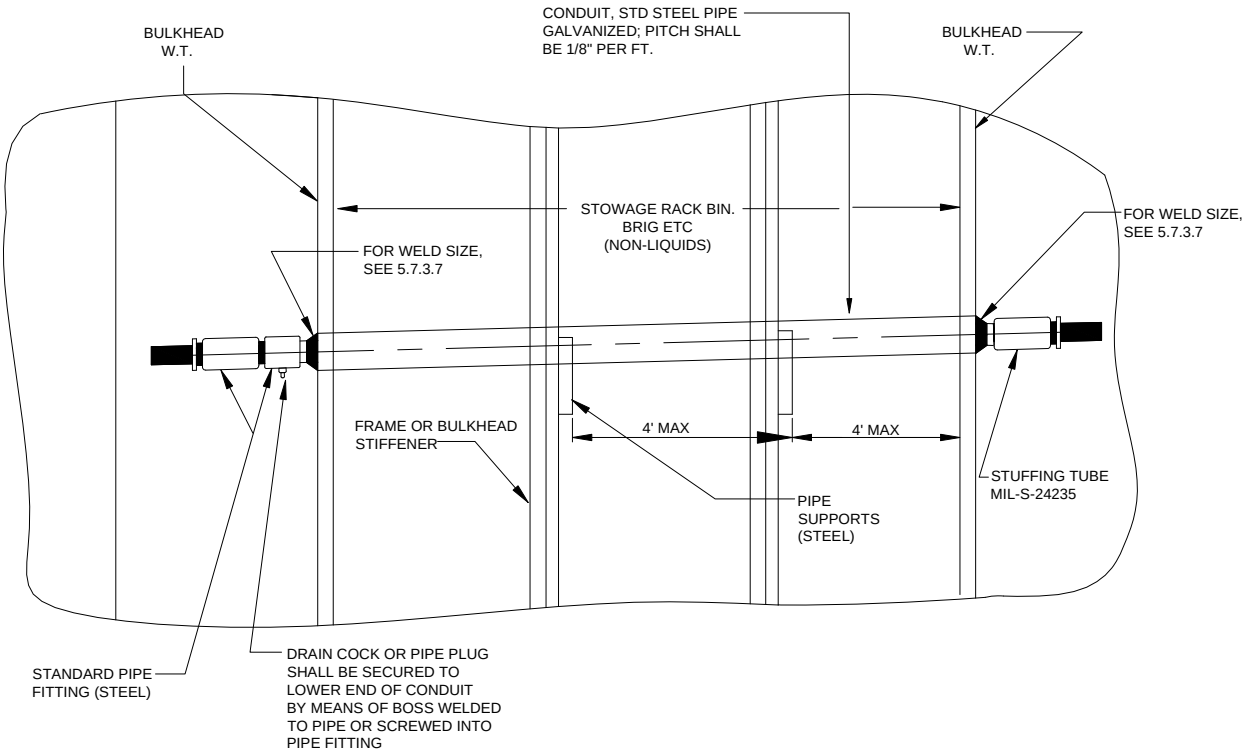


FIGURE 4B45. Cableways through storage spaces and tanks (surface ships).

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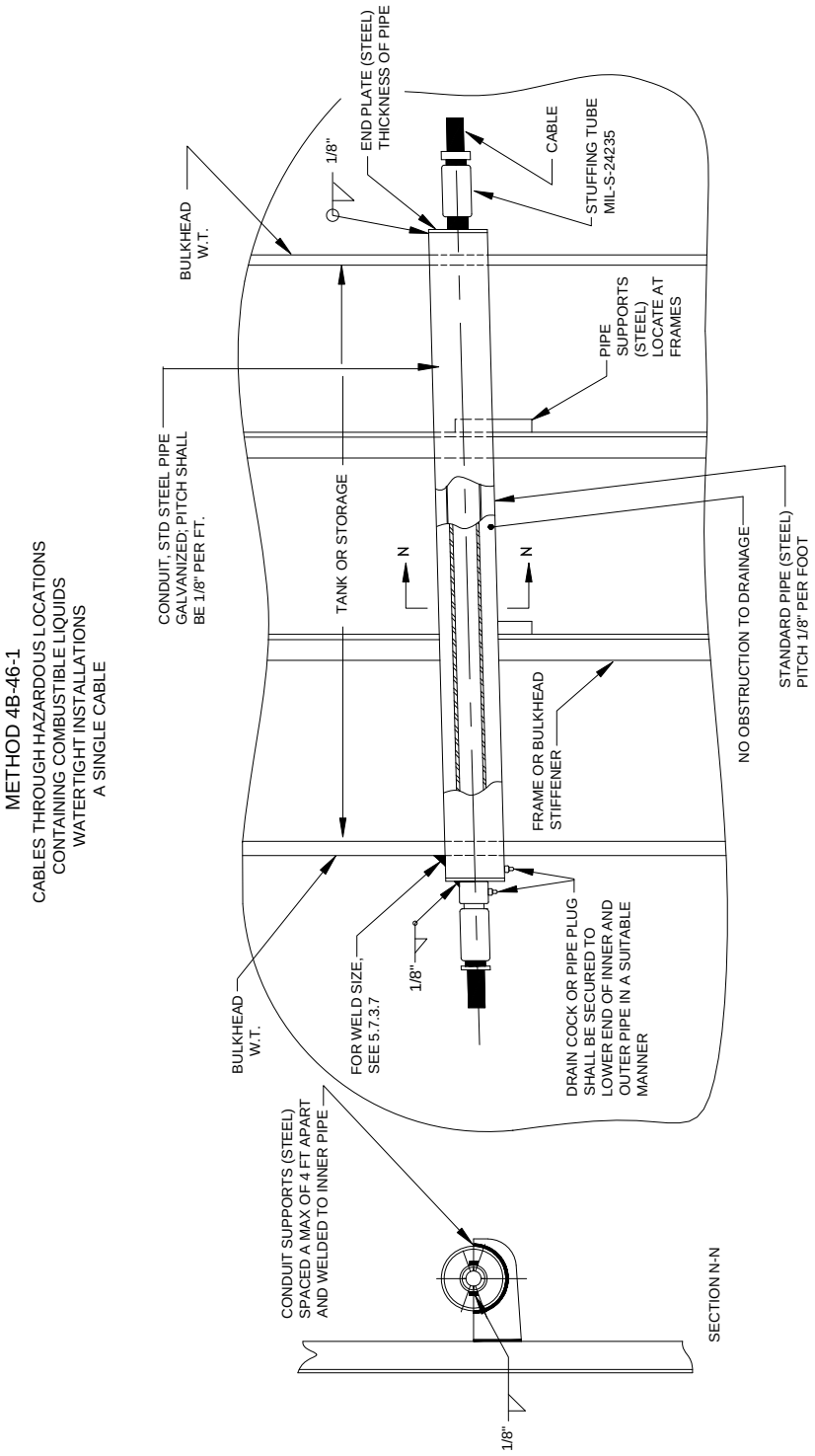


FIGURE 4B46. Cableways through storage spaces and tanks (surface ships).

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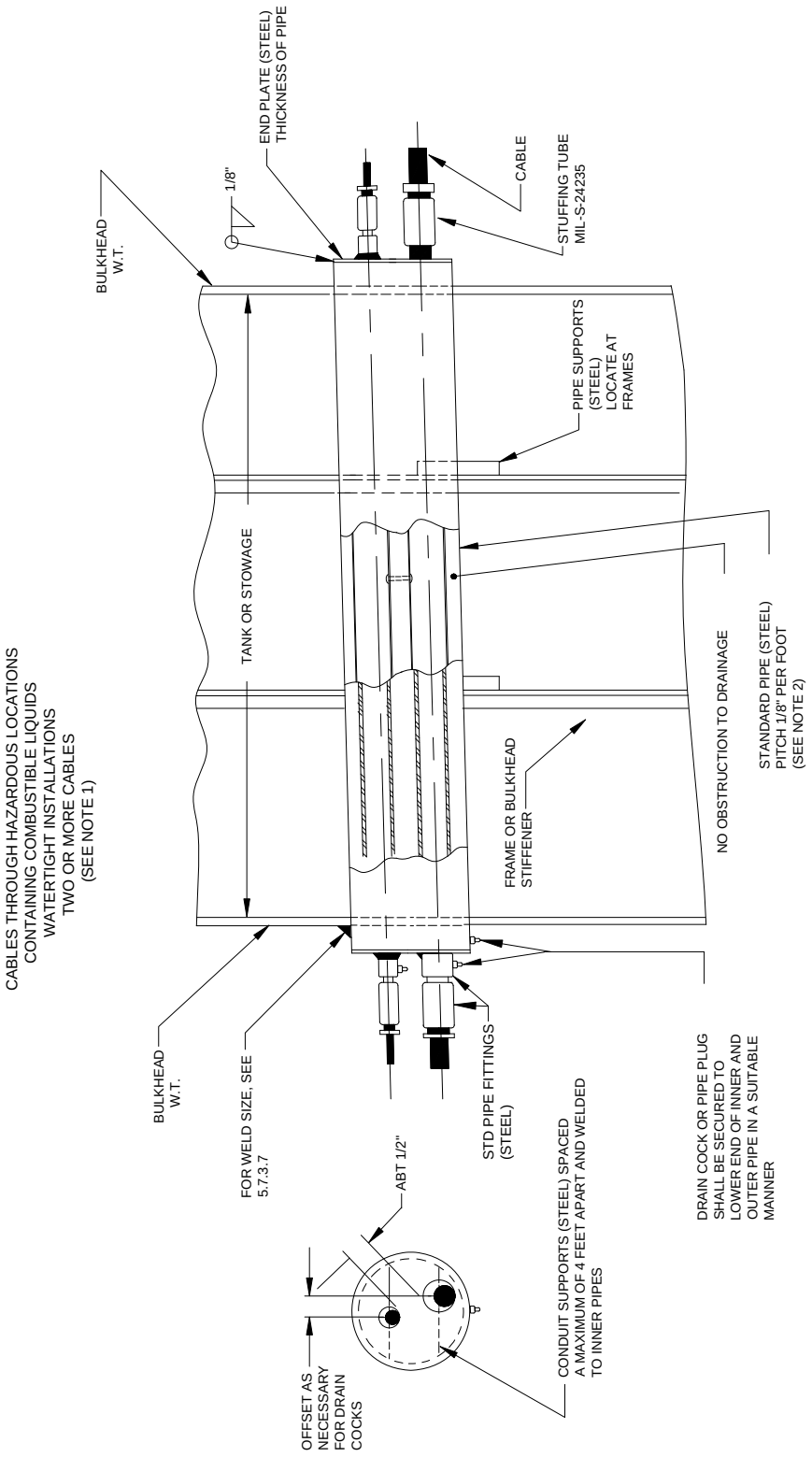


FIGURE 4B47. Cableways through storage spaces and tanks (surface ships).

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NOTES:

1. For figure 4B70, up to three cables may be used in the inner pipe providing no more than two simultaneously loaded cables shall be continuous to each other. This is an alternative method and will require an end plate on the protruding inner pipe with the riser tops to be welded directly to the end plate.
2. The standard pipe for this figure shall be clean and free of all foreign contamination.

FIGURE 4B47. Cableways through storage spaces and tanks (surface ships) - Continued.

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METHOD 4B-48-1
CABLES THROUGH CONDUIT; NON-WATERTIGHT INSTALLATIONS IN NON-HAZARDOUS LOCATIONS

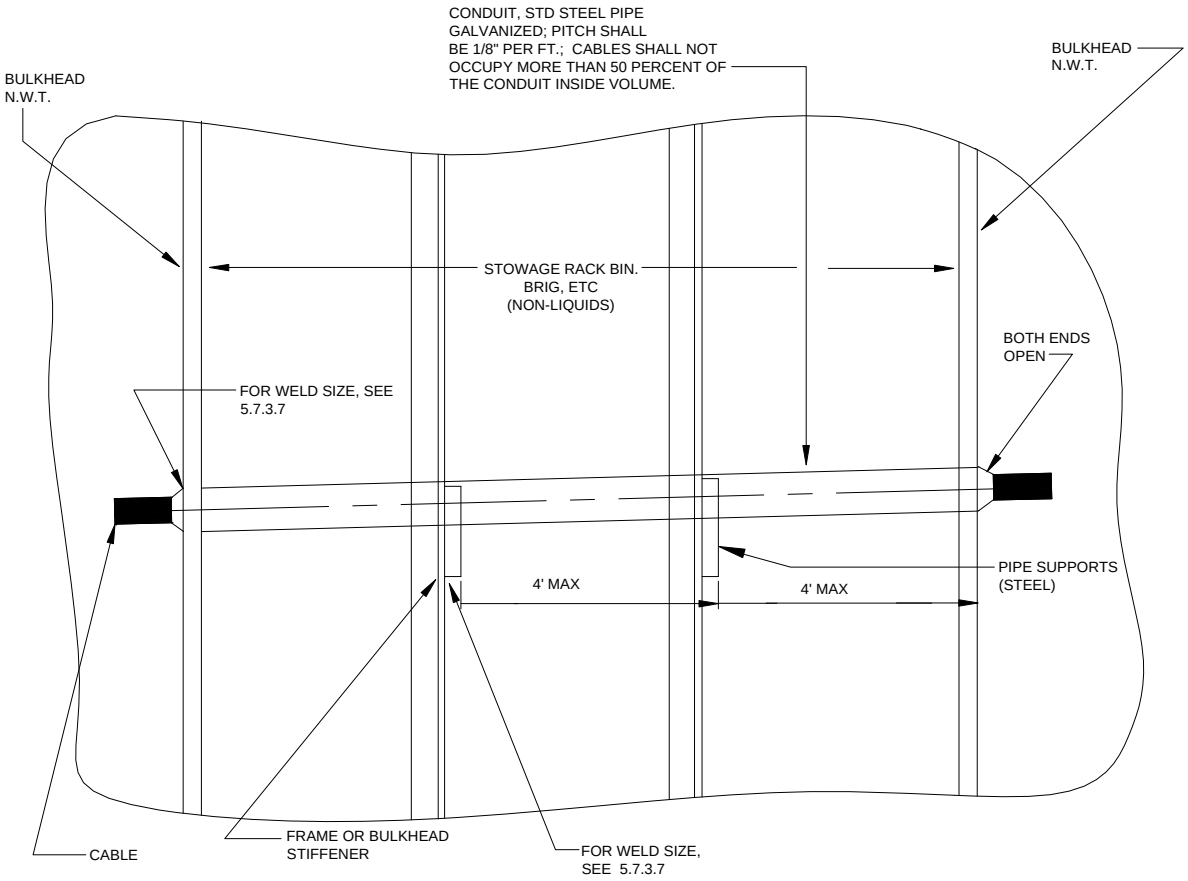


FIGURE 4B48. Cableways through storage spaces and tanks (surface ships).

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METHOD 4B-49-1
CABLES IN TRUNKS (W.T. OR O.T.)
MAY BE USED FOR FIRE PROTECTION

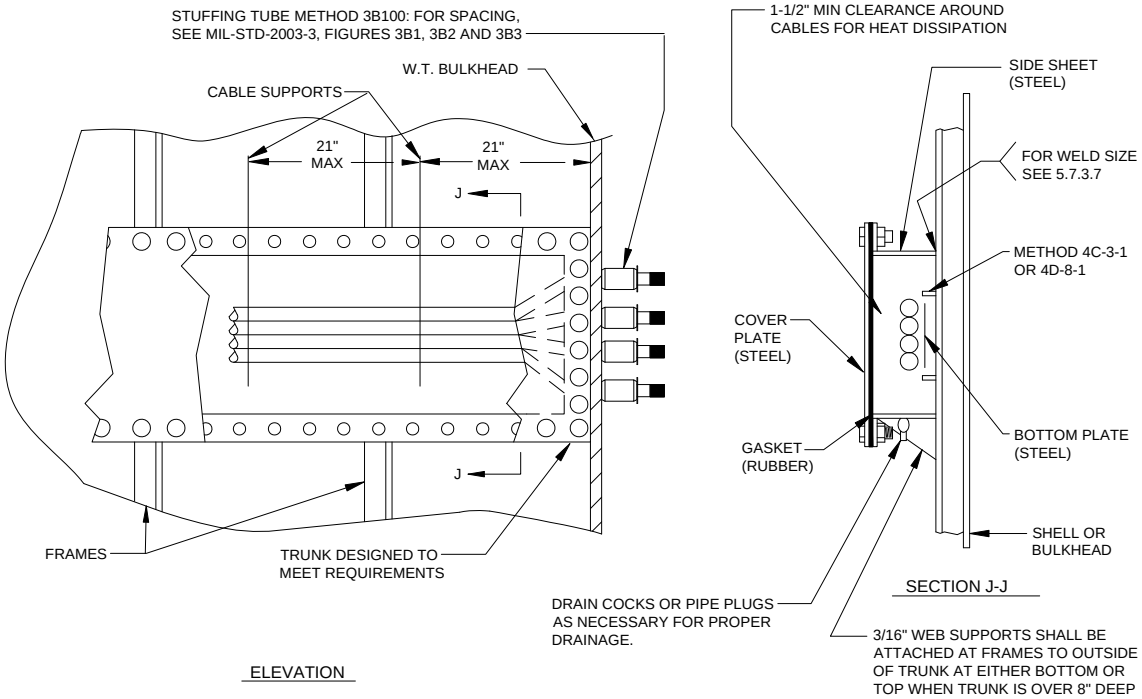


FIGURE 4B49. Cableways through trunks and channels (surface ships).

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METHOD 4B-50-1
CABLES IN CHANNEL

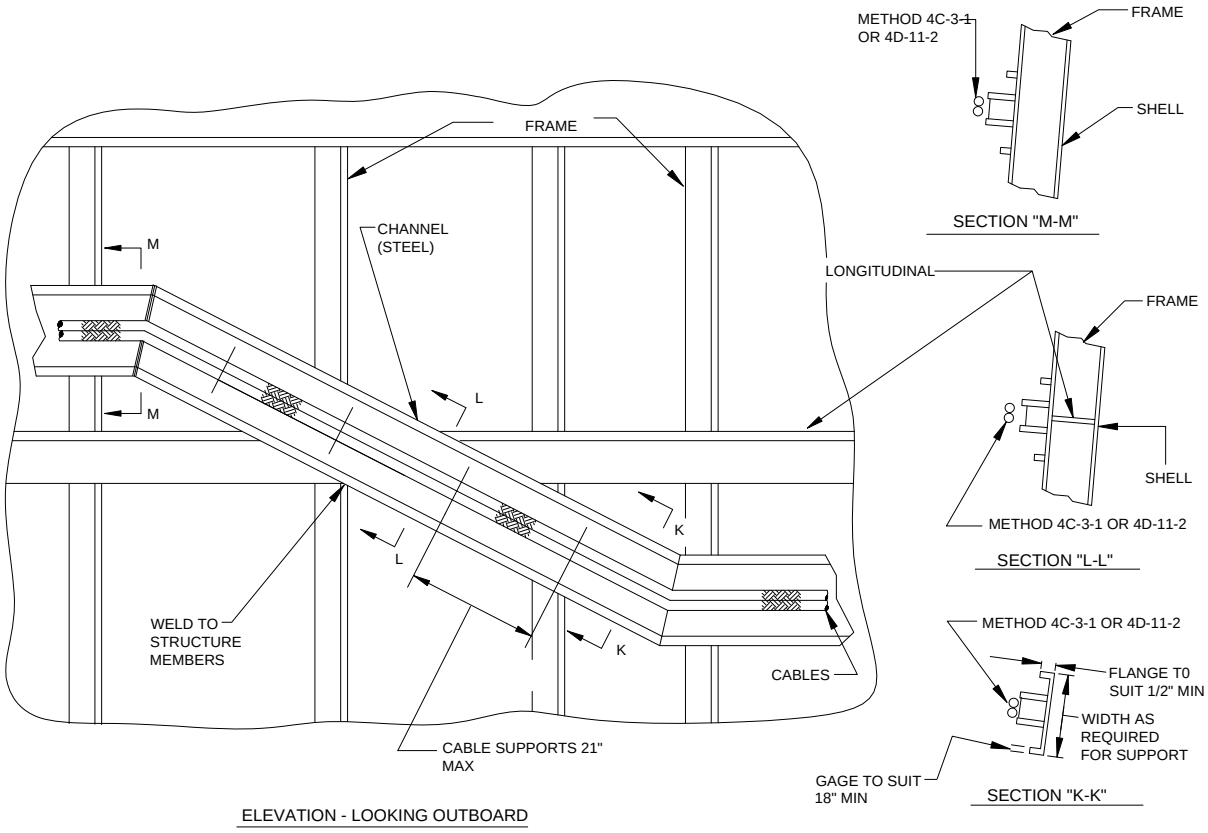


FIGURE 4B50. Cableways through trunks and channels (surface ships).

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METHOD 4B-51-1
CABLES THROUGH VOID SPACES NOT SUBJECT TO FLOODING

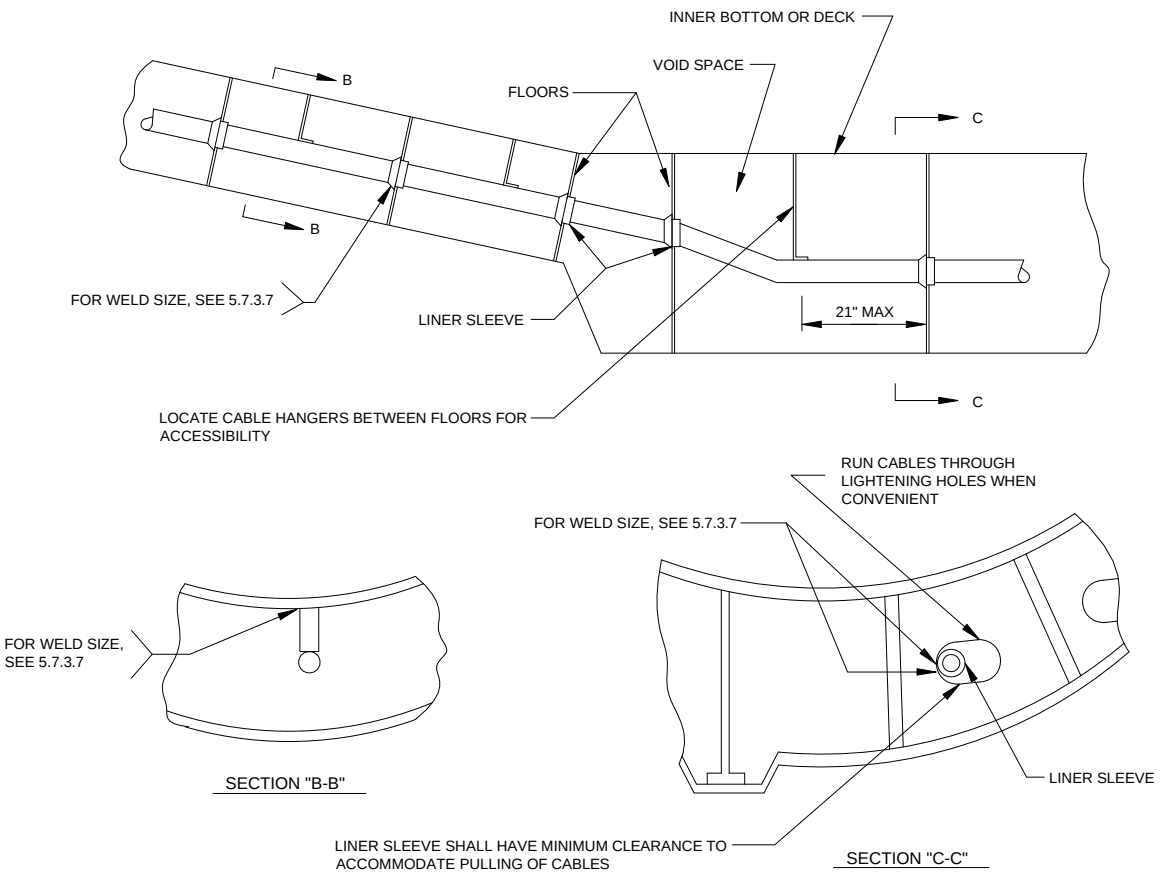
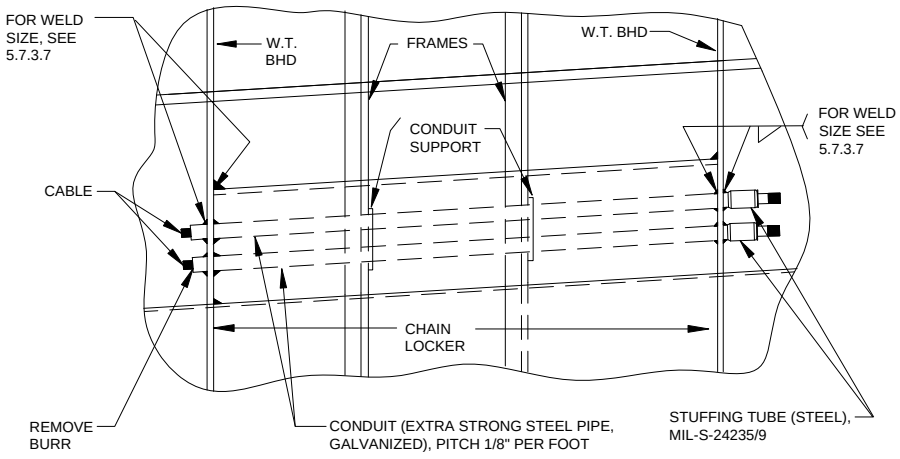


FIGURE 4B51. Cables through trunks and channels (surface ships).

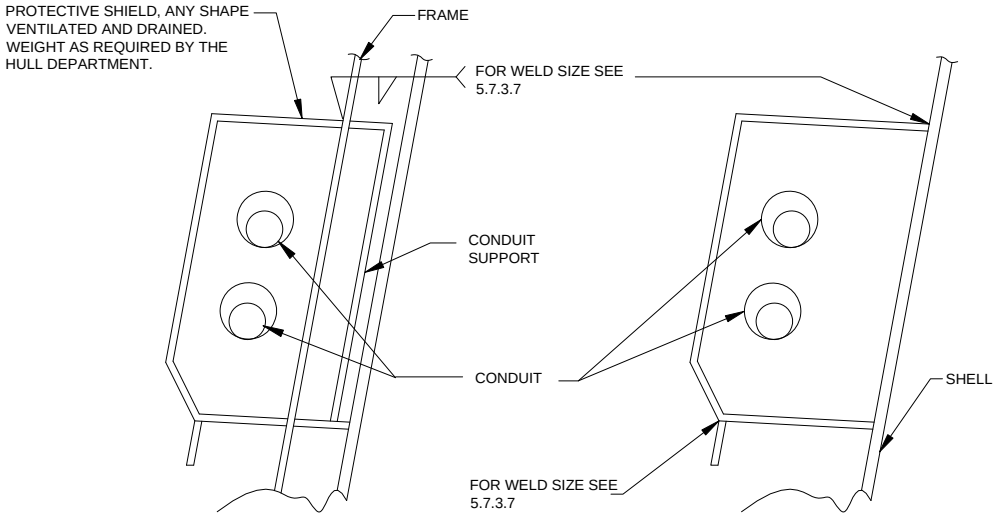
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METHOD 4B-52-1
CABLES THROUGH CHAIN LOCKERS
(SEE NOTE 1)



ELEVATION LOOKING OUTBOARD

SECTION VIEWS
CABLES THROUGH CHAIN LOCKERS



SECTION AT FRAME

SECTION BETWEEN FRAMES

ELEVATION LOOKING OUTBOARD

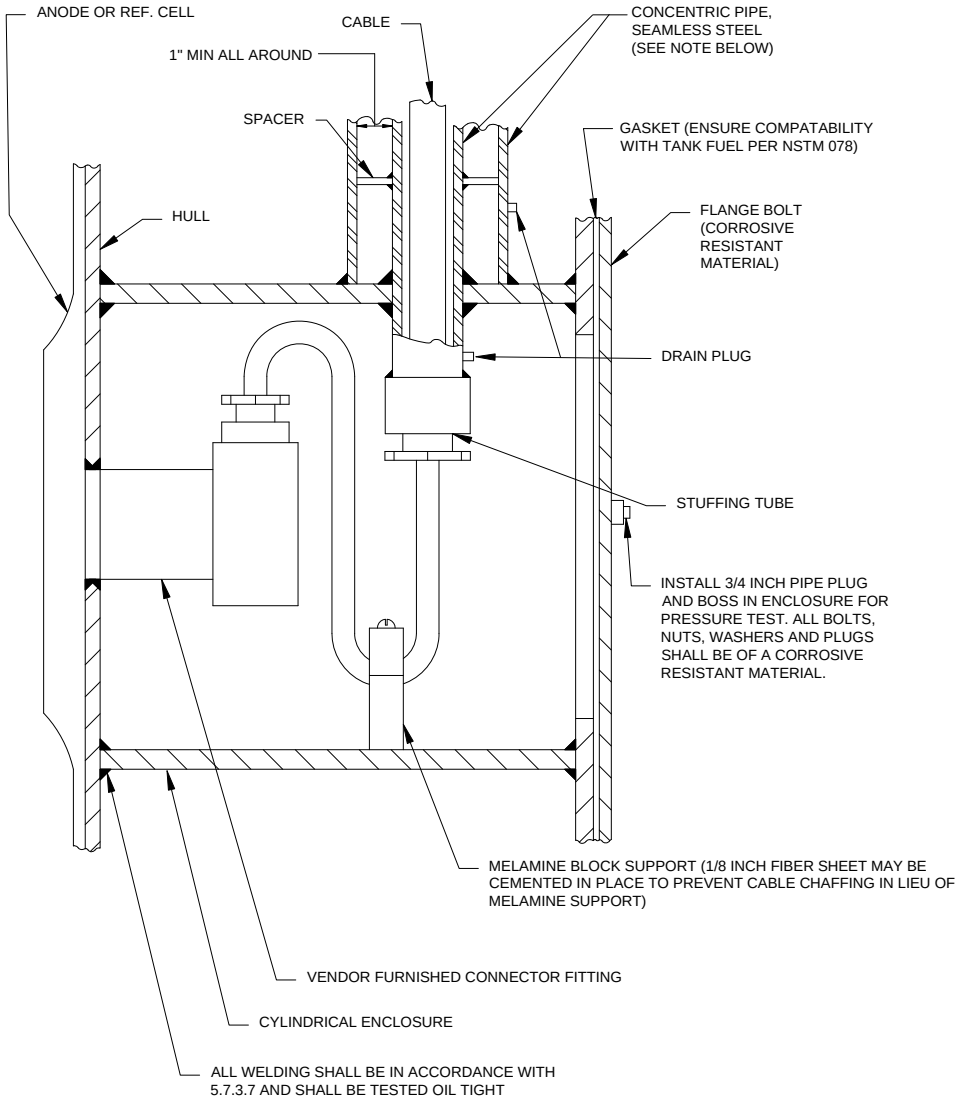
NOTE:

1. A maximum of three cables may be used in the inner pipe providing no more than two simultaneously loaded cables shall be contiguous to each other. This is an alternative method and will require an end plate on the protruding inner pipe with the riser tops to be welded directly to the end plate.

FIGURE 4B52. Cableways through trunks and channels (surface ships).

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METHOD 4B-53-1
CABLE THROUGH CONDUIT IN FUEL OIL TANKS
WATERTIGHT INSTALLATION - SINGLE CABLE
(SEE NOTE 1)



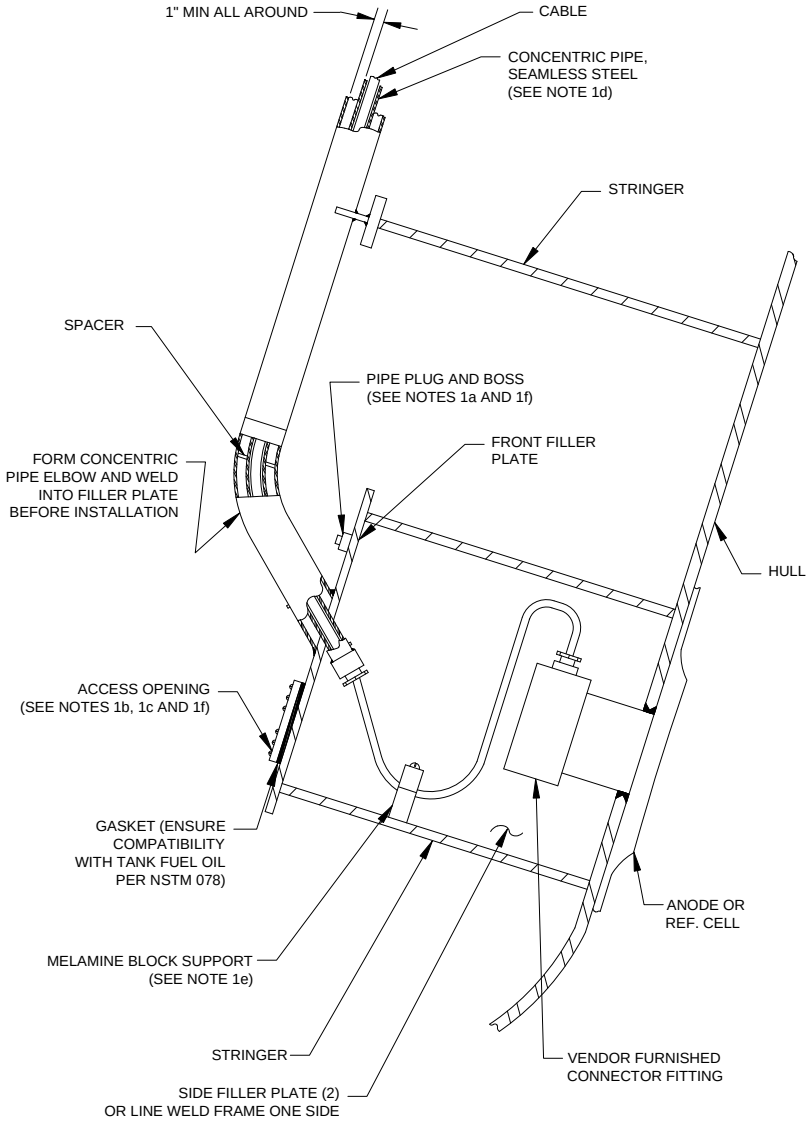
NOTES:

1. For cableways through fuel oil tanks for cathodic protection system:
   a. Install ¾" pipe plug and boss in enclosure for pressure test.
   b. All welding shall be in accordance with 5.7.3.7 and shall be tested oil tight.
   c. Alternate location for access opening will be on side filler plate requiring a heavier plate.
   d. Access cover shall be 2" greater than access opening all around.
   e. The inner concentric pipe shall be galvanized. The outer concentric pipe shall be painted to conform to the tank paint system.
   f. Fiber sheet (⅛" thick) may be cemented in place to prevent cable chafing in lieu of melamine support.
   g. All bolts, nuts, washers and plugs shall be of a corrosive resistant material.
2. The inner concentric pipe shall be galvanized. The outer concentric pipe shall be painted to conform to the tank paint system.

FIGURE 4B53. Cableways through fuel oil tanks for cathodic protection system (surface ships).

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METHOD 4B-54-1
CABLE THROUGH CONDUIT IN FUEL OIL TANKS
WATERTIGHT INSTALLATION - SINGLE CABLE
ALTERNATE ENCLOSURE METHOD
(USING STRINGERS OF LONGITUDINALS)



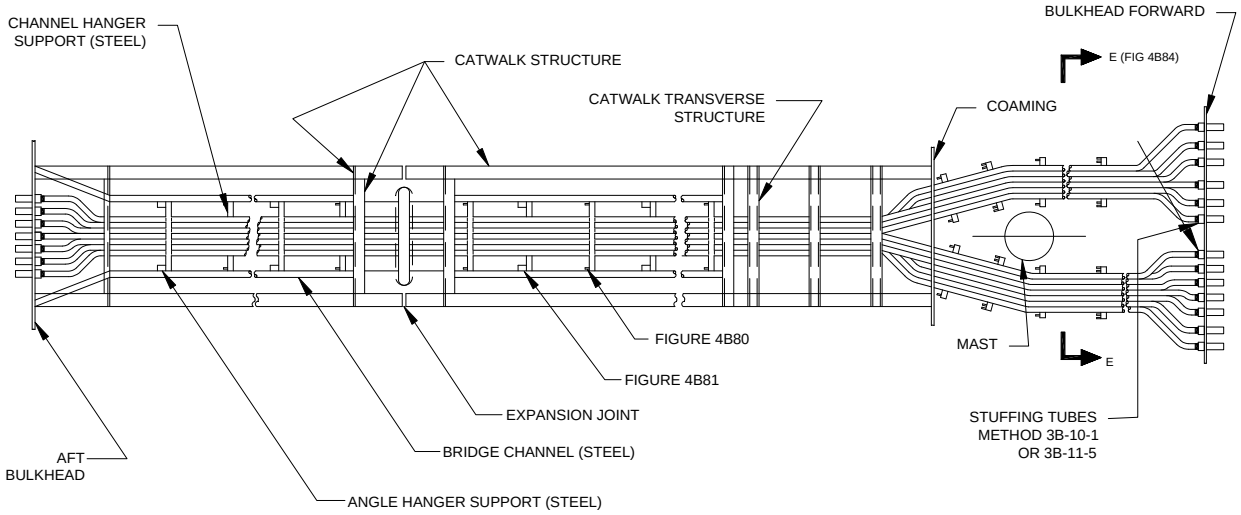
NOTE:

1. For cableways through fuel oil tanks for cathodic protection system:
   a. Install ¾" pipe plug and boss in enclosure for pressure test.
   b. Alternate location for access opening will be on side filler plate requiring a heavier plate.
   c. Access cover shall be 2" greater than access opening all around.
   d. The inner concentric pipe shall be galvanized. The outer concentric pipe shall be painted to conform to the tank paint system.
   e. Fiber sheet (⅛" thick) may be cemented in place to prevent cable chafing in lieu of melamine support.
   f. All bolts, nuts, washers and plugs shall be of a corrosive resistant material.

FIGURE 4B54. Cableways through fuel oil tanks for cathodic protection system (surface ships) – alternate method.

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METHOD 4B-55-1
CABLES UNDER CATWALKS
FREE OF PIPING
SEE NOTES 1 THROUGH 3



PLAN VIEW
PROTECTIVE PLATES REMOVED
(FOR SECTION AND DETAIL VIEWS
SEE FIGURES 4B57 THRU 4B61)

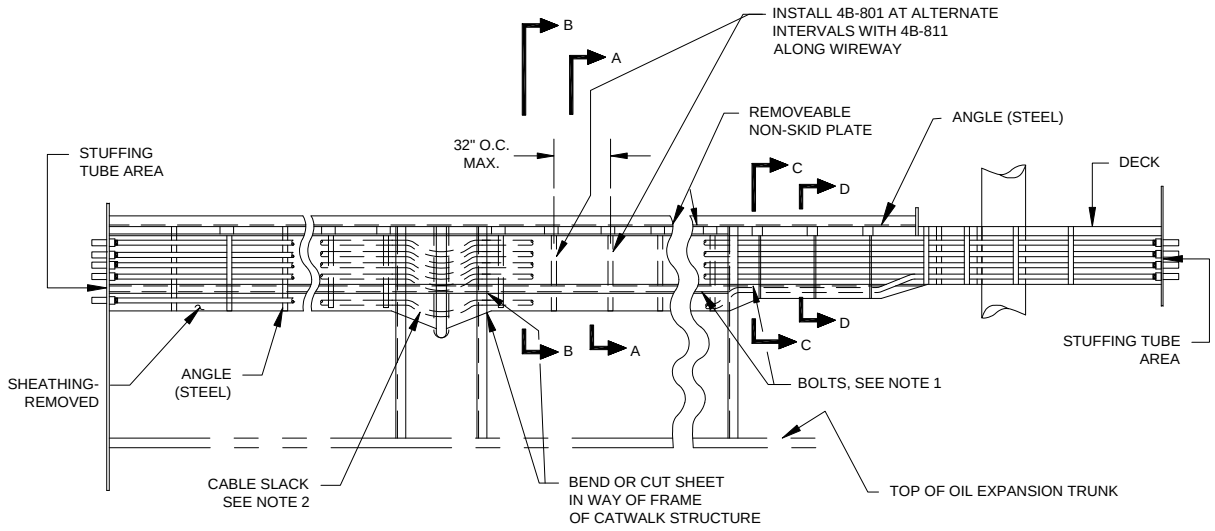
NOTES:

1. All mild steel material used for supporting and protecting cables under catwalk shall be zinc coated after drilling and prior to installation. The complete installation shall be painted.
2. Bolts used to secure protective sheath shall be spaced approximately 6" on center. Sections of protective sheathing shall be five-pound plate shall not exceed approximately 64" in length (side and bottom of wireway).
3. Provide sufficient slack in cables for working of expansion joint. Allow 2" minimum clearance between cables and bottom framing.
4. The fire installation detail on figure 4B55 shall be installed in accordance with NAVSEA drawing 803-5184182.

FIGURE 4B55. Topside cableways on tankers.

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ELEVATION VIEW FOR METHOD 4B-56-1
CABLES UNDER CATWALKS
FREE OF PIPING
SEE FIGURE 4B55 FOR PLAN VIEW



ELEVATION

FOR SECTION AND DETAIL VIEWS
SEE FIGURES 4B57 THRU 4B60

NOTES:

1. Bolts used to secure protective sheath shall be spaced approximately 6" on center. Sections of protective sheathing shall be five-pound plate shall not exceed approximately 64" in length (side and bottom of wireway).
2. Provide sufficient slack in cables for working of expansion joint. Allow 2" minimum clearance between cables and bottom framing.

FIGURE 4B56. Topside cableways on tankers – elevation view.

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SECTION A-A FOR FIGURE 4B56
VIEW SHOWING SUPPORT OF CABLES AND SHEATHING
USING SQUARE TUBE HANGERS
ATTACHED TO BRIDGE CHANNELS

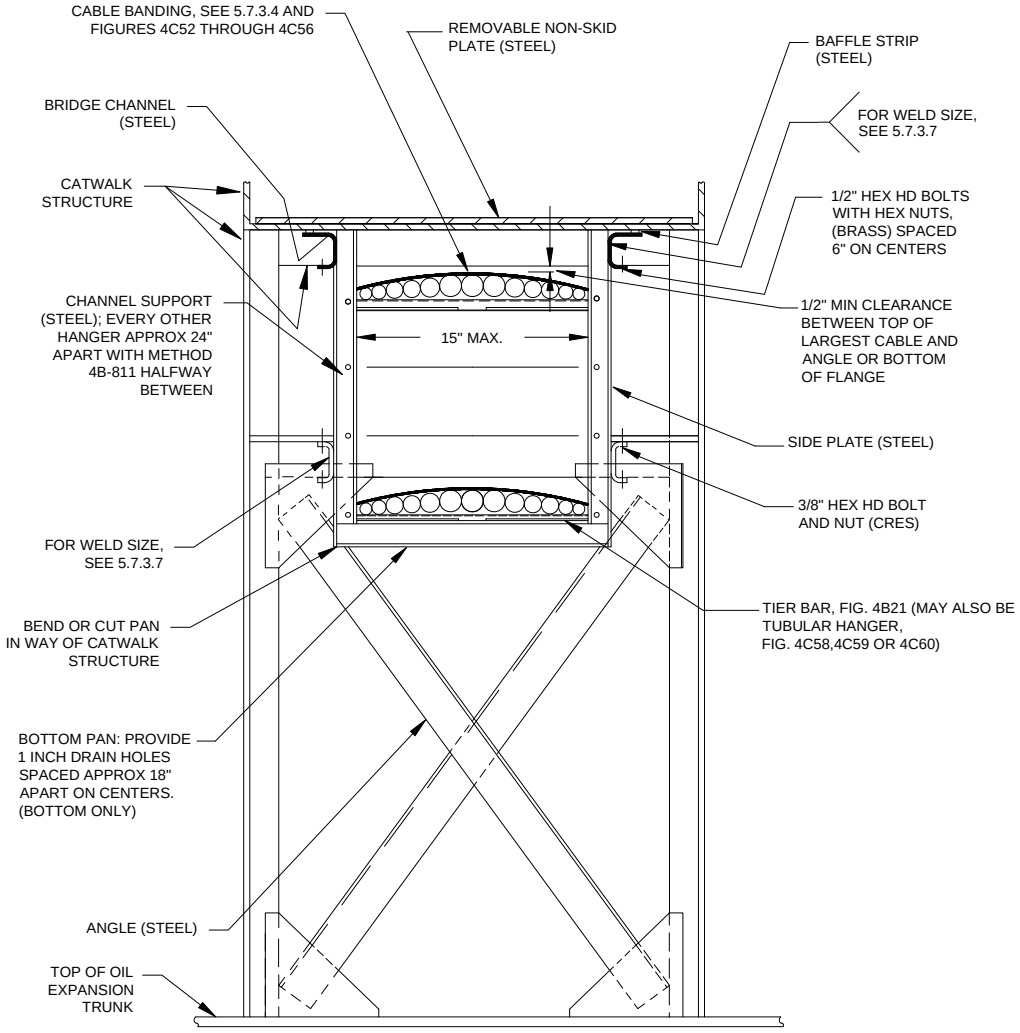


FIGURE 4B57. Topside cableways on tankers – section "A-A".

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SECTION B-B FOR FIGURE 4B56
VIEW SHOWING SUPPORT OF CABLES
USING ANGLE HANGER SUPPORT
SHEATHING NOT ATTACHED TO HANGER

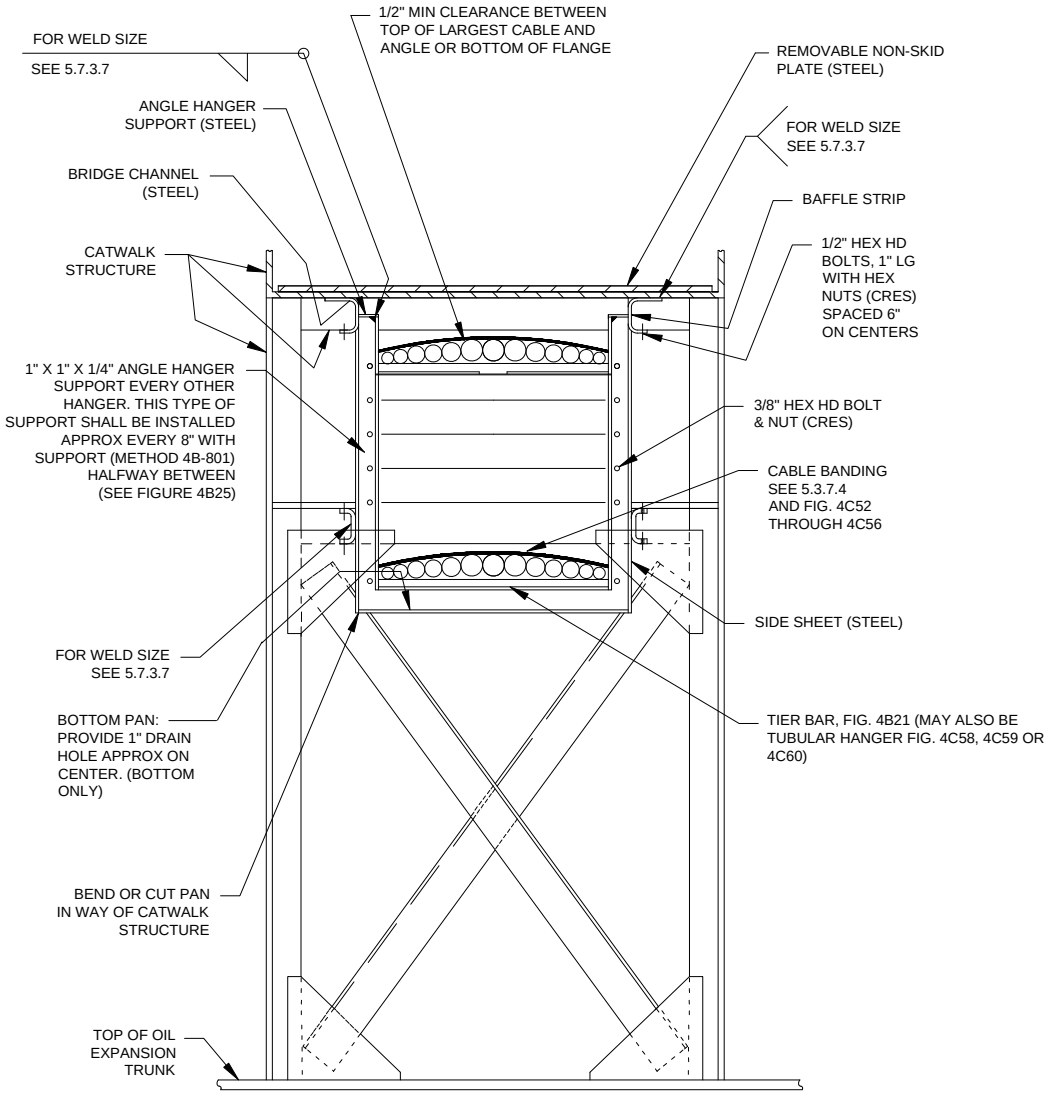


FIGURE 4B58. Topside cableways on tankers – section "B-B".

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SECTION C-C FOR FIGURE 4B56
VIEW SHOWING CHANNEL HANGER SUPPORT
SUSPENDED FROM CATWALK TRANSVERSE STRUCTURE

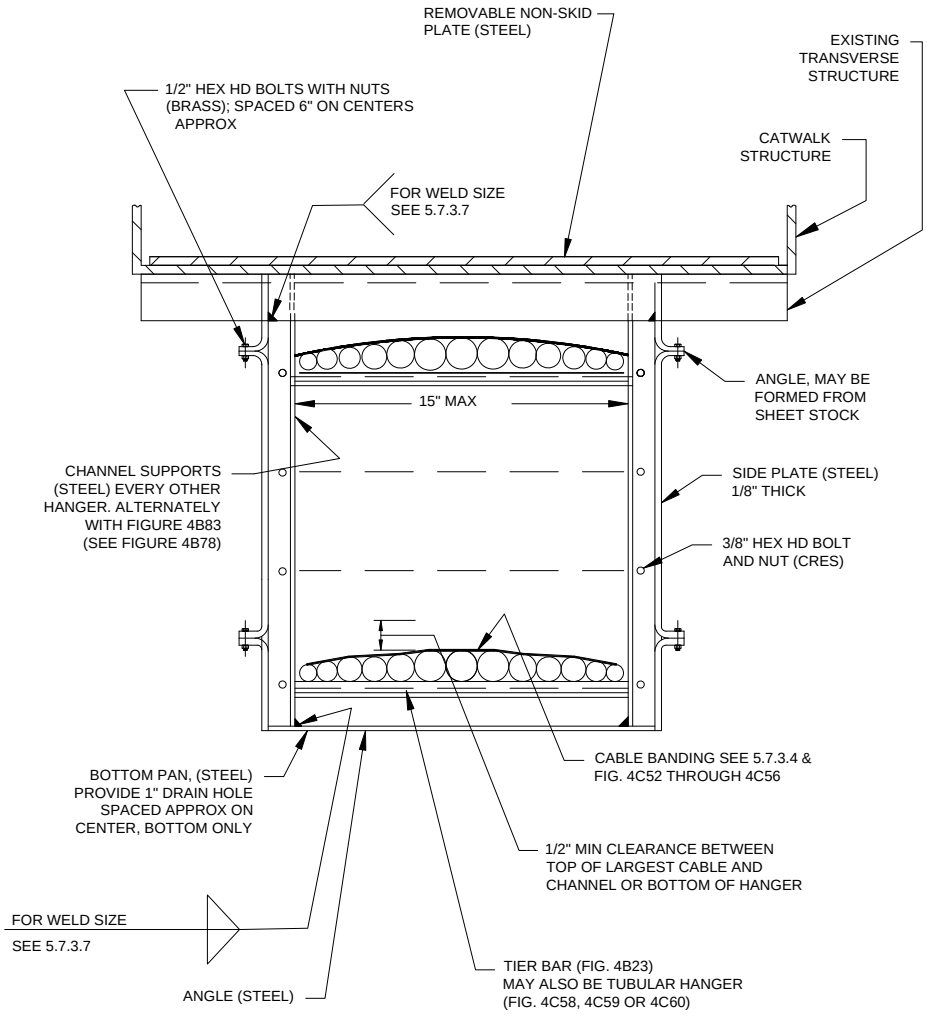


FIGURE 4B59. Topside cableways on tankers – section "C-C".

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SECTION D-D FOR FIGURE 4B56
VIEW SHOWING CHANNEL HANGER SUPPORT
SUSPENDED FROM CATWALK TRANSVERSE STRUCTURE

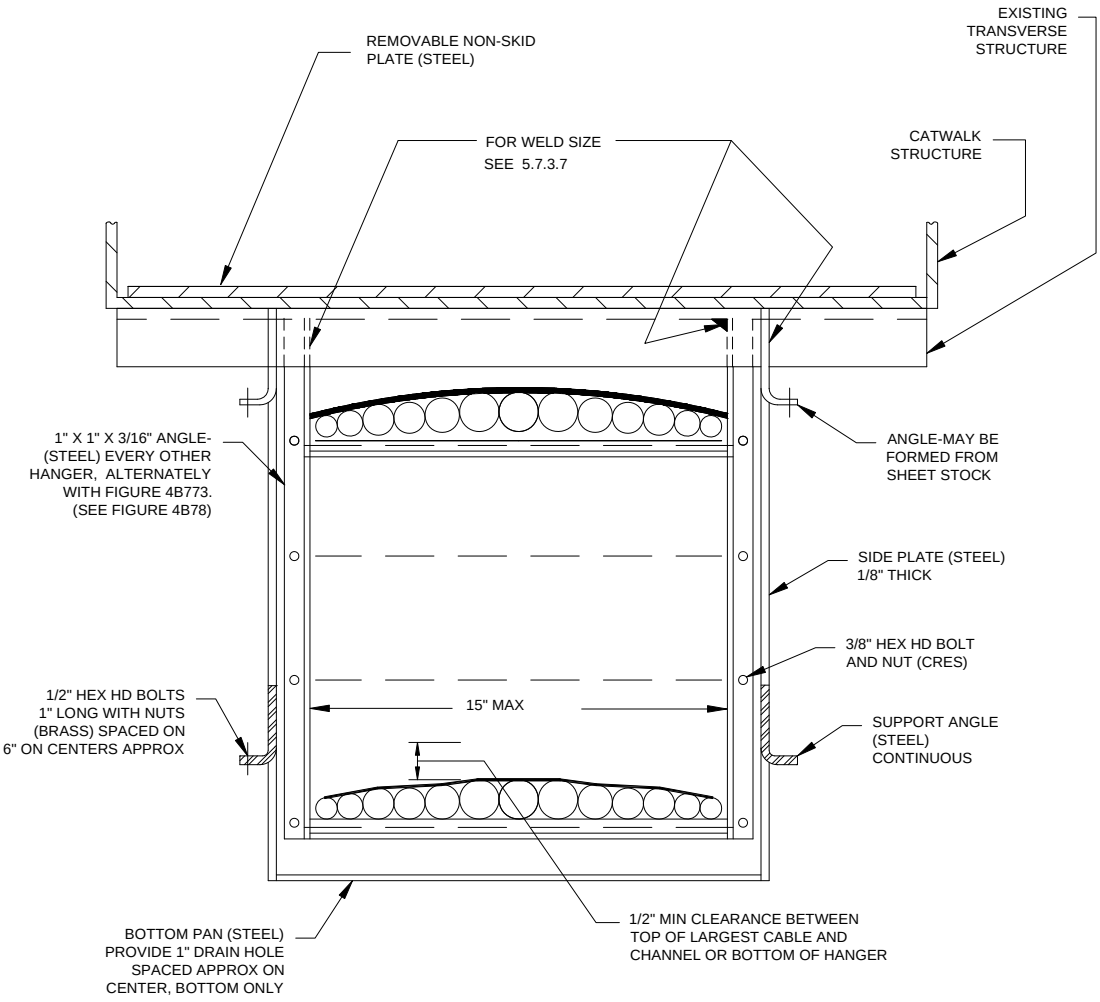


FIGURE 4B60. Topside cableways on tankers – section "D-D".

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SECTION E-E FOR FIGURE 4B55
VIEW SHOWING CABLEWAY
DIVIDED TO PASS AN OBSTRUCTION

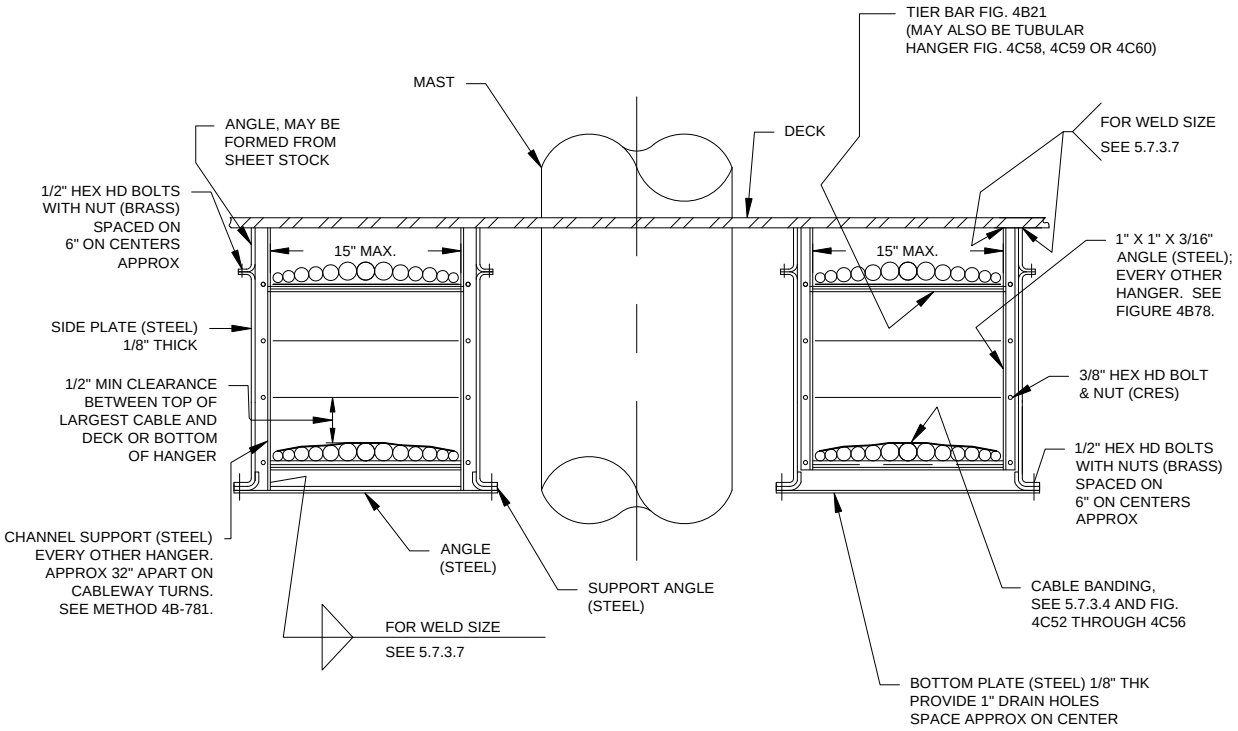


FIGURE 4B61. Topside cableways on tankers – divided to pass obstruction.

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Check the source to verify that this is the current version before use.

NOTE: Figures 4B62 through 4B64 are alternate methods and may be used in lieu of those methods shown on figures 4B57 through 4B61.

METHOD 4B-62-1
VIEW SHOWING HANGER SUPPORT
ATTACHED TO TRANSVERSE MEMBER
SEE METHOD 4B-55-1

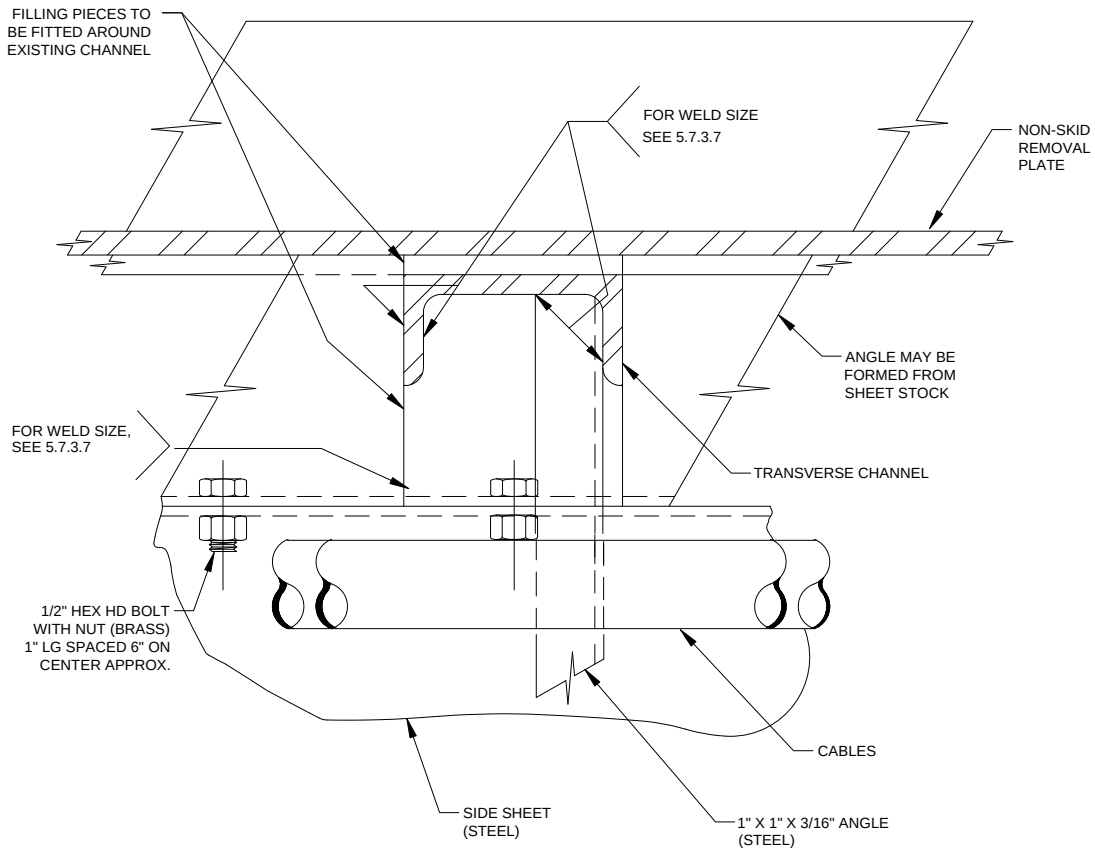
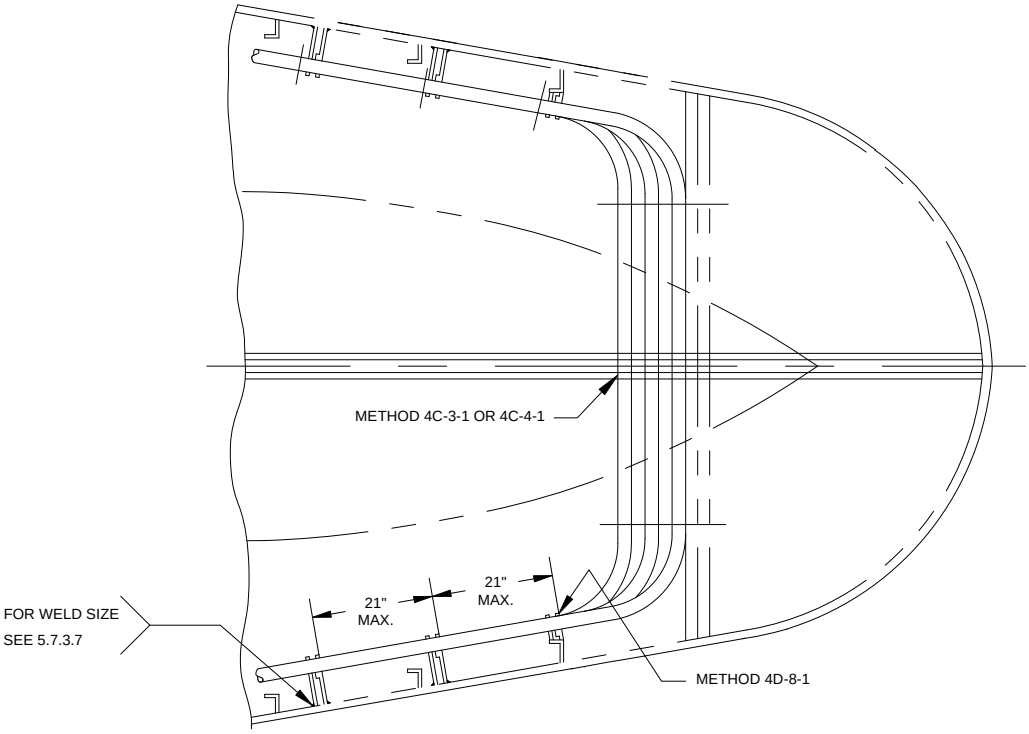


FIGURE 4B62. Topside cableways on tankers.

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Check the source to verify that this is the current version before use.

NOTE: Figures 4B62 through 4B64 are alternate methods and may be used in lieu of those methods shown on figures 4B57 through 4B61.

METHOD 4B-63-1
CABLES IN TURN OF BOW ATTACHMENT TO GIRDER
METHOD MAY BE USED FOR GENERAL INSTALLATION
WHERE APPLICABLE (SEE NOTE 1)



PLAN VIEW
SEE FIGURES 4B64 AND 4B65 FOR ELEVATION VIEWS

NOTE:

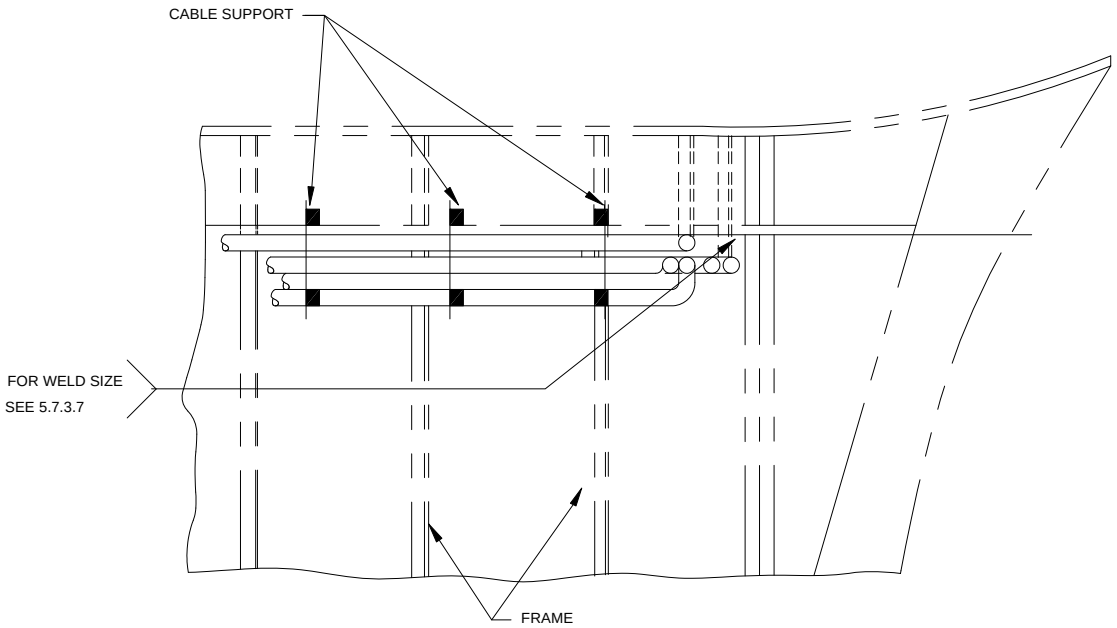
1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B63. Cable support for degaussing cables (surface ships) – plan view.

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NOTE: Figures 4B62 through 4B64 are alternate methods and may be used in lieu of those methods shown on figures 4B57 through 4B61.

ELEVATION FOR METHOD 4B-64-1
CABLES IN TURN OF BOW ATTACHMENT TO GIRDER
METHOD MAY BE USED FOR GENERAL INSTALLATION
WHERE APPLICABLE (SEE NOTE 1)



ELEVATION LOOKING OUTBOARD
(SEE FIGURE 4B63 FOR PLAN VIEW)

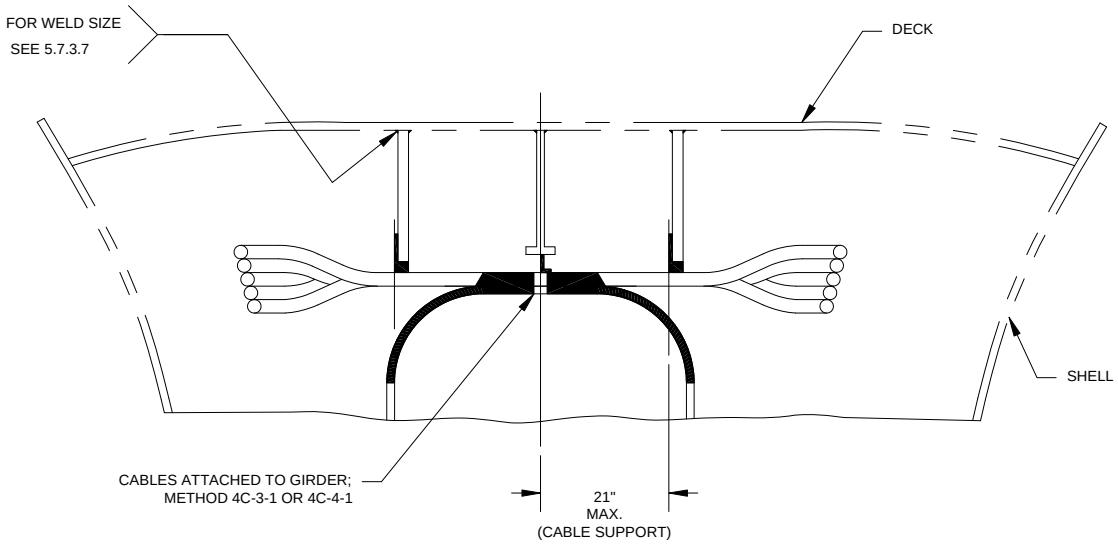
NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B64. Cable support for degaussing cables (surface ships) – elevation looking outboard.

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TYPICAL FRAME FOR METHOD 4B-65-1
CABLES IN TURN OF BOW
ATTACHMENT TO GIRDER
(SEE NOTE 1)



TYPICAL FRAME - LOOKING FORWARD
(SEE FIGURE 4B63 FOR PLAN VIEW)

NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B65. Cable support for degaussing cables (surface ships) – typical frame.

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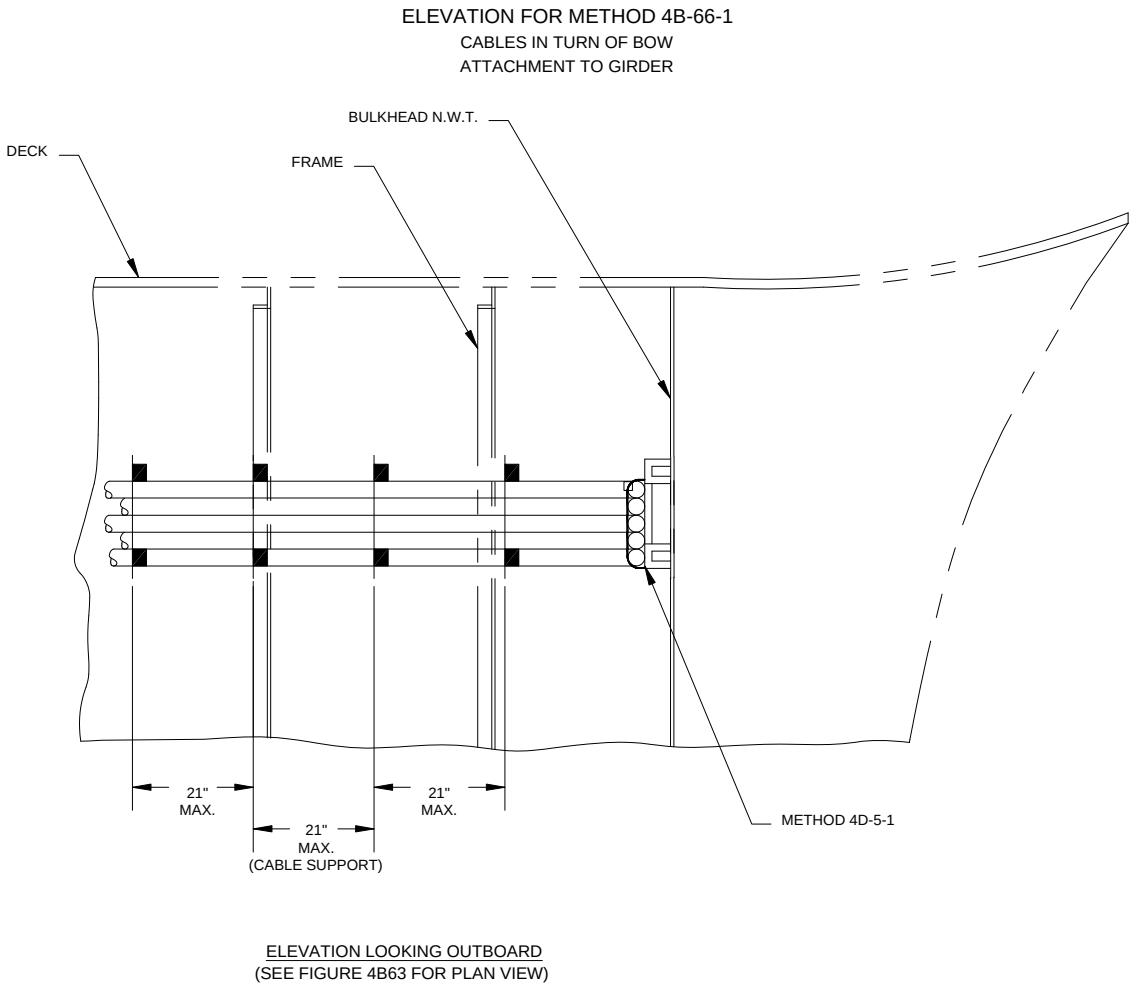
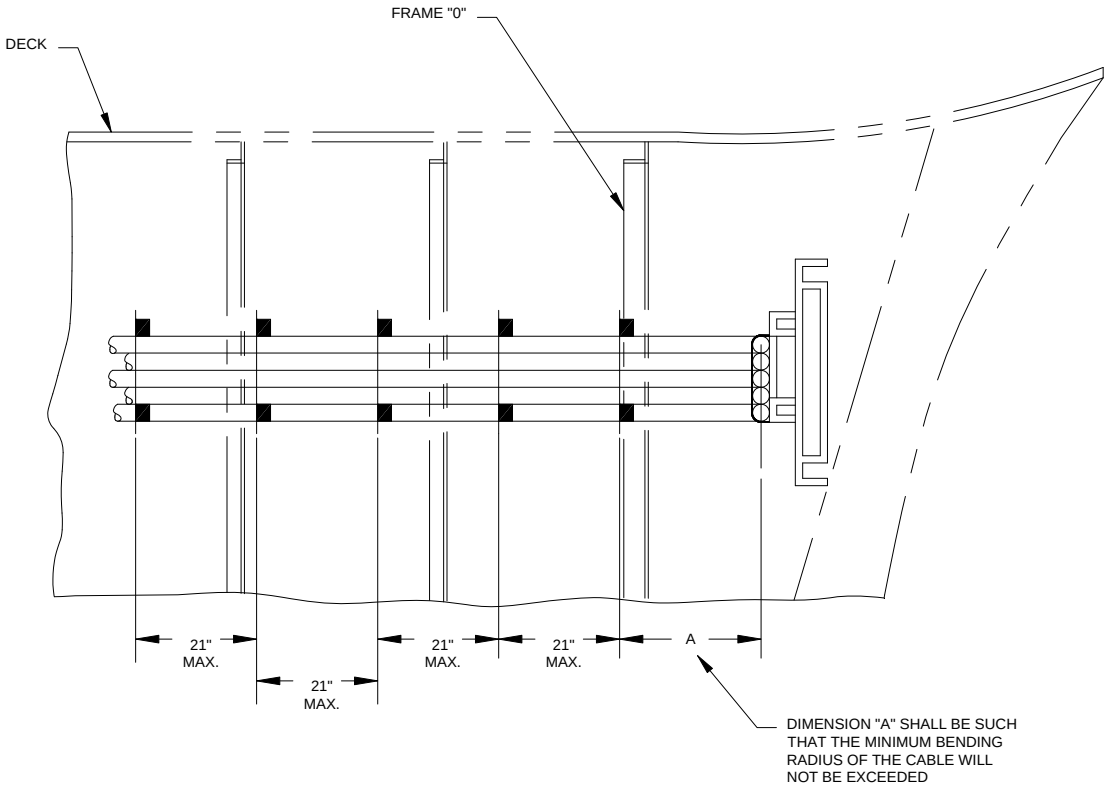


FIGURE 4B66. Cable support for degaussing cables (surface ships) – elevation looking outboard.

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METHOD 4B-67-1
CABLES IN TURN OF BOW
EXTENDING FORWARD OF FRAME "0"
(SEE NOTE 1)



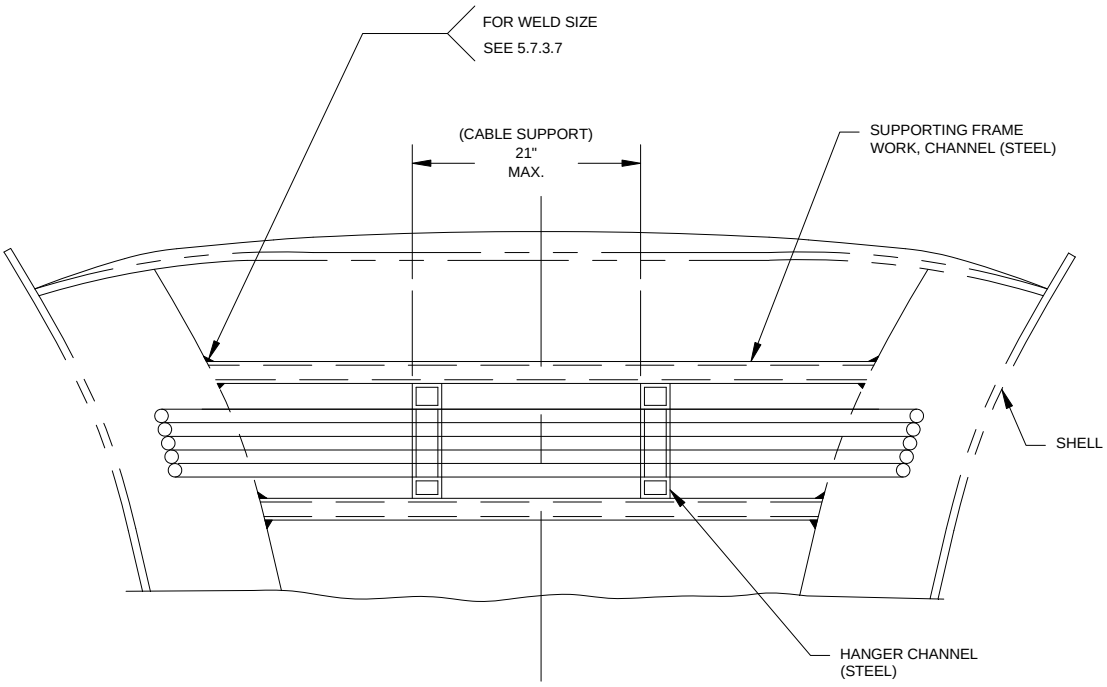
NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B67. Cable support for degaussing cables (surface ships) – elevation looking outboard.

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TYPICAL FRAME (LOOKING FORWARD) FOR METHOD 4B-71-1
CABLES IN TURN OF BOW
ATTACHMENT TO BULKHEAD

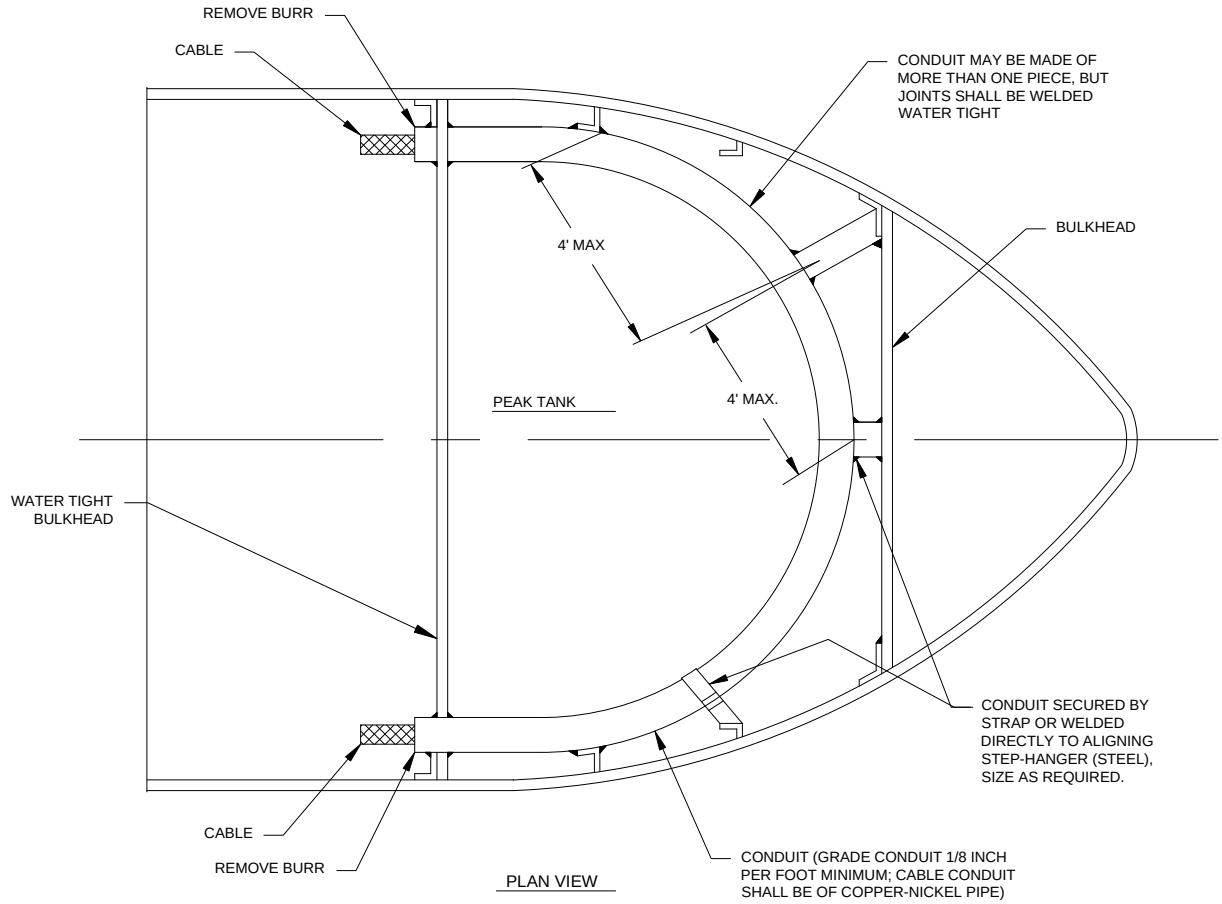


TYPICAL FRAME LOOKING FORWARD

FIGURE 4B68. Cable support for degaussing cables (surface ships) – typical frame.

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METHOD 4B-69-1
CABLES IN TURN OF BOW THROUGH CONDUIT
METHOD MAY BE USED FOR GENERAL INSTALLATION WHERE APPLICABLE
(SEE NOTE 1)



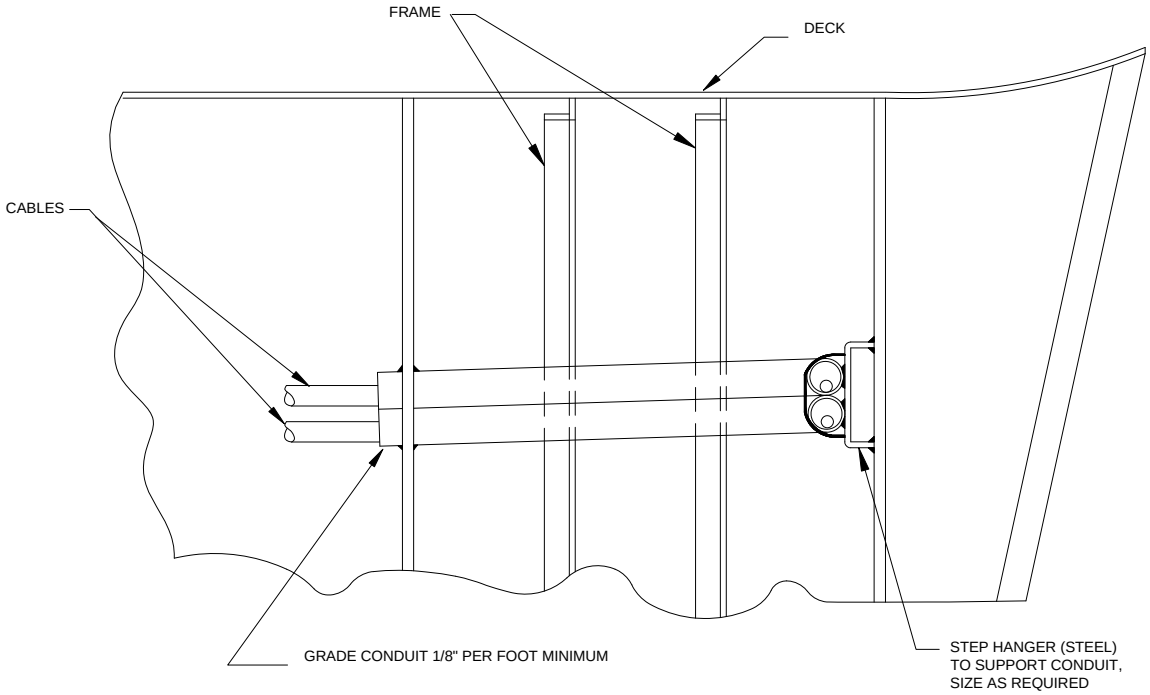
NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B69. Cable support for degaussing cables (surface ships).

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ELEVATION VIEW (LOOKING OUTBOARD) FOR METHOD 4B-69-1
CABLES IN TURN OF BOW THROUGH CONDUIT

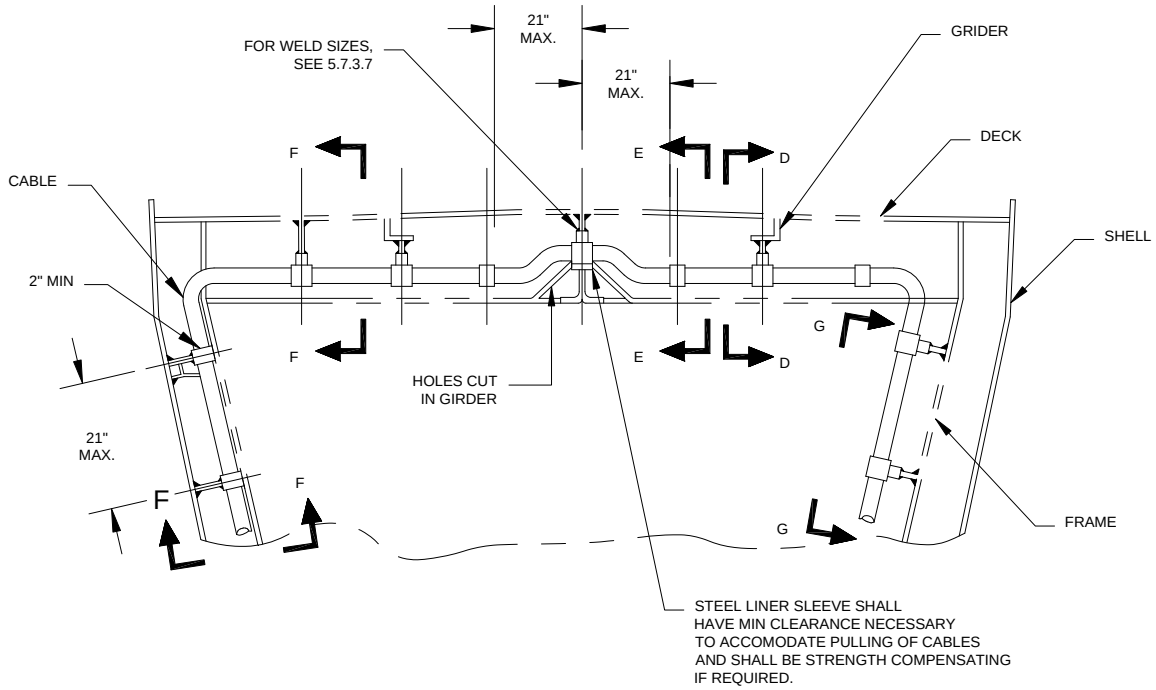


ELEVATION LOOKING OUTBOARD
(SEE 4B69 FOR PLAN VIEW)

FIGURE 4B70. Cable support for degaussing cables (surface ships).

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METHOD 4B-71-1
CABLE RUNS ATHWART SHIP - OVERHEAD
(SEE NOTE 1)



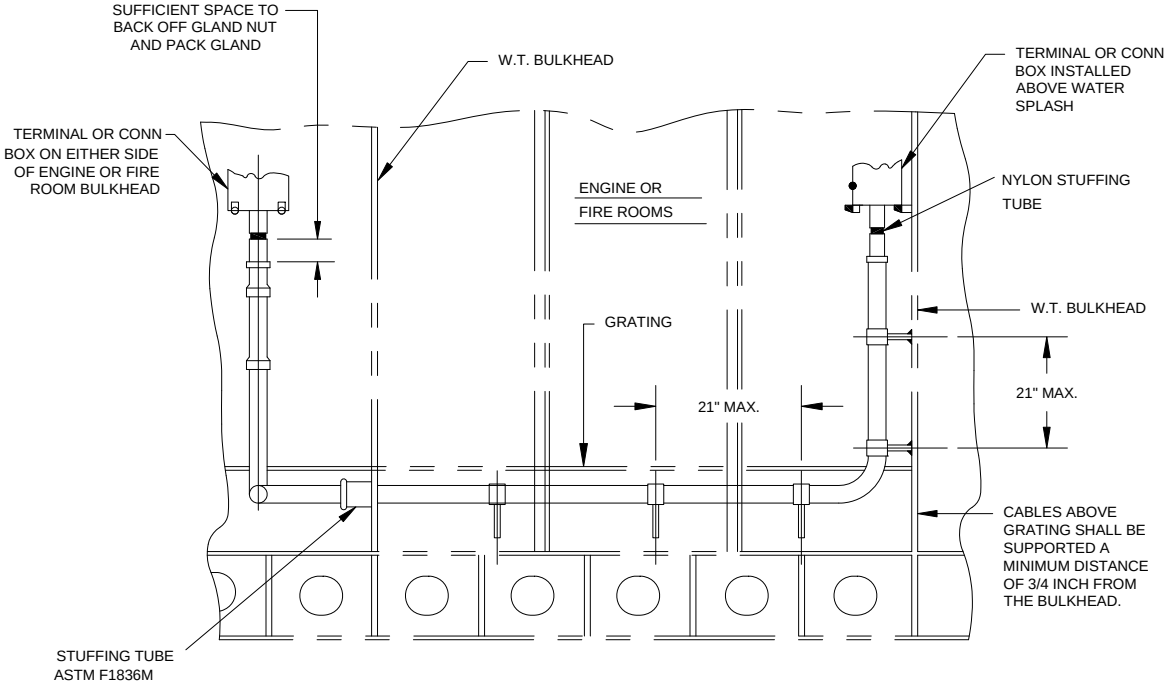
NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B71. Cable support for degaussing cables (surface ships).

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METHOD 4B-72-1
CABLE IN BILGES - LEAD SHEATHED
(SEE NOTE 1)



ELEVATION LOOKING OUTBOARD

NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B72. Cable support for degaussing cables (surface ships).

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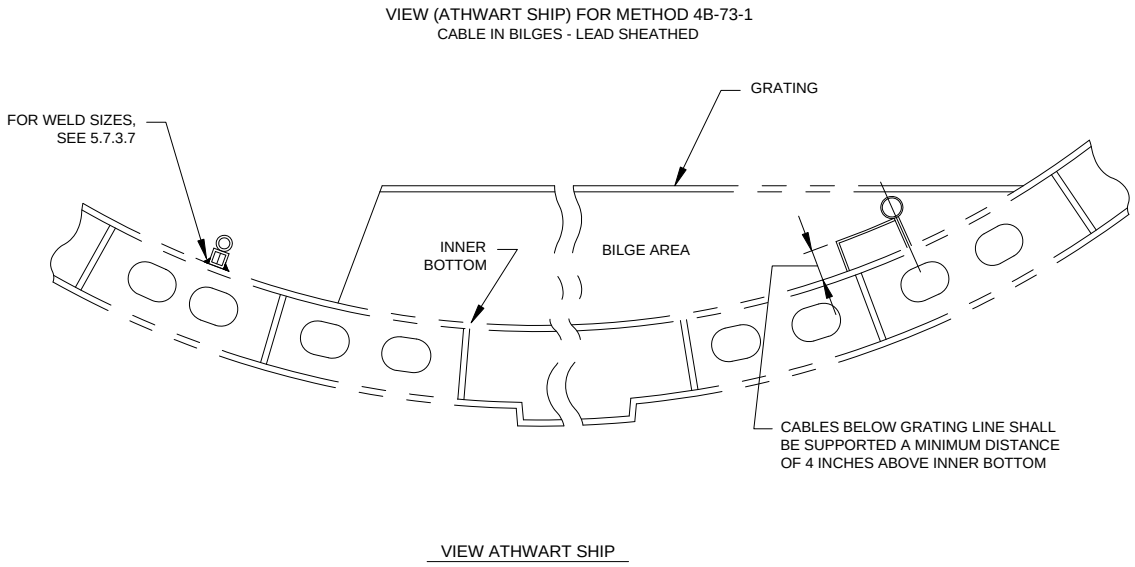


FIGURE 4B73. Cable support for degaussing cables (surface ships).

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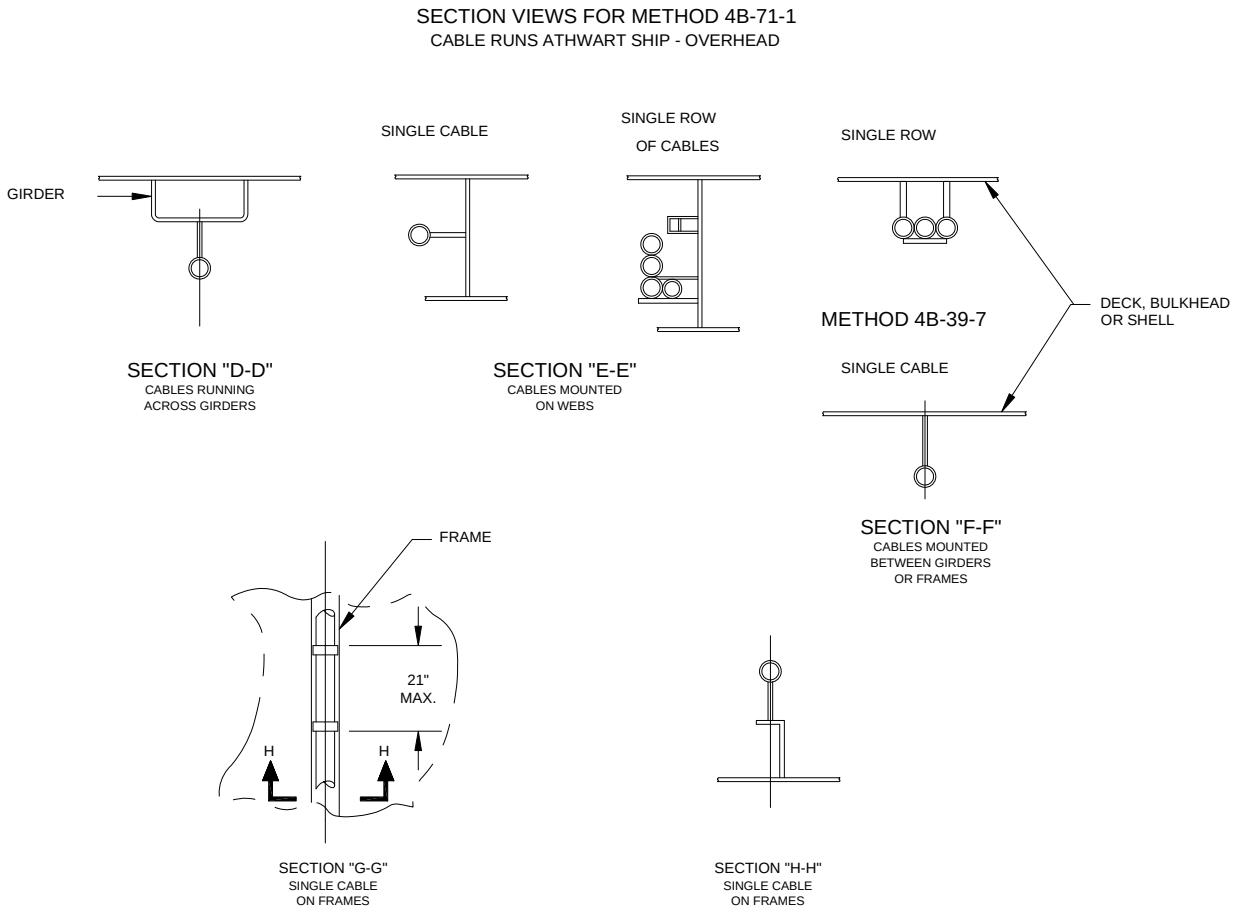


FIGURE 4B74. Cable support for degaussing cables (surface ships) – section views.

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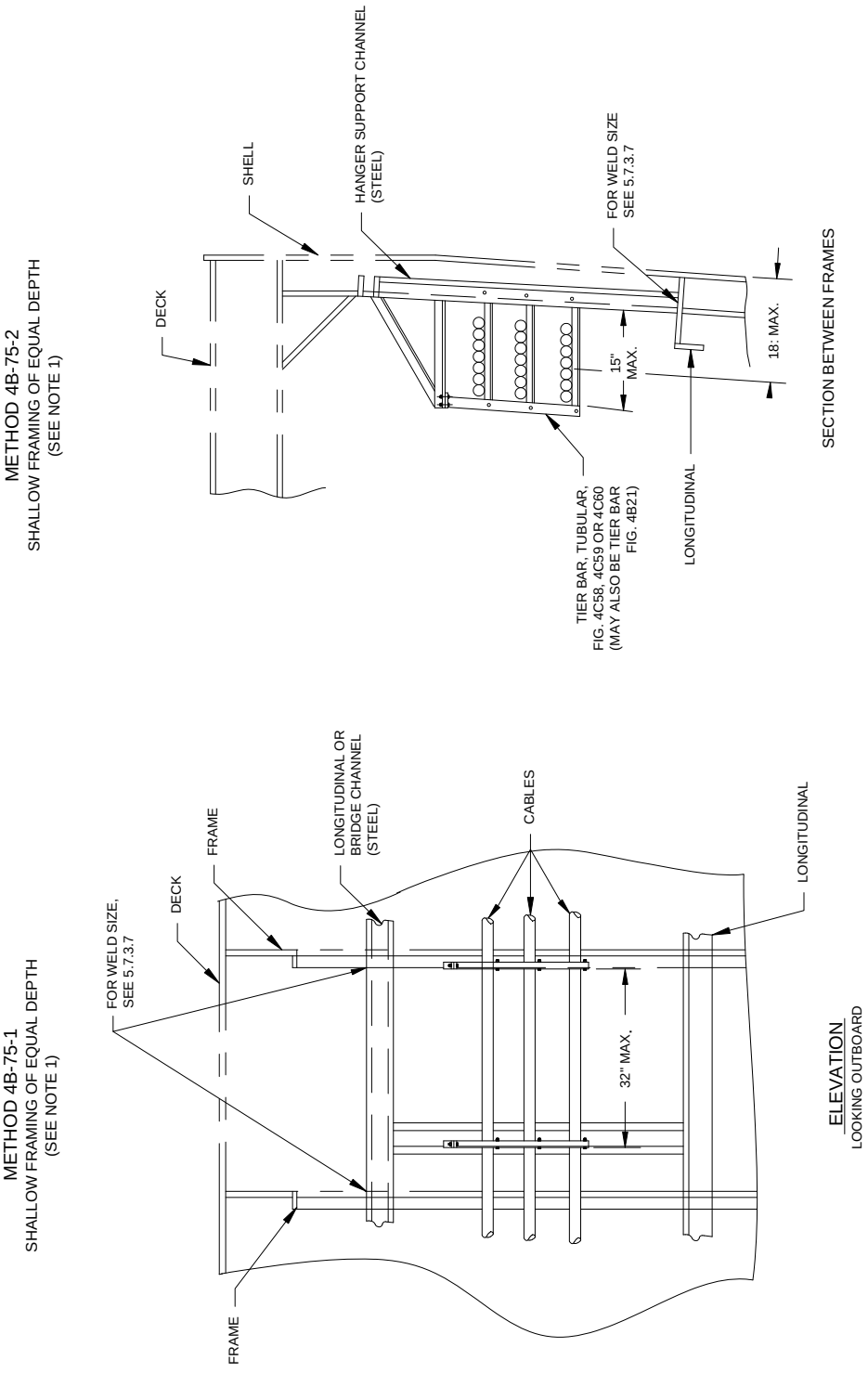


FIGURE 4B75. Cable support for degaussing cables (surface ships).

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NOTE:

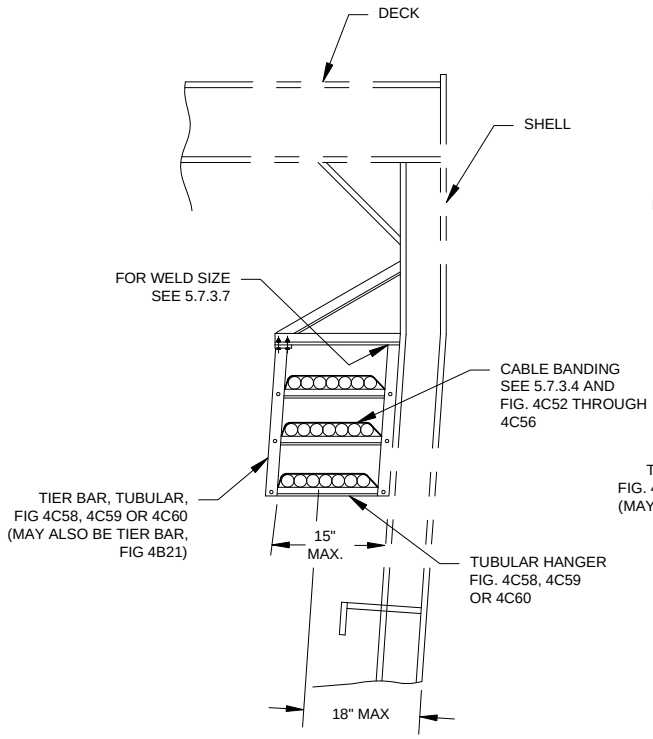
1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B75. Cable support for degaussing cables (surface ships) - Continued.

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METHOD 4B-76-1
SHALLOW FRAMING OF EQUAL DEPTH

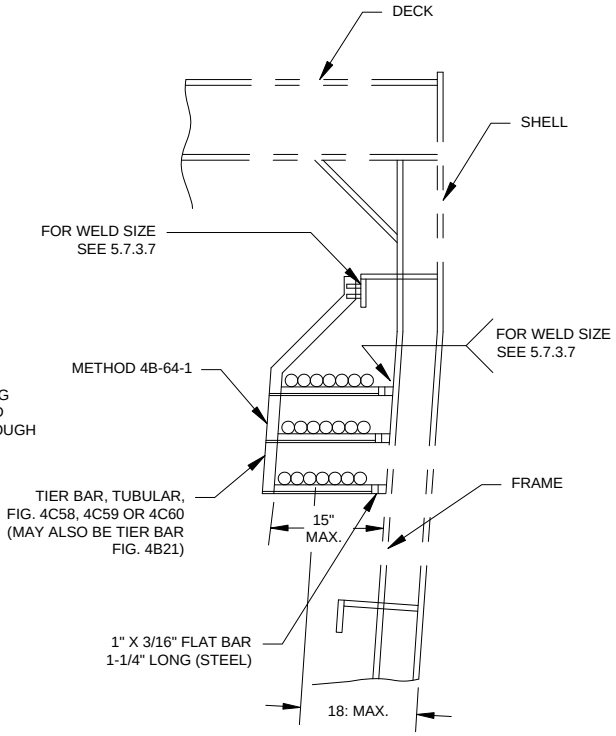
METHOD SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATIONS WHERE APPLICABLE



SECTION AT FRAMES
LARGE VESSEL

METHOD 4B-76-2
SHALLOW FRAMING OF EQUAL DEPTH
(SEE NOTE 1)

METHOD SHOWN ON THIS SHEET MAY BE USED FOR GENERAL INSTALLATIONS WHERE APPLICABLE



SECTION AT FRAMES
SMALL VESSELS

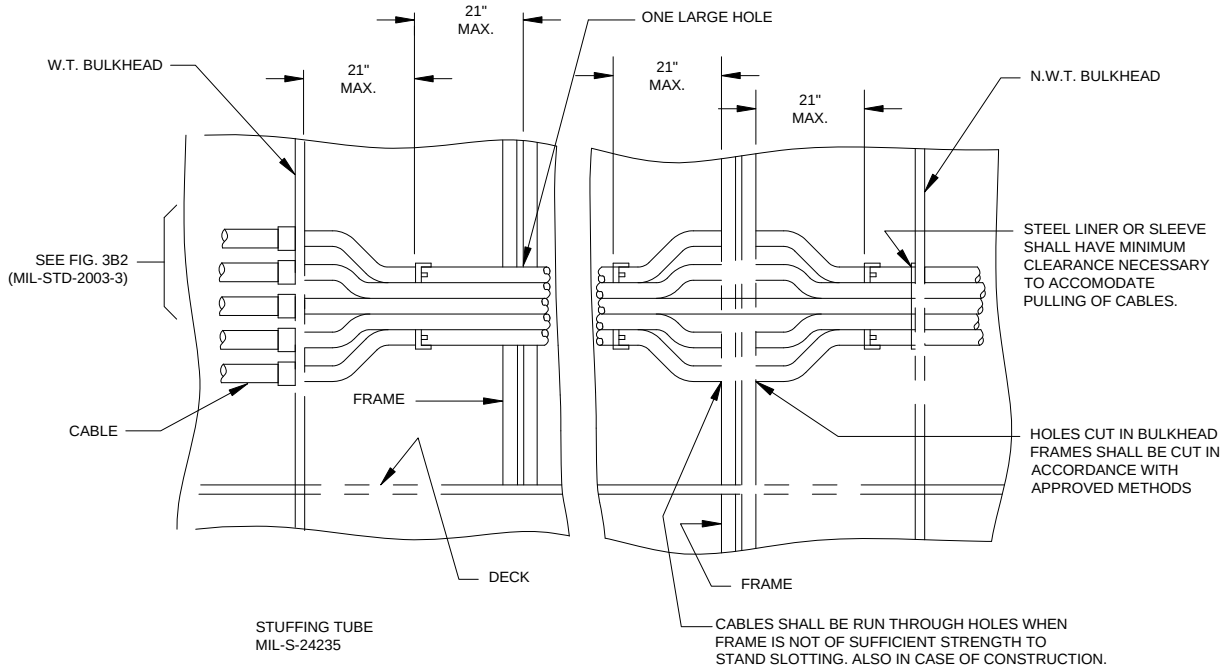
NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B76. Cable support for degaussing cables (surface ships).

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METHOD 4B-77-1
CABLES RUN LONGITUDINALLY
THROUGH WEB FRAMING
(SEE NOTE 1)



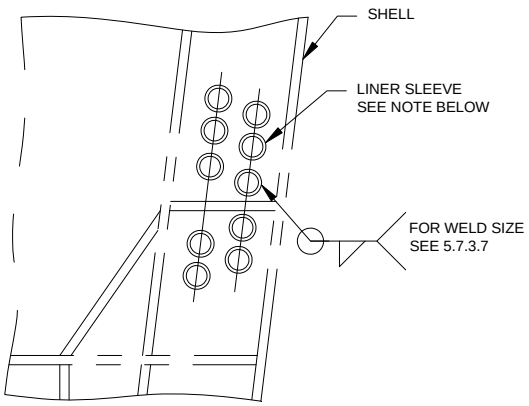
NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B77. Cable support for degaussing cables (surface ships).

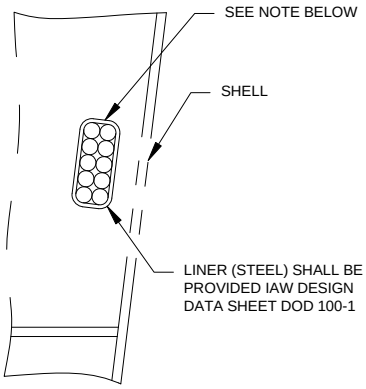
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Check the source to verify that this is the current version before use.

METHOD 4B-78-1
CABLES RUN LONGITUDINALLY THROUGH WEB FRAMING
TWO TIERS - USING LINER SLEEVES



SECTION AT FRAME

METHOD 4B-78-2
CABLES RUN LONGITUDINALLY THROUGH WEB FRAMING
TWO TIERS - USING STEEL LINER

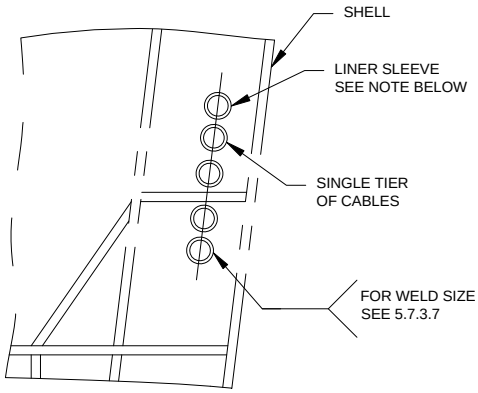


SECTION AT N.W.T.
BULKHEAD OR FRAME

NOTE: Where light-tightness is required, cables passing through N.W.T. bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer per MIL-I-3064, Type HF.

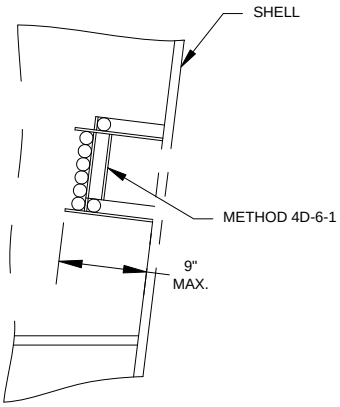
FIGURE 4B78. Cable support for degaussing cables (surface ships).

METHOD 4B-79-1
CABLES RUN LONGITUDINALLY THROUGH WEB FRAMING
SINGLE TIER - USING LINER SLEEVE



SECTION AT FRAME

METHOD 4B-79-2
CABLES RUN LONGITUDINALLY THROUGH WEB FRAMING
SINGLE TIER - USING CABLE HANGER



SECTION BETWEEN FRAMES
WHEN CABLE TIER IS 9 INCHES
OR UNDER FROM SHELL

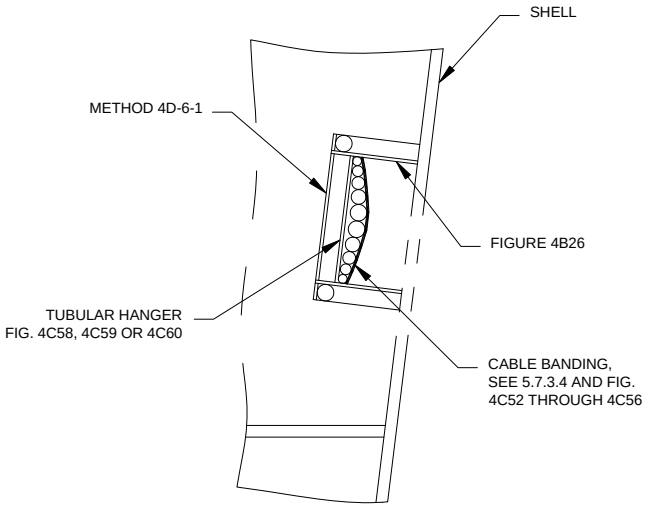
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. Where light-tightness is required, cables passing through N.W.T. bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer per MIL-I-3064, Type HF.

FIGURE 4B79. Cable support for degaussing cables (surface ships).

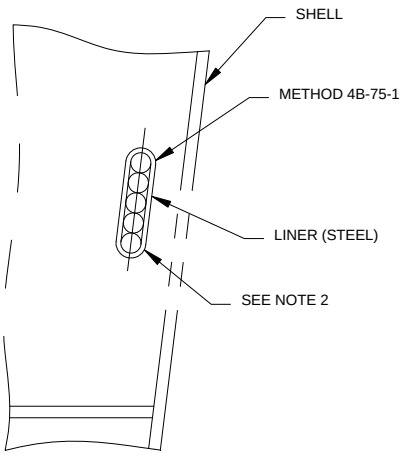
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METHOD 4B-80-1
CABLES RUN LONGITUDINALLY THROUGH WEB FRAMING
SINGLE TIER - USING CABLE HANGER
(SEE NOTE 1)



SECTION BETWEEN FRAMES
WHEN CABLE TIER IS OVER
9 INCHES FROM SHELL
BALLISTIC AND NON-BALLISTIC

METHOD 4B-80-2
CABLES RUN LONGITUDINALLY THROUGH WEB FRAMING
SINGLE TIER - USING STEEL LINER
(SEE NOTE 1)



SECTION AT N.W.T.
BULKHEAD OR FRAME

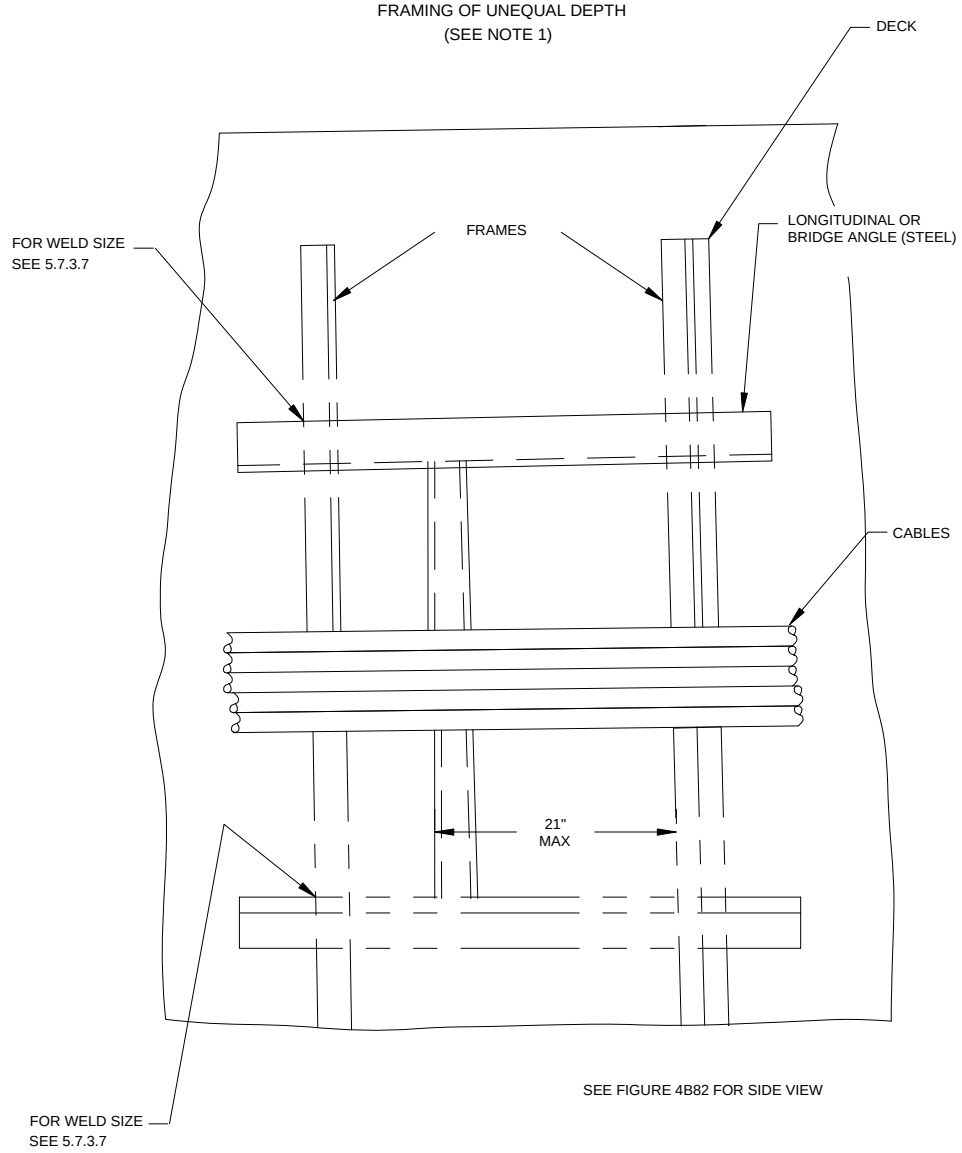
NOTES:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.
2. Where light-tightness is required, cables passing through N.W.T. bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer per MIL-I-3064, Type HF.

FIGURE 4B80. Cable support for degaussing cables (surface ships).

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METHOD 4B-81-1
FRAMING OF UNEQUAL DEPTH
(SEE NOTE 1)

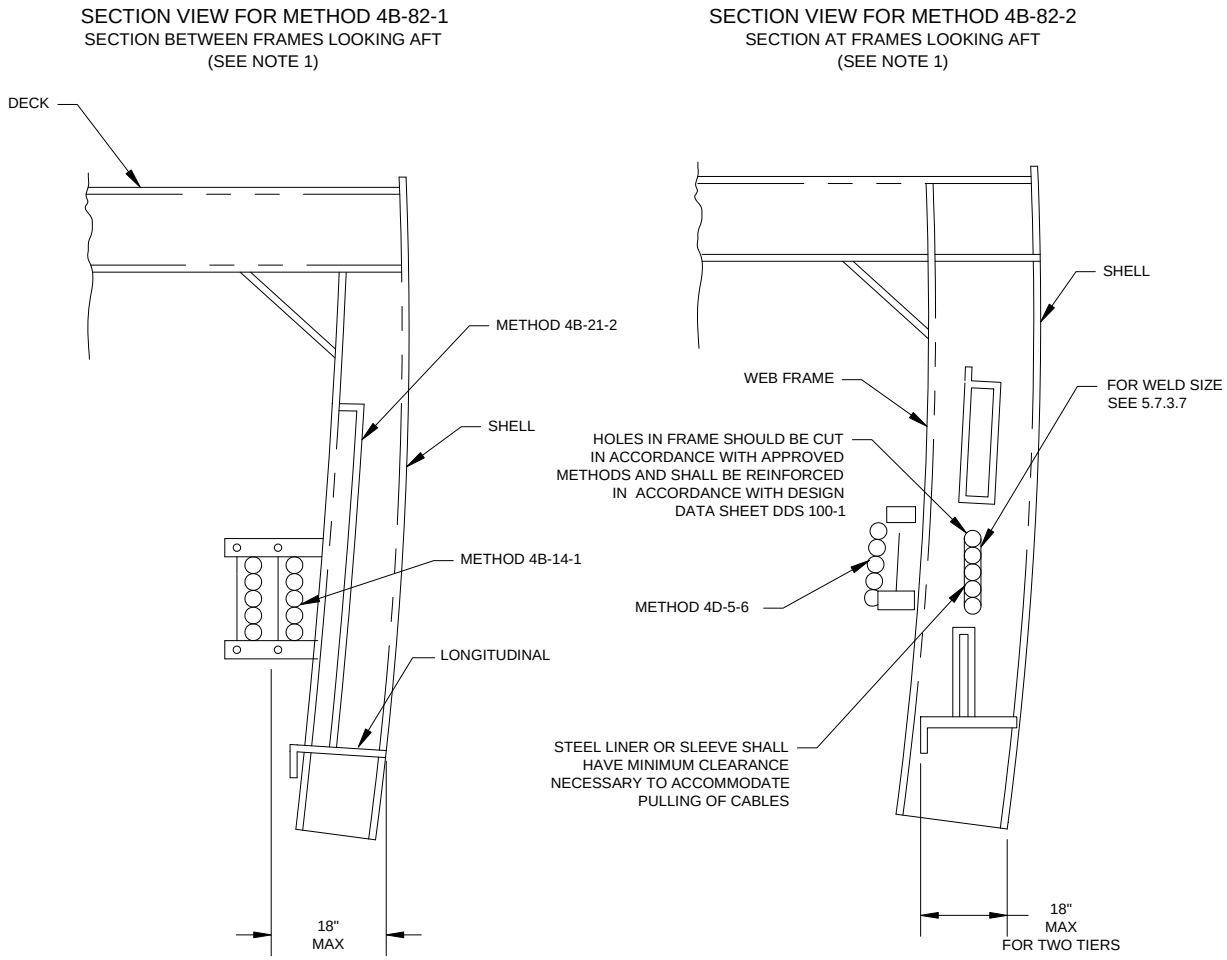


NOTE:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.

FIGURE 4B81. Cable support for degaussing cables (surface ships).

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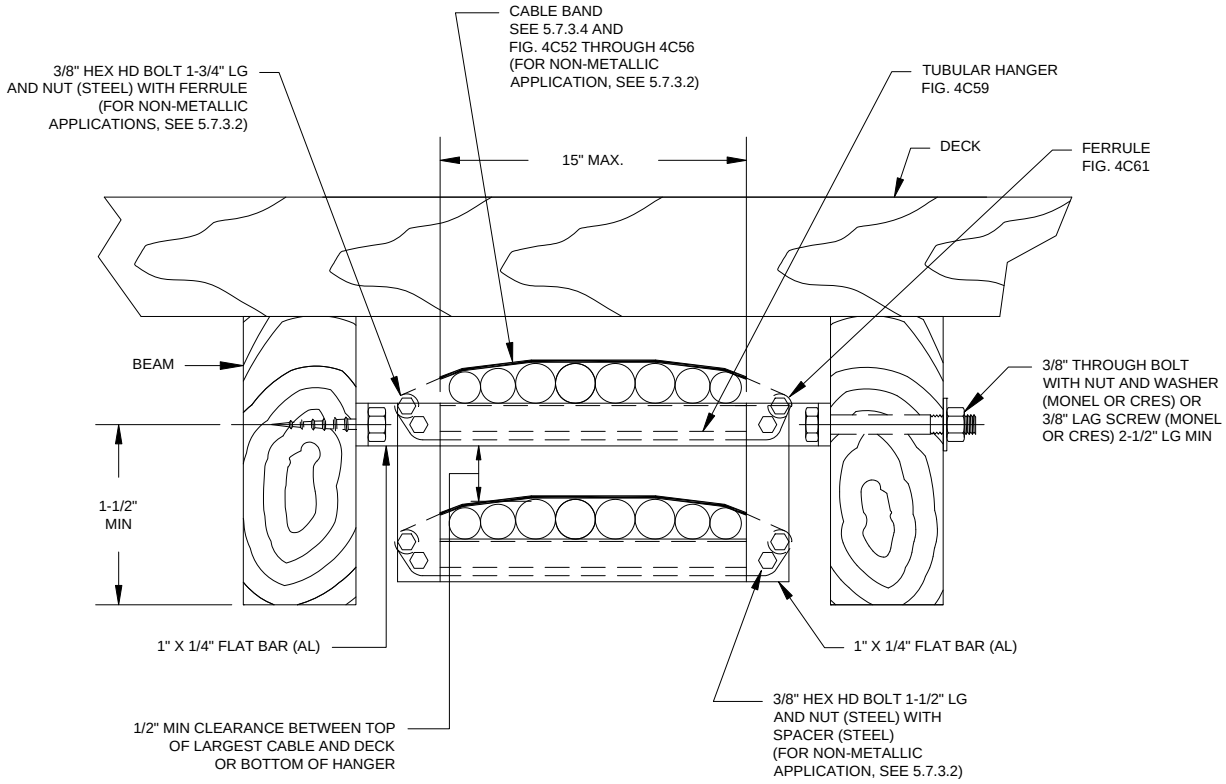
NOTES:

1. This figure provides cable support details for degaussing cables. Methods shown may be used for general installation where applicable.
   a. When insulation is used with degaussing cables, use corresponding insulation methods for cable hanger methods listed.
   b. Use appropriate ballistic methods for degaussing cable installation as required.
   c. Where light-tightness is required, cables passing through non-watertight bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer in accordance with MIL-I-3064, Type HF.
   d. Conduit shall be graded at ⅛" per foot minimum.
   e. Cable conduit shall be of copper-nickel pipe.
2. Where light-tightness is required, cables passing through N.W.T. bulkheads shall be run through individual liner sleeves and then carefully packed with plastic sealer per MIL-I-3064, Type HF.

FIGURE 4B82. Cable support for degaussing cables (surface ships).

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Check the source to verify that this is the current version before use.

METHOD 4B-83-1
CABLE RUNS BETWEEN BEAMS
(CABLE BANDS NOT REQUIRED)
(SEE NOTES)



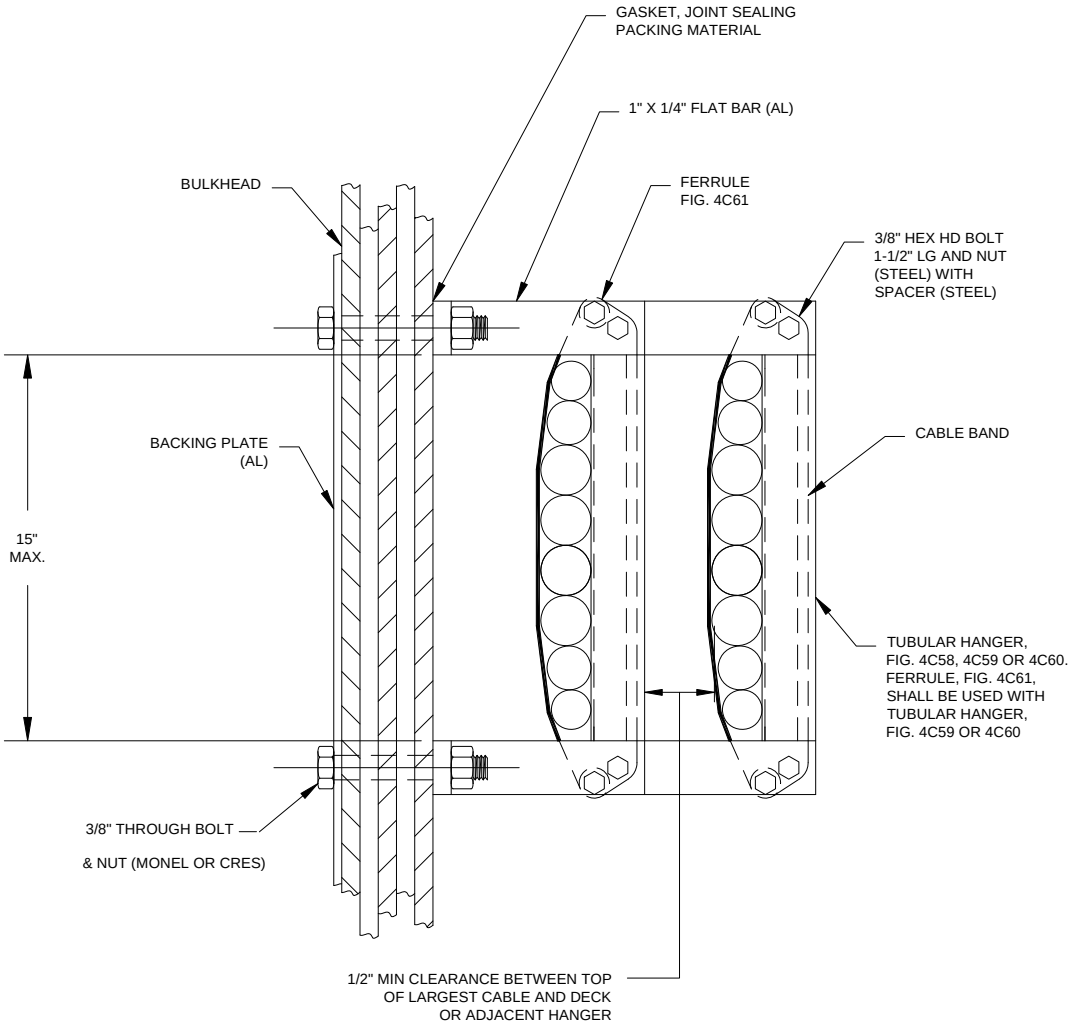
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.

FIGURE 4B83. Cableways on wooden ships (surface ships)
(*Guidance only*)

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METHOD 4B-84-1
WATERTIGHT BULKHEADS
UNDER ONE INCH THICK
(SEE NOTES)



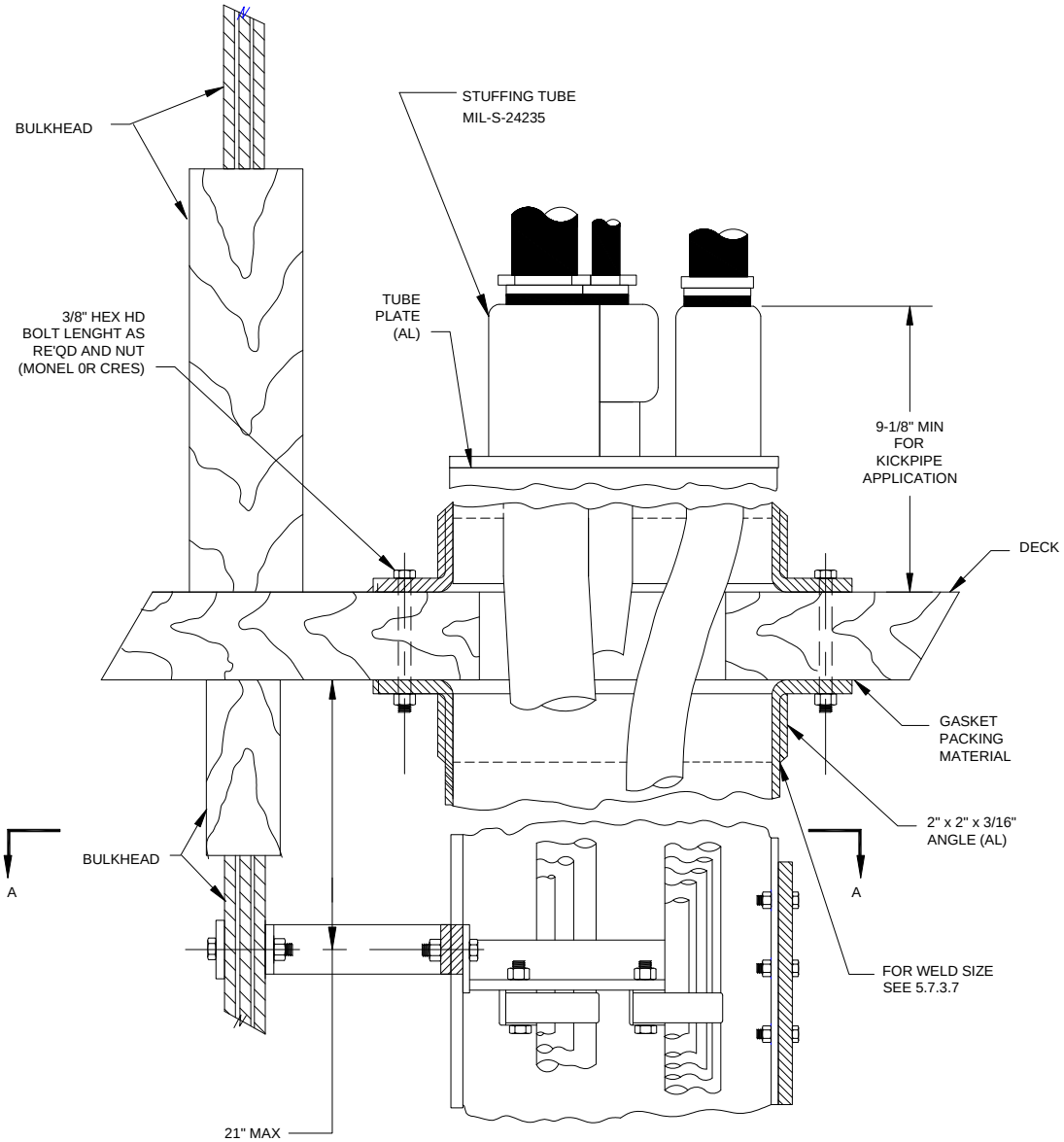
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.

FIGURE 4B84. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-85-1
VERTICAL TIERS OF CABLES IN TRUNKS
FOR SECTION "A-A" VIEW, SEE FIGURE 4B86
(SEE NOTES)



ELEVATION
PART IN SECTION

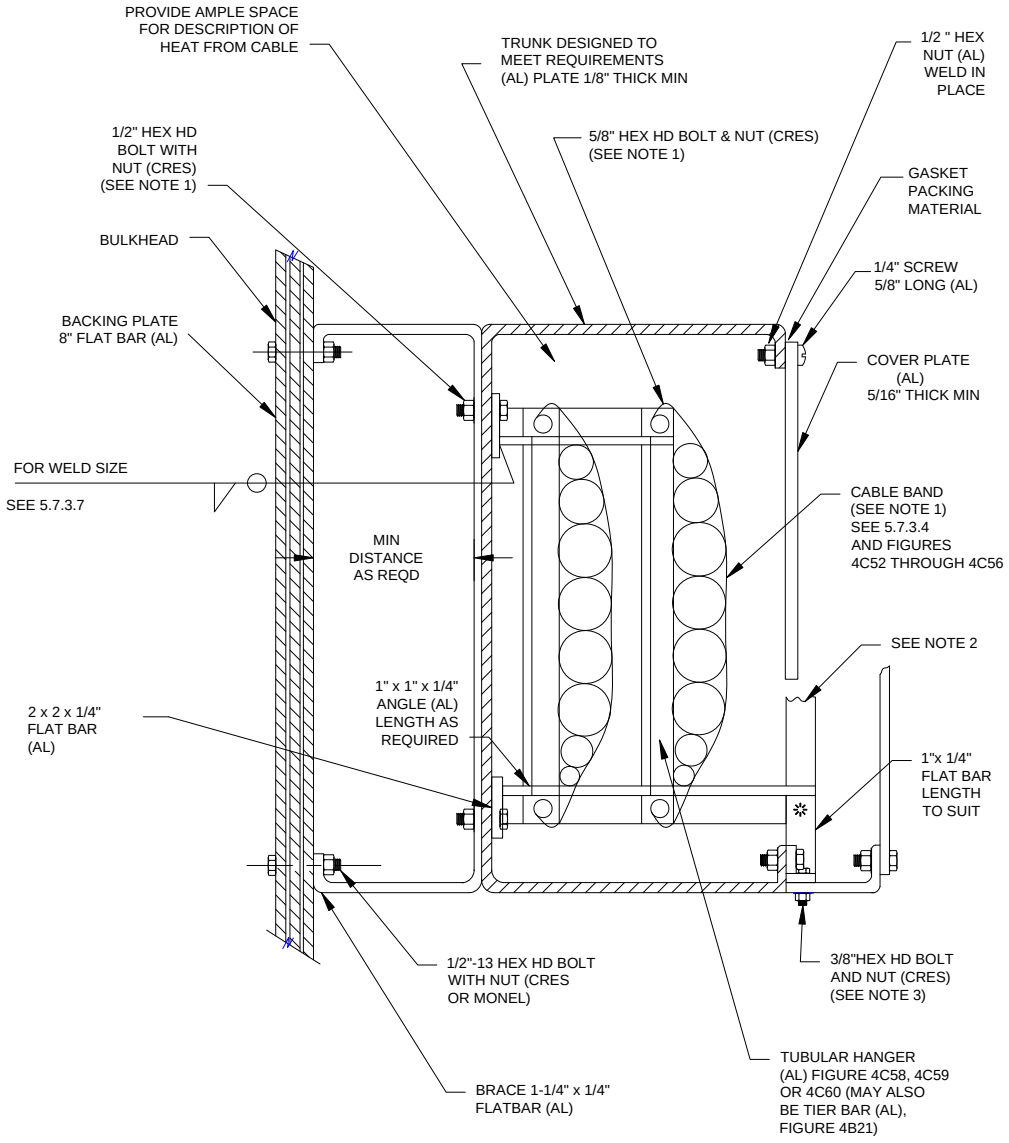
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.

FIGURE 4B85. Cableways on wooden ships (surface ships).
(*Guidance only*)

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SECTION A-A VIEW FOR METHOD 4B-85-1
VERTICAL TIERS OF CABLES IN TRUNKS
(SEE NOTES)



PLAN VIEW
AT SECTION A-A

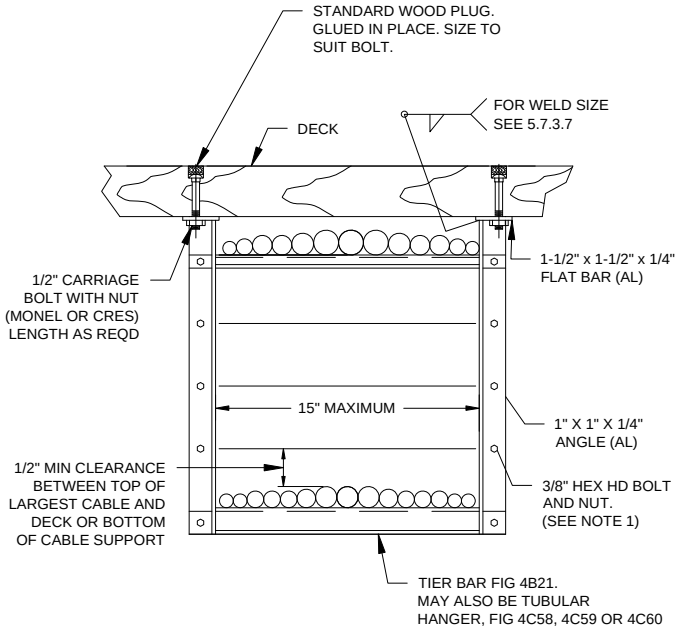
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. Two rows of cables with a 9" length of cable span shall be considered normal. If a longer span or three rows of cables are required, secure the outer end of hanger support to the wall of the trunk.

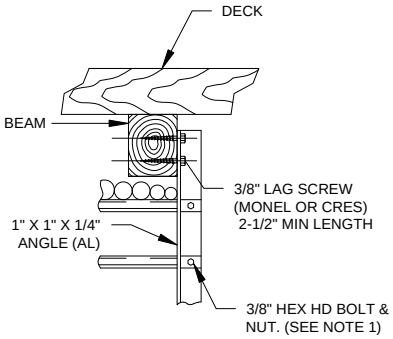
FIGURE 4B86. Cableways on wooden ships (surface ships).
(*Guidance only*)

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Check the source to verify that this is the current version before use.

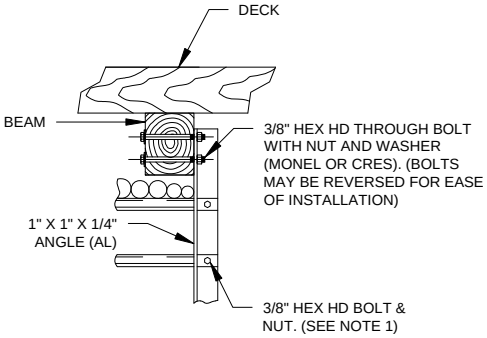
METHOD 4B-87-1
HANGER SUPPORTS SECURED TO WATERTIGHT DECKS
(SEE NOTES)



METHOD 4B-87-2
HANGER SUPPORTS SECURED TO BEAMS
(SEE NOTES)



METHOD 4B-87-3
HANGER SUPPORTS SECURED TO BEAMS
(SEE NOTES)



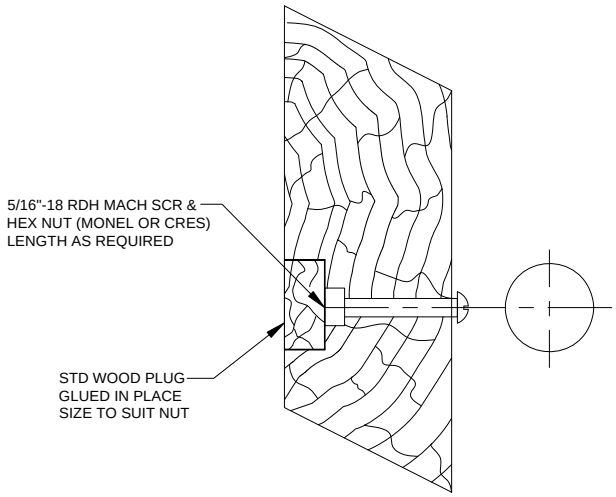
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. The methods shown are for use with cables up to 2" in diameter. For cables greater than 2" in diameter, use methods shown on figures 4B118, 4B119, 4B120, and 4B121.

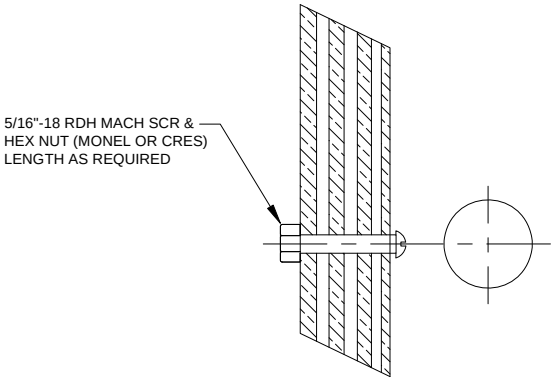
FIGURE 4B87. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-88-1
WATERTIGHT BULKHEADS OVER ONE INCH THICK
WHERE MOISTURE OR DRIP IS PRESENT
CABLE SUPPORTED VERTICALLY
(SEE NOTES)



METHOD 4B-88-2
WATERTIGHT BULKHEADS ONE INCH MAXIMUM THICKNESS
WHERE MOISTURE OR DRIP IS PRESENT
CABLE SUPPORTED HORIZONTALLY
(SEE NOTES)



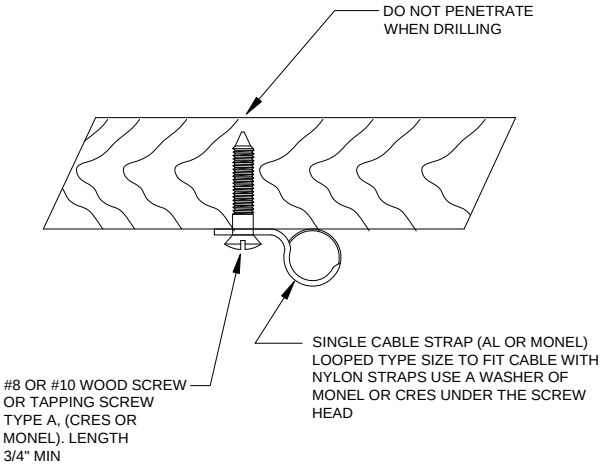
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.

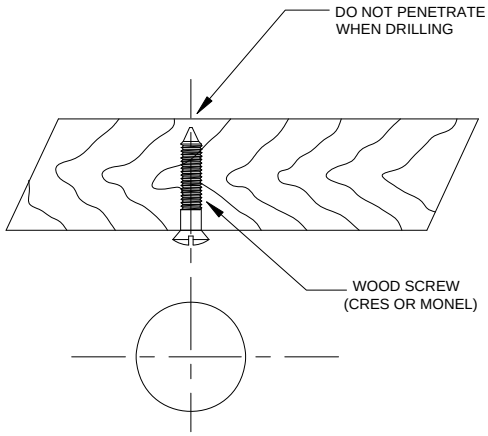
FIGURE 4B88. Cableways on wooden ships (surface ships).
(*Guidance only*)

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Check the source to verify that this is the current version before use.

METHOD 4B-89-1
DECKS OR BULKHEADS OVER ONE INCH THICK
ALSO FOR BEAMS
CABLES WITH A MAXIMUM DIAMETER OF 5/8"
SINGLE SUPPORT - ALUMINUM CABLE STRAPS
(SEE NOTES)



METHOD 4B-89-2
DECKS OR BULKHEADS OVER ONE INCH THICK
ALSO FOR BEAMS
(SEE NOTES)



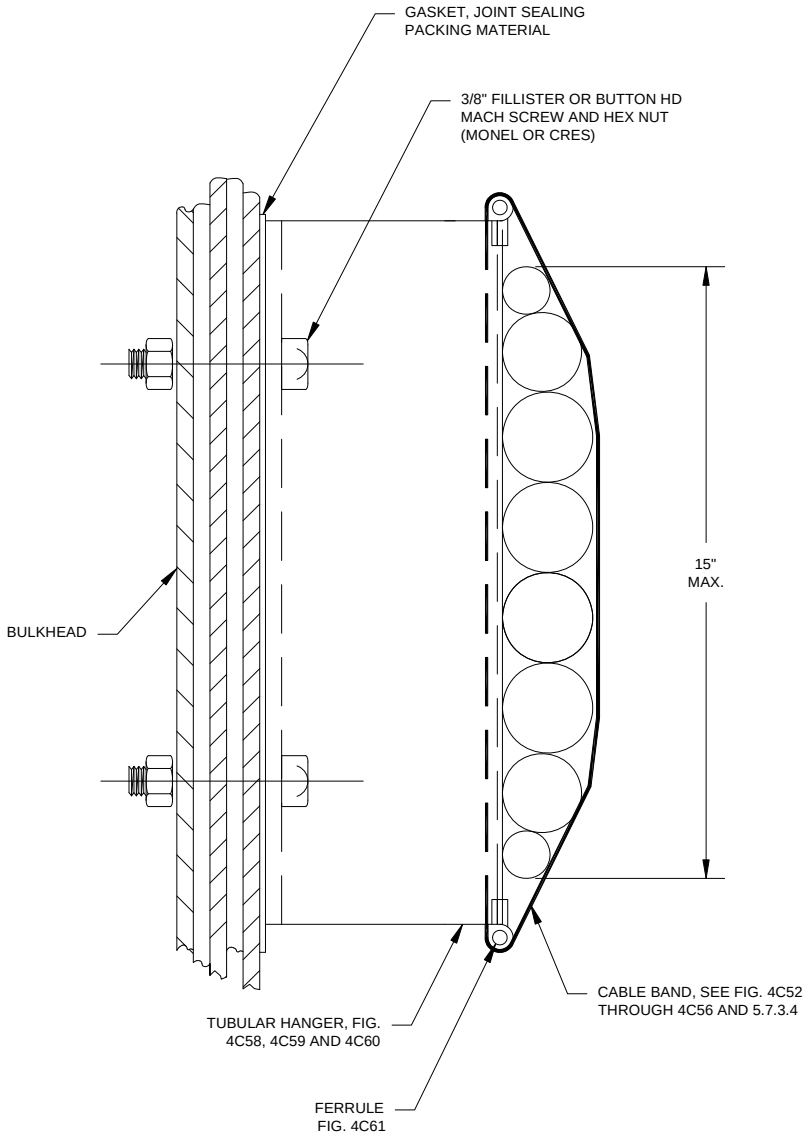
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.

FIGURE 4B89. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-90-1
SINGLE TIER OF CABLES
SUPPORTED FROM WATERTIGHT BULKHEADS
UNDER ONE INCH THICKNESS
(SEE NOTES)



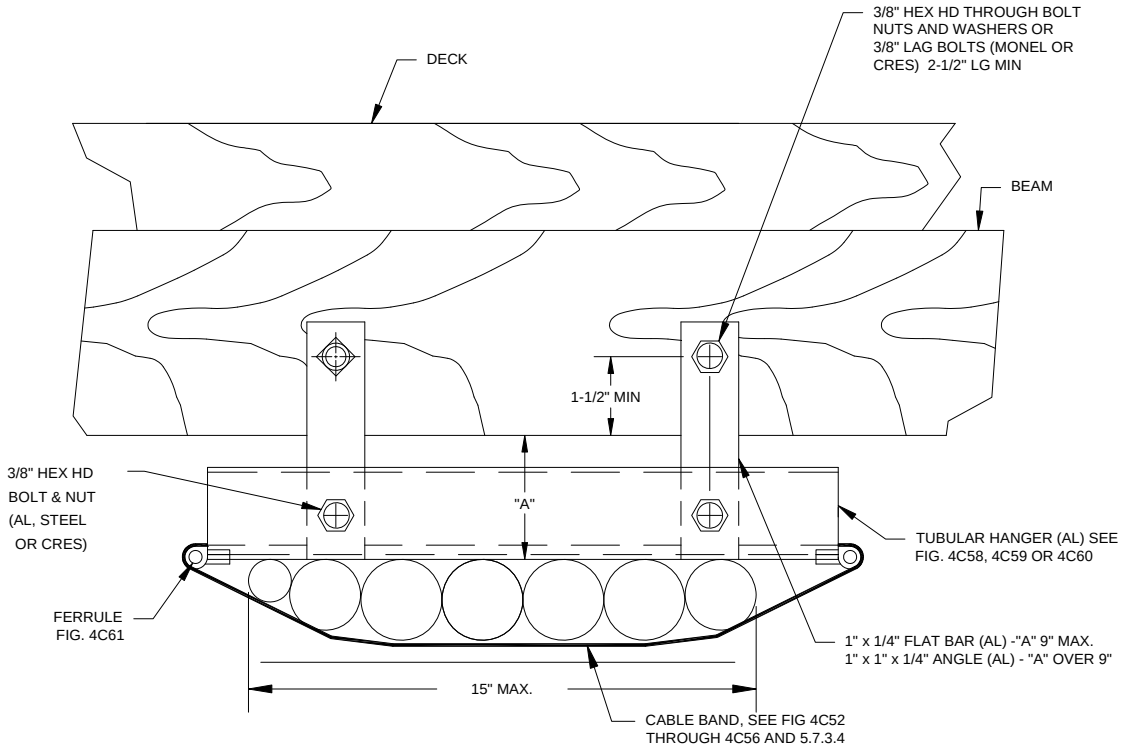
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. The method shown here may be used in lieu of those shown on figures 4B92 through 4B95.

FIGURE 4B90. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-91-1
SINGLE TIER OF CABLES
SUPPORTED FROM BEAMS
(SEE NOTES)



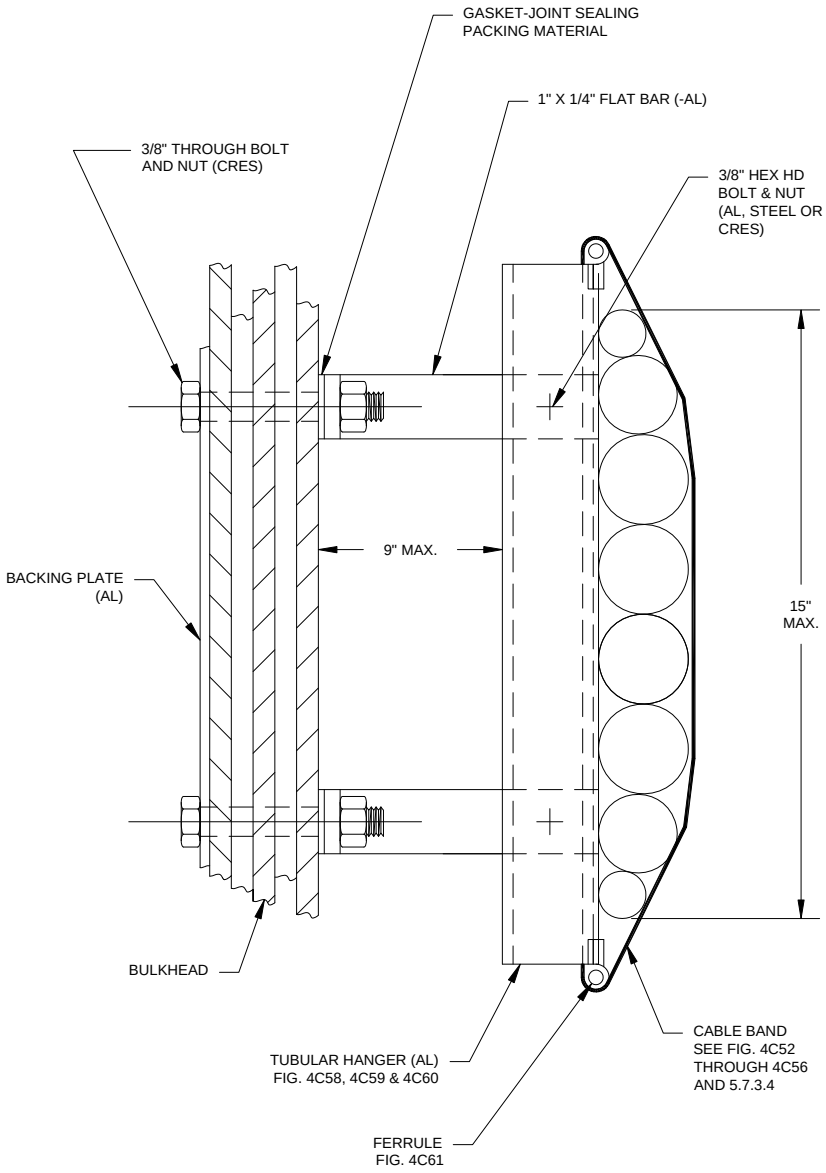
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. For cables greater than 2" in diameter, use methods shown on figures 4B92 through 4B95.
4. The method shown on figure 4B91 may be used in lieu of those shown on figures 4B92 through 4B95.

FIGURE 4B91. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-92-1
SINGLE TIER OF CABLES
SUPPORTED FROM WATERTIGHT BULKHEADS
UNDER ONE INCH THICKNESS
(SEE NOTES)



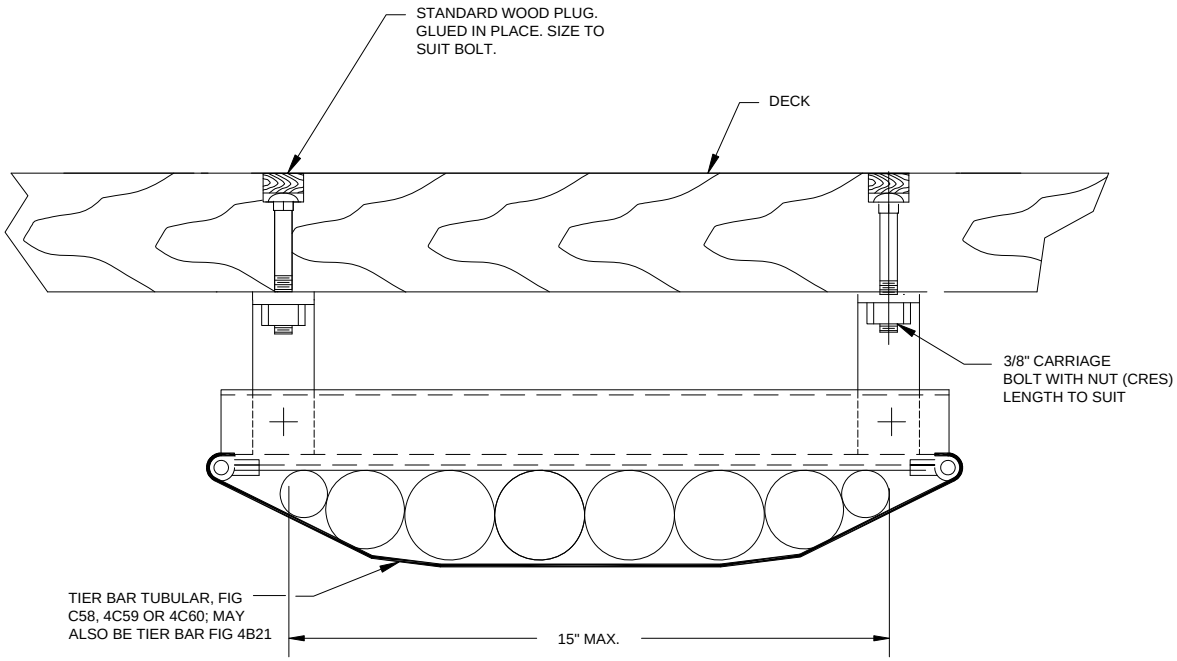
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. For cables greater than 2" in diameter, use methods shown on figures 4B92 through 4B95.
4. The method shown on figure 4B91 may be used in lieu of those shown on figures 4B92 through 4B95.

FIGURE 4B92. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-93-1
SINGLE TIER OF CABLES
SUPPORTED FROM WATERTIGHT DECKS
TWO INCH MINIMUM THICKNESS
(SEE NOTES)



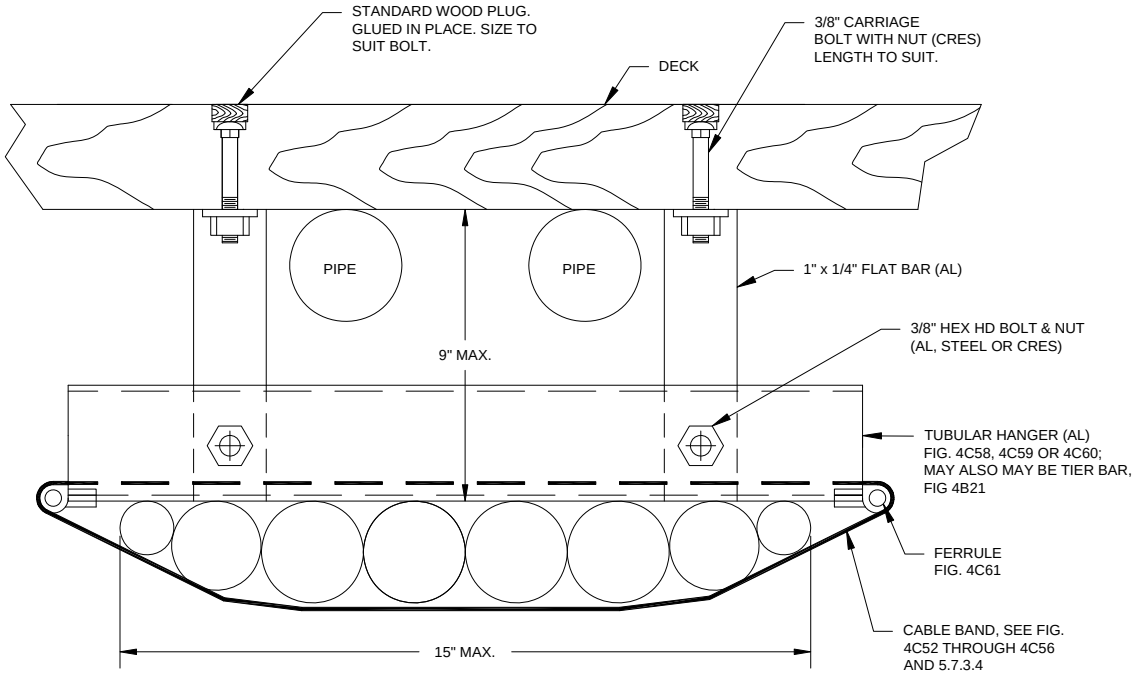
NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. For cables greater than 2" in diameter, use methods shown on figures 4B92 through 4B95.
4. The method shown on figure 4B91 may be used in lieu of those shown on figures 4B92 through 4B95.

FIGURE 4B93. Cableways on wooden ships (surface ships).
(*Guidance only*)

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METHOD 4B-94-1
SINGLE TIER OF CABLES
SUPPORTED AWAY FROM WOOD DECKS
(SEE NOTES)

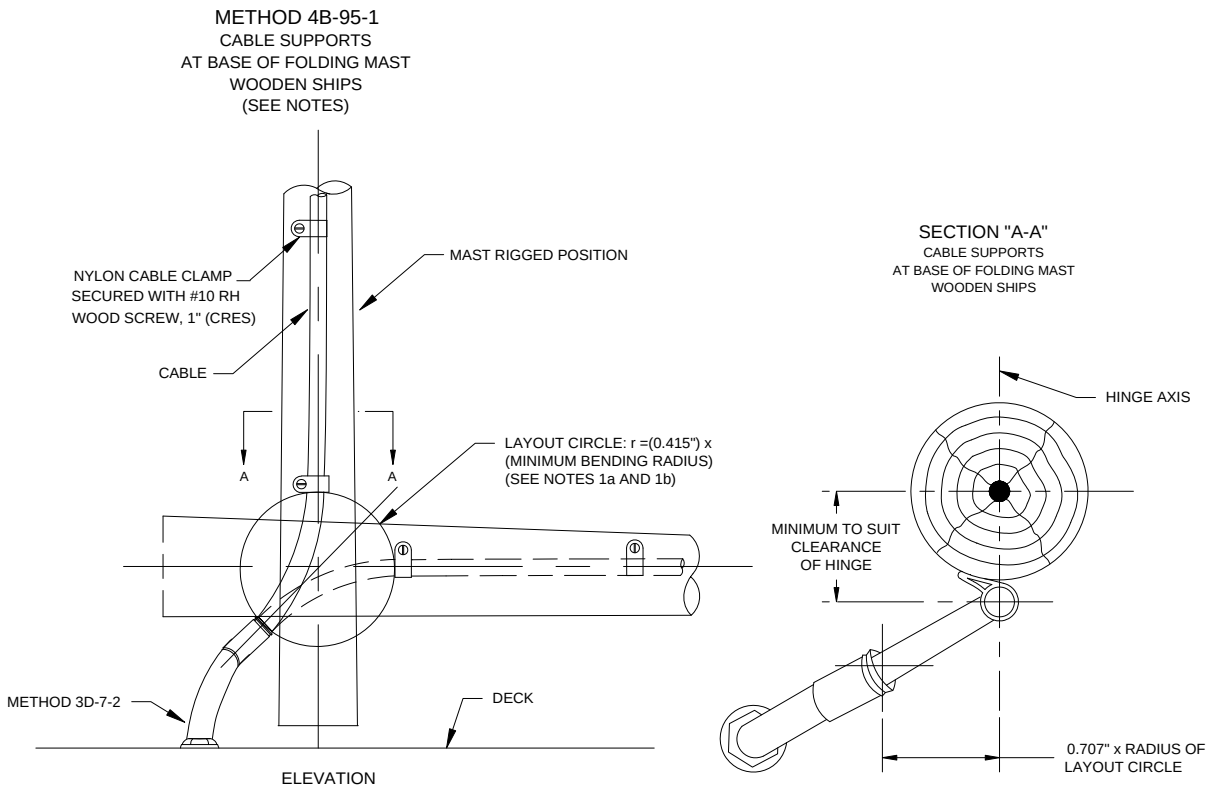


NOTES:

1. Methods shown may be used for general installations where applicable.
2. For non-magnetic application, see 5.7.3.2.
3. For cables greater than 2" in diameter, use methods shown on figures 4B92 through 4B95.
4. The method shown on figure 4B91 may be used in lieu of those shown on figures 4B92 through 4B95.

FIGURE 4B94. Supporting a single row of cables to avoid an obstruction or preserve alignment of cable runs on wooden decks.
(*Guidance only*)

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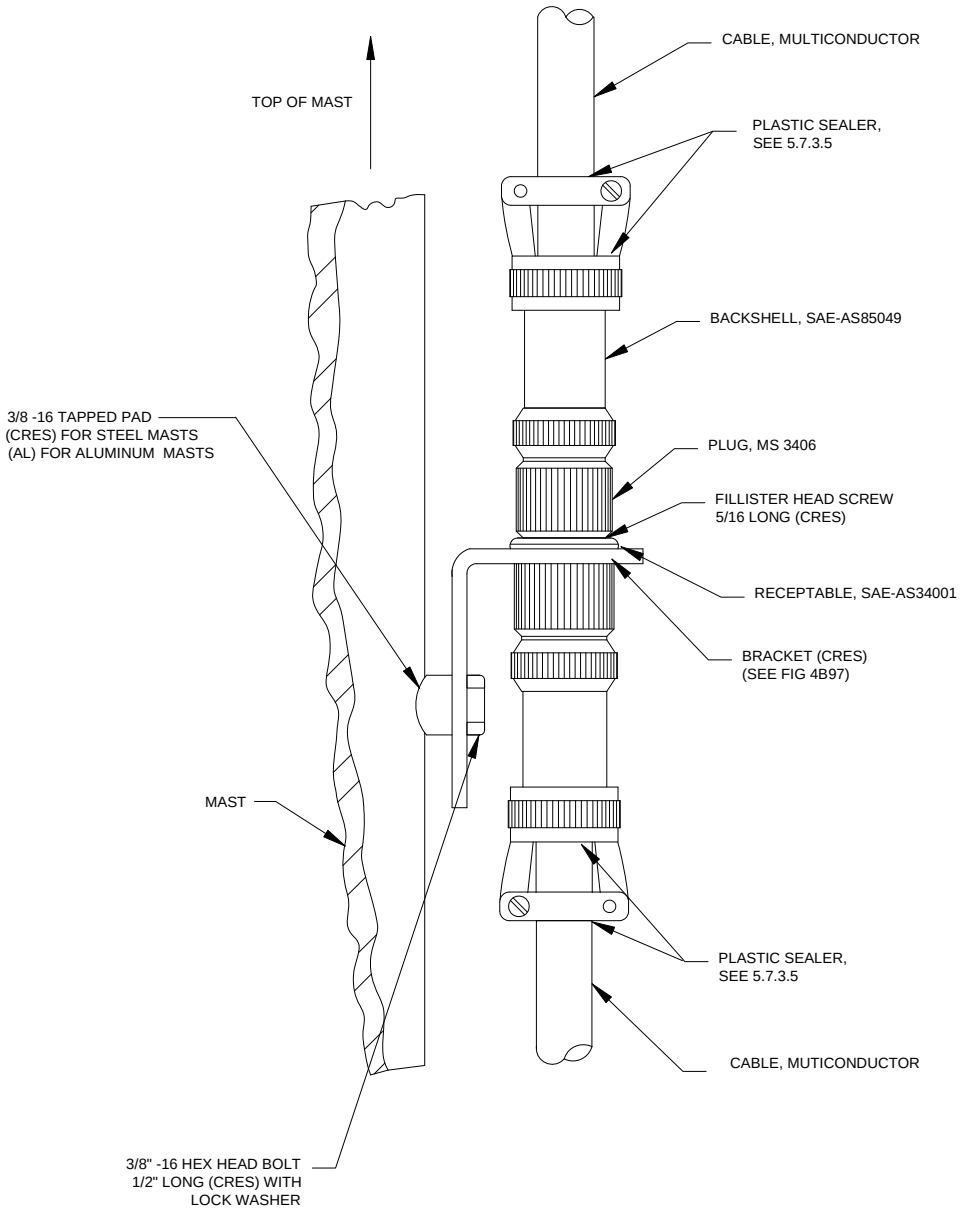
NOTE:

1. For folding masts on wooden ships:
   a. The dimension used for “r” shall not be less than 4".
   b. Where more than one cable is involved, “r” shall be determined by using the bending radius of the largest cable.
   c. Cable shall be firmly fastened with mast in 45° position to ensure proper length for flexing.

FIGURE 4B95. Cableways on folding masts (surface ships).

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METHOD 4B-96-1
COUPLING OF MULTIPLE CABLES
(CONNECTORS PER MIL-DTL-5015)



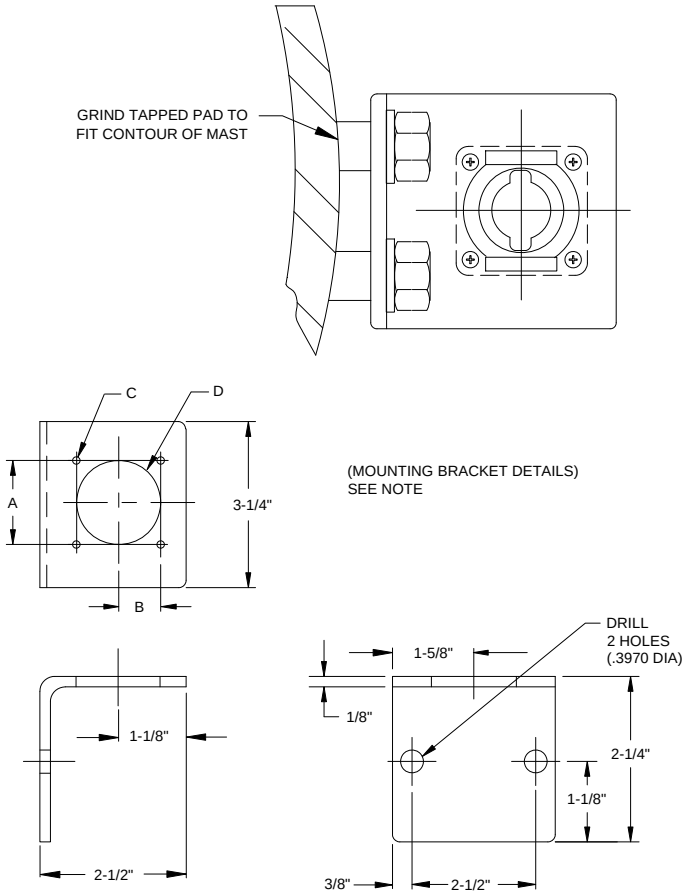
NOTE:

Water intrusion into connectors which are installed topside poses a significant problem. It is recommended that installers take additional precautions to protect these connectors. Various options are available. Local approved methods should be used when available. If a local method is not selected, the following is recommended: The connector shall be wrapped with a suitable electrical tape using a one half lap wrap followed with an application of heat shrink tubing in accordance with SAE-AMS-DTL-23053/5 or SAE-AMS-DTL-23053/15. This procedure shall be accomplished on each portion of the connector above and below the bracket. The tape shall extend 3 inches beyond the cable clamps and the heat shrink tubing shall extend 4 inches beyond the cable clamps.

FIGURE 4B96. Cableways on masts with connectors (surface ships).

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DETAILS FOR METHOD 4B-97-1
COUPLING OF MULTIPLE CABLES
(CONNECTORS PER MIL-DTL-5015)



NOTE:

1. Dimensions for the mounting bracket are as follows:

Shell size	Dimensions (inches)			
	A ±0.010	**B ±0.010**	**C (dia) ±0.031**	**D (dia) +0.010 -0.005**
18	1.062	0.531	0.120	1.125
20	1.156	0.578	0.120	1.125
22	1.250	0.625	0.120	1.375
24	1.375	0.688	0.147	1.500
28	1.562	0.781	0.147	1.813
32	1.750	0.875	0.173	2.063
36	1.938	0.967	0.173	2.313

FIGURE 4B97. Cableways on masts with connectors (surface ships).

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METHOD 4B-98-1
CABLES THROUGH ENGINE AND BOILER ROOMS
SUPPORTED BETWEEN SHIPS LONGITUDINALS

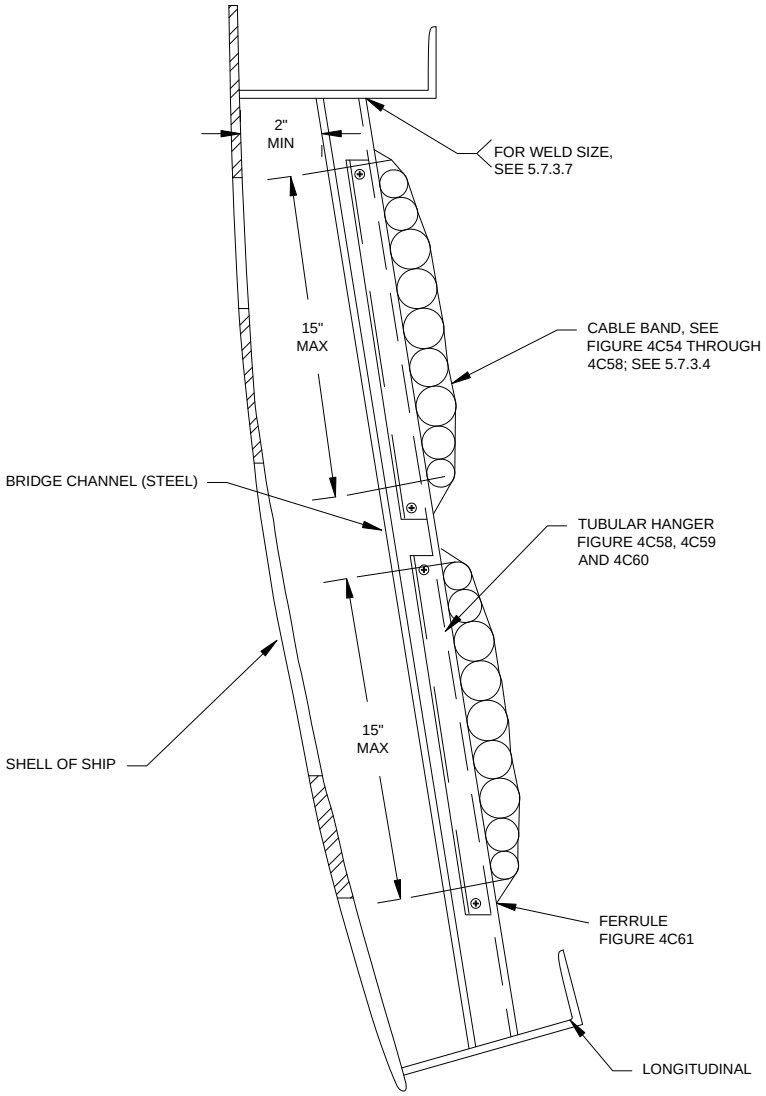


FIGURE 4B98. Cableways of main cable runs on small vessels (surface ships).

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METHOD 4B-99-1
CABLES THROUGH ENGINE AND BOILER ROOMS
SUPPORTED BETWEEN SHIPS LONGITUDINALS
(FOR SECTION "A-A", SEE FIGURE 4B100)

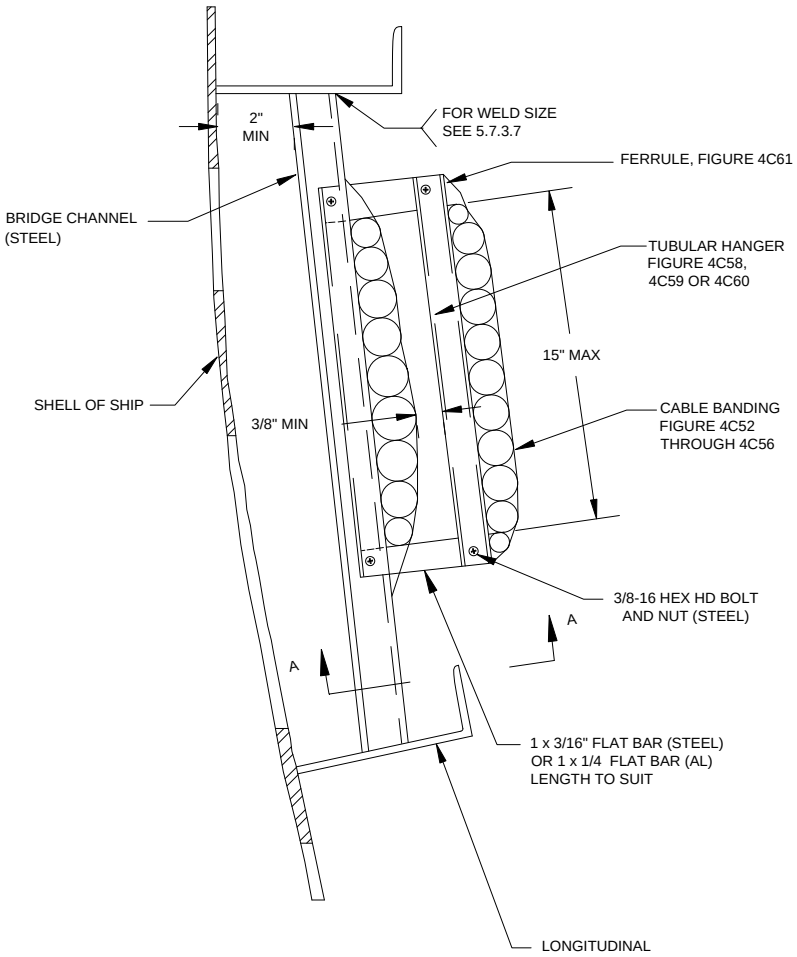
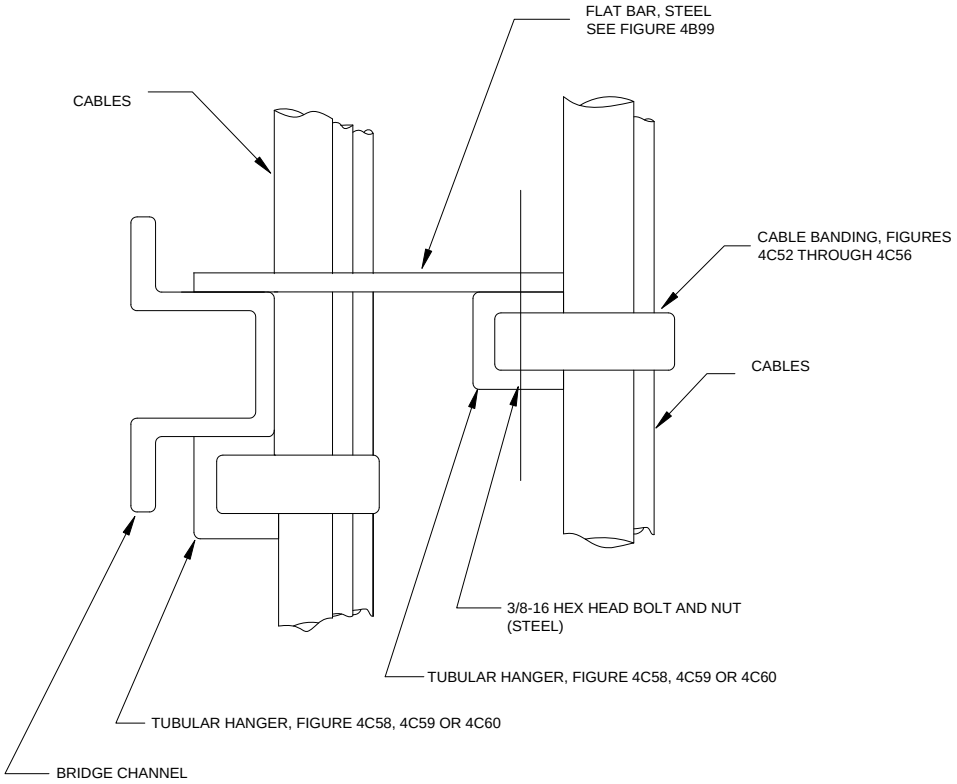


FIGURE 4B99. Cableways of main cable runs on small vessels (surface ships).

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SECTION "A-A" FOR METHOD 4B-100-1
CABLES THROUGH ENGINE AND BOILER ROOMS
SUPPORTED BETWEEN SHIPS LONGITUDINALS



SECTION "A - A"

FIGURE 4B100. Cableways of main cable runs on small vessels (surface ships) – section A-A.

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GROUP 4C – CABLEWAYS (GENERAL)

C.1 SCOPE

C.1.1 Scope. This appendix describes general procedures for the installation of cableways on surface ships and submarines.

C.2. APPLICABLE DOCUMENTS

C.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

C.2.2 Government documents.

C.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-S-698	-	Steel, Sheet and Strip, Low Carbon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-I-3064	-	Insulation, Electrical, Plastic-Sealer
MIL-PRF-6855	-	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-S-24149	-	Studs, Welding and Arc Shields (Ferrules), General Specification for
MIL-S-24149/1	-	Stud, Welding, and Arc Shields (Ferrules); Type I, Class 1, 2, 3, and Type II, Class 1, 4, 5, 5A, 6, Carbon Steel, for Direct Energy, Arc Welding
MIL-S-24149/2	-	Studs, Welding, and Arc Shields (Ferrules); Type III, Class 1, 2, 3, and Type IV, Class 1, 2, 3, 4, 5, 6, Aluminum Alloy, for Direct Energy Arc Welding
MIL-S-24149/3	-	Studs, Welding, & Arc Shields (Ferrules); Type V, Class 1, 4, 5, 5A, Corrosion-Resistant Steel, for Direct Energy ARC Welding

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-167-2	-	Mechanical Vibrations of Shipboard Equipment (Reciprocating Machinery and Propulsion System and Shafting) Types III, IV, and V

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-299	-	Cable Comparison Handbook Data Pertaining to Electrical Shipboard Cable

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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Check the source to verify that this is the current version before use.

C.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A240/A240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A276	-	Standard Specification for Stainless Steel Bars and Shapes
ASTM A666	-	Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar
ASTM A693	-	Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM A713	-	Standard Specification for Steel Wire, High-Carbon Spring, for Heat-Treated Components
ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B241/B241M	-	Standard Specification for Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube
ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D3953	-	Standard Specification for Strapping, Flat Steel and Seals
ASTM F1836M	-	Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

C.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

C.3 REQUIRED EQUIPMENT AND MATERIALS

C.3.1 Required equipment and materials. The equipment and materials in the tables and referenced in the text of this section shall be used for the installation of cableways on surface ships and submarines. For fiber optic cable, refer to MIL-STD-2042-4 for additional information and guidance.

C.4 NOTES AND PROCEDURES

C.4.1 Dimensions. For figures and tables in this section, all dimensions are in inches unless otherwise noted.

C.4.2 Figures. Table 4CI provides information for the figures in this group. Figures 4C2 through 4C12 and 4C35 are for local runs only.

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TABLE 4CI. Figures for the installation of cableways on surface ships and submarines.

Figure number	Title	Page
4C1	Single cableway on steel stanchions	203
4C2	Single cableway on steel stanchion secured by angle collar studs or flat bar	204
4C3	Single cableway on steel structure secured with angle collar studs	206
4C4	Single cableway on insulated steel structure secured with angle collar studs	207
4C5	Single cableway tier secured by bolting on steel structure	208
4C6	Single cableway secured with flat bar or angle on steel structure	209
4C7	Single cableway secured with flat bar or angle on insulated decks or bulkheads	210
4C8	Single cableway on steel structure secured with flat bar (or angle) and weld stud	211
4C9	Single cableway on steel structure secured with flat bar (or angle) and weld stud	212
4C10	Single cableway on steel structure secured with female welding stud	213
4C11	Single cableway on insulated steel structure secured with female welding stud	214
4C12	Single cableway on steel structure secured with welding stud	216
4C13	Bottom loading hanger on steel and aluminum decks and bulkheads (one cable)	218
4C14	Bottom loading hanger on steel and aluminum decks and bulkheads (two cables)	220
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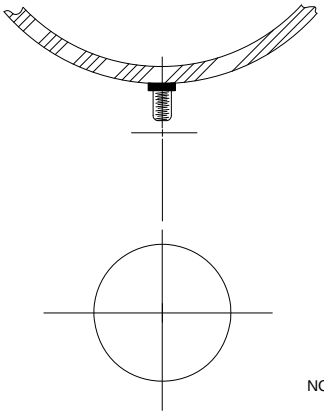
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TABLE 4CI. Figures for the installation of cableways on surface ships and submarines – Continued.

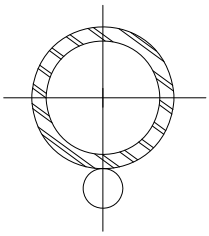
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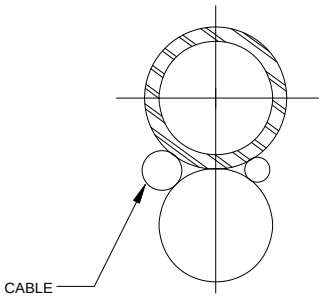
METHOD 4C-1-1
SINGLE CABLES ON STEEL STANCHIONS
HANGER SECURED WITH STUD



METHOD 4C-1-2
SINGLE CABLES ON SMALL STANCHIONS



METHOD 4C-1-3
TWO TO THREE CABLES ON
SMALL STANCHIONS

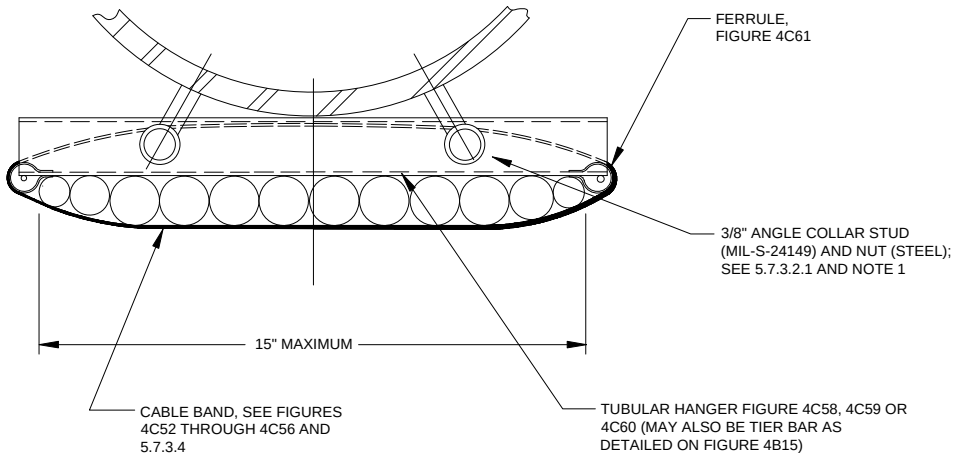


NOTE: METHODS SHOWN ON THIS SHEET ARE FOR USE WITH CABLES UP TO TWO INCHES IN DIAMETER. FOR CABLES GREATER THAN TWO INCHES IN DIAMETER, USE METHODS SHOWN ON FIGURE 4C2.

FIGURE 4C1. Single cableway on steel stanchions.

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METHOD 4C-2-1
SINGLE TIER OF CABLES ON STEEL STANCHIONS SECURED
BY ANGLE COLLAR STUDS
(FOR LOCAL RUNS ONLY)



METHOD 4C-2-2
SINGLE TIER OF CABLES ON STEEL STANCHIONS SECURED
BY FLAT BAR
(FOR LOCAL RUNS ONLY)

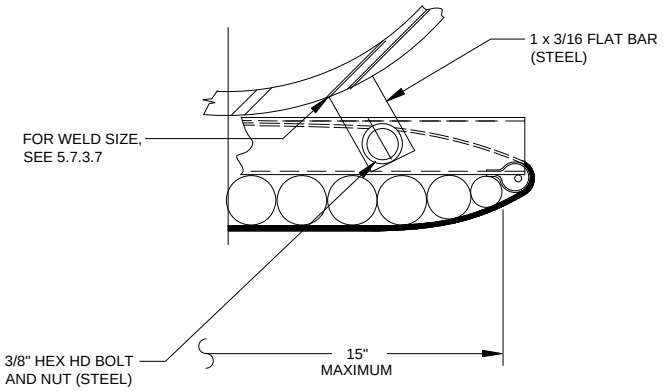


FIGURE 4C2. Single cableway on steel stanchion secured by angle collar studs or flat bar.

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NOTE:

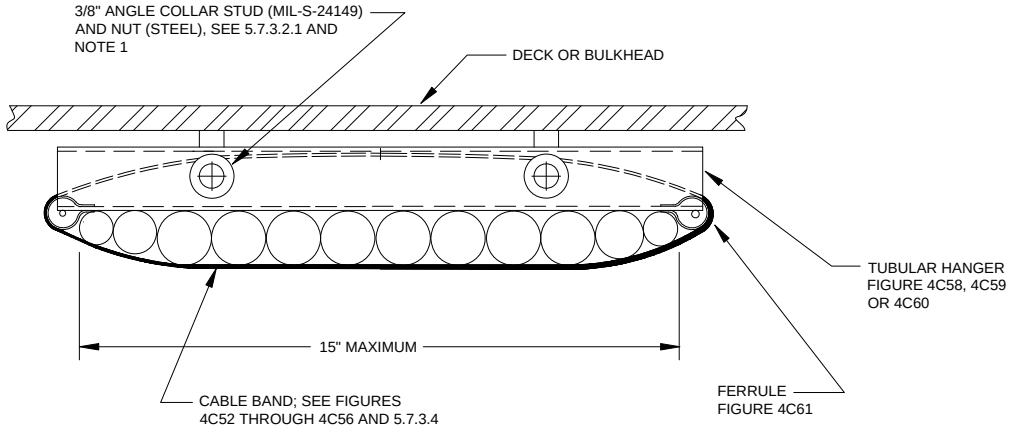
1. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, Class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4C2. Single cableway on steel stanchion secured by angle collar studs or flat bar - Continued.

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METHOD 4C-3-1
SINGLE TIER SECURED
WITH ANGLE COLLAR STUDS
(LOCAL RUNS ONLY)



NOTE:

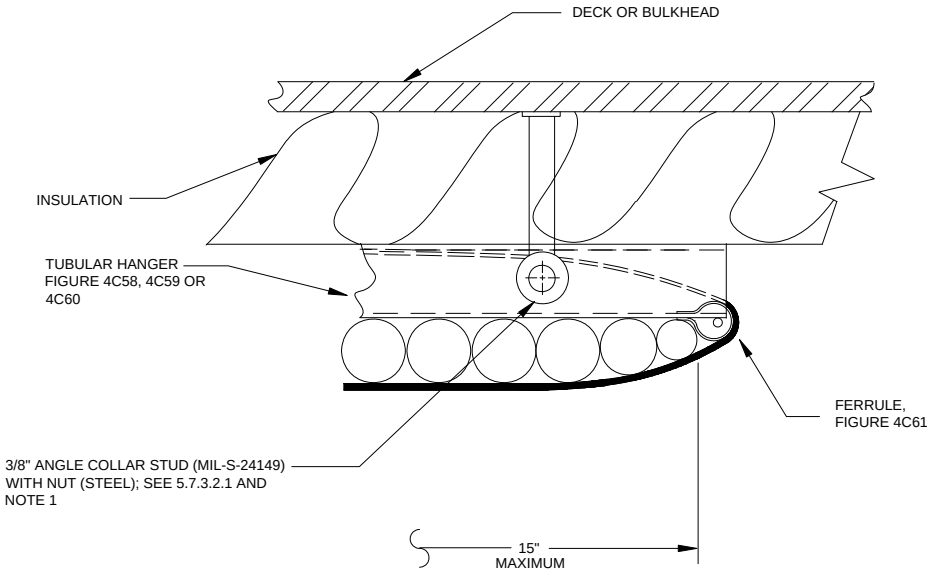
1. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4C3. Single cableway on steel structure secured with angle collar studs.

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METHOD 4C-4-1
SINGLE TIER SECURED WITH ANGLE COLLAR STUDS
ON INSULATED DECKS & BULKHEADS
(LOCAL RUNS ONLY)



NOTE:

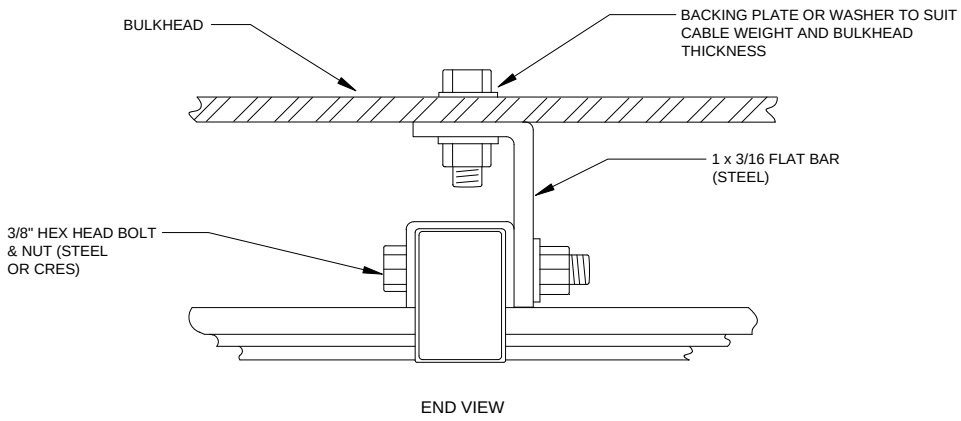
1. National stock numbers for ⅜-16 angled threaded collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 6	MIL-S-24149/3 CRES type V, class 8	MIL-S-24149/2 aluminum type IV, class 4
0.675	5307-01-148-8237	5307-01-293-9583	
0.750	5307-01-293-1815	5307-01-293-9584	
0.875	5307-01-148-3720	5307-01-293-9585	
1.000	5307-01-293-6161	5307-01-293-9586	5307-01-293-9596
1.125	5307-01-293-6162	5307-01-293-9587	5307-01-293-9597
1.250	5307-01-293-1816	5307-01-293-9588	5307-01-294-4372
1.350			5307-01-294-1750
1.500	5307-01-293-1074	5307-01-293-9589	5307-01-294-1751
1.625			5307-01-294-0267
1.750	5307-01-293-6163	5307-01-293-9590	5307-01-294-3147
1.875			5307-01-294-0268
2.000	5307-01-293-1073	5307-01-293-9591	5307-01-294-1752
2.250			5307-01-294-1097
2.500	5307-01-293-1817	5307-01-293-9592	
3.000	5307-01-293-6164	5307-01-293-9593	
3.500	5307-01-293-1074	5307-01-293-9594	
4.000	5307-01-293-9598	5307-01-293-9595	

FIGURE 4C4. Single cableway on insulated steel structure secured with angle collar studs.

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METHOD 4C-5-1
SINGLE TIER SECURED BY BOLTING
(LOCAL RUNS ONLY)



END VIEW

FIGURE 4C5. Single cableway tier secured by bolting on steel structure.

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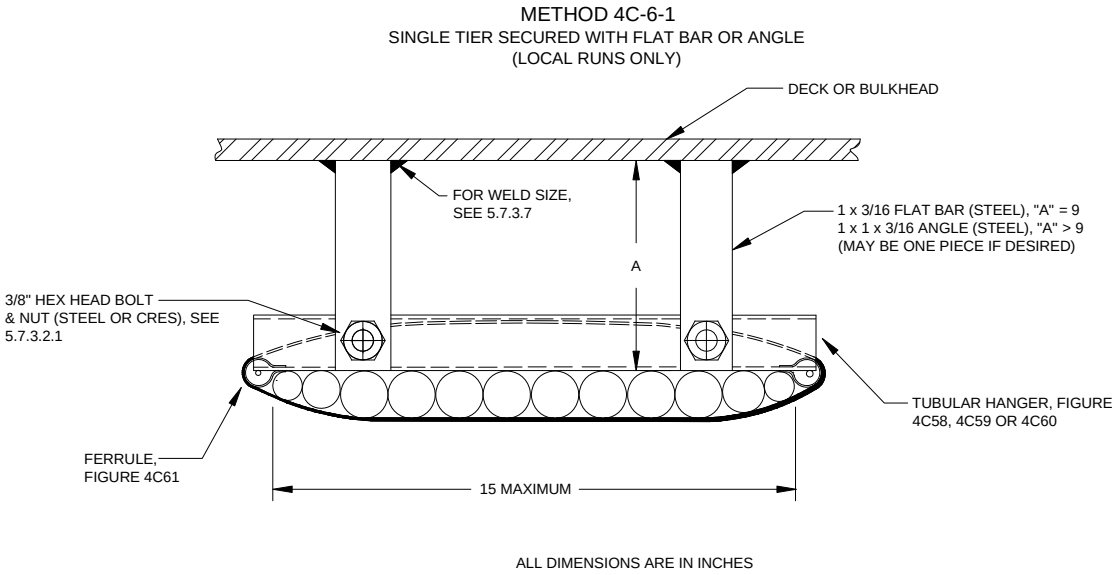


FIGURE 4C6. Single cableway secured with flat bar or angle on steel structure.

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METHOD 4C-7-1
SINGLE TIER SECURED WITH FLAT BAR OR ANGLE
ON INSULATED DECKS AND BULKHEADS
(LOCAL RUNS ONLY)

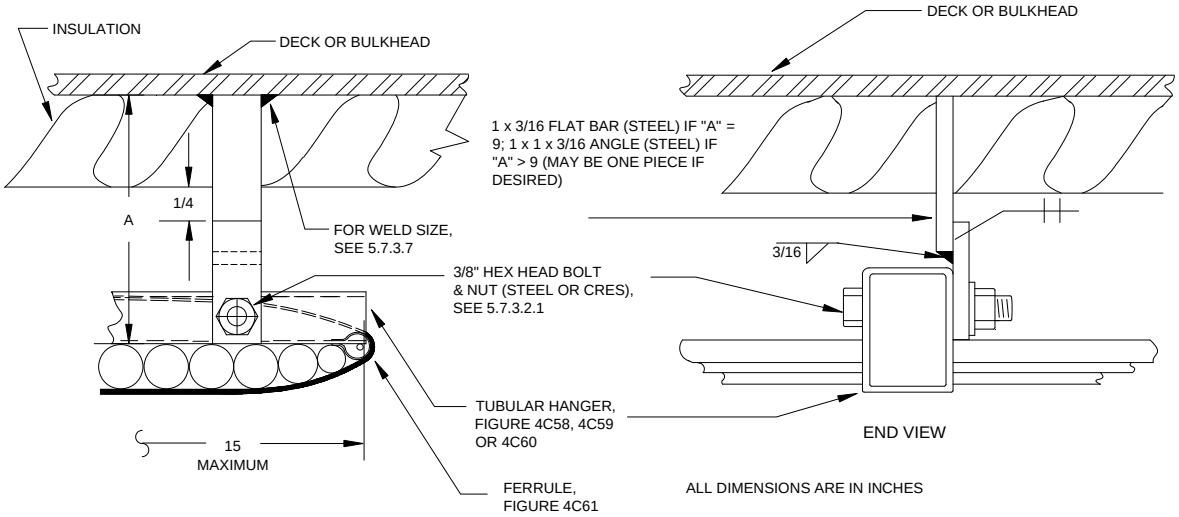


FIGURE 4C7. Single cableway secured with flat bar or angle on insulated decks or bulkheads.

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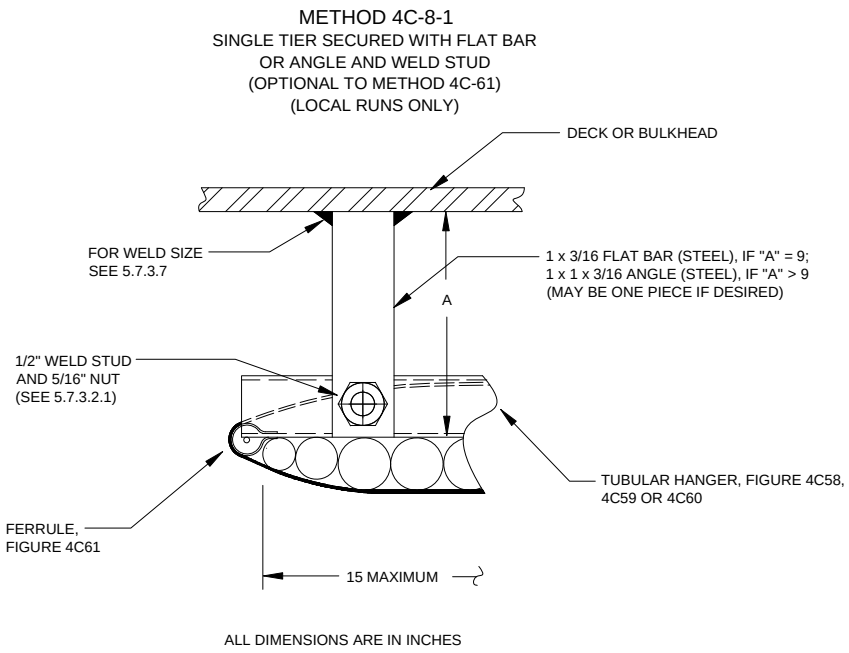


FIGURE 4C8. Single cableway on steel structure secured with flat bar (or angle) and weld stud.

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METHOD 4C-9-1
SINGLE TIER SECURED WITH FLAT BAR OR
ANGLE AND WELD STUDS
(OPTIONAL TO METHOD 4C-7-1)
(LOCAL RUNS ONLY)

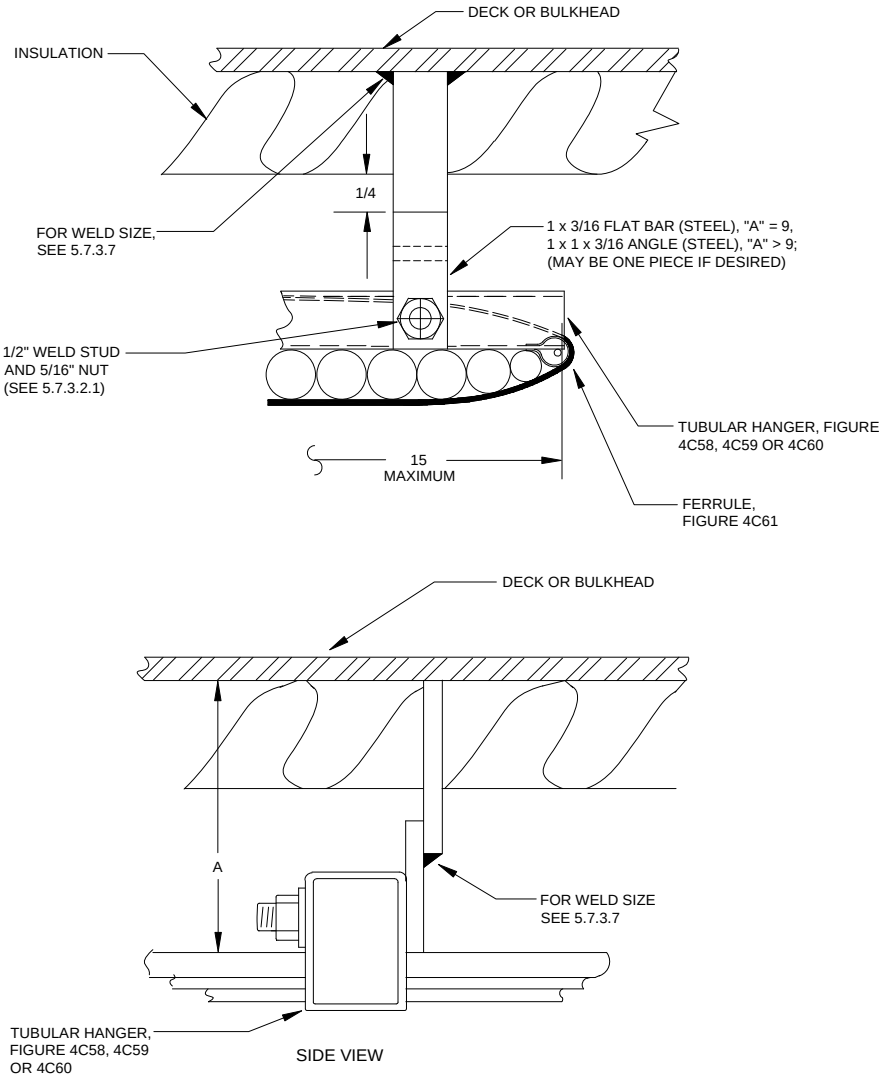
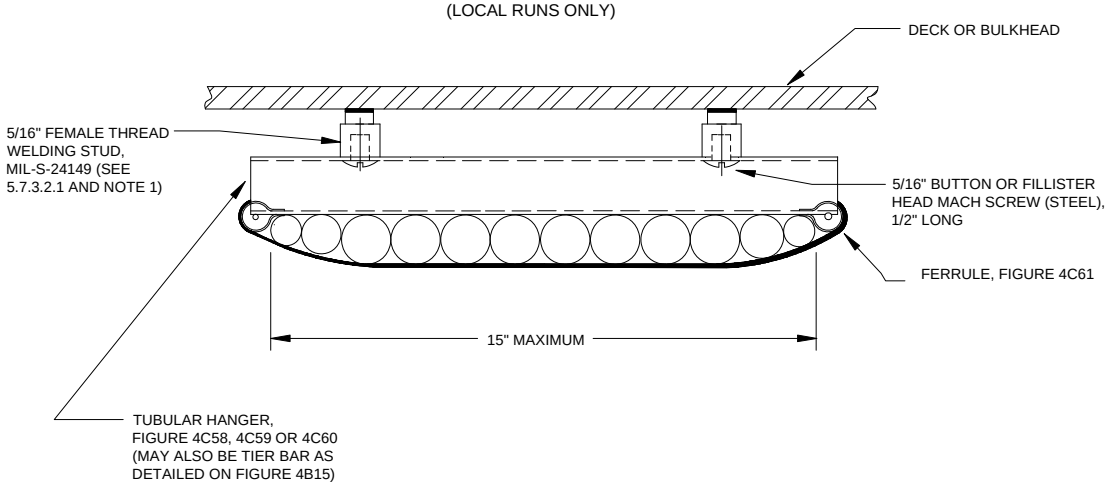


FIGURE 4C9. Single cableway on steel structure secured with flat bar (or angle) and weld stud.

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METHOD 4C-10-1
SINGLE TIER SECURED
WITH FEMALE THREAD WELDING STUDS
ON STEEL DECKS AND BULKHEADS
(LOCAL RUNS ONLY)



NOTE:

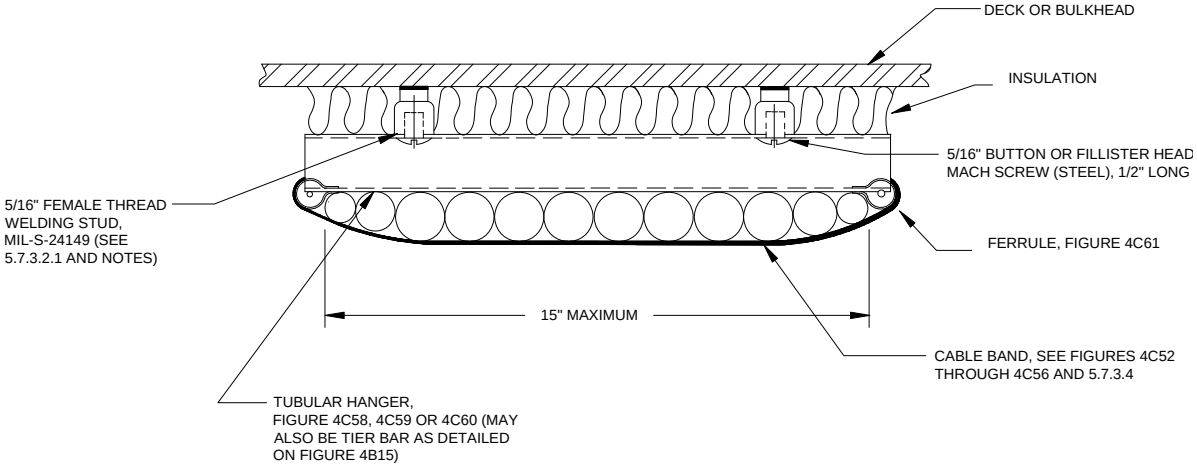
1. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

FIGURE 4C10. Single cableway on steel structure secured with female welding stud.

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METHOD 4C-11-1
SINGLE TIER SECURED
WITH FEMALE THREAD WELDING STUDS
ON INSULATED DECKS AND BULKHEADS
(LOCAL RUNS ONLY)



NOTES:

1. National stock numbers for $\frac{5}{16}$-17 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

FIGURE 4C11. Single cableway on insulated steel structure secured with female welding stud.

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NOTES (continued):

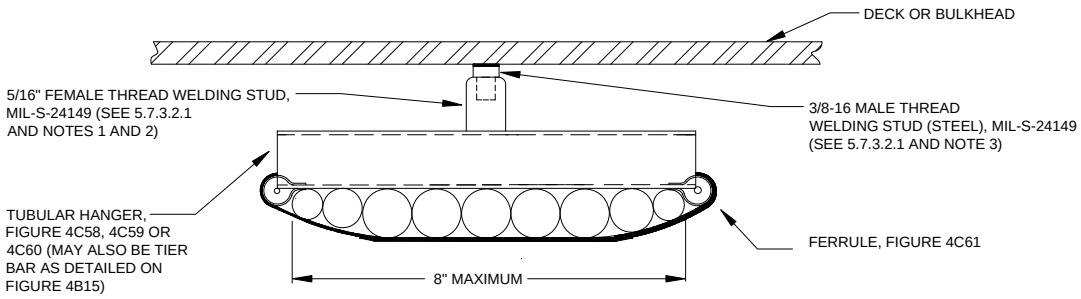
2. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

FIGURE 4C11. Single cableway on insulated steel structure secured with female welding stud - Continued.

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METHOD 4C-12-1
SINGLE TIER SECURED WITH WELD STUD
ON STEEL DECKS AND BULKHEADS
(MAXIMUM CABLE WEIGHT: 10 POUNDS PER SUPPORT)
(LOCAL RUNS ONLY)



NOTES:

1. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

FIGURE 4C12. Single cableway on steel structure secured with welding stud.

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Check the source to verify that this is the current version before use.

NOTES (continued):

2. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

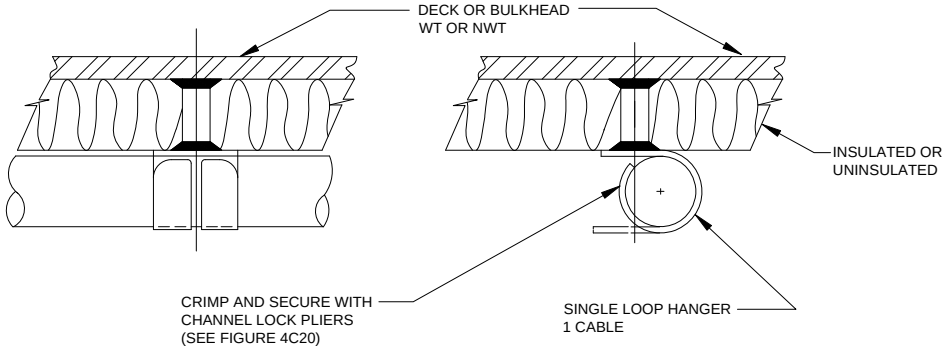
3. National stock numbers for 3/8-16 external thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 1	MIL-S-24149/3 CRES 316 type V, class 1	MIL-S-24149/2 aluminum type IV, class 1
0.750	5307-01-293-7981	5307-01-293-8797	5307-01-298-5394
0.875	5307-01-294-3148	5307-01-294-9538	5307-01-293-5428
1.000	5307-01-293-1818	5307-01-297-4107	5307-01-293-5427
1.250	5307-01-294-2149	5307-01-294-4356	5307-01-293-5428
1.500	5307-01-244-6967	5307-01-294-4357	5307-01-293-5429
1.750	5307-01-280-9436	5307-01-294-4358	5307-01-293-5430
2.000	5307-01-293-1819	5307-01-294-4359	5307-01-294-4370
2.500	5307-01-292-9609	5307-01-294-4360	5307-01-293-5431
3.000	5307-01-292-9610	5307-01-294-4361	5307-01-293-5432
3.500	5307-01-294-3150	5307-01-294-4362	5307-01-294-4371
4.000	5307-01-295-1098	5307-01-294-4363	5307-01-293-7980

FIGURE 4C12. Single cableway on steel structure secured with welding stud - Continued.

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METHOD 4C-13-1
SINGLE LOOP HANGER - BOTTOM LOADING (ONE CABLE)
(SEE FIGURE 4C21 FOR ALTERNATE METHOD)



NOTES:

1. The crimp-on weld-stud type cable hanger may be used as an alternate to cable clamps (PCs 4C-221 and 4C-222) shown on figure 4C22.
2. The weight of the cables supported shall not exceed two pounds per stud for the single loop hanger and four pounds per stud for the dual-loop hanger. The spacing between studs shall not exceed 32".
3. The make of hangers procured shall have passed a high-impact shock test, MIL-S-901, and vibration test, MIL-STD-167-1 and MIL-STD-167-2, using the maximum hanger loadings and spacing.
4. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
5. Studs with aluminum male threads shall be coated.
6. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
7. Loop hangers installed topside and in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C13. Bottom loading hanger on steel and aluminum decks and bulkheads (one cable).

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NOTES (continued):

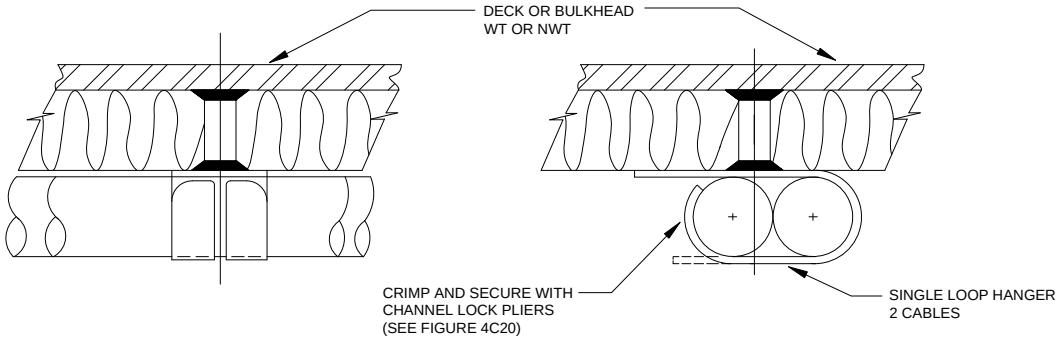
8. National stock numbers for loop hangers are as follows:

Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

FIGURE 4C13. Bottom loading hanger on steel and aluminum decks and bulkheads (one cable) - Continued.

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METHOD 4C-14-1
SINGLE LOOP HANGER - BOTTOM LOADING (TWO CABLES)



NOTES:

1. The crimp-on weld-stud type cable hanger may be used as an alternate to cable clamps (PCs 4C-221 and 4C-222) shown on figure 4C22.
2. The weight of the cables supported shall not exceed two pounds per stud for the single loop hanger and four pounds per stud for the dual-loop hanger. The spacing between studs shall not exceed 32".
3. The make of hangers procured shall have passed a high-impact shock test, MIL-S-901, and vibration test, MIL-STD-167-1 and MIL-STD-167-2, using the maximum hanger loadings and spacing.
4. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
5. Studs with aluminum male threads shall be coated.
6. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
7. Loop hangers installed topside and in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C14. Bottom loading hanger on steel and aluminum decks and bulkheads (two cables).

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NOTES (continued):

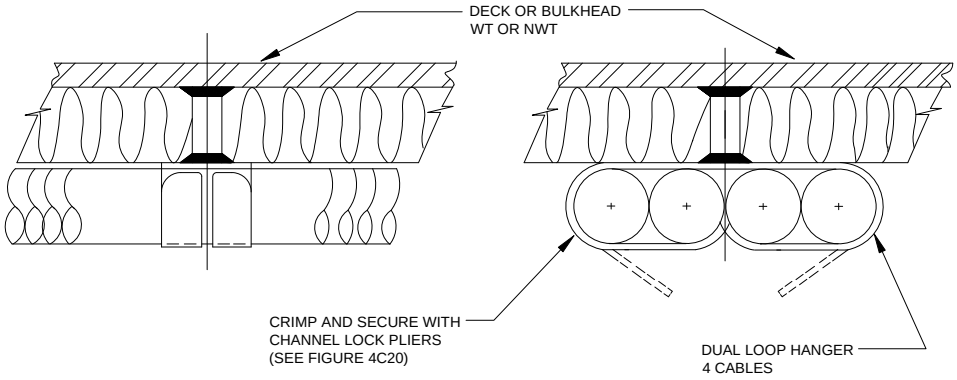
8. National stock numbers for loop hangers are as follows:

Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

FIGURE 4C14. Bottom loading hanger on steel and aluminum decks and bulkheads (two cables) - Continued.

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METHOD 4C-15-1
SINGLE LOOP HANGER - BOTTOM LOADING (FOUR CABLES)



NOTES:

1. The crimp-on weld-stud type cable hanger may be used as an alternate to cable clamps (PCs 4C-221 and 4C-222) shown on figure 4C22.
2. The weight of the cables supported shall not exceed two pounds per stud for the single loop hanger and four pounds per stud for the dual-loop hanger. The spacing between studs shall not exceed 32".
3. The make of hangers procured shall have passed a high-impact shock test, MIL-S-901, and vibration test, MIL-STD-167-1 and MIL-STD-167-2, using the maximum hanger loadings and spacing.
4. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
5. Studs with aluminum male threads shall be coated.
6. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
7. Loop hangers installed topside and in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C15. Bottom loading hanger on steel and aluminum decks and bulkheads (four cables).

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NOTES (continued):

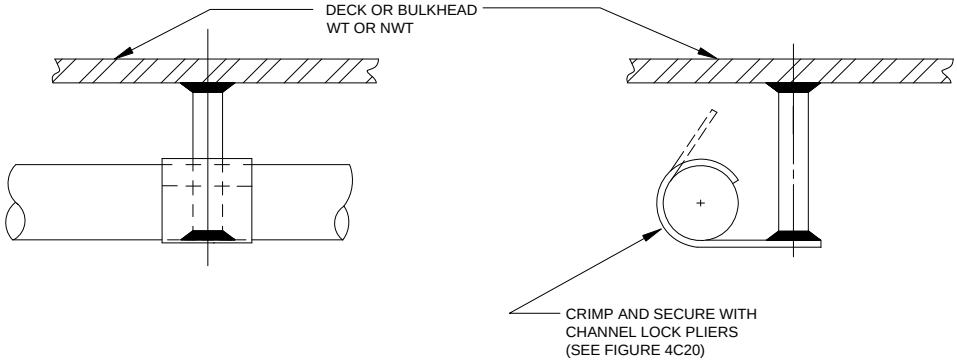
8. National stock numbers for loop hangers are as follows:

Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

FIGURE 4C15. Bottom loading hanger on steel and aluminum decks and bulkheads (four cables) - Continued.

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METHOD 4C-16-1
SINGLE LOOP HANGER-TOP LOADING (ONE CABLE)
(SEE FIGURE 4C21 FOR ALTERNATE METHOD)



NOTES:

1. The crimp-on weld-stud type cable hanger may be used as an alternate to cable clamps (PCs 4C-221 and 4C-222) shown on figure 4A22.
2. The weight of the cables supported shall not exceed two pounds per stud for the single loop hanger and four pounds per stud for the dual-loop hanger. The spacing between studs shall not exceed 32".
3. The make of hangers procured shall have passed a high-impact shock test, MIL-S-901, and vibration test, MIL-STD-167-1 and MIL-STD-167-2, using the maximum hanger loadings and spacing.
4. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
5. Studs with aluminum male threads shall be coated.
6. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
7. Loop hangers installed topside and in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C16. Top loading hanger on steel and aluminum decks and bulkheads (one cable).

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NOTES (continued):

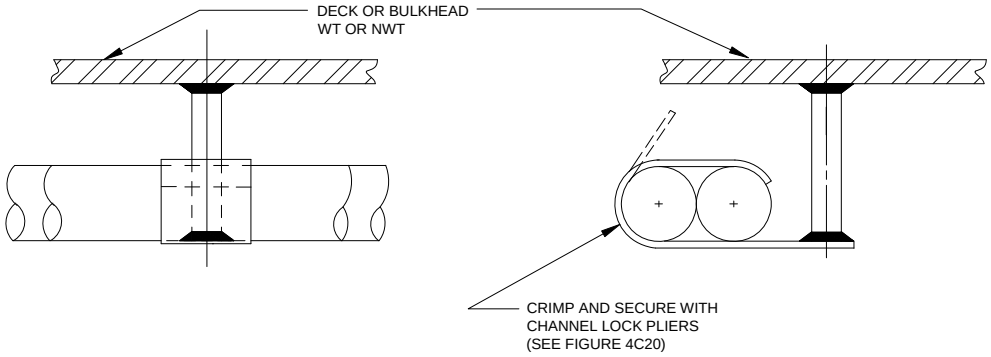
8. National stock numbers for loop hangers are as follows:

Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

FIGURE 4C16. Top loading hanger on steel and aluminum decks and bulkheads (one cable) - Continued.

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METHOD 4C-17-1
SINGLE LOOP HANGER-TOP LOADING (TWO CABLES)
I



NOTES:

1. The crimp-on weld-stud type cable hanger may be used as an alternate to cable clamps (PCs 4C-221 and 4C-222) shown on figure 4C22.
2. The weight of the cables supported shall not exceed two pounds per stud for the single loop hanger and four pounds per stud for the dual-loop hanger. The spacing between studs shall not exceed 32".
3. The make of hangers procured shall have passed a high-impact shock test, MIL-S-901, and vibration test, MIL-STD-167-1 and MIL-STD-167-2, using the maximum hanger loadings and spacing.
4. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
5. Studs with aluminum male threads shall be coated.
6. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
7. Loop hangers installed topside and in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C17. Top loading hanger on steel and aluminum decks and bulkheads (two cables).

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NOTES (continued):

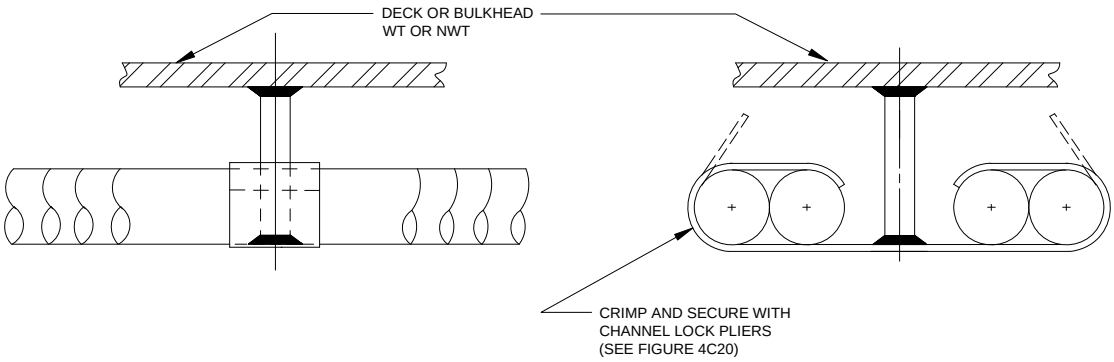
8. National stock numbers for loop hangers are as follows:

Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

FIGURE 4C17. Top loading hanger on steel and aluminum decks and bulkheads (two cables) - Continued.

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METHOD 4C-18-1
SINGLE LOOP HANGER-TOP LOADING (FOUR CABLES)



NOTES:

1. The crimp-on weld-stud type cable hanger may be used as an alternate to cable clamps (PCs 4C-221 and 4C-222) shown on figure 4C22.
2. The weight of the cables supported shall not exceed two pounds per stud for the single loop hanger and four pounds per stud for the dual-loop hanger. The spacing between studs shall not exceed 32".
3. The make of hangers procured shall have passed a high-impact shock test, MIL-S-901, and vibration test, MIL-STD-167-1 and MIL-STD-167-2, using the maximum hanger loadings and spacing.
4. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
5. Studs with aluminum male threads shall be coated.
6. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
7. Loop hangers installed topside and in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C18. Top loading hanger on steel and aluminum decks and bulkheads (four cables).

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NOTES (continued):

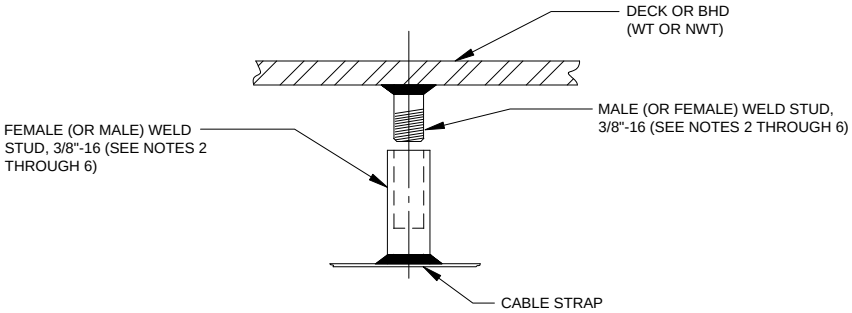
8. National stock numbers for loop hangers are as follows:

Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

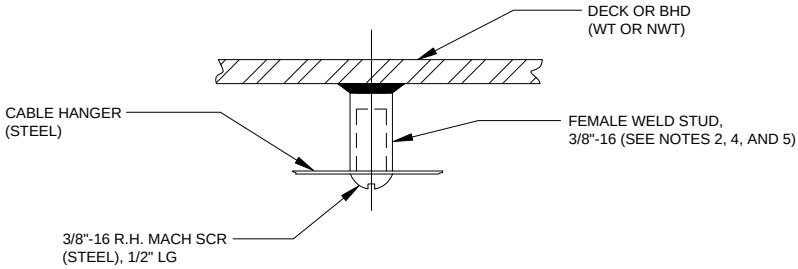
FIGURE 4C18. Top loading hanger on steel and aluminum decks and bulkheads (four cables) - Continued.

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DETAIL "1"
FEMALE WELD STUD METHOD
(MAY BE USED AS AN ALTERNATE TO METHODS 4C-13-1 THROUGH 4C-18-1)
SEE NOTE 1



DETAIL "2"
FEMALE WELD STUD METHOD
(MAY BE USED AS ALTERNATE TO METHODS 4C-13-1 THROUGH 4C-18-1
AND 4B-24-1 AND 4B-24-2)
SEE NOTE 1



NOTES:

1. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
2. Studs with aluminum male threads shall be coated.
3. The contact of steel female threads with aluminum male threads shall be coated with anti-seize compound.
4. National stock numbers for $\frac{5}{16}$-17 internal thread welding studs are as follows:

Length	MIL-S-24149/2 aluminum type IV, class 5	Length	MIL-S-24149/2 aluminum type IV, class 5
1.000	5307-01-293-5417	1.875	5307-01-293-5424
1.125	5307-01-293-5418	2.000	5307-01-293-5425
1.250	5307-01-293-5419	2.250	5307-01-293-6712
1.375	5307-01-293-5420	2.500	5307-01-294-6829
1.500	5307-01-293-5421	2.750	5307-00-306-4912
1.625	5307-01-293-5422	3.000	5307-01-294-1749
1.750	5307-01-293-5423		

FIGURE 4C19. Single cableway on steel and aluminum decks and bulkheads, weld stud.

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NOTES (continued):

5. National stock numbers for 5/16-18 internal thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, class 5A	MIL-S-24149/3 CRES 316 type V, class 5A
0.875	5307-01-295-7614	5307-01-293-5411
1.000	5307-01-294-3146	5307-01-293-5412
1.125	5307-01-294-2273	5307-01-295-0179
1.250	5307-01-293-2911	5307-01-293-9578
1.375	5307-01-293-2912	5307-01-293-9579
1.500	5307-01-293-1073	5307-01-293-5380
1.625	5307-01-293-2913	5307-01-294-1747
1.750	5307-01-293-2914	5307-01-293-5413
1.875	5307-01-293-1814	5307-01-293-9521
2.000	5307-01-294-1748	5307-01-293-9532
2.250	5307-01-293-6710	5307-01-294-4367
2.500	5307-01-293-6711	5307-01-293-5414
2.750	5307-01-294-4368	5307-01-293-5415
3.000	5307-01-294-4369	5307-01-293-5416

6. National stock numbers for 3/8-16 external thread welding studs are as follows:

Length	MIL-S-24149/1 steel type II, Class 1	MIL-S-24149/3 CRES 316 type V, class 1	MIL-S-24149/2 aluminum type IV, class 1
0.750	5307-01-293-7981	5307-01-293-8797	5307-01-298-5394
0.875	5307-01-294-3148	5307-01-294-9538	5307-01-293-5428
1.000	5307-01-293-1818	5307-01-297-4107	5307-01-293-5427
1.250	5307-01-294-2149	5307-01-294-4356	5307-01-293-5428
1.500	5307-01-244-6967	5307-01-294-4357	5307-01-293-5429
1.750	5307-01-280-9436	5307-01-294-4358	5307-01-293-5430
2.000	5307-01-293-1819	5307-01-294-4359	5307-01-294-4370
2.500	5307-01-292-9609	5307-01-294-4360	5307-01-293-5431
3.000	5307-01-292-9610	5307-01-294-4361	5307-01-293-5432
3.500	5307-01-294-3150	5307-01-294-4362	5307-01-294-4371
4.000	5307-01-295-1098	5307-01-294-4363	5307-01-293-7980

FIGURE 4C19. Single cableway on steel and aluminum decks and bulkheads, weld stud - Continued.

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CRIMPING METHODS

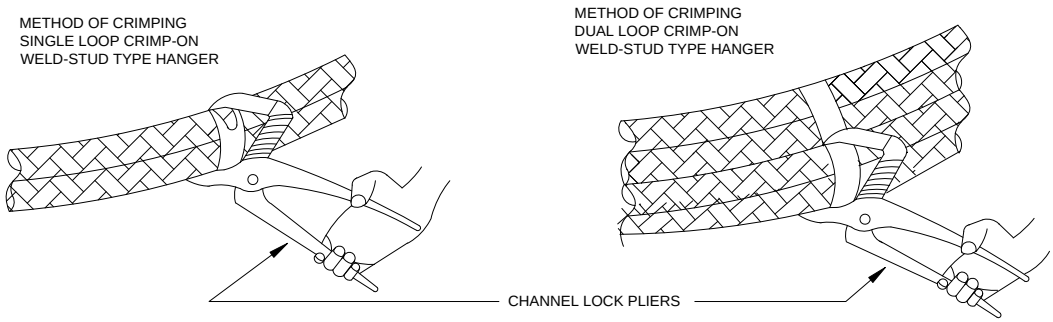
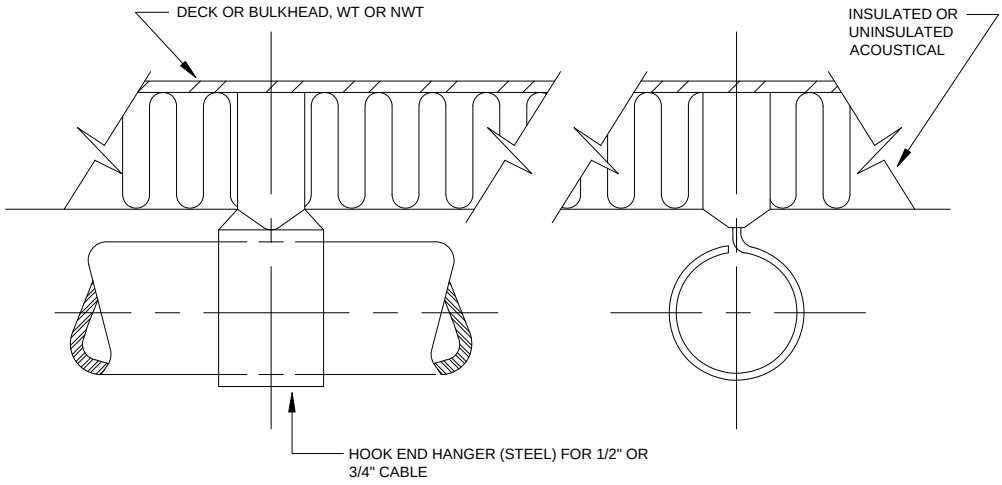


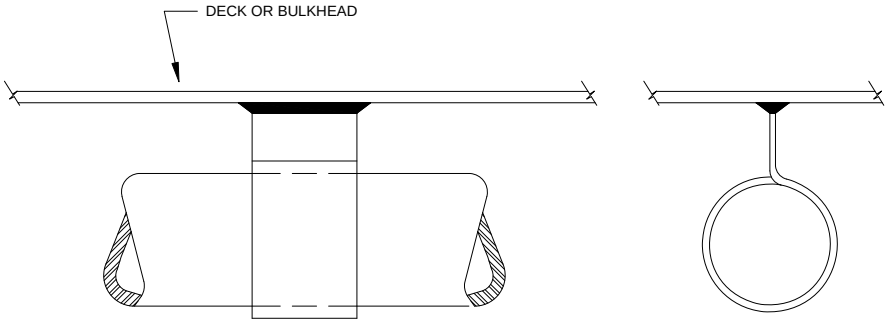
FIGURE 4C20. Single cableway on steel and aluminum decks and bulkheads, crimping methods.

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METHOD 4C-21-1
SINGLE CABLEWAY ON STEEL AND ALUMINUM DECKS, INSULATED
(FOR ALTERNATE METHODS 4C-13-1 AND 4C-16-1,
SEE FIGURES 4C13 AND 4C16)
SEE NOTE 1



METHOD 4C-21-2
SINGLE CABLEWAY ON STEEL AND ALUMINUM DECKS, UNINSULATED
(FOR ALTERNATE METHODS 4C-13-1 AND 4C-16-1,
SEE FIGURES 4C13 AND 4C16)
SEE NOTES 1 THROUGH 4



NOTES:

1. The female weld stud method (attached to either the cable strap or the ship's structure) shown in Detail 1 on figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1, and the female weld stud method shown in Detail 2 of figure 4C19 may be used as an alternate to Methods 4C-13-1 through 4C-18-1 and 4C-21-1 and 4C-21-2.
2. Cable clamp hangers detailed as PCs 4C-221 through 4C-223 shall be unplated unless otherwise specified (see 5.7.3.2.1).
3. Hangers (4C-22-1 through 4C-22-3) on figure 4C22 shall be suitable for welding by the metal arc stud welding process.

FIGURE 4C21. Single cableway on aluminum decks and bulkheads.

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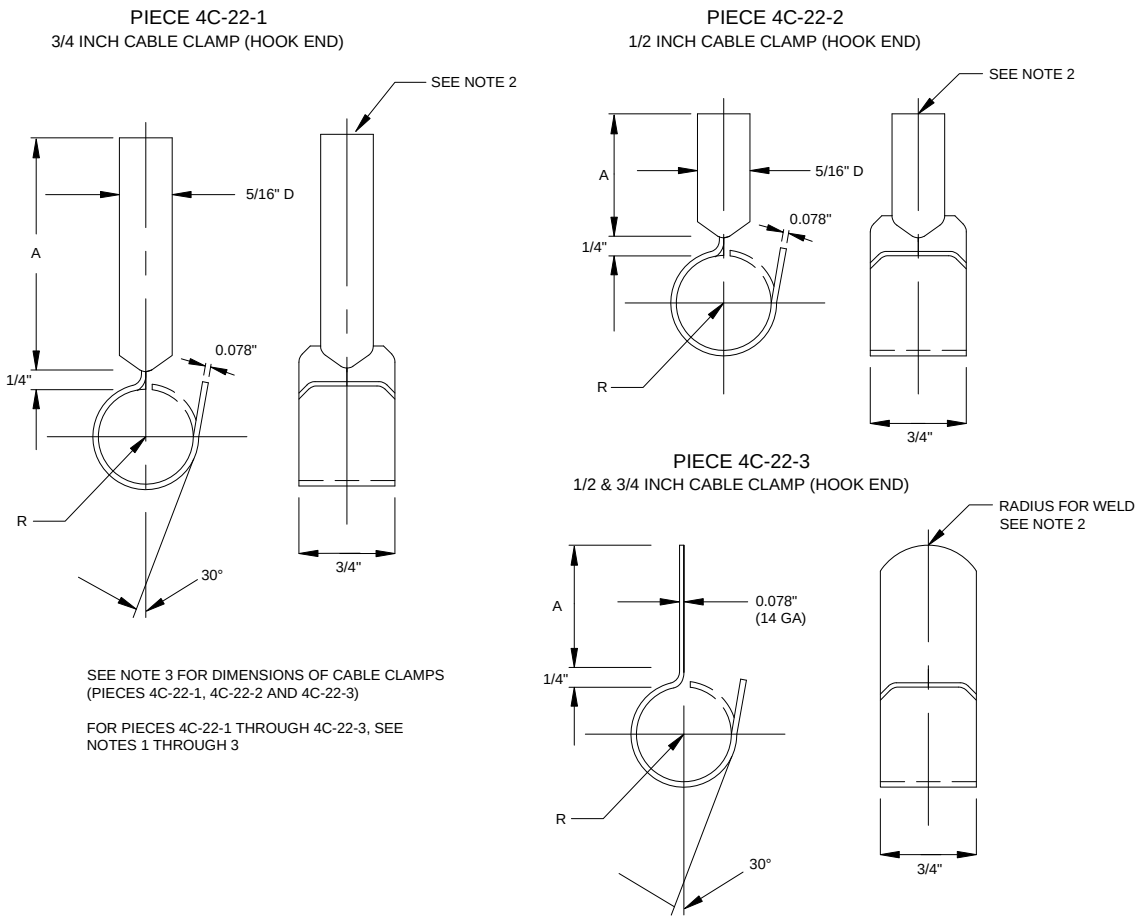
NOTES (continued):

4. The applicable piece number and size (e.g., 4C-222-H) shall be stamped on each hook end hanger in ⅛" letters. See table below.

PC no.	Size	Style	Dimension (inches)	
			A	R
4C-221 4C-222 4C-223	A	½	⅜	¼
	B	½	⅝	¼
	C	½	1	¼
	D	½	2	¼
	E	½	3	¼
	F	¾	⅜	⅜
	G	¾	⅝	⅜
	H	¾	1	⅜
	J	¾	2	⅜
	K	¾	3	⅜

FIGURE 4C21. Single cableway on aluminum decks and bulkheads - Continued.

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NOTES:

1. Cable clamp hangers detailed as PCs 4C-22-1 through 4C-22-3 shall be unplated unless otherwise specified (see 5.7.3.2.1).
2. Hangers (4C-22-1 through 4C-22-3) shall be suitable for welding by the metal arc stud welding process.

FIGURE 4C22. Cable clamps for single cableway on steel and aluminum decks and bulkheads.

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NOTES (continued):

3. National stock numbers for loop hangers are as follows:

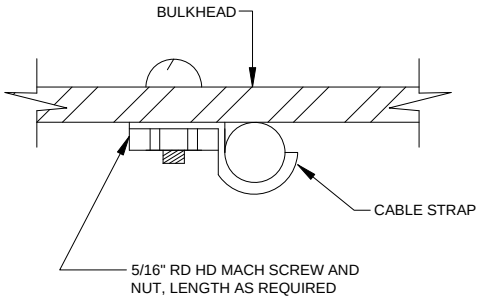
Type	Size	Material	National stock number
Single loop	0.531	CRES 316	5975-01-294-2133
	0.358	CRES 316	5975-01-294-2134
	0.680	CRES 316	5975-01-294-2135
	0.900	CRES 316	5975-01-294-2136
	0.750	CRES 316	5975-01-294-2137
	1.224	CRES 316	5975-01-294-2138
	0.531	Steel	5975-01-030-9018
	0.680	Steel	5975-01-020-9171
	1.224	Steel	5975-01-114-8071
	0.750	Steel	5975-01-294-9092
	0.358	Steel	5975-01-294-9093
Double loop	0.531	CRES 316	5975-01-020-9174
	0.358	CRES 316	5975-01-294-6558
	0.680	CRES 316	5975-01-294-2132
	0.900	CRES 316	5975-01-239-2353
	0.531	Steel	
	0.680	Steel	5975-01-020-9176
	0.900	Steel	
	0.358	Steel	
Triple loop	0.531	CRES 316	5975-01-294-9091
	0.531	Steel	5975-01-020-9183
Double loop	1.000	CRES 316	5975-01-294-9090
	1.000	Steel	5975-01-082-0894

FIGURE 4C22. Cable clamps for single cableway on steel and aluminum decks and bulkheads - Continued.

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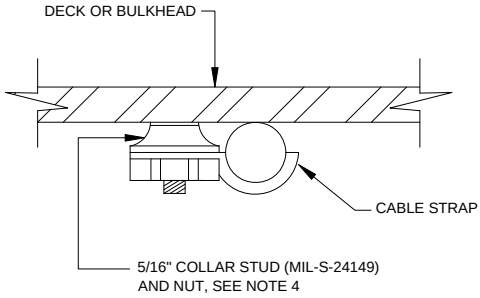
METHOD 4C-23-1

CABLES SECURED TO NON-WATER-TIGHT BULKHEADS
(FOR CABLES 5/8" DIA MAXIMUM)
(SEE NOTES 1 THROUGH 3)



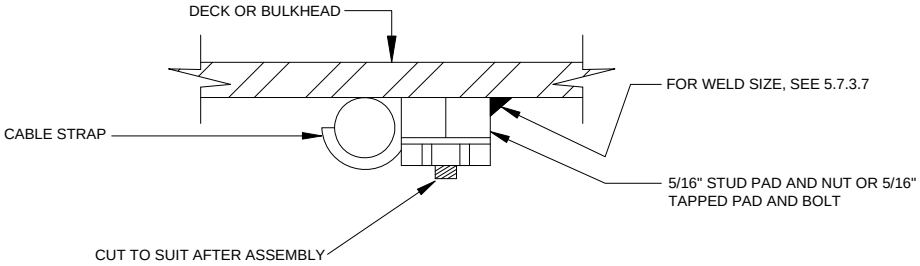
METHOD 4C-23-2

CABLES SECURED WITH COLLAR STUDS
(FOR CABLES 5/8" DIA MAXIMUM)
(SEE NOTES 1 THROUGH 3)



METHOD 4C-23-3

CABLES SECURED WITH PADS
(CABLES 5/8" DIA MAXIMUM)
(SEE NOTES 1 THROUGH 3)



NOTES:

1. All hanger material shall be the same as bulkhead material.
2. Cable strap shall not dent or otherwise exert pressure on the semi-solid dielectric coaxial cables in any of these methods. Add spacer washers if necessary to keep cable free of pressure.
3. Collar studs, stud pads, and tapped pads installed topside or in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C23. Single cableway for coaxial cable.

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NOTES (continued):

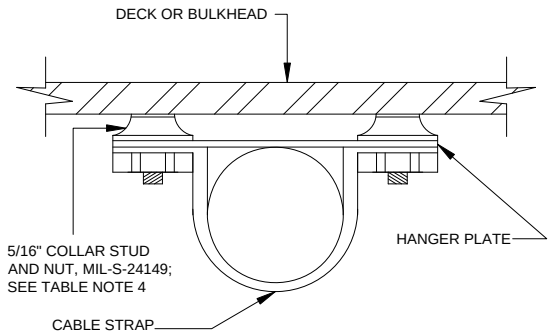
4. Dimensions and national stock numbers (NSN) for collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 4	MIL-S-24149/3 CRES type V, class 4	MIL-S-24149/2 aluminum type IV, class 3
1.000	5307-01-294-3145	5307-01-294-1746	
1.125	5307-01-293-5406	5307-01-294-4363	
1.250	5307-01-293-1806	5307-01-294-3144	
1.350	5307-01-293-1807	5307-01-294-4365	-
1.375	-	-	5307-01-293-6708
1.500	5307-01-293-1808	5307-01-295-0178	5307-01-294-4366
1.625	5307-01-293-1809	5307-01-295-2586	5307-01-293-6709
1.750	5307-01-293-6158	5307-01-294-8129	5307-01-293-5410
1.875	5307-01-293-6159	5307-01-293-6703	
2.000	5307-01-293-6160	5307-01-293-6704	
2.250	5307-01-292-9602		
2.500	5307-01-293-1810	5307-01-293-7979	
2.750	5307-01-293-1811		
3.000	5307-01-293-5409	5307-01-293-6705	
3.250	5307-01-293-1812		
3.500	5307-01-292-9603	5307-01-293-6706	
3.750	5307-01-292-9604		
4.000	5307-01-292-9605	5307-01-293-6707	
4.250	5307-01-292-9606		
4.500	5307-01-292-9607		
4.750	5307-01-293-1813		
5.000	5307-01-292-9608		
5.250	5307-01-293-1072		
5.500	5307-01-293-9577		

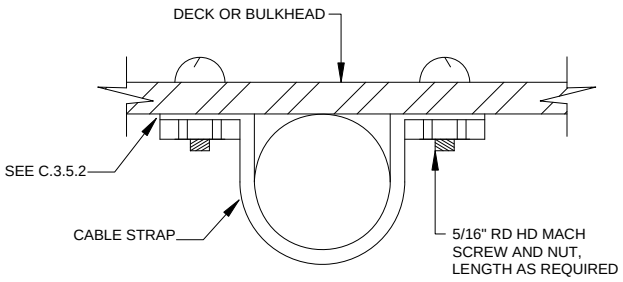
FIGURE 4C23. Single cableway for coaxial cable - Continued.

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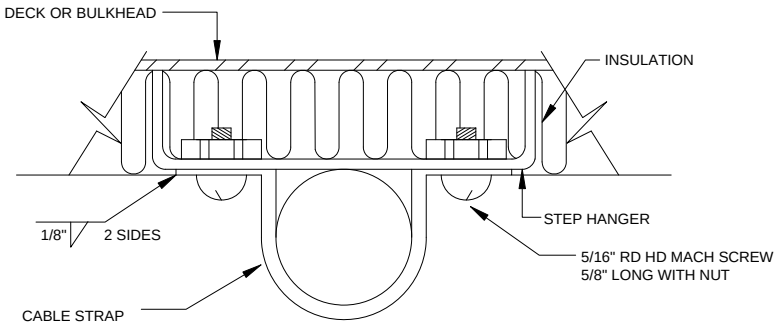
METHOD 4C-24-1
CABLES SECURED WITH COLLAR STUDS
(SEE NOTES 1 THROUGH 3)



METHOD 4C-24-2
CABLES SECURED TO NON-WATERTIGHT BULKHEADS
(SEE NOTES 1 THROUGH 3)



METHOD 4C-24-3
CABLES SECURED WITH STEP-HANGER ON INSULATED DECKS AND BULKHEADS
(SEE NOTES 1 THROUGH 3)



NOTES:

1. All hanger material shall be the same as bulkhead material.
2. Cable strap shall not dent or otherwise exert pressure on the semi-solid dielectric coaxial cables in any of these methods. Add spacer washers if necessary to keep cable free of pressure.
3. Collar studs, stud pads, and tapped pads installed topside or in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C24. Single cableway for coaxial cable.

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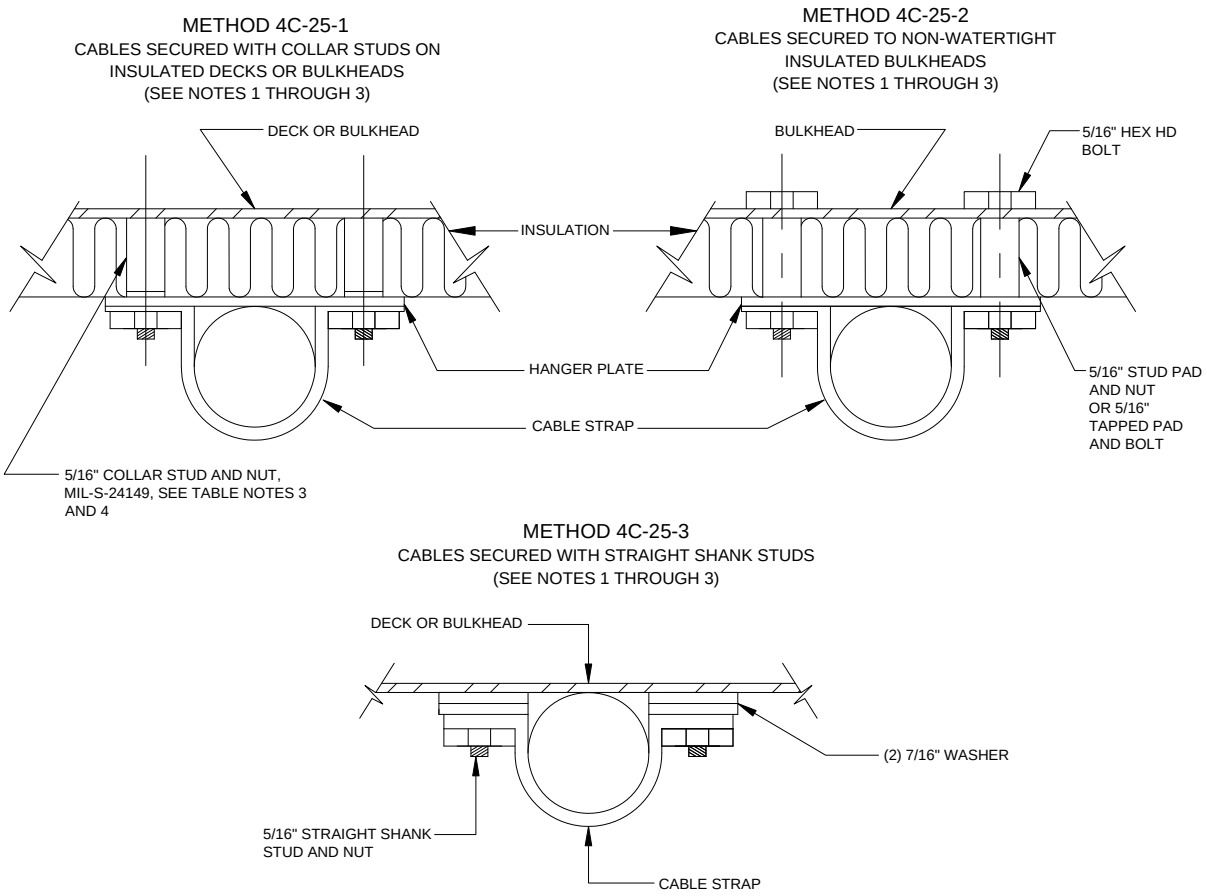
NOTES (continued):

4. Dimensions and national stock numbers (NSN) for collar studs are as follows:

Length	MIL-S-24149/1 steel type H, class 4	MIL-S-24149/3 CRES type V, class 4	MIL-S-24149/2 aluminum type IV, class 3
1.000	5307-01-294-3145	5307-01-294-1746	
1.125	5307-01-293-5406	5307-01-294-4363	
1.250	5307-01-293-1806	5307-01-294-3144	
1.350	5307-01-293-1807	5307-01-294-4365	-
1.375	-	-	5307-01-293-6708
1.500	5307-01-293-1808	5307-01-295-0178	5307-01-294-4366
1.625	5307-01-293-1809	5307-01-295-2586	5307-01-293-6709
1.750	5307-01-293-6158	5307-01-294-8129	5307-01-293-5410
1.875	5307-01-293-6159	5307-01-293-6703	
2.000	5307-01-293-6160	5307-01-293-6704	
2.250	5307-01-292-9602		
2.500	5307-01-293-1810	5307-01-293-7979	
2.750	5307-01-293-1811		
3.000	5307-01-293-5409	5307-01-293-6705	
3.250	5307-01-293-1812		
3.500	5307-01-292-9603	5307-01-293-6706	
3.750	5307-01-292-9604		
4.000	5307-01-292-9605	5307-01-293-6707	
4.250	5307-01-292-9606		
4.500	5307-01-292-9607		
4.750	5307-01-293-1813		
5.000	5307-01-292-9608		
5.250	5307-01-293-1072		
5.500	5307-01-293-9577		

FIGURE 4C24. Single cableway for coaxial cable - Continued.

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NOTES:

1. All hanger material shall be the same as bulkhead material.
2. Cable strap shall not dent or otherwise exert pressure on the semi-solid dielectric coaxial cables in any of these methods. Add spacer washers if necessary to keep cable free of pressure.
3. Collar studs, stud pads, and tapped pads installed topside or in locations subject to a corrosive environment shall be in accordance with 5.7.3.2.1.

FIGURE 4C25. Single cableway for coaxial cable.

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NOTES (continued):

4. Dimensions and national stock numbers (NSN) for collar studs are as follows:

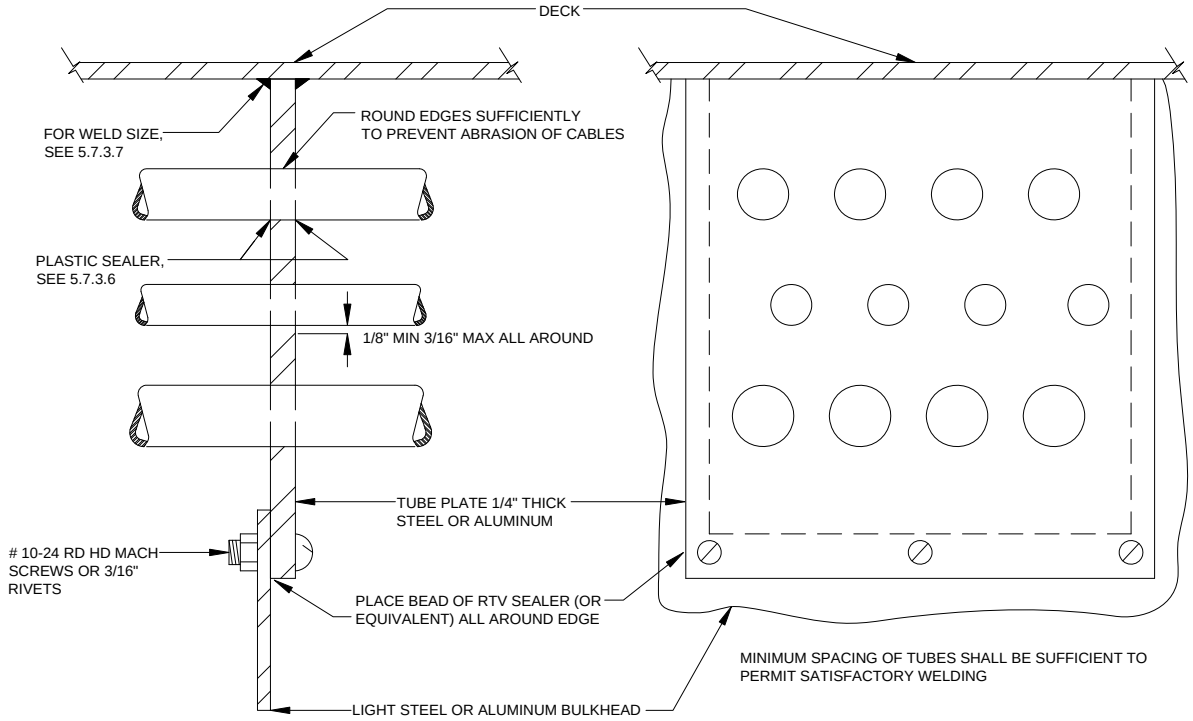
Length	MIL-S-24149/1 steel type H, class 4	MIL-S-24149/3 CRES type V, class 4	MIL-S-24149/2 aluminum type IV, class 3
1.000	5307-01-294-3145	5307-01-294-1746	
1.125	5307-01-293-5406	5307-01-294-4363	
1.250	5307-01-293-1806	5307-01-294-3144	
1.350	5307-01-293-1807	5307-01-294-4365	-
1.375	-	-	5307-01-293-6708
1.500	5307-01-293-1808	5307-01-295-0178	5307-01-294-4366
1.625	5307-01-293-1809	5307-01-295-2586	5307-01-293-6709
1.750	5307-01-293-6158	5307-01-294-8129	5307-01-293-5410
1.875	5307-01-293-6159	5307-01-293-6703	
2.000	5307-01-293-6160	5307-01-293-6704	
2.250	5307-01-292-9602		
2.500	5307-01-293-1810	5307-01-293-7979	
2.750	5307-01-293-1811		
3.000	5307-01-293-5409	5307-01-293-6705	
3.250	5307-01-293-1812		
3.500	5307-01-292-9603	5307-01-293-6706	
3.750	5307-01-292-9604		
4.000	5307-01-292-9605	5307-01-293-6707	
4.250	5307-01-292-9606		
4.500	5307-01-292-9607		
4.750	5307-01-293-1813		
5.000	5307-01-292-9608		
5.250	5307-01-293-1072		
5.500	5307-01-293-9577		

FIGURE 4C25. Single cableway for coaxial cable - Continued.

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METHOD 4C-26-1

MULTIPLE TIERS OF CABLES THROUGH LIGHT BULKHEAD WITH AIR AND FUME TIGHTNESS REQUIREMENTS



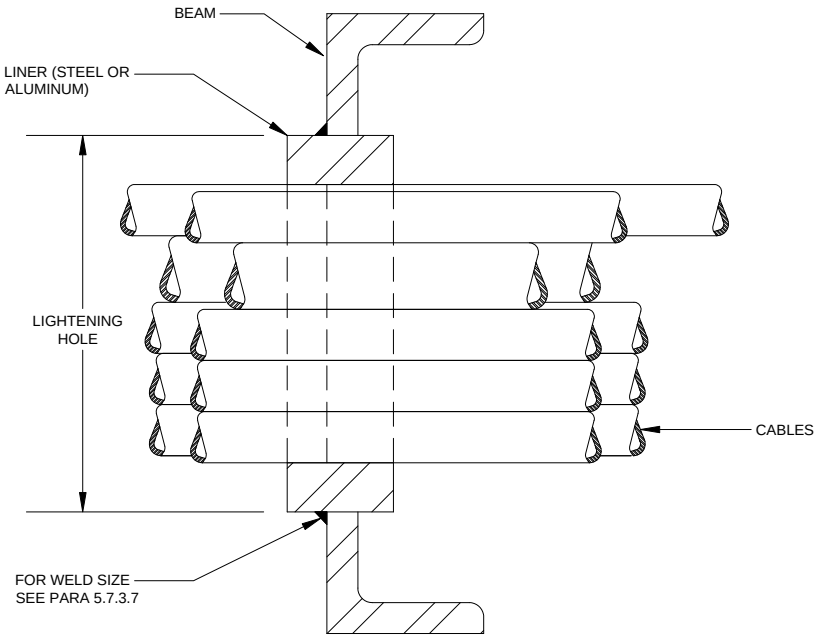
NOTES:

1. When holes are cut in the ship's structure solely for the passage of cables, the liner shall be of sufficient weight to compensate for the loss of strength due to the metal removed.
2. Liners 1/16" thick can be used as an alternate to half round bar.
3. Installation of liner in beams may be made in shop or on the ship.
4. Where no lightening holes are provided, cables may be run outside of beam when no interference with either equipment or normal headroom results.
5. When cables pierce non-watertight bulkheads required to be light-tight, the space between the cable and the inside of the bushing tubing or drilled hole shall be thoroughly packed with plastic sealer (MIL-I-3064, Type HF) to form a permanent light-tight seal.

FIGURE 4C26. Cableway through non-watertight bulkheads and beams without firestops.

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METHOD 4C-27-1
CABLES THROUGH LIGHTENING HOLES OR CABLEWAYS
IN STRUCTURAL MEMBERS (N0N-LIGHT TIGHT)



NOTES:

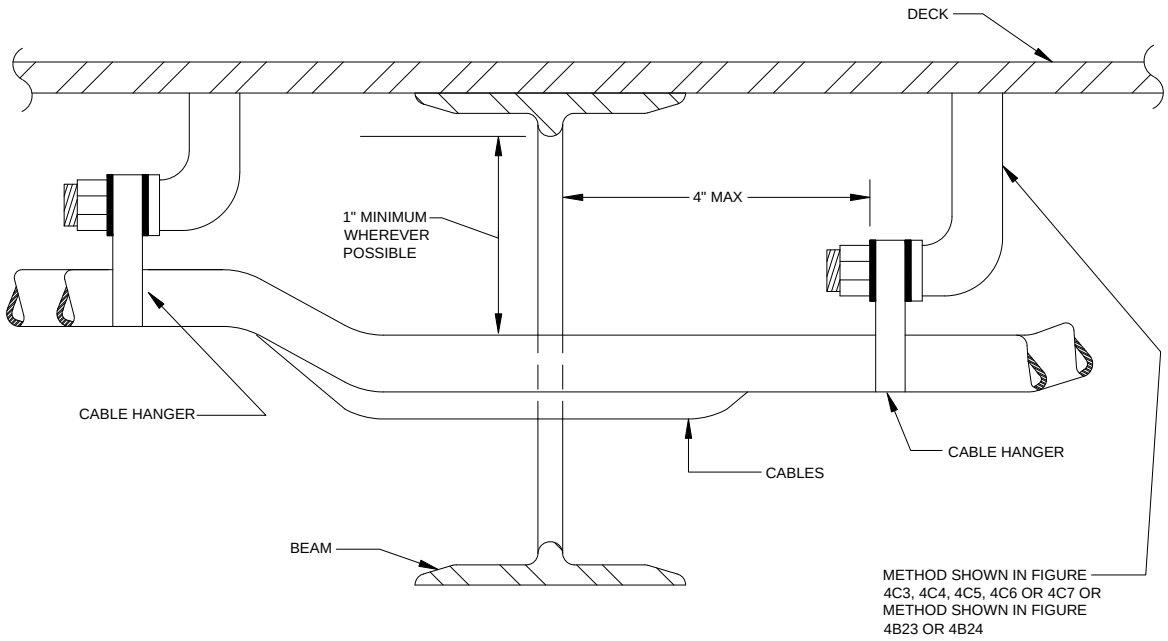
1. When holes are cut in the ship's structure solely for the passage of cables, the liner shall be of sufficient weight to compensate for the loss of strength due to the metal removed.
2. Liners ⅛" thick can be used as an alternate to half round bar.
3. Installation of liner in beams may be made in shop or on the ship.
4. Where no lightening holes are provided, cables may be run outside of beam when no interference with either equipment or normal headroom results.

FIGURE 4C27. Cableway through non-watertight bulkheads and beams without firestops.

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METHOD 4C-28-1

A GROUP OF CABLES
PASSING THROUGH LIGHTENING HOLES
(LIGHTENING HOLES NOT USED AS SUPPORT)



NOTES:

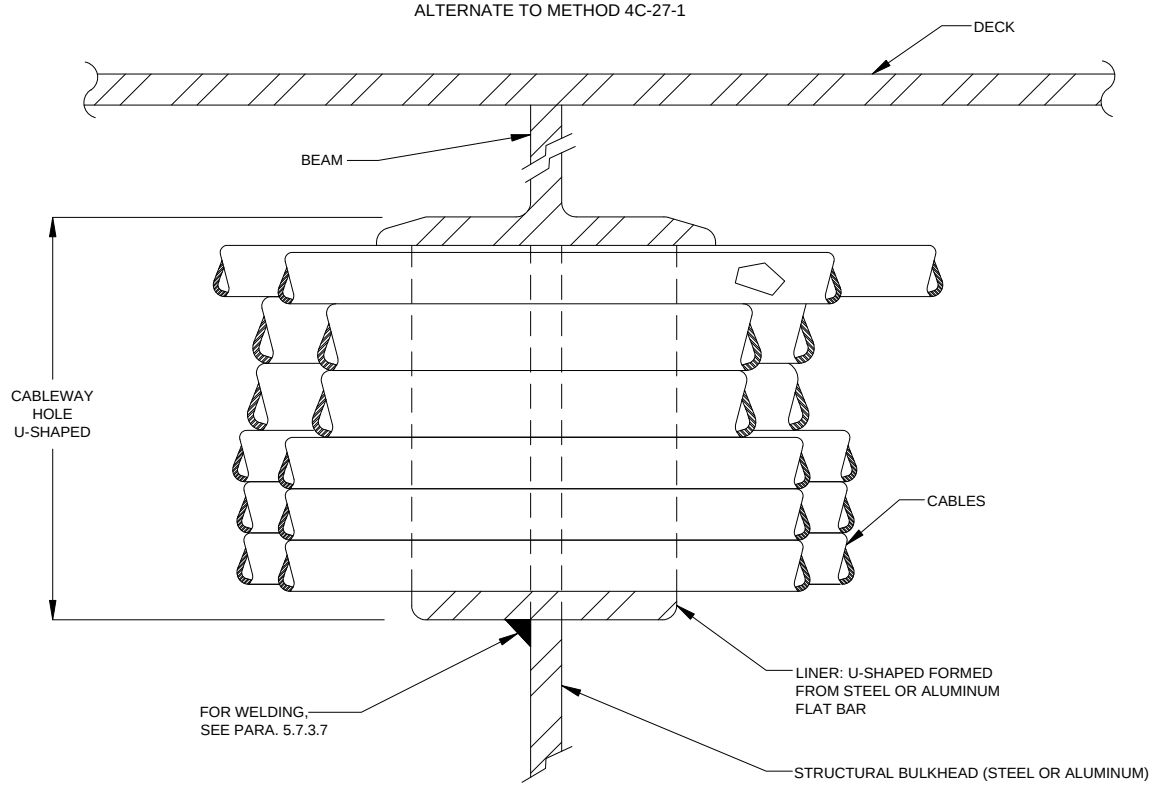
1. When holes are cut in the ship's structure solely for the passage of cables, the liner shall be of sufficient weight to compensate for the loss of strength due to the metal removed.
2. Liners 1/16" thick can be used as an alternate to half round bar.
3. Installation of liner in beams may be made in shop or on the ship.
4. Where no lightening holes are provided, cables may be run outside of beam when no interference with either equipment or normal headroom results.

FIGURE 4C28. Cableway through non-watertight bulkheads and beams without firestops.

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METHOD 4C-29-1

CABLEWAY IN STRUCTURAL BULKHEADS (NON-LIGHT TIGHT)
ALTERNATE TO METHOD 4C-27-1



NOTES:

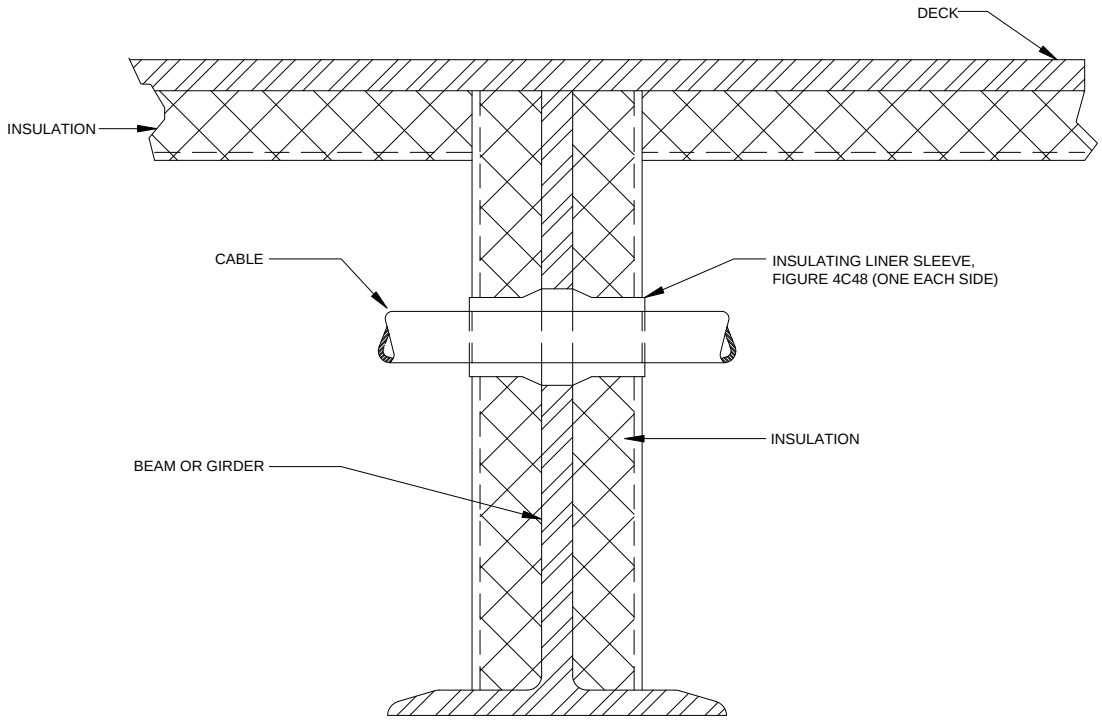
1. When holes are cut in the ship's structure solely for the passage of cables, the liner shall be of sufficient weight to compensate for the loss of strength due to the metal removed.
2. Liners 1/16" thick can be used as an alternate to half round bar.
3. Installation of liner in beams may be made in shop or on the ship.
4. Where no lightening holes are provided, cables may be run outside of beam when no interference with either equipment or normal headroom results.

FIGURE 4C29. Cableway through non-watertight bulkheads and beams without firestops.

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METHOD 4C-30-1

SINGLE CABLE THROUGH BEAMS AND GIRDERS WITH INSULATION WITHOUT METAL SHEATHING (ALTERNATE TO METHOD 4C-31-2)



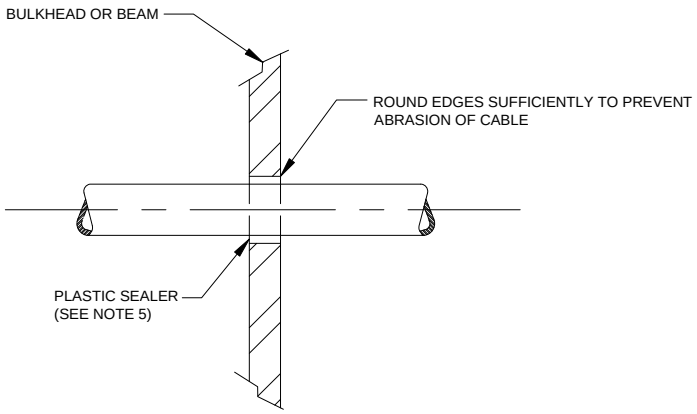
NOTES:

1. When holes are cut in the ship's structure solely for the passage of cables, the liner shall be of sufficient weight to compensate for the loss of strength due to the metal removed.
2. Liners 1/16" thick can be used as an alternate to half round bar.
3. Installation of liner in beams may be made in shop or on the ship.
4. Where no lightening holes are provided, cables may be run outside of beam when no interference with either equipment or normal headroom results.

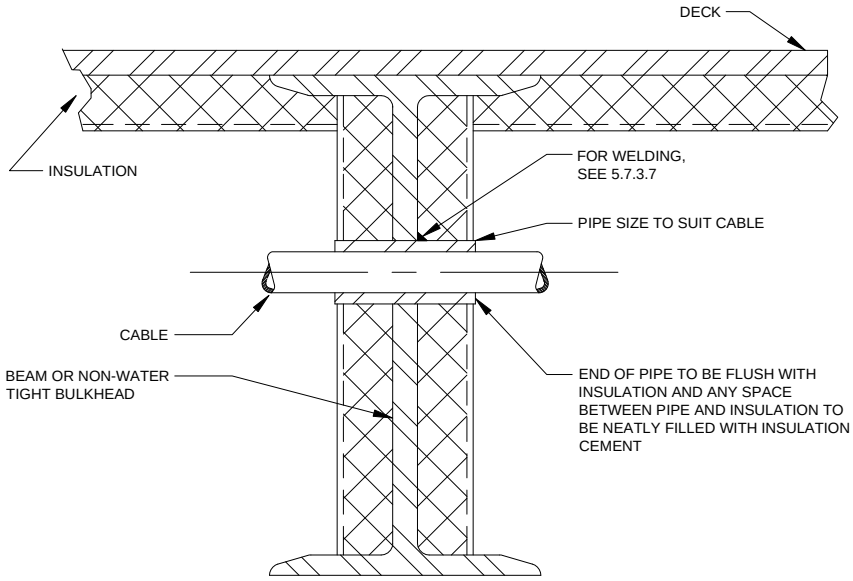
FIGURE 4C30. Cableway through non-watertight bulkheads and beams without firestops.

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METHOD 4C-31-1
CABLES THROUGH BULKHEADS AND
BEAMS 1/4" THICKNESS AND OVER
SEE NOTES 1 THROUGH 5



METHOD 4C-31-2
INSULATED BULKHEAD OR BEAM CABLE THROUGH TUBING
SEE NOTES 1 THROUGH 5



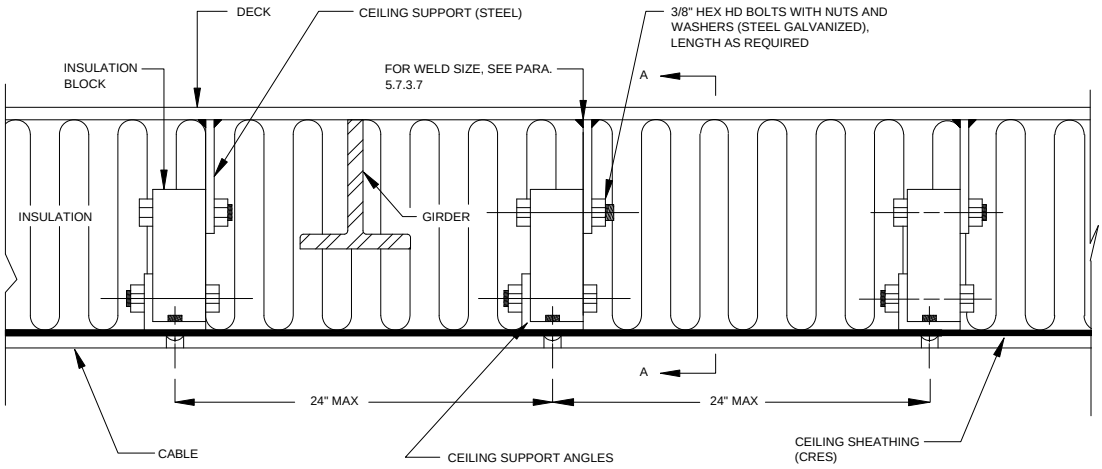
NOTES:

1. When holes are cut in the ship's structure solely for the passage of cables, the liner shall be of sufficient weight to compensate for the loss of strength due to the metal removed.
2. Liners ⅛" thick can be used as an alternate to half round bar.
3. Installation of liner in beams may be made in shop or on the ship.
4. Where no lightening holes are provided, cables may be run outside of beam when no interference with either equipment or normal headroom results.
5. When cables pierce non-watertight bulkheads required to be light-tight, the space between the cable and the inside of the bushing tubing or drilled hole shall be thoroughly packed with plastic sealer (MIL-I-3064, Type HF) to form a permanent light-tight seal.

FIGURE 4C31. Cableway through non-watertight bulkheads and beams without firestops.

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METHOD 4C-32-1
CABLE RUNS ACROSS CEILING SHEATHING SUPPORT FRAMING



FOR SECTION A-A, SEE FIGURE 4C33

FIGURE 4C32. Single cableway in refrigerated spaces.

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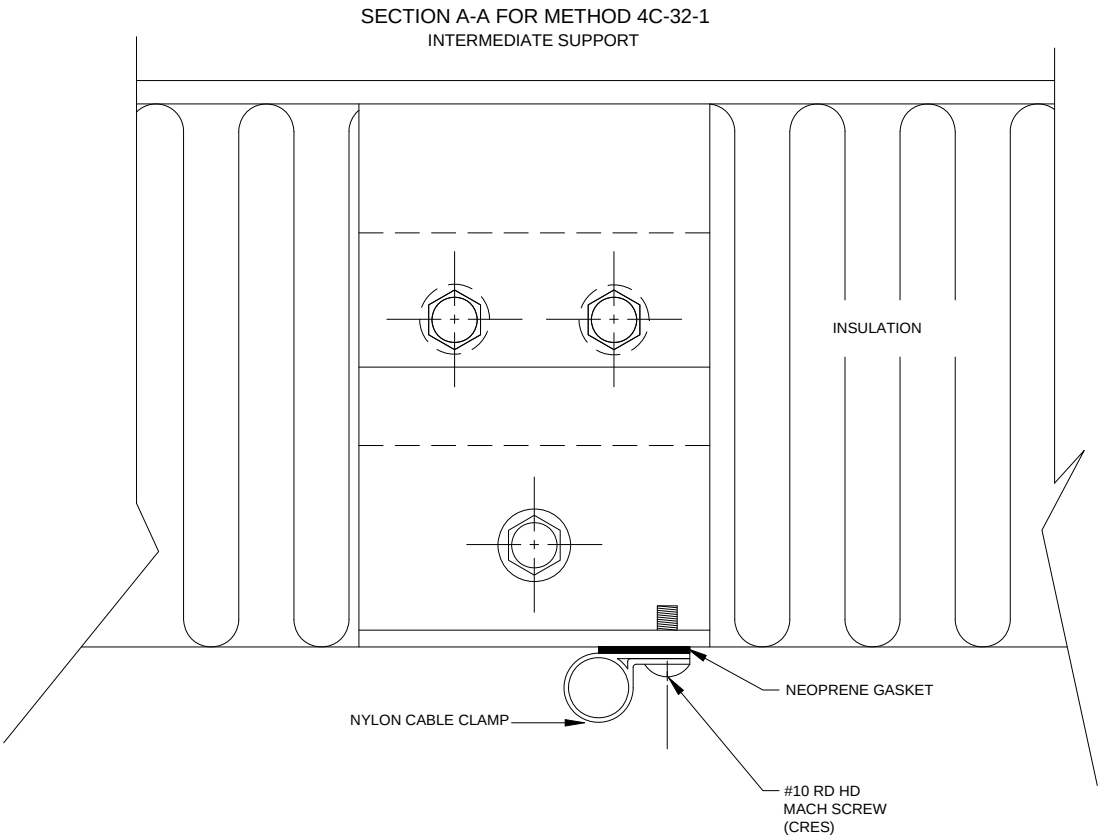
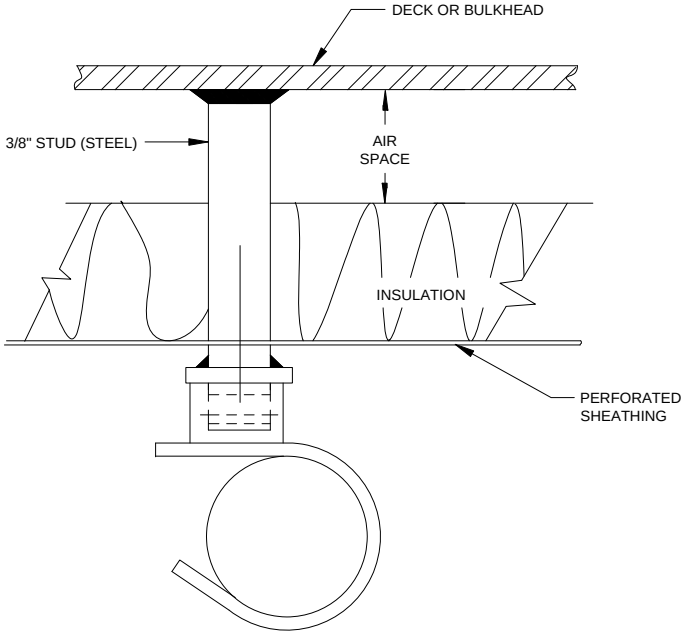


FIGURE 4C33. Single cableway in refrigerated spaces – section A-A for method 4C-32-1.

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METHOD 4C-34-1

WATERTIGHT DECK OR BULKHEAD
WITH INSULATION AND SHEATHING
HANGER SECURED WITH COLLAR STUD



NOTE: THE METHOD SHOWN ON THIS FIGURE IS FOR USE WITH CABLE UP TO TWO INCHES IN DIAMETER. FOR CABLE GREATER THAN TWO INCHES IN DIAMETER, USE THE METHOD SHOWN ON FIGURE 4C35.

FIGURE 4C34. Single cableways in acoustical spaces secured with collar stud.

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METHOD 4C-35-1
DECKS OR BULKHEADS WITH INSULATION AND SHEATHING
CABLES SECURED TO STUD PADS OR TAPPED PADS
(LOCAL RUNS ONY)

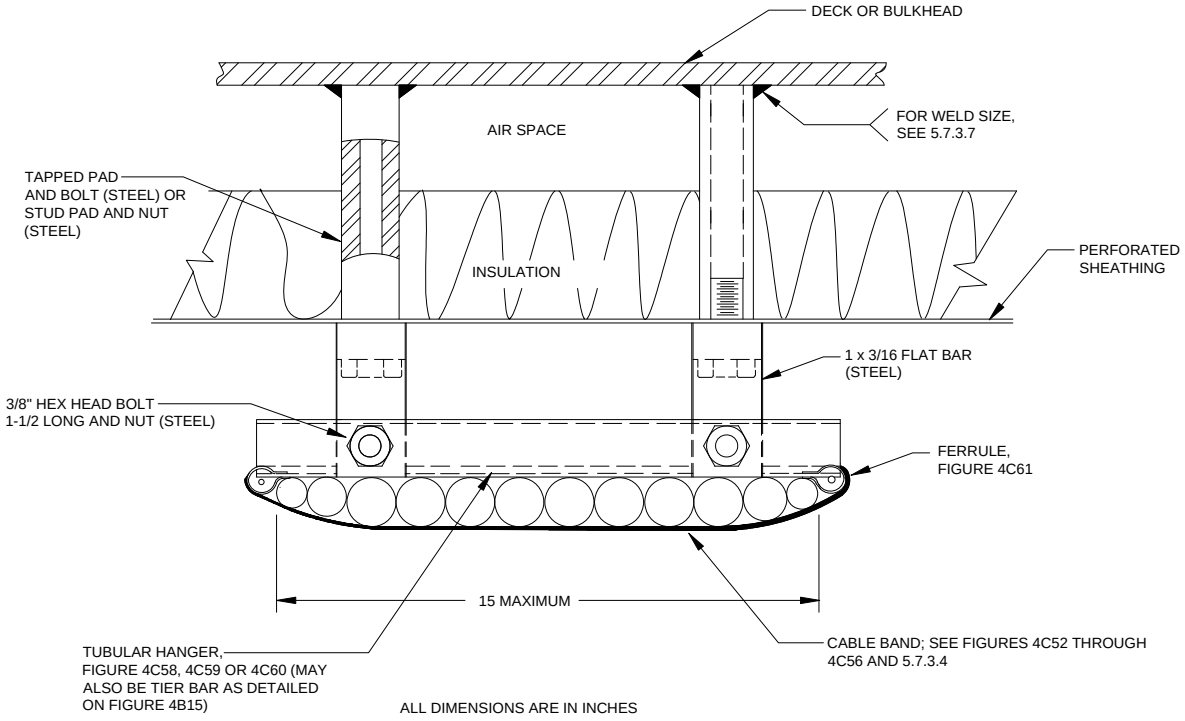


FIGURE 4C35. Single cableway in acoustical spaces secured to stud pads or tapped pads.

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METHOD 4C-36-1
CABLE SECURED TO MOTOR SUPPORT

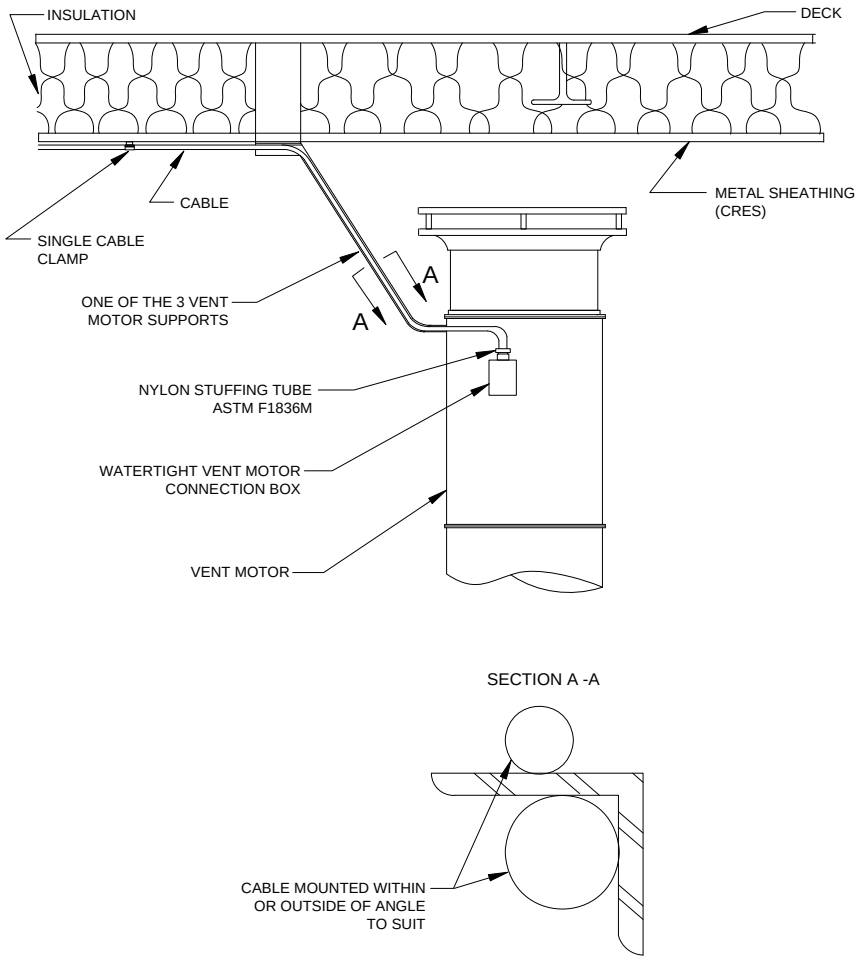


FIGURE 4C36. Single cableway to motors – cable secured to motor support.

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Check the source to verify that this is the current version before use.

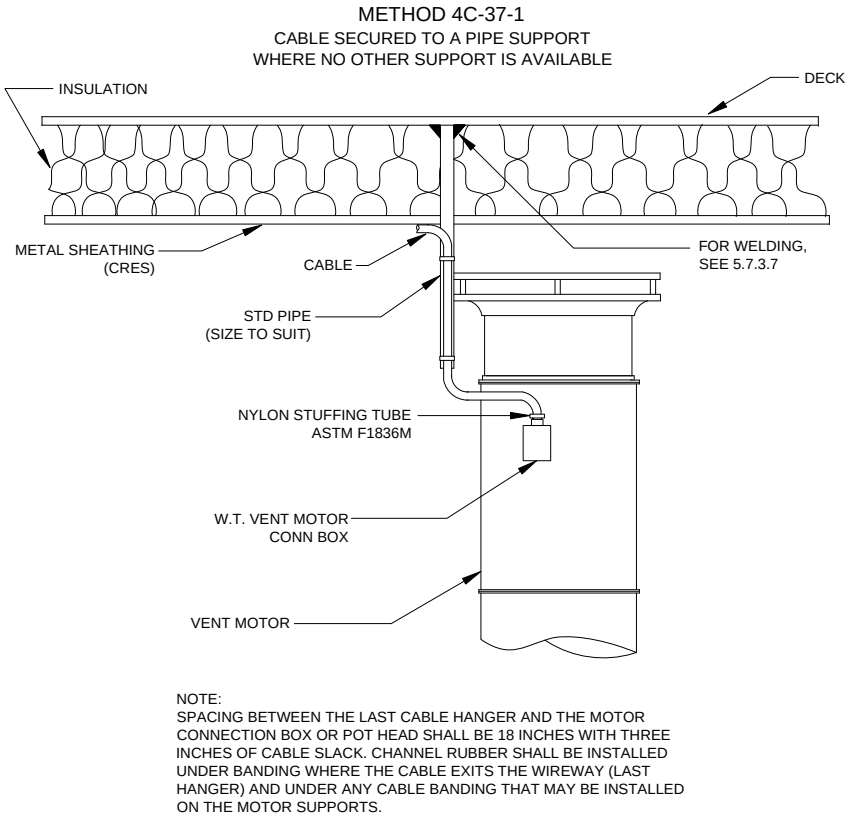
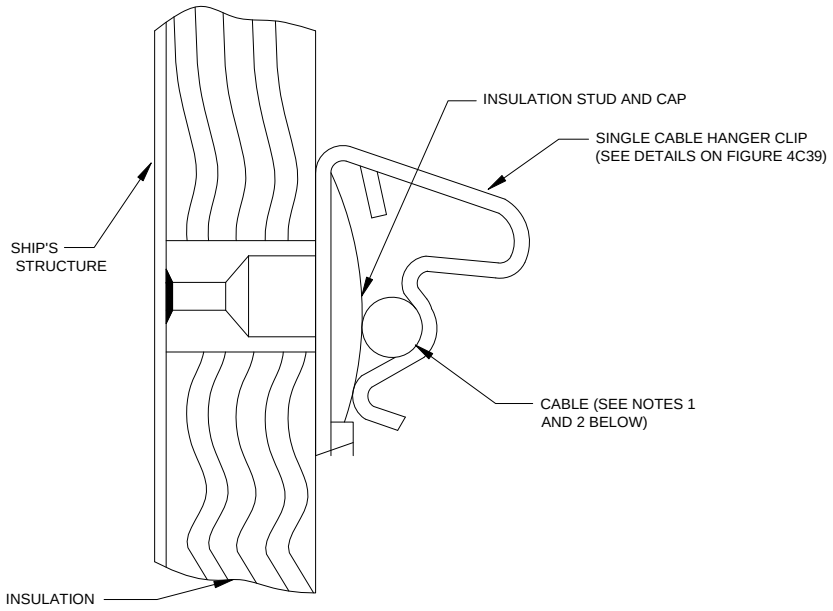


FIGURE 4C37. Single cableway to motors – cable secured to pipe support.

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METHOD 4C-38-1

SINGLE CABLE HANGER CLIP USED IN CONJUCTION WITH INSULATION STUD CAP



NOTES:

1. This hanger clip is for cable ½" in diameter or less and weighing ¼ pound per foot or less.
2. Where a maximum length of the cable is installed along the surface of the hull board insulation and it is necessary to originate or terminate the run outside the insulation boundary, this hanger may be used for continuation of the run if advantageous to do so.
3. The material shall be spring steel, spheroidized, ASTM A713, Rockwell hardness 45-55, zinc electro-plated in accordance with ASTM B633, Type I, thickness class FE-ZN 8 for service condition (SC) 2.

FIGURE 4C38. Single cableway on hullboard insulation.

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CLIP DETAILS FOR METHOD 4C-38-1
SINGLE CABLE HANGER CLIP USED IN CONJUCTION WITH INSULATION STUD CAP

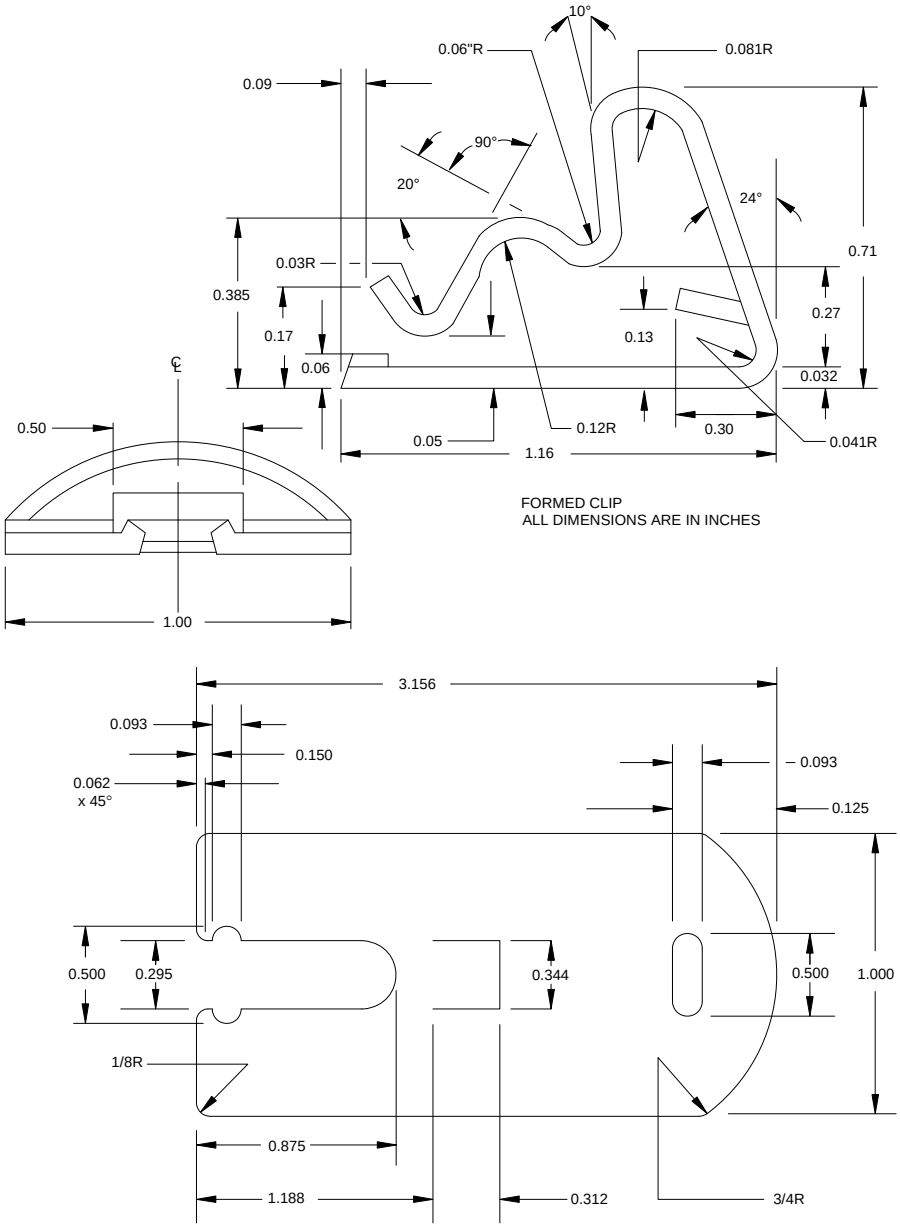
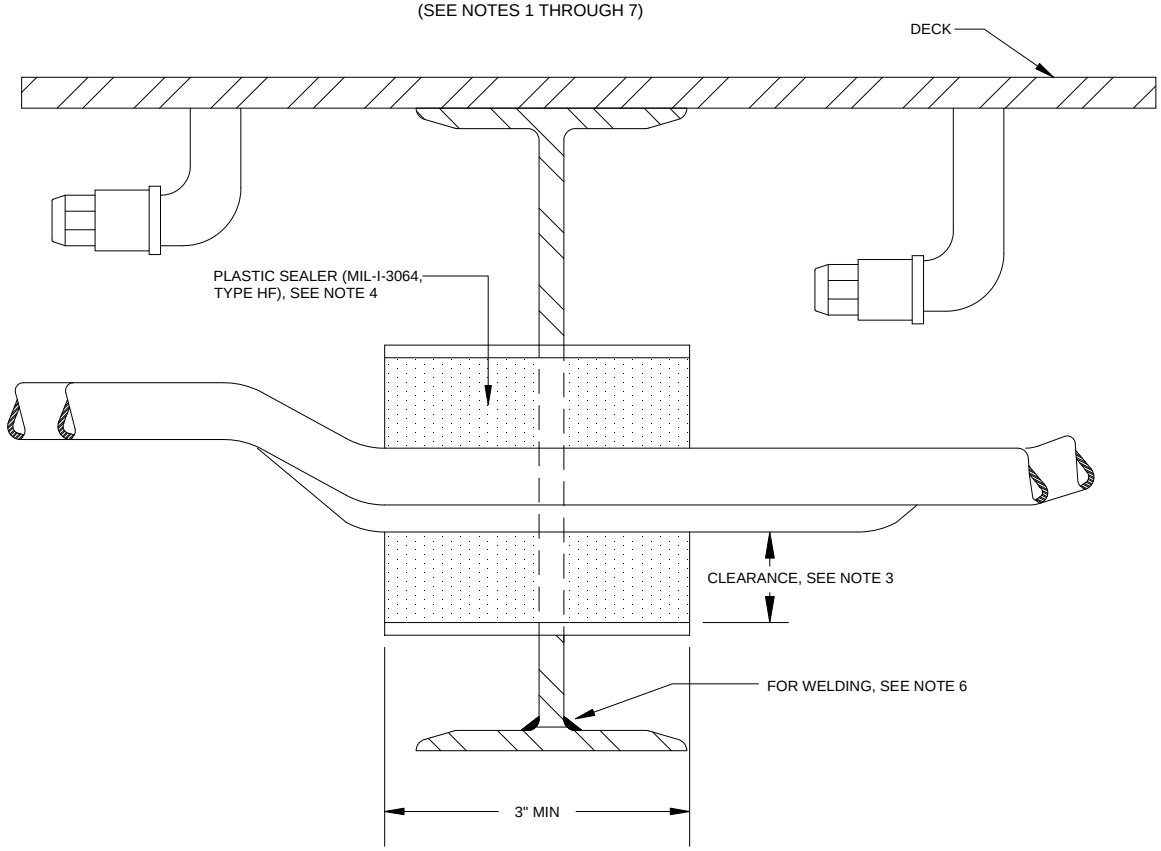


FIGURE 4C39. Clip details for single cableway on hullboard insulation.

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METHOD 4C-40-1

CABLES PASSING THROUGH LIGHTENING HOLES WITH FIRE STOP
(FOR NON-WATERTIGHT CABLE PENETRATIONS AND FOR AIRTIGHT AND
FLAMETIGHT SPACES WHERE APPLICABLE)
(SEE NOTES 1 THROUGH 7)



NOTES:

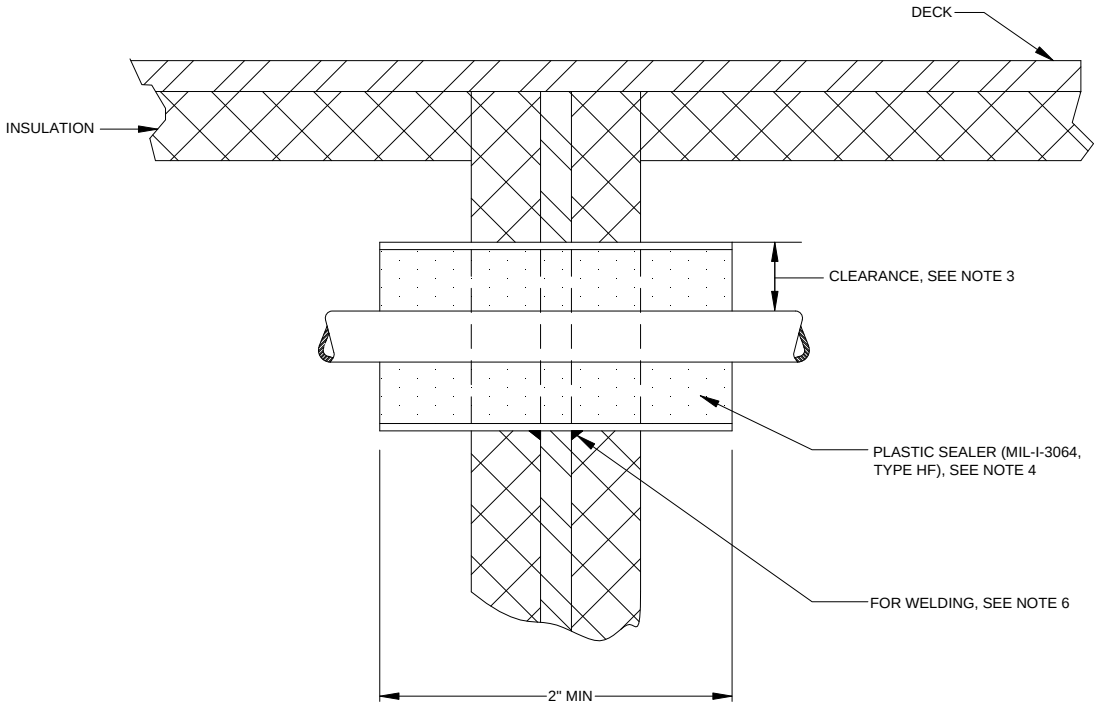
1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.
7. For existing installations, stuffing units may be split lengthwise for installing over cable.

FIGURE 4C40. Cableway through non-watertight bulkheads and beams with firestops.

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METHOD 4C-41-1

SINGLE CABLE PENETRATION THRU NON-WATERTIGHT
BULKHEADS OR STRUCTURAL MEMBERS WITH FIRE STOP
(SEE NOTES 1 THROUGH 7)



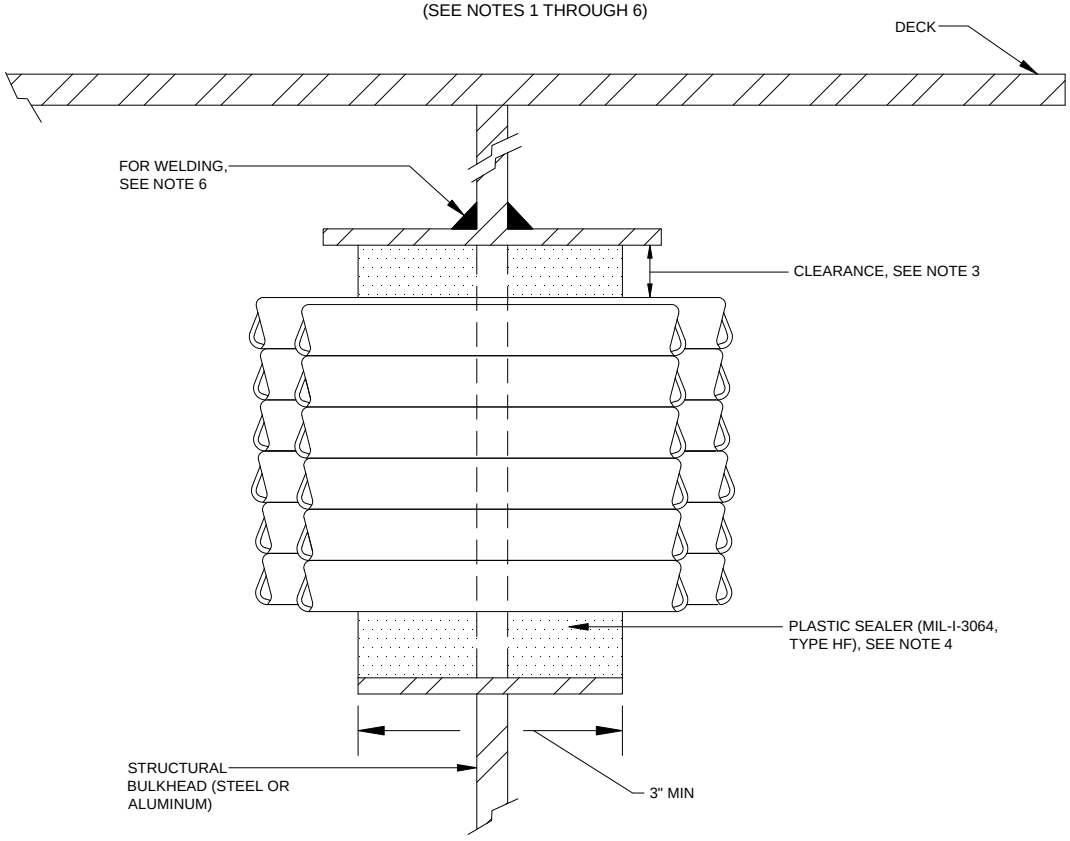
NOTES:

1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.
7. For existing installations, stuffing units may be split lengthwise for installing over cable.

FIGURE 4C41. Cableway through non-watertight bulkheads and beams with firestops.

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METHOD 4C-42-1
MULTIPLE CABLE PENETRATION THRU NON-WATERTIGHT
BULKHEADS OR STRUCTURAL MEMBERS WITH FIRE STOP
(SEE NOTES 1 THROUGH 6)



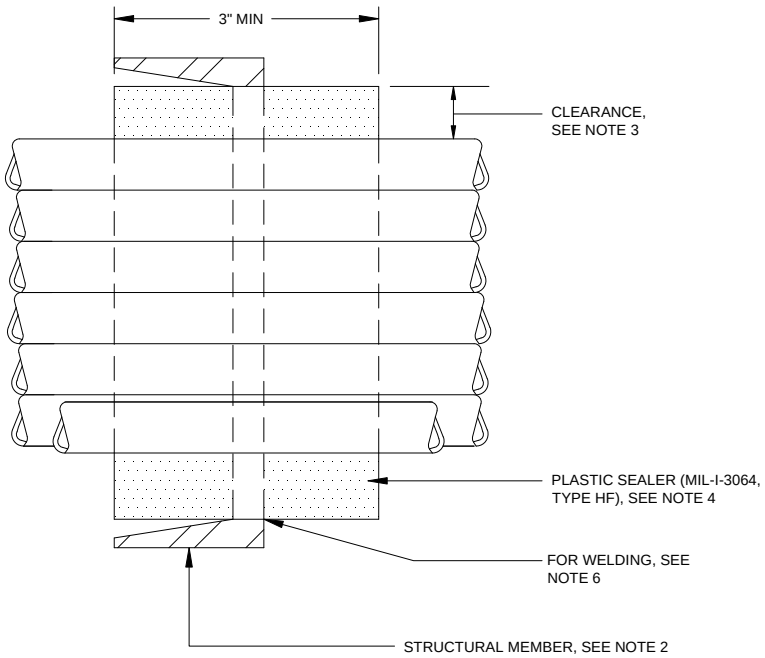
NOTES:

1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.

FIGURE 4C42. Cableway through non-watertight bulkheads and beams with firestops.

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METHOD 4C-43-1
MULTIPLE CABLE PENETRATION THRU STRUCTURAL MEMBER WITH FIRE STOP
(SEE NOTES 1 THROUGH 6)



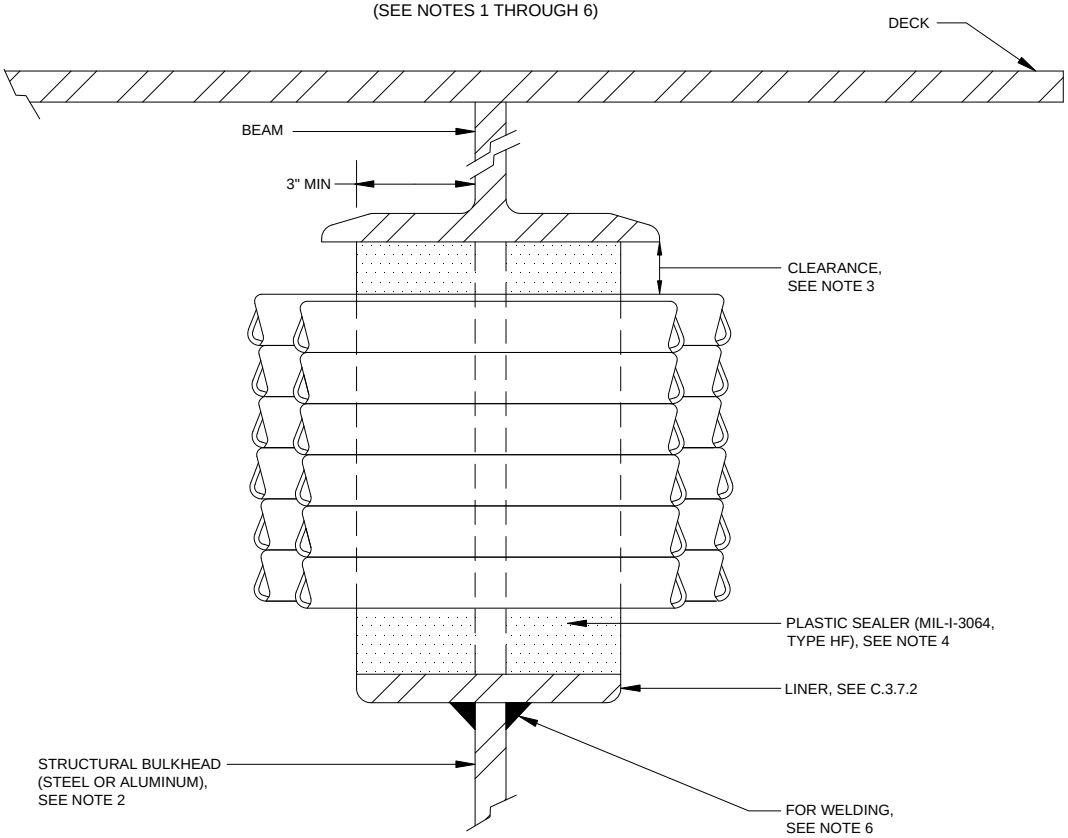
NOTES:

1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.

FIGURE 4C43. Cableway through non-watertight bulkheads and beams with firestops.

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METHOD 4C-44-1
CABLEWAY IN STRUCTURAL BULKHEADS
(SEE NOTES 1 THROUGH 6)



NOTES:

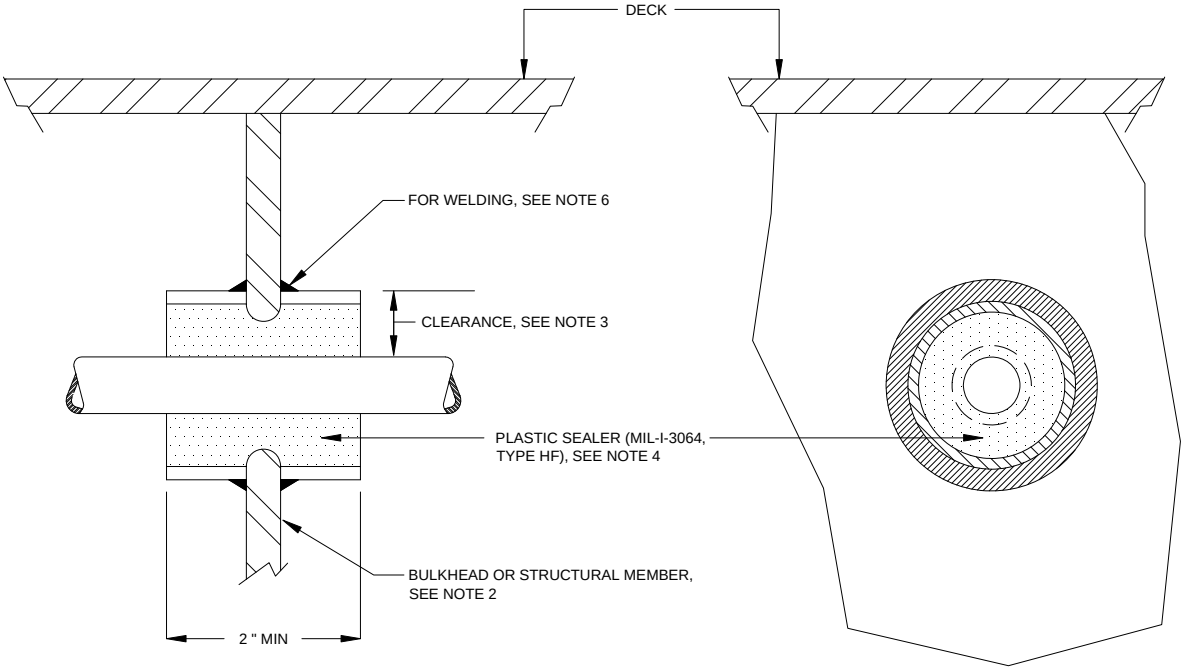
1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.

FIGURE 4C44. Cableway through non-watertight bulkheads and beams with firestops.

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METHOD 4C-45-1

SINGLE CABLE PENETRATION THRU NON-WATERTIGHT
BULKHEADS OR STRUCTURAL MEMBERS WITH FIRE STOPS
(EXISTING CABLE PENETRATIONS)
(SEE NOTES 1 THROUGH 7)



NOTES:

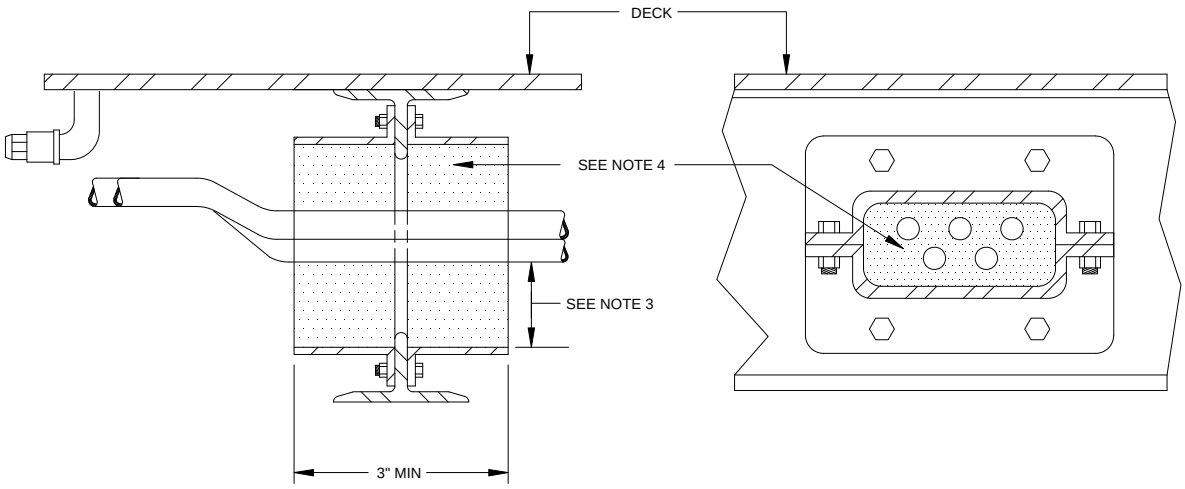
1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.
7. For existing installations, stuffing units may be split lengthwise for installing over cable.

FIGURE 4C45. Cableway through non-watertight bulkheads with firestops, existing penetration.

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METHOD 4C-46-1

CABLE PASSING THRU LIGHTENING HOLES WITH FIRE STOPS - MULTIPLE CABLE PENETRATIONS
(EXISTING CABLE PENETRATIONS)
(SEE NOTES 1 THROUGH 7)



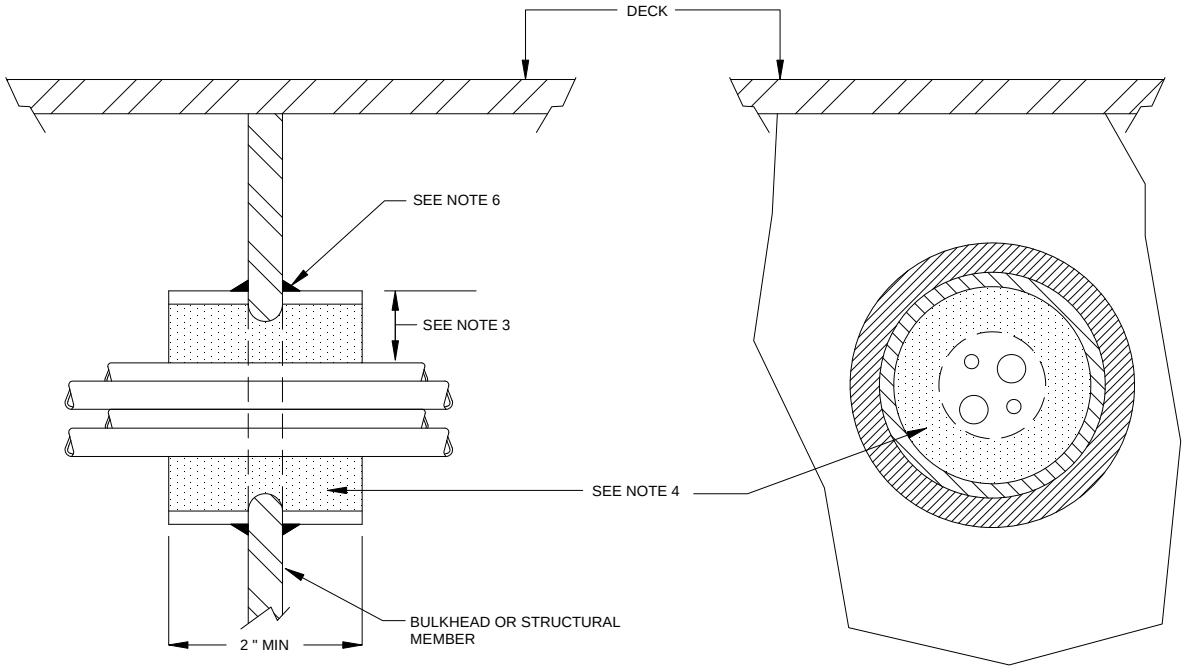
NOTES:

1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.
7. For existing installations, stuffing units may be split lengthwise for installing over cable.

FIGURE 4C46. Cableway through non-watertight bulkheads with firestops, existing penetration.

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METHOD 4C-47-1
MULTIPLE CABLE PENETRATION THRU NON-WATERTIGHT
BULKHEADS OR STRUCTURAL MEMBERS WITH FIRE STOPS
(EXISTING CABLE PENETRATIONS)
(SEE NOTES 1 THROUGH 7)



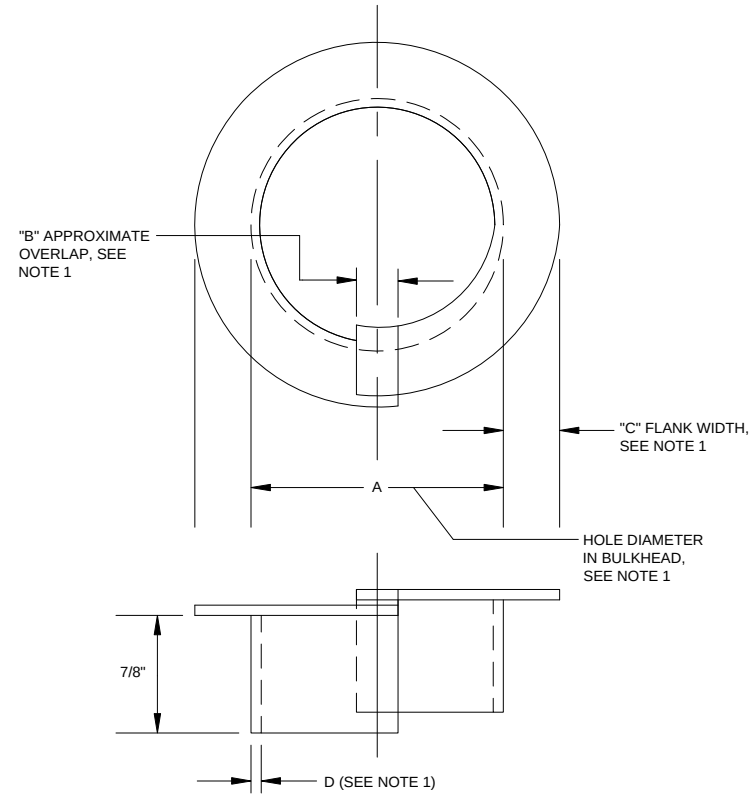
NOTES:

1. Fire stop methods are for non-watertight cable penetrations and also for airtight and flametight spaces where applicable.
2. Stuffing tubes (tubing, collars, and liners) shall be steel for steel structures and aluminum for aluminum structures.
3. The minimum clearance space around cables before packing material is applied shall be ¼" for a single cable and 1" for multiple cables.
4. Pack thoroughly around and between cables with plastic sealer, MIL-I-3064, Type HF.
5. Stuffing unit may be attached by all-around weld, tack weld, or fasteners provided that the attachment conforms to the structural and tightness requirements of the bulkhead or member to which it is attached.
6. All welding shall be in accordance with 5.7.3.7.
7. For existing installations, stuffing units may be split lengthwise for installing over cable.

FIGURE 4C47. Cableway through non-watertight bulkheads with firestops, existing penetration.

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INSULATING LINERS FOR CABLE HOLES
(LINERS MAY BE COMMERCIAL SIZES. MATERIAL SHALL BE VULCANIZED FIBER OR OTHER NON-TOXIC TYPE PLASTIC MATERIAL)
SEE NOTE 1 FOR DIMENSIONS



NOTE: FOR DETAILS OF THE USE OF THIS LINER, REFER TO METHOD 4C-172, FIGURE 4C17.

NOTES:

1. Dimensions (in inches) for insulating liners are as follows:

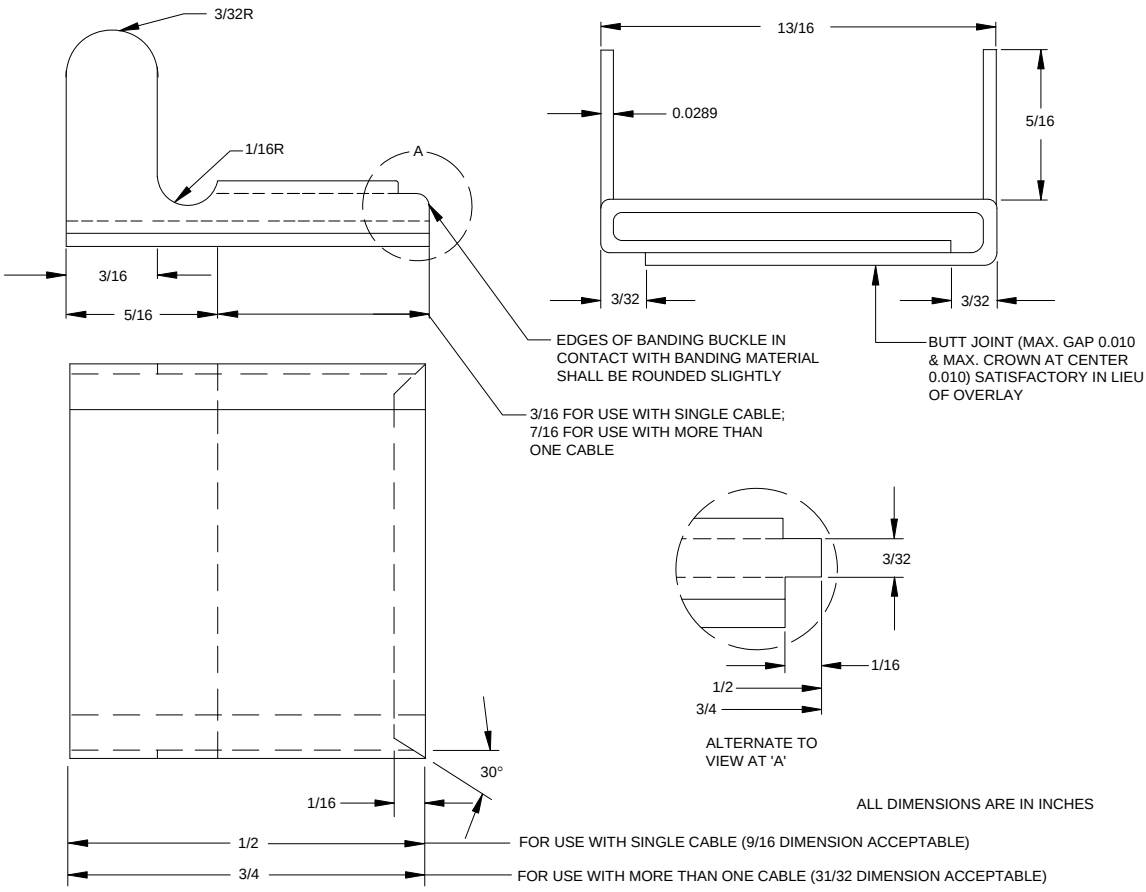
A	B	C	D
3/4	3/16	1/8	0.025
1 5/32	3/16	1/8	0.025
1 7/16	3/16	5/32	0.025
1 13/16	1/4	5/32	0.030
2 1/16	1/4	5/32	0.030
2 19/32	5/16	5/32	0.030
3 5/16	3/8	7/32	0.040
3 3/8	3/8	7/32	0.040
4 1/4	7/16	7/32	0.040

2. For details of the use of this liner, refer to Method 4C-17-2, figure 4C17.

FIGURE 4C48. Insulating liners for cable holes in beams and girders.

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METHOD 4C-49-1
BANDING BUCKLE
(SEE NOTES 1 THROUGH 3)



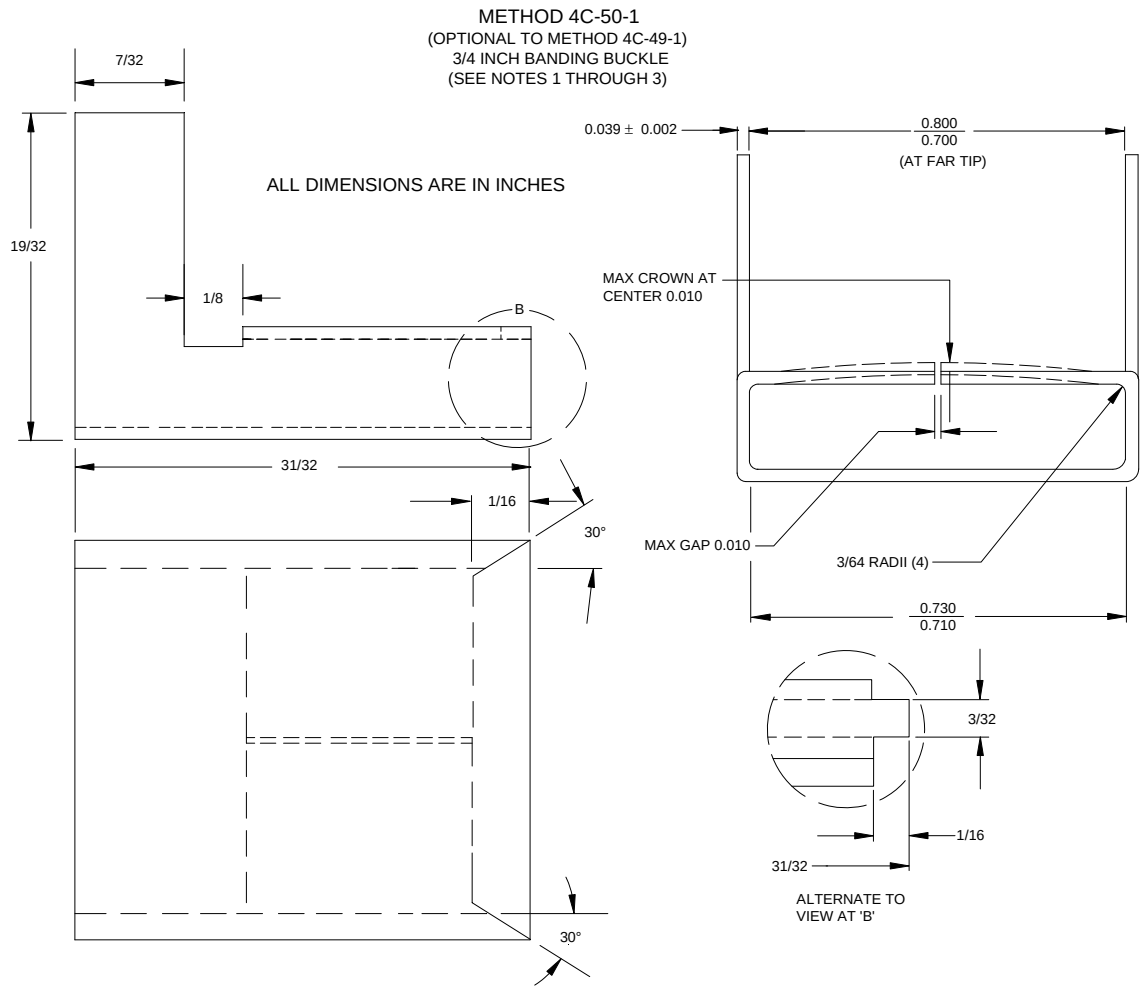
NOTES:

1. Banding buckles shall be made of corrosion-resistant steel, ASTM A240/A240M, ASTM A666, and ASTM A693.
2. The edges of the banding buckle in contact with the banding material shall be rounded slightly.
3. Specifications, vendor information, and part numbers for selected banding buckles are as follows:

Buckle length (inch)	Vendor	Description/part number
½	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	"Inverted Wing Seal" 202SS (stainless steel)
¾	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	"Wing Seal" 204SS (stainless steel)
¾	Independent Metal Strap Company, Inc. 34 Lumber Road, Roslyn, NY 11576-0334 or equal	"Royal Wing Seal" (stainless steel)
⅝ x ½	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	213SS (stainless steel); NSN 9Q8135-01-030-1632
⅝ x ¾	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	203SS (stainless steel); NSN 9Q8135-01-120-8754

FIGURE 4C49. Cable banding buckle and miscellaneous details.

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NOTES:

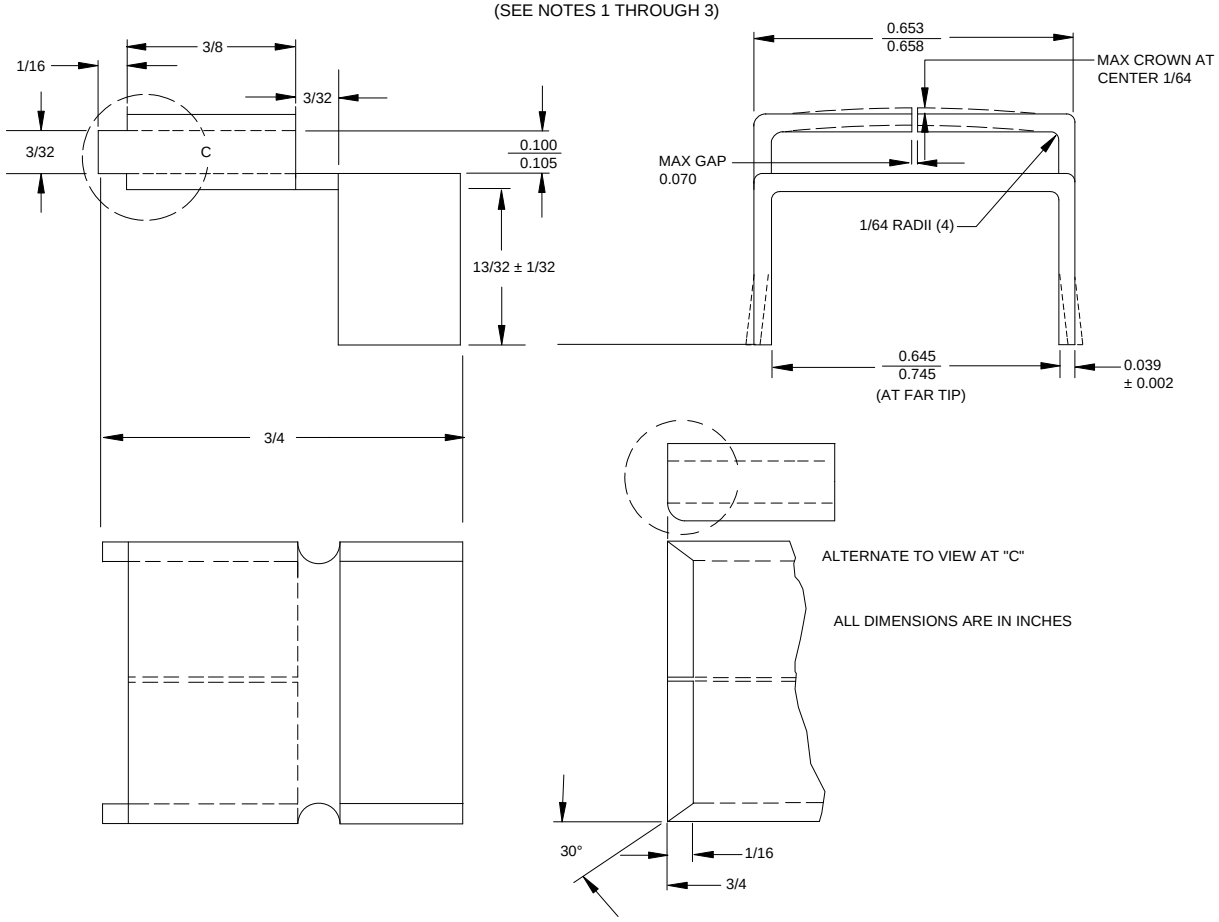
1. Banding buckles shall be made of corrosion-resistant steel, ASTM A240/A240M, ASTM A666, and ASTM A693.
2. The edges of the banding buckle in contact with the banding material shall be rounded slightly.
3. Specifications, vendor information, and part numbers for selected banding buckles are as follows:

Buckle length (inch)	Vendor	Description/part number
½	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	"Inverted Wing Seal" 202SS (stainless steel)
¾	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	"Wing Seal" 204SS (stainless steel)
¾	Independent Metal Strap Company, Inc. 34 Lumber Road, Roslyn, NY 11576-0334 or equal	"Royal Wing Seal" (stainless steel)
⅝ x ½	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	213SS (stainless steel); NSN 9Q8135-01-030-1632
⅝ x ¾	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	203SS (stainless steel); NSN 9Q8135-01-120-8754

FIGURE 4C50. Cable banding buckle and miscellaneous details.

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METHOD 4C-51-1
OPTIONAL TO METHOD 4C-49-1
BANDING BUCKLE (INVERTED), 1/2 INCH
(SEE NOTES 1 THROUGH 3)



NOTES:

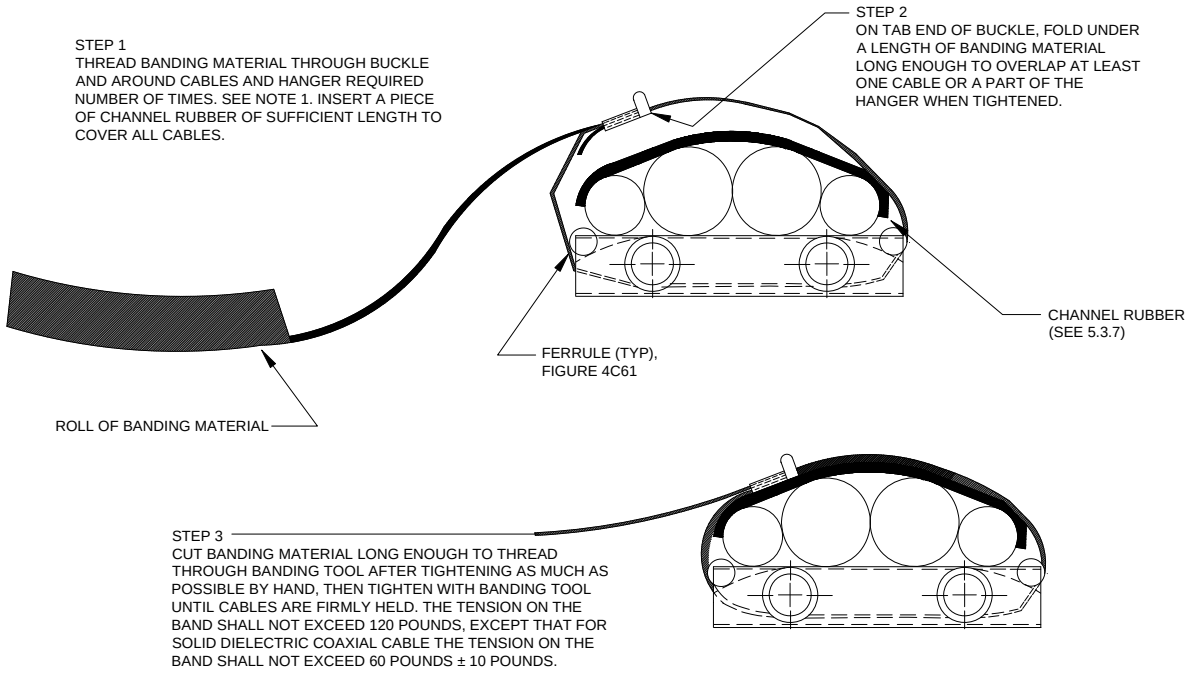
1. Banding buckles shall be made of corrosion-resistant steel, ASTM A240/A240M, ASTM A666, and ASTM A693.
2. The edges of the banding buckle in contact with the banding material shall be rounded slightly.
3. Specifications, vendor information, and part numbers for selected banding buckles are as follows:

Buckle length (inch)	Vendor	Description/part number
½	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	"Inverted Wing Seal" 202SS (stainless steel)
¾	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	"Wing Seal" 204SS (stainless steel)
¾	Independent Metal Strap Company, Inc. 34 Lumber Road, Roslyn, NY 11576-0334 or equal	"Royal Wing Seal" (stainless steel)
⅝ x ½	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	213SS (stainless steel); NSN 9Q8135-01-030-1632
⅝ x ¾	Gerrard Packaging 2451 South Wolf Road, Des Plaines, IL 60018-2608 or equal	203SS (stainless steel); NSN 9Q8135-01-120-8754

FIGURE 4C51. Cable banding buckle (inverted) and miscellaneous details.

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APPLICATION OF BANDING MATERIAL
(SEE NOTES 1 THROUGH 13)



NOTES:

1. For the application of banding material, follow steps 1 through 8 on figures 4C52 and 4C53. Band requirements are as follows:

Single cables 1" diameter and under	0.015 thick, one turn
Single cables over 1" diameter	0.015 thick, two turns
Multiple cables 2" diameter and under	0.015 thick, two turns
Multiple cables over 2" diameter	0.015 thick, three turns or 0.020 thick, two turns

2. Information for cable banding strapping material is as follows:

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

FIGURE 4C52. Applying banding material.

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NOTES (continued):

3. Information for channel rubber size, material, and specification is as follows:

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	⅞	⅝	5/16	MIL-PRF-6855, Class 2, Group 60

4. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. "56" series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for "56" series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

WARNING: Marine snaplock cable clamps are prohibited for use in reactor compartments. Justification: Portsmouth Naval Shipyard Liaison Action Request (LAR) #PNS725-51 of 9/29/98 issued to request determination on use of "Marine Snaplock Cable Clamps" in reactor compartments. In response, Nuclear Liaison Inquiry #NLI-2535 issued by EB and approved by NAVSEA 08 prohibited their use in reactor compartments.

5. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
6. Cable straps shall be applied in conjunction with channel rubber.
7. The maximum bundle diameter shall not exceed 3".
8. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
9. The minimum radius of bend shall be as specified in MIL-HDBK-299.
10. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
11. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C52. Applying banding material - Continued.

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NOTES (continued):

12. Part numbers, sizes, and stock numbers for Panduit® banding are as follows:

Single wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4SH-LP	17.1	⅝	0.015	50
MLT8SH-LP	29.7	⅝	0.015	50
MLT12SH-Q	42.2	⅝	0.015	50
Single wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4EH15-LP316	17.1	½	0.015	50
MLT6EH15-LP316	23.4	½	0.015	50
MLT8EH15-LP316	29.7	½	0.015	50
Double wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DSH-Q	29.5	⅝	0.015	25
MLT6DSH-Q	41.5	⅝	0.015	25
MLT8DSH-Q	53.5	⅝	0.015	25
Double wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DEH15-Q316	29.5	½	0.015	25
MLT6DEH15-Q316	41.5	½	0.015	25
MLT8DEH15-Q316	53.5	½	0.015	25

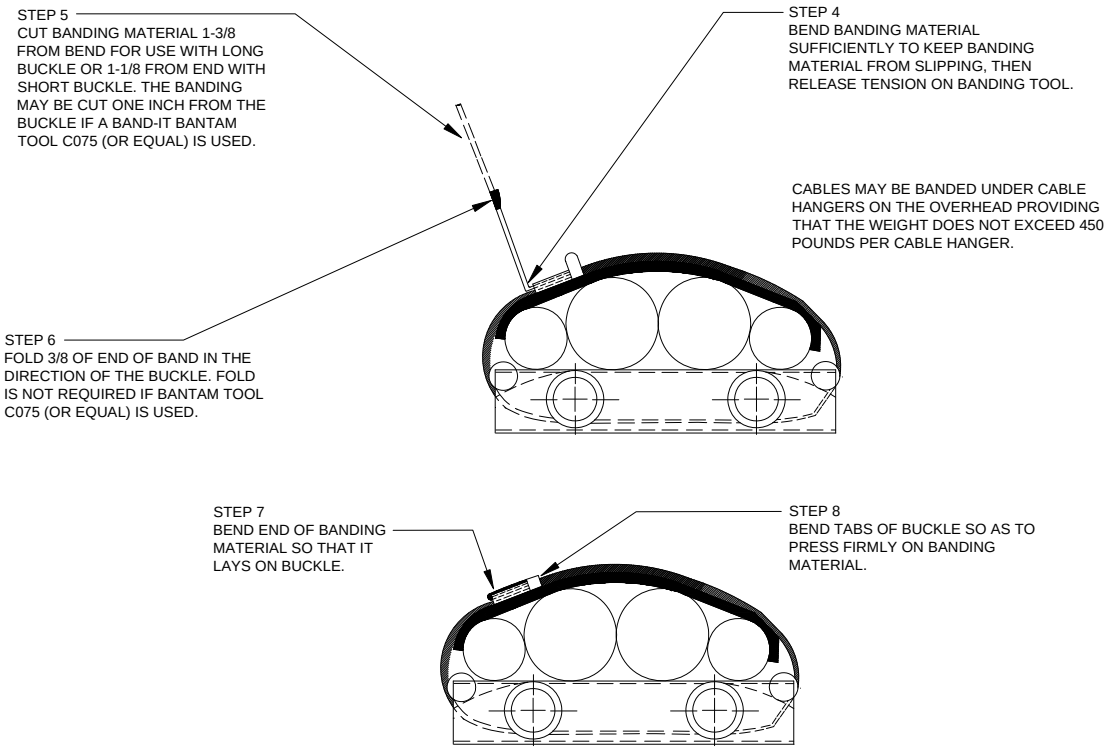
Panduit part number	**National stock number (NSN)**	**Panduit part number**	**National stock number (NSN)**
MLT4EH15-LP316	5975-01-526-3677	MLT12SH-LP316	N/A
MLT6EH15-LP316	5975-01-526-3679	MLT4DSH-Q316	5975-01-526-4437
MLT8EH15-LP316	5975-01-526-3678	MLT6DSH-Q316	5975-01-526-3685
MLT10EH15-LP316	N/A	MLT8DSH-Q316	5975-01-526-3684
MLT12EH15-LP316	N/A	MS4W50T15-L6	N/A
MLT4DEH15-Q316	N/A	MS6W50T15-L6	N/A
MLT6DEH15-Q316	N/A	MS8W50T15-L6	N/A
MLT8DEH15-Q316	N/A	MS10W50T15-L6	N/A
MLT4SH-LP316	5975-01-526-3680	MS4W63T15-L6	N/A
MLT6SH-LP316	N/A	MS6W63T15-L6	N/A
MLT8SH-LP316	5975-01-526-3683	MS8W63T15-L6	N/A
MLT10SH-LP316	N/A		

13. When installing and cutting banding material, follow the instructions of the manufacturer of the banding tool.

FIGURE 4C52. Applying banding material - Continued.

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APPLICATION OF BANDING MATERIAL (CONTINUED)
(SEE NOTES 1 THROUGH 13)



NOTES:

1. For the application of banding material, follow steps 1 through 8 on figures 4C52 and 4C53. Band requirements are as follows:

Single cables 1" diameter and under	0.015 thick, one turn
Single cables over 1" diameter	0.015 thick, two turns
Multiple cables 2" diameter and under	0.015 thick, two turns
Multiple cables over 2" diameter	0.015 thick, three turns or 0.020 thick, two turns

2. Information for cable banding strapping material is as follows:

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

FIGURE 4C53. Applying banding material.

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NOTES (continued):

3. Information for channel rubber size, material, and specification is as follows.

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	7/8	5/8	5/16	MIL-PRF-6855, Class 2, Group 60

4. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. “56” series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for “56” series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

5. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
6. Cable straps shall be applied in conjunction with channel rubber.
7. The maximum bundle diameter shall not exceed 3".
8. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
9. The minimum radius of bend shall be as specified in MIL-HDBK-299.
10. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
11. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C53. Applying banding material - Continued.

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12. Part numbers, sizes, and stock numbers for Panduit® banding are as follows:

Single wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4SH-LP	17.1	⅝	0.015	50
MLT8SH-LP	29.7	⅝	0.015	50
MLT12SH-Q	42.2	⅝	0.015	50
Single wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4EH15-LP316	17.1	½	0.015	50
MLT6EH15-LP316	23.4	½	0.015	50
MLT8EH15-LP316	29.7	½	0.015	50
Double wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DSH-Q	29.5	⅝	0.015	25
MLT6DSH-Q	41.5	⅝	0.015	25
MLT8DSH-Q	53.5	⅝	0.015	25
Double wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DEH15-Q316	29.5	½	0.015	25
MLT6DEH15-Q316	41.5	½	0.015	25
MLT8DEH15-Q316	53.5	½	0.015	25

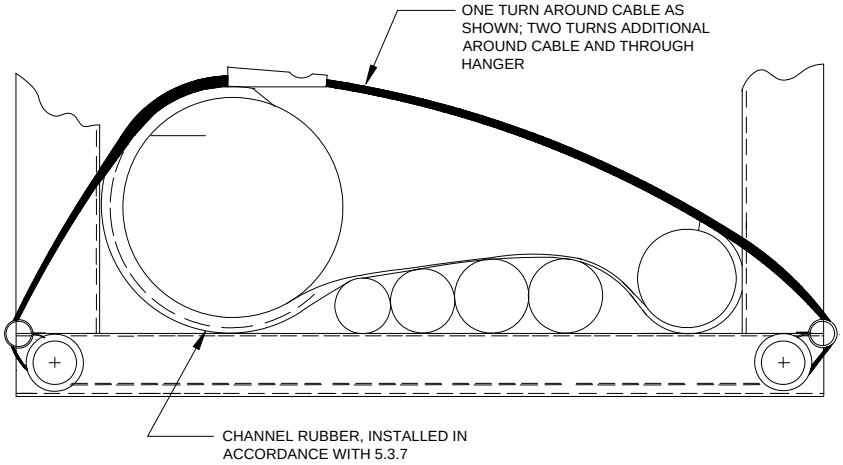
Panduit part number	**National stock number (NSN)**	**Panduit Part Number**	**National stock number (NSN)**
MLT4EH15-LP316	5975-01-526-3677	MLT12SH-LP316	N/A
MLT6EH15-LP316	5975-01-526-3679	MLT4DSH-Q316	5975-01-526-4437
MLT8EH15-LP316	5975-01-526-3678	MLT6DSH-Q316	5975-01-526-3685
MLT10EH15-LP316	N/A	MLT8DSH-Q316	5975-01-526-3684
MLT12EH15-LP316	N/A	MS4W50T15-L6	N/A
MLT4DEH15-Q316	N/A	MS6W50T15-L6	N/A
MLT6DEH15-Q316	N/A	MS8W50T15-L6	N/A
MLT8DEH15-Q316	N/A	MS10W50T15-L6	N/A
MLT4SH-LP316	5975-01-526-3680	MS4W63T15-L6	N/A
MLT6SH-LP316	N/A	MS6W63T15-L6	N/A
MLT8SH-LP316	5975-01-526-3683	MS8W63T15-L6	N/A
MLT10SH-LP316	N/A		

13. When installing and cutting banding material, follow the instructions of the manufacturer of the banding tool.

FIGURE 4C53. Applying banding material - Continued.

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METHOD 4C-54-1
HANGERS WITH LARGE & SMALL
CABLES INTERMIXED
(SEE NOTES 1 THROUGH 11)



NOTES:

1. For fully loaded hangers (where hanger width does not exceed the width of a single cable or a single row of cables by more than ½"), use Method 4C-55-2.
2. For partially loaded hangers (where hanger width exceeds the width of a single cable or a single rows of cables by more than ½"), use Method 4C-56-1.
3. The length of banding hanger shall not be less than the overall width of cables for a single cable or a row of cables.
4. Band requirements are as follows:

Single cables, 1" diameter and under	0.015 thick, one turn
Single cables over 1" diameter	0.015 thick, two turns
Multiple cables 2" diameter and under	0.015 thick, two turns
Multiple cables over 2" diameter	0.015 thick, three turns or 0.020 thick, two turns

5. Cables shall be arranged so that the larger cables are in the center of the hanger, if possible.
6. In non-magnetic minesweepers, use CRES band except that an aluminum band (0.025 thick, see table below for strapping material) may be used where cables are under 2" in diameter and hanger spacing is 21" on center, maximum.

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

FIGURE 4C54. Details of partially and fully loaded banding hangers.

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NOTES (continued):

7. For weather decks or highly corrosive areas, use CRES band, size 8.
8. The banding method shall be used to support coaxial cables except coaxial cables of the semi-solid dielectric type. Semi-solid dielectric coaxial cables shall be supported by the contour strap methods such as those shown on figures 4C23, 4C24, and 4C25.
9. Channel rubber shall be installed in accordance with 5.3.7. Information for channel rubber size, material, and specification is as follows:

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	7/8	5/8	5/16	MIL-PRF-6855, Class 2, Group 60

10. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. “56” series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for “56” series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

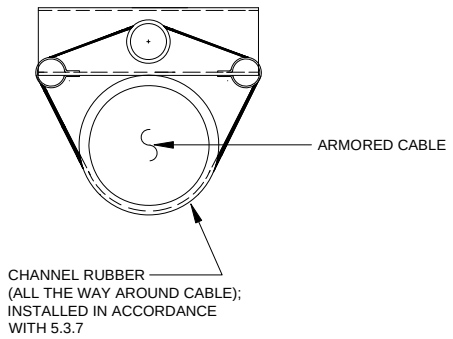
11. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
    a. Cable straps shall be applied in conjunction with channel rubber.
    b. The maximum bundle diameter shall not exceed 3".
    c. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
    d. The minimum radius of bend shall be as specified in MIL-HDBK-299.
    e. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
    f. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C54. Details of partially and fully loaded banding hangers - Continued.

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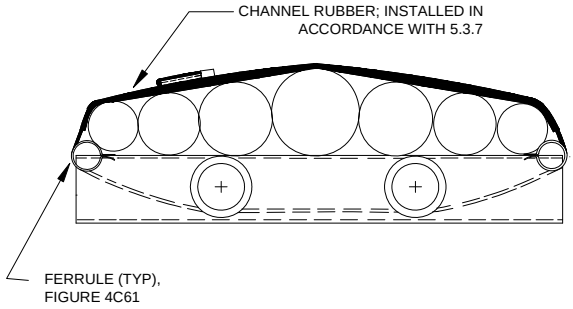
METHOD 4C-55-1

USE OF CHANNEL RUBBER FOR SOUND ISOLATION



METHOD 4C-55-2

FULLY LOADED HANGERS



NOTES:

1. For the application of banding material, follow steps 1 through 8 on figures 4C52 and 4C53. Band requirements are as follows:

Single cables, 1" diameter and under	0.015 thick, one turn
Single cables over 1" diameter	0.015 thick, two turns
Multiple cables 2" diameter and under	0.015 thick, two turns
Multiple cables over 2" diameter	0.015 thick, three turns or 0.020 thick, two turns

2. Information for cable banding strapping material is as follows:

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

3. Information for channel rubber size, material, and specification is as follows:

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	⅞	⅝	5⁄16	MIL-PRF-6855, Class 2, Group 60

FIGURE 4C55. Details of partially and fully loaded banding hangers.

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NOTES (continued):

4. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. “56” series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for “56” series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

5. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
   a. Cable straps shall be applied in conjunction with channel rubber.
   b. The maximum bundle diameter shall not exceed 3".
   c. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
   d. The minimum radius of bend shall be as specified in MIL-HDBK-299.
   e. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
   f. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C55. Details of partially and fully loaded banding hangers - Continued.

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Check the source to verify that this is the current version before use.

NOTES (continued):

6. Part numbers, sizes, and stock numbers for Panduit® banding are as follows:

Single wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4SH-LP	17.1	⅝	0.015	50
MLT8SH-LP	29.7	⅝	0.015	50
MLT12SH-Q	42.2	⅝	0.015	50
Single wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4EH15-LP316	17.1	½	0.015	50
MLT6EH15-LP316	23.4	½	0.015	50
MLT8EH15-LP316	29.7	½	0.015	50
Double wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DSH-Q	29.5	⅝	0.015	25
MLT6DSH-Q	41.5	⅝	0.015	25
MLT8DSH-Q	53.5	⅝	0.015	25
Double wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DEH15-Q316	29.5	½	0.015	25
MLT6DEH15-Q316	41.5	½	0.015	25
MLT8DEH15-Q316	53.5	½	0.015	25

Panduit part number	**National stock number (NSN)**	**Panduit Part Number**	**National stock number (NSN)**
MLT4EH15-LP316	5975-01-526-3677	MLT12SH-LP316	N/A
MLT6EH15-LP316	5975-01-526-3679	MLT4DSH-Q316	5975-01-526-4437
MLT8EH15-LP316	5975-01-526-3678	MLT6DSH-Q316	5975-01-526-3685
MLT10EH15-LP316	N/A	MLT8DSH-Q316	5975-01-526-3684
MLT12EH15-LP316	N/A	MS4W50T15-L6	N/A
MLT4DEH15-Q316	N/A	MS6W50T15-L6	N/A
MLT6DEH15-Q316	N/A	MS8W50T15-L6	N/A
MLT8DEH15-Q316	N/A	MS10W50T15-L6	N/A
MLT4SH-LP316	5975-01-526-3680	MS4W63T15-L6	N/A
MLT6SH-LP316	N/A	MS6W63T15-L6	N/A
MLT8SH-LP316	5975-01-526-3683	MS8W63T15-L6	N/A
MLT10SH-LP316	N/A		

FIGURE 4C55. Details of partially and fully loaded banding hangers - Continued.

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NOTES (continued):

7. When installing and cutting banding material, follow the instructions of the manufacturer of the banding tool.
8. For fully loaded hangers (where hanger width does not exceed the width of a single cable or a single row of cables by more than ½"), use Method 4C-55-2.
9. For partially loaded hangers (where hanger width exceeds the width of a single cable or a single rows of cables by more than ½"), use Method 4C-56-1.
10. The length of banding hanger shall not be less than the overall width of cables for a single cable or a row of cables.
11. Cables shall be arranged so that the larger cables are in the center of the hanger, if possible.
12. In non-magnetic minesweepers, use CRES band except that an aluminum band (0.025 thick, see table below for strapping material) may be used where cables are under 2" in diameter and hanger spacing is 21" on center, maximum.

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

13. For weather decks or highly corrosive areas, use CRES band, size 8.
14. The banding method shall be used to support coaxial cables except coaxial cables of the semi-solid dielectric type. Semi-solid dielectric coaxial cables shall be supported by the contour strap methods such as those shown on figures 4C23, 4C24, and 4C25.
15. Channel rubber shall be installed in accordance with 5.3.7. Information for channel rubber size, material, and specification is as follows:

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	⅞	⅝	5/16	MIL-PRF-6855, Class 2, Group 60

FIGURE 4C55. Details of partially and fully loaded banding hangers - Continued.

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NOTES (continued):

16. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. “56” series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for “56” series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

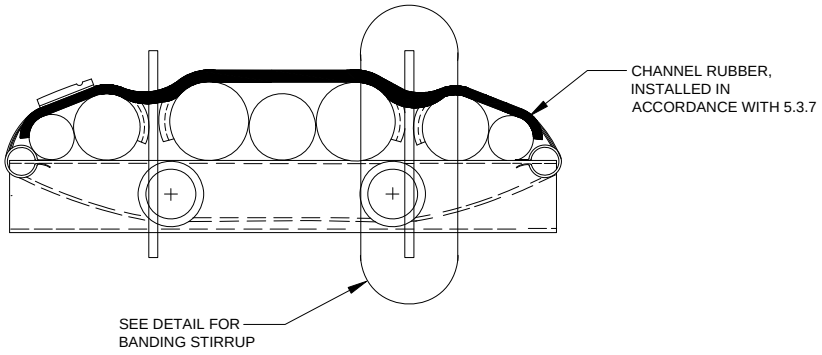
17. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
    a. Cable straps shall be applied in conjunction with channel rubber.
    b. The maximum bundle diameter shall not exceed 3".
    c. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
    d. The minimum radius of bend shall be as specified in MIL-HDBK-299.
    e. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
    f. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C55. Details of partially and fully loaded banding hangers - Continued.

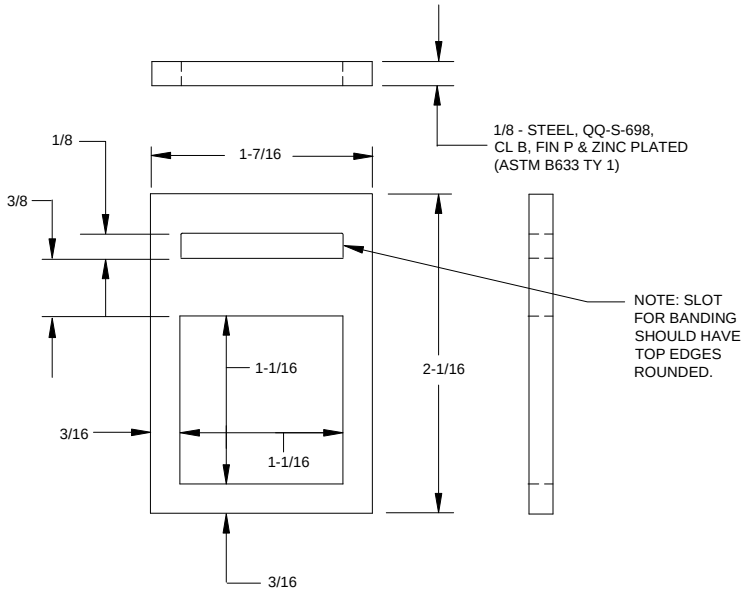
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METHOD 4C-56-1

PARTIALLY LOADED HANGERS



BANDING STIRRUP



NOTES:

1. For the application of banding material, follow steps 1 through 8 on figures 4C52 and 4C53. Band requirements are as follows:

Single cables, 1" diameter and under	0.015 thick, one turn
Single cables over 1" diameter	0.015 thick, two turns
Multiple cables 2" diameter and under	0.015 thick, two turns
Multiple cables over 2" diameter	0.015 thick, three turns or 0.020 thick, two turns

FIGURE 4C56. Details of partially and fully loaded banding hangers.

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NOTES (continued):

2. Information for cable banding strapping material is as follows:

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

3. Information for channel rubber size, material, and specification is as follows:

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	⅞	⅝	5⁄16	MIL-PRF-6855, Class 2, Group 60

4. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. “56” series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for “56” series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

5. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
   a. Cable straps shall be applied in conjunction with channel rubber.
   b. The maximum bundle diameter shall not exceed 3".
   c. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
   d. The minimum radius of bend shall be as specified in MIL-HDBK-299.
   e. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
   f. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C56. Details of partially and fully loaded banding hangers - Continued.

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NOTES (continued):

6. Part numbers, sizes, and stock numbers for Panduit® banding are as follows:

Single wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4SH-LP	17.1	⅝	0.015	50
MLT8SH-LP	29.7	⅝	0.015	50
MLT12SH-Q	42.2	⅝	0.015	50
Single wrap, 316 stainless (topside)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4EH15-LP316	17.1	½	0.015	50
MLT6EH15-LP316	23.4	½	0.015	50
MLT8EH15-LP316	29.7	½	0.015	50
Double wrap, 304 stainless (inside hull)				
Part number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DSH-Q	29.5	⅝	0.015	25
MLT6DSH-Q	41.5	⅝	0.015	25
MLT8DSH-Q	53.5	⅝	0.015	25
Double wrap, 316 stainless (topside)				
Part Number	**Length**	**Width**	**Thickness**	**Quantity**
MLT4DEH15-Q316	29.5	½	0.015	25
MLT6DEH15-Q316	41.5	½	0.015	25
MLT8DEH15-Q316	53.5	½	0.015	25

Panduit part number	**National stock number (NSN)**	**Panduit part number**	**National stock number (NSN)**
MLT4EH15-LP316	5975-01-526-3677	MLT12SH-LP316	N/A
MLT6EH15-LP316	5975-01-526-3679	MLT4DSH-Q316	5975-01-526-4437
MLT8EH15-LP316	5975-01-526-3678	MLT6DSH-Q316	5975-01-526-3685
MLT10EH15-LP316	N/A	MLT8DSH-Q316	5975-01-526-3684
MLT12EH15-LP316	N/A	MS4W50T15-L6	N/A
MLT4DEH15-Q316	N/A	MS6W50T15-L6	N/A
MLT6DEH15-Q316	N/A	MS8W50T15-L6	N/A
MLT8DEH15-Q316	N/A	MS10W50T15-L6	N/A
MLT4SH-LP316	5975-01-526-3680	MS4W63T15-L6	N/A
MLT6SH-LP316	N/A	MS6W63T15-L6	N/A
MLT8SH-LP316	5975-01-526-3683	MS8W63T15-L6	N/A
MLT10SH-LP316	N/A		

FIGURE 4C56. Details of partially and fully loaded banding hangers - Continued.

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NOTES (continued):

7. When installing and cutting banding material, follow the instructions of the manufacturer of the banding tool.
8. For fully loaded hangers (where hanger width does not exceed the width of a single cable or a single row of cables by more than ½"), use Method 4C-55-2.
9. For partially loaded hangers (where hanger width exceeds the width of a single cable or a single rows of cables by more than ½"), use Method 4C-56-1.
10. The length of banding hanger shall not be less than the overall width of cables for a single cable or a row of cables.
11. Cables shall be arranged so that the larger cables are in the center of the hanger, if possible.
12. In non-magnetic minesweepers, use CRES band except that an aluminum band (0.025 thick, see table below for strapping material) may be used where cables are under 2" in diameter and hanger spacing is 21" on center, maximum.

Material	Dimensions (inches)		Specifications	Finish
	Width	Thickness		
Mild steel	⅝	0.015	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.015	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D
Aluminum	⅝	0.025	ASTM B209	None
Mild steel	⅝	0.020	ASTM D3953, Type 1, Class B, Grade 1	Zinc Electro-plated
CRES	⅝	0.020	ASTM A240/A240M, ASTM A666, ASTM A693, ASTM A276	2D

13. For weather decks or highly corrosive areas, use CRES band, size 8.
14. The banding method shall be used to support coaxial cables except coaxial cables of the semi-solid dielectric type. Semi-solid dielectric coaxial cables shall be supported by the contour strap methods such as those shown on figures 4C23, 4C24, and 4C25.
15. Channel rubber shall be installed in accordance with 5.3.7. Information for channel rubber size, material, and specification is as follows:

Material	Size (inches)			Specification
	Width	Channel	Height	
Rubber	⅞	⅝	5/16	MIL-PRF-6855, Class 2, Group 60

FIGURE 4C56. Details of partially and fully loaded banding hangers - Continued.

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NOTES (continued):

16. Marine snaplock type cable clamps (hose-clamps), Ideal Corporation or equal, with a maximum diameter of 22" may be used in lieu of buckle type cable clamps. “56” series cable retention devices of a length not greater than 22" are approved as an alternate to conventional banding materials. Dimensions for “56” series snaplock cable clamps are as follows:

Part no.	Clamp length	Clamp diameter (max.)	Stock number
5612	4	1.250	9B 4730-01-298-5296
5620	5½	1.750	9B 4730-00-864-4299
5628	7	2.250	9B 4730-00-020-4270
5636	8½	2.750	9B 4730-00-720-1274
5648	11	3.500	9B 4730-00-988-8390
5656	12½	4.000	9B 4730-00-132-9948
5672	16	5.120	9B 4730-00-880-3539
5688	18½	6.000	9B 4730-00-617-6921
56104	22	7.120	9B 4730-01-037-1374

17. Panduit® Pan-Steel™ strapping is acceptable for use on surface ships and submarines subject to the following conditions:
    a. Cable straps shall be applied in conjunction with channel rubber.
    b. The maximum bundle diameter shall not exceed 3".
    c. The maximum static weight of the cable bundle shall not exceed six pounds per metal locking tie (MLT).
    d. The minimum radius of bend shall be as specified in MIL-HDBK-299.
    e. In horizontal installations, the cable straps shall be no greater than 16" on-center apart. In vertical cable installations, the cable straps shall be no greater than 12" apart.
    f. For the installation of Panduit® banding, use the appropriate tool as specified by the manufacturer.

FIGURE 4C56. Details of partially and fully loaded banding hangers - Continued.

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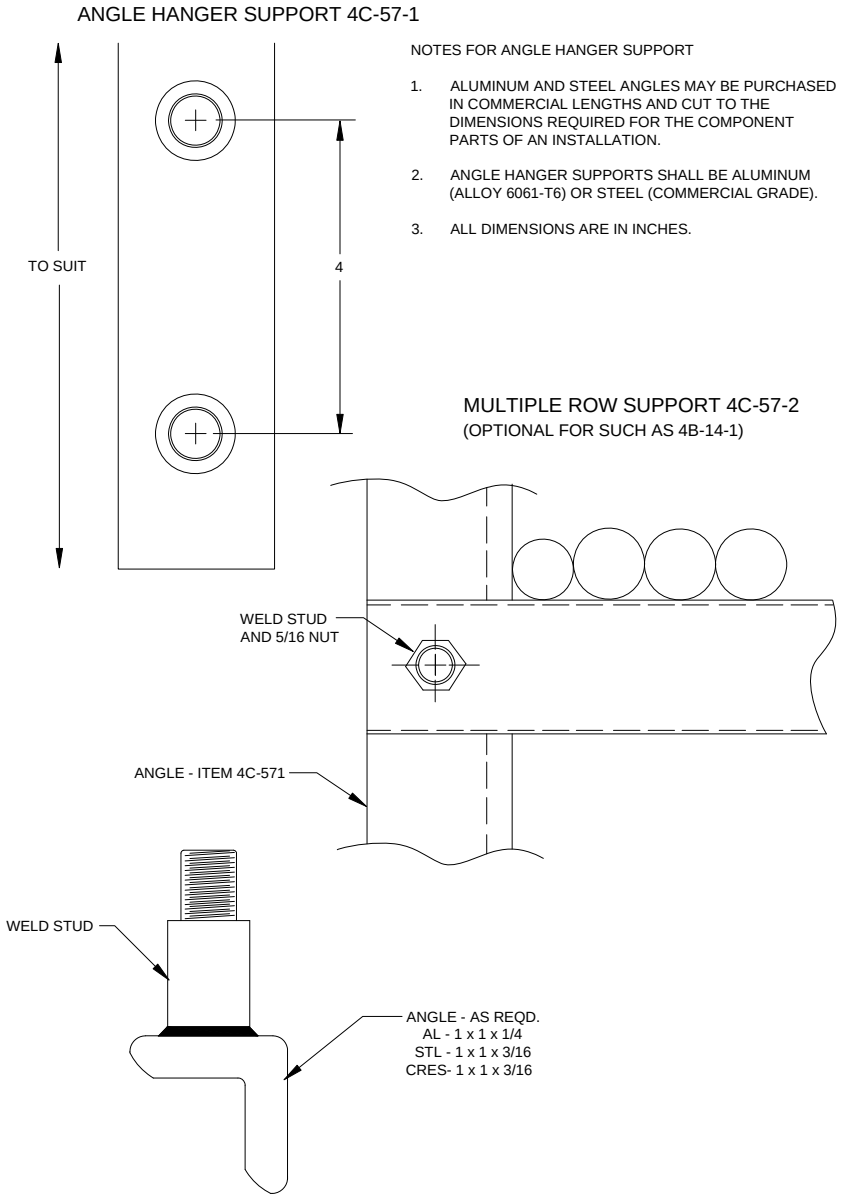
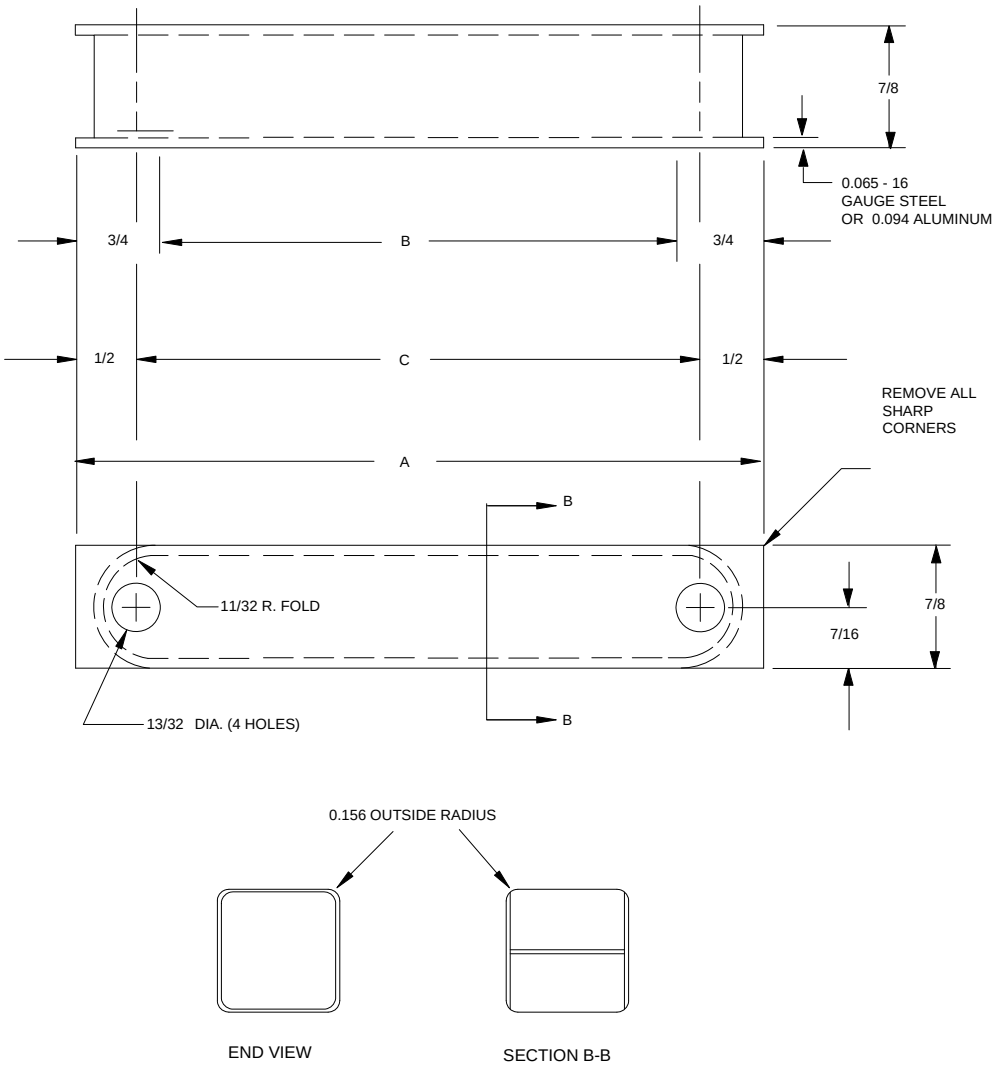


FIGURE 4C57. Angle hanger support details.

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SIDE MOUNTED TUBULAR HANGER
4C-58-1 - ALUMINUM
4C-58-2 - STEEL
4C-58-3 - CRES

(CUT TUBULAR HANGERS TO LENGTHS REQUIRED IN TABULATIONS)
TUBULAR HANGER DETAILED BELOW MAY BE USED AS ALTERNATE TO
ITEMS 4C-59-1, 4C-59-2 AND 4C-59-3
FERRULES ARE NOT REQUIRED FOR ITEMS 4C-58-1, 4C-58-2 AND 4C-58-3
(SEE NOTES 1 THROUGH 4)



NOTES:

1. Square tubing may be purchased in commercial lengths. Cut tubular hangers to the lengths required in tabulations.

FIGURE 4C58. Details of side mounted tubular hangers.

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NOTES (continued):

2. Dimensions for tubular hangers are as follows:

Item no.	Dimensions (inches)		
	A	B	C
4C-611 4C-612 4C-613	3	1½	2
	4	2½	3
	5	3½	4
	6	4½	5
	7	5½	6
	8	6½	7
	9	7½	8
	10	8½	9
	11	9½	10
	12	10½	11
	13	11½	12
	14	12½	13
	15	13½	14
	16	14½	15
	17	15½	16

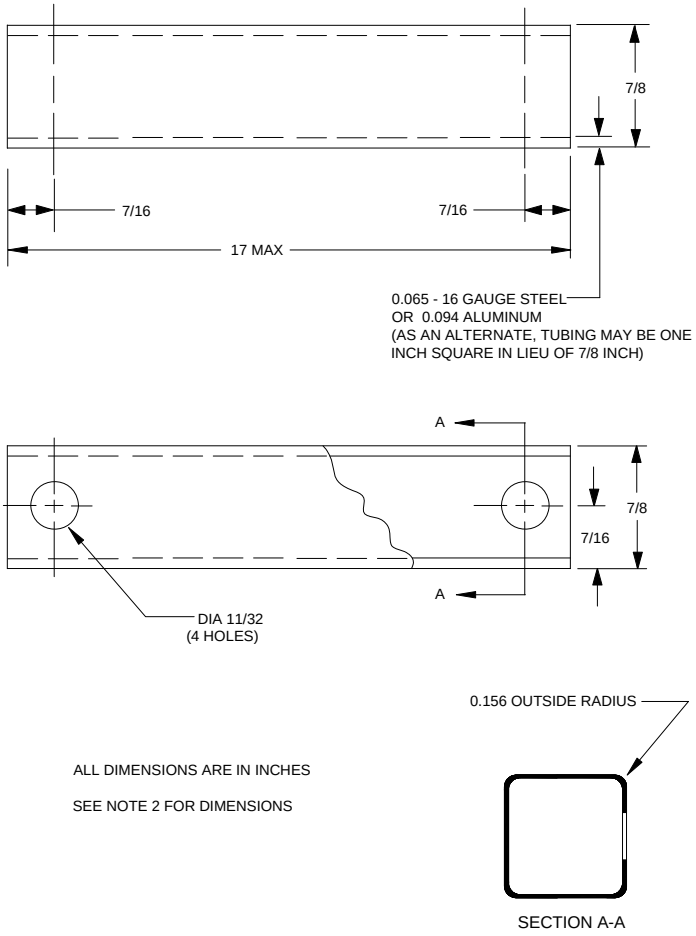
Dimensions (inches)	
Hanger length A	Hole spacing B
3	2
4	2
5	2
6	2
7	6½
8	6½
9	5½
10 to 17 maximum	Dim A minus 7/16

3. Aluminum tubing shall be seamless drawn and in accordance with ASTM B241/B241M, Temper 6.
4. Steel tubing shall be hot rolled, weld seamflash in commercial grade.

FIGURE 4C58. Details of side mounted tubular hangers - Continued.

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SIDE MOUNTED TUBULAR HANGER
4C-59-1 - ALUMINUM
4C-59-2 - STEEL
4C-59-3 - CRES
(4C-59-1, 4C-59-2 AND 4C-59-3 TO BE USED AS
ALTERNATE TO 4C-58-1, 4C-58-2 AND 4C-58-3)
(SEE NOTES 1 THROUGH 4)



NOTES:

1. Square tubing may be purchased in commercial lengths. Cut tubular hangers to the lengths required in tabulations.

FIGURE 4C59. Details of side mounted tubular hanger – alternate design.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

NOTES (continued):

2. Dimensions for tubular hangers are as follows:

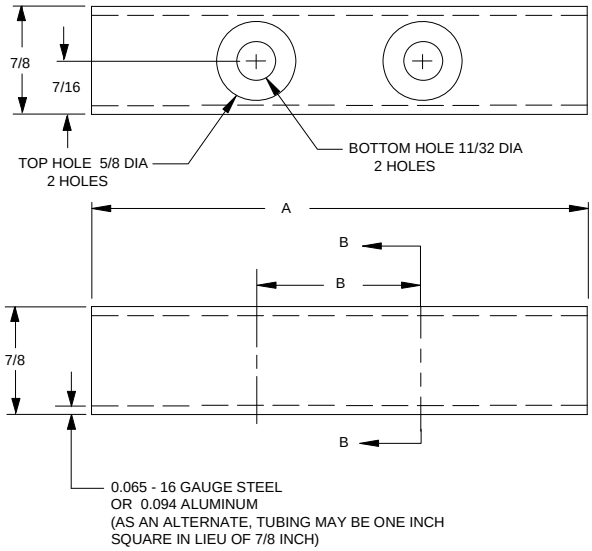
Dimensions (inches)	
Hanger length A	**Hole spacing B**
3	2
4	2
5	2
6	2
7	6½
8	6½
9	5½
10 to 17 maximum	Dim A minus 7/16

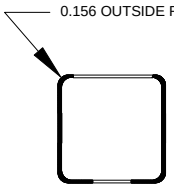
3. Aluminum tubing shall be seamless drawn and in accordance with ASTM B241/B241M, Temper 6.
4. Steel tubing shall be hot rolled, weld seamflash in commercial grade.

FIGURE 4C59. Details of side mounted tubular hanger – alternate design - Continued.

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BOTTOM MOUNTED TUBULAR HANGER
4C-60-1 - ALUMINUM
4C-60-2 - STEEL
4C-60-3 - CRES
(SEE NOTES 1 THROUGH 4)





SECTION B-B

ALL DIMENSIONS ARE IN INCHES

SEE NOTE 2 FOR DIMENSIONS

NOTES:

1. Square tubing may be purchased in commercial lengths. Cut tubular hangers to the lengths required in tabulations.
2. Dimensions for tubular hangers are as follows:

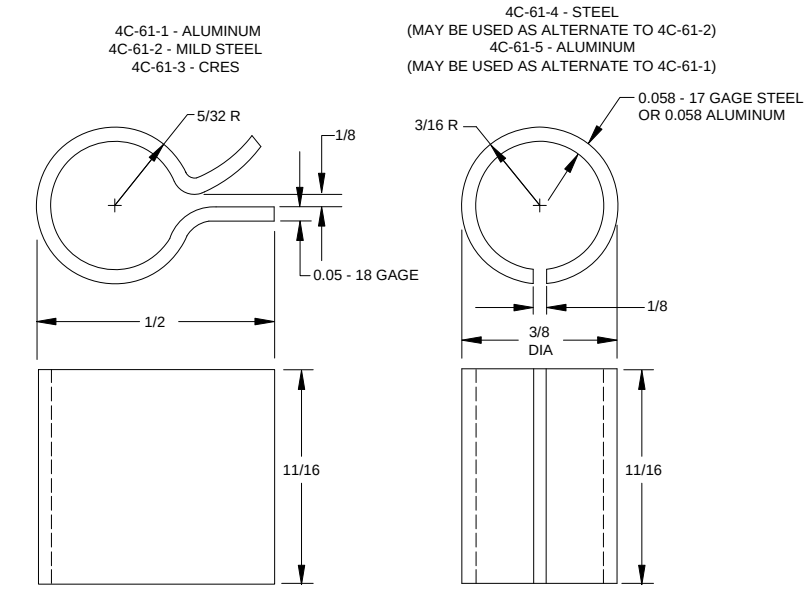
Dimensions (inches)	
Hanger length A	**Hole spacing B**
3	2
4	2
5	2
6	2
7	6½
8	6½
9	5½
10 to 17 maximum	Dim A minus 7/16

3. Aluminum tubing shall be seamless drawn and in accordance with ASTM B241/B241M, Temper 6.
4. Steel tubing shall be hot rolled, weld seamflash in commercial grade.

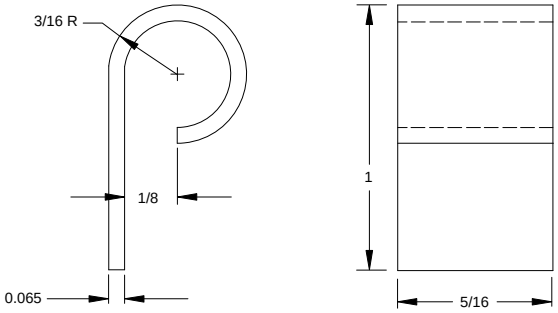
FIGURE 4C60. Details of bottom mounted tubular hanger.

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FERRULE DETAILS
(SEE NOTES 1 AND 2 BELOW)



4C-61-6 - STEEL
(MAY BE USED AS ALTERNATE TO 4C-61-2)



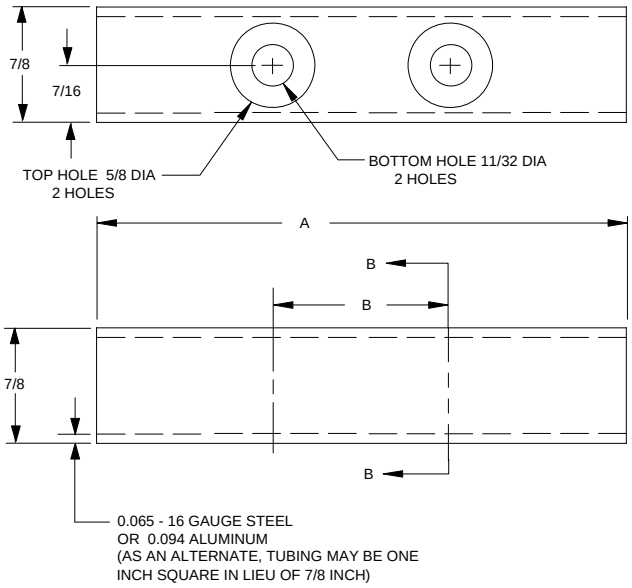
NOTES:

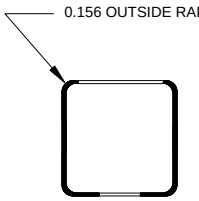
1. Ferrules shall be formed or extruded steel, aluminum, or nylon, commercial grade.
2. All dimensions are in inches.

FIGURE 4C61. Tubular banding hangers ferrule details.

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BOTTOM MOUNTED TUBULAR HANGER
4C-62-1 - ALUMINUM
4C-622- - STEEL
4C-62-3 - CRES
(SEE NOTES 1 THROUGH 4)





SECTION B-B

ALL DIMENSIONS ARE IN INCHES

SEE NOTE 2 FOR DIMENSIONS

NOTES:

1. Square tubing may be purchased in commercial lengths. Cut tubular hangers to the lengths required in tabulations.
2. Dimensions for tubular hangers are as follows:

Dimensions (inches)	
Hanger length A	**Hole spacing B**
3	2
4	2
5	2
6	2
7	6½
8	6½
9	5½
10 to 17 maximum	Dim A minus 7/16

FIGURE 4C62. Details of bottom mounted tubular hanger.

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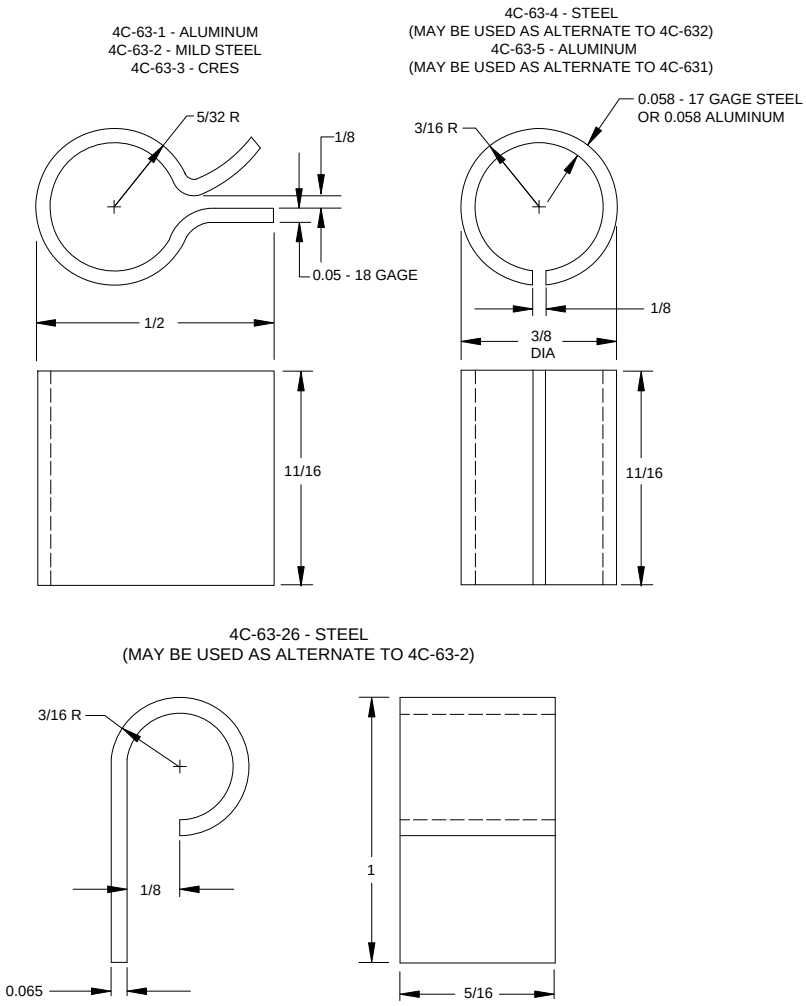
NOTES (continued):

3. Aluminum tubing shall be seamless drawn and in accordance with ASTM B241/B241M, Temper 6.
4. Steel tubing shall be hot rolled, weld seamflash in commercial grade.

FIGURE 4C62. Details of bottom mounted tubular hanger - Continued.

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FERRULE DETAILS
(SEE NOTES 1 AND 2 BELOW)



NOTES:

1. Ferrules shall be formed or extruded steel, aluminum, or nylon, commercial grade.
2. All dimensions are in inches.

FIGURE 4C63. Tubular banding hangers ferrule details.

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GROUP 4D – CABLE PROTECTION

D.1 SCOPE

D.1.1 Scope. This appendix describes cable protection procedures related to the installation of cableways on submarines and surface ships.

D.2. APPLICABLE DOCUMENTS

D.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

D.2.2 Government documents.

D.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-3041	-	Wrench, Open End Ratchet (TAC Pattern) for Tube Fitting, Electrical Cable Terminals, and Stuffing Tube Gland Nuts

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-Y-1140	-	Yarn, Cord, Sleeving, Cloth and Tape-Glass
MIL-R-21607	-	Resins, Polyester, Low Pressure Laminating, Fire-Retardant
MIL-P-21929	-	Plastic Material, Cellular Polyurethane, Foam-in-Place, Rigid (2 Pounds Per Cubic Foot)
MIL-S-24235	-	Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2003-3	-	Electric Plant Installation Standard Methods for Surface Ship and Submarines (Penetrations)
MIL-STD-2042-4	-	Fiber Optic Cable Topology Installation Standard Methods for Naval Ships (Cableways)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

D.2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

D.3 REQUIRED EQUIPMENT AND MATERIALS

D.3.1 Required equipment and materials. The equipment and materials in the tables and referenced in the text of this section shall be used for the protection of cables. For fiber optic cable, refer to MIL-STD-2042-4 for additional information and guidance.

D.4 NOTES AND PROCEDURES

D.4.1 Dimensions. For tables and figures in this section, all dimensions are in inches unless otherwise noted. All welding shall be in accordance with 5.7.3.7 and 5.7.3.2.1.

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D.4.2 Figures. Table 4DI provides information for the figures in this group. The details and methods shown on figures 4D1 through 4D29 specify protection requirements and procedures for cables installed on surface ships and submarines as specified on the figures.

TABLE 4DI. Figures for cable protection surface ships and submarines.

Figure number	Requirements/procedures	Page
4D1	Protection of cables penetrating watertight decks	300
4D2	Protection of cables penetrating watertight decks – wet spaces	302
4D3	Protection of cables penetrating watertight decks – dry areas	303
4D4	Cableway adjacent to bulkhead penetration (submarines)	304
4D5	Protection of cables penetrating watertight decks (alternate method) – section A-A	306
4D6	Protection of cables penetrating non-watertight (NWT) decks	308
4D7	Protection of cables from mechanical injury	309
4D8	Protection of degaussing cables from mechanical injury – horizontal run	310
4D9	Protection of degaussing cables from mechanical injury – vertical run	311
4D10	Protection of degaussing cables from mechanical injury	312
4D11	Protection of degaussing cables from mechanical injury	313
4D12	Protection of cables on mast (surface ships) – forward view	314
4D13	Protection of cables on mast (surface ships) – port view	316
4D14	Protection of cables on mast (surface ships) – section “A-A”	317
4D15	Protection of cables on mast (surface ships) – section “B-B”	318
4D16	Protection of cables on mast (surface ships) – section “C-C”	319
4D17	Protection of cables on mast (surface ships) – section “D-D”	320
4D18	Protection of cables on mast (surface ships) – section “E-E”	321
4D19	Protection of cables on mast (surface ships) – section “F-F”	322
4D20	Protection of cables on mast (surface ships) – section “G-G”	323
4D21	Protection of cables on mast (surface ships) – section “J-J”	324
4D22	Protection of cables on mast (surface ships) – section “K-K”	325
4D23	Protection of cables on mast (surface ships) – detail “CS”	326
4D24	Protection of cables on mast (surface ships) – detail “CT”	327
4D25	Protection of cables on mast (surface ships) – detail “FT”	328
4D26	Arrangement of duct and double tier of cables on mast (surface ships)	329
4D27	Protection of cables in hangar spaces (surface ships)	330
4D28	Protection of cables in hangar spaces (surface ships) – elevation	331
4D29	Protection of cables in hangar spaces (surface ships) – side view	332
4D24	Protection of cables on mast (surface ships) – detail “CT”	327
4D25	Protection of cables on mast (surface ships) – detail “FT”	328
4D26	Arrangement of duct and double tier of cables on mast (surface ships)	329

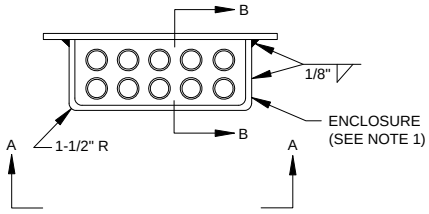
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TABLE 4DI. Figures for cable protection surface ships and submarines - Continued.

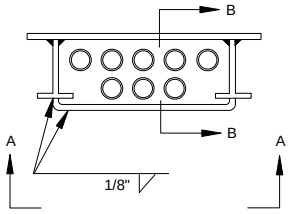
Figure number	Requirements/procedures	Page
4D27	Protection of cables in hangar spaces (surface ships)	330
4D28	Protection of cables in hangar spaces (surface ships) – elevation	331
4D29	Protection of cables in hangar spaces (surface ships) – side view	332

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NOTE: Figures 4D1 through 4D3 provide information for the protection of cables penetrating watertight decks. The length and width of the enclosure shall suit the cable group. The size of the stuffing tube group in decks and bulkheads shall be limited such as to permit re-tightening of all gland nuts in the group by ships force using a stuffing tube wrench set (NSN 5120-00-293-0013), A-A-3041.

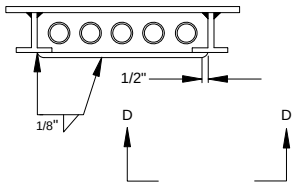


PLAN OF TYPICAL INSTALLATION

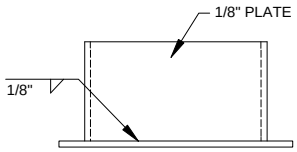


PLAN IN WAY OF DEEP STIFFENERS

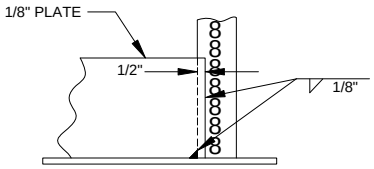
FOR SECTION B-B, SEE FIGURE 4D2



PLAN IN WAY OF SHALLOW STIFFENERS



SECTION A - A



SECTION D - D

SEE METHODS 4D-2-1 AND 4D-2-2 ON FIGURE 4D2

NOTES:

1. Surfaces in contact with plastics shall be thoroughly cleaned and free of oil or grease.
2. The finished 15° surface shall be smooth.

FIGURE 4D1. Protection of cables penetrating watertight decks.

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NOTES (continued):

3. Material shall be in accordance with the following:

Material for protection of cables penetrating watertight decks.

Material	Specification	Commercial source
Polyester resin	MIL-R-21607	Owens Corning or similar
Chopped fiberglass	MIL-Y-1140	Owens Corning or similar
Fire retardant polyurethane foam	MIL-P-21929	Gold Bond Zero-0-CEL, National Gypsum Co. or similar

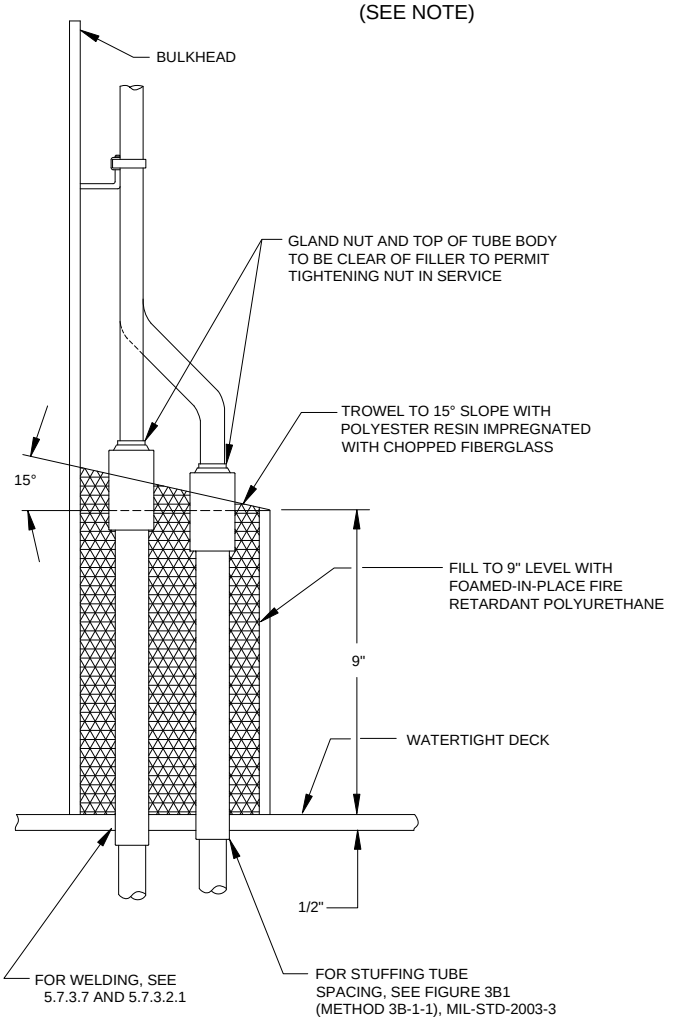
4. These enclosures may be used on steel or aluminum alloy structures using ⅛" steel or ⅛" aluminum to suit.
5. These enclosures shall not be installed until after all cables are installed and compartments are checked for tightness.
6. These enclosures shall not be installed on the weather decks.
7. Collar height shall be 3" in dry areas below decks.
8. Collar height shall be 9" in wet deck spaces below decks such as washrooms and sculleries.
9. A glass-reinforced collar may be used in lieu of a metal collar.

FIGURE 4D1. Protection of cables penetrating watertight decks - Continued.

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NOTE: Figures 4D1 through 4D3 provide information for the protection of cables penetrating watertight decks. The length and width of the enclosure shall suit the cable group. The size of the stuffing tube group in decks and bulkheads shall be limited such as to permit re-tightening of all gland nuts in the group by ships force using a stuffing tube wrench set (NSN 5120-00-293-0013), A-A-3041.

METHOD 4D-2-1
WET DECK SPACES BELOW DECKS
(SEE NOTE)



SECTION B - B
(SEE FIGURE 4D1)

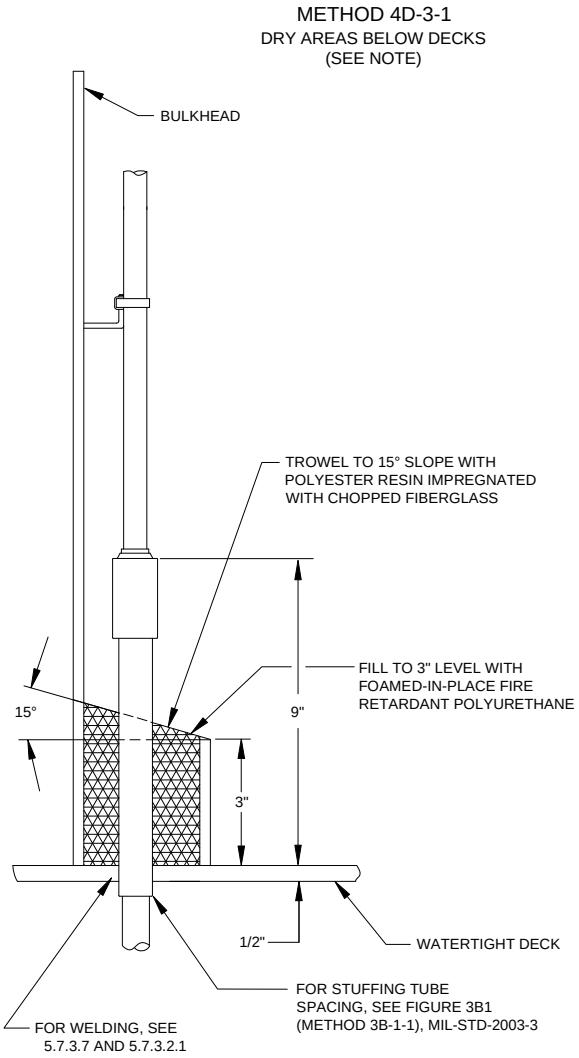
NOTES:

1. This method is not for use in weather applications.
2. Collar height shall be 9" in wet deck spaces below decks such as washrooms and sculleries.

FIGURE 4D2. Protection of cables penetrating watertight decks – wet spaces.

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NOTE: Figures 4D1 through 4D3 provide information for the protection of cables penetrating watertight decks. The length and width of the enclosure shall suit the cable group. The size of the stuffing tube group in decks and bulkheads shall be limited such as to permit re-tightening of all gland nuts in the group by ships force using a stuffing tube wrench set (NSN 5120-00-293-0013), A-A-3041.



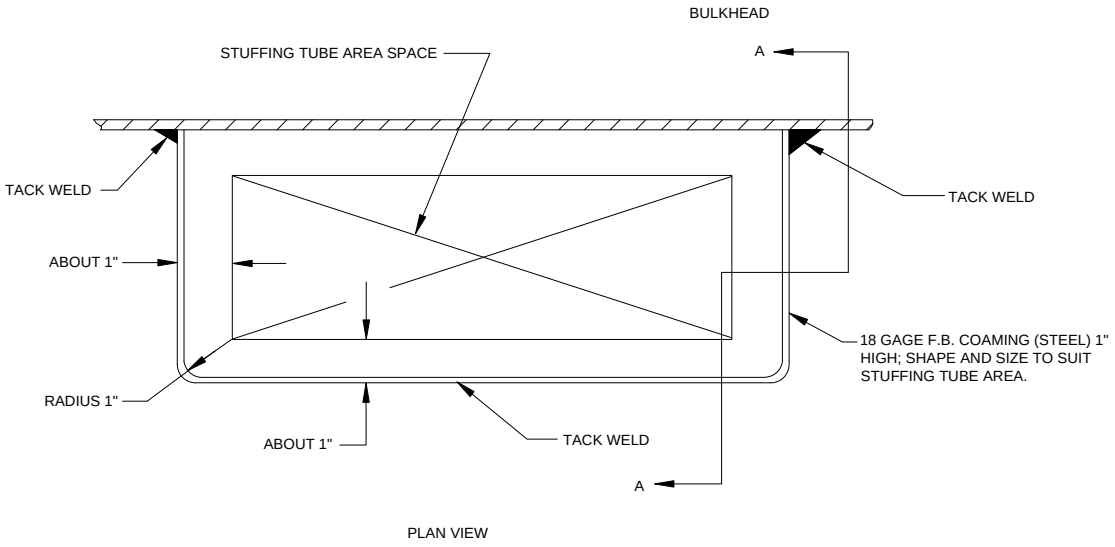
NOTES:

1. This method is not for use in weather applications.
2. Collar height shall be 3" in dry areas below decks.

FIGURE 4D3. Protection of cables penetrating watertight decks – dry areas.

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METHOD 4D-4-1
ALTERNATE METHOD FOR WATERTIGHT DECKS



TYPICAL DECK STUFFING TUBE AREA
SEE FIGURE 4D5 FOR SECTION A-A

NOTES:

1. Figure 4D4 provides an alternate method for protecting cables penetrating watertight decks, and the following procedure applies.
2. For stuffing tube areas other than in the weather, a method using an epoxy compound resin on the deck around the kick-pipes shall be accomplished as follows:
   a. After the kick-pipe and stuffing tubes have been installed and the cables pulled in, the 16-gauge steel coaming shall be fabricated to suit and tack-welded. Where decks have camber/shear, the coaming shall be fabricated to match the slope of the deck so that the epoxy compound will be level with the top edge of the entire coaming. This may necessitate that the coaming be made and tack welded in three separate pieces in lieu of one continuous flat bar as shown in plan view. The high side of this type of coaming shall not exceed 1½", and the corners shall be rounded.
   b. The deck around the kick-pipes within the coaming enclosure and the deck around the outside of the coaming shall be thoroughly cleaned as follows (it is not necessary to remove paint):
      (1) Blow out all dirt and loose material.
      (2) Scrub with a detergent solution.
      (3) Rinse with fresh water.
      (4) Blow out excess water.
   c. Put a small amount of Philadelphia Resin Co. adhesive resin (or equivalent) into a shallow container or on a flat plate. Add an equal amount of Philadelphia Resin Co. adhesive hardener (or equivalent) and mix thoroughly until the mixture is uniform in color (three tablespoons each of resin and hardener will provide enough adhesive for about three lineal feet of coaming).
   d. Apply this adhesive "putty" around the outside of the coaming with a rounded-end ¾" wide spatula (or equivalent) and shape it to a fillet with the end of the spatula. Dip the spatula in water to provide a smooth surface.

FIGURE 4D4. Protection of cables penetrating watertight decks – alternate method.

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NOTES (continued):

e. Pour the Philadelphia Resin Co. (or equivalent) PRS10FR hardener (black labeled can) into the white labeled can containing PRS10FR resin and mix thoroughly for about five minutes until the mixture is a uniform light gray color and any solid material in the bottom of the can has dispersed. Do not thin the mixture. (Mixing can be accomplished by hand or with a small electric drill with a paint mixer attachment.)

f. Pour this mixture into the kick-pipe area enclosure until it begins to overflow the coaming. Wipe excess off outside of coaming. Allow to cure hard before painting.

2. The PRS10FR material is packaged in six pound units (½ gallon size) and twelve pound units (one gallon size). The table below provides a guide to unit sizes for the resin. The OV adhesive material is packaged in ½-gallon size units (consisting of one quart size resin and one quart size hardener).

Unit size	Resin size	Hardener size	Coverage
Six pound	½ gallon	One pint	9" x 12" tube area
12 pound	One gallon	One quart	12" x 18" tube area

3. The table below provides cure times for the resin. For temperatures below 60 °F, heat lamps shall be used for curing the epoxy resin. The pot life for the resin is 45 minutes, which means that the epoxy-resin compound should be used within 45 minutes after mixing. Shelf-life of the material before mixing is two years.

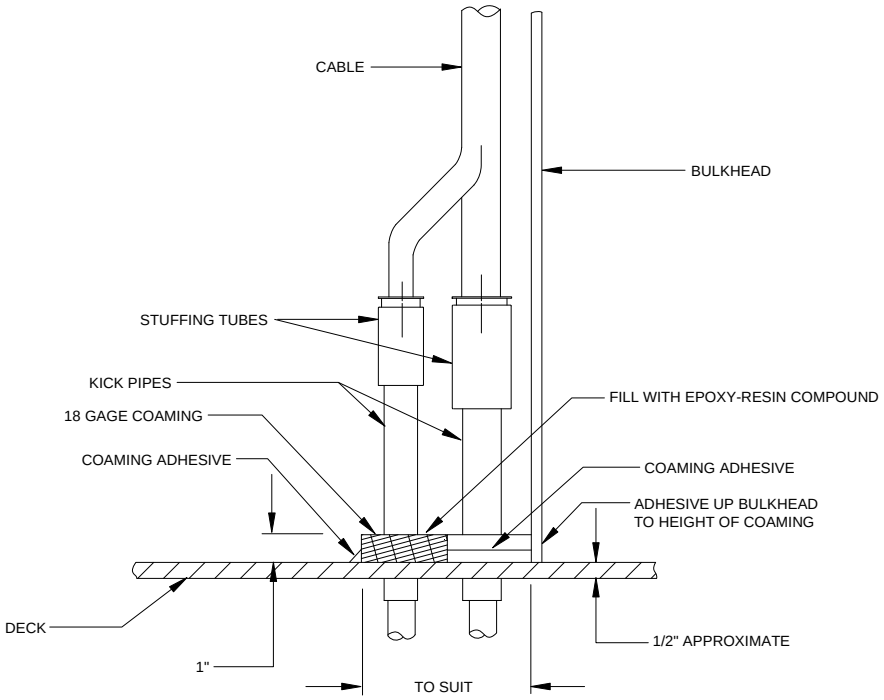
Temperature	Cure time (hours)
85 °F	4
75 °F	8
65 °F	12

4. Local exhaust ventilation is not necessary when using these epoxy materials, however, good personnel hygiene habits should be used and contact with the skin should be avoided. After using the epoxy material, wash with borax soap or waterless skin cleaner. Solvents should be avoided as a skin cleaner, but, if necessary, alcohol should be used (denatured ethyl or isopropyl are preferred). In no case should petroleum solvents or chlorinated hydrocarbons such as gasoline, kerosene, benzene, degreasing solvents, or dry cleaning fluids be used for cleaning the skin.

FIGURE 4D4. Protection of cables penetrating watertight decks – alternate method - Continued.

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SECTION A-A FOR METHOD 4D-4-1
ALTERNATE METHOD FOR WATERTIGHT DECKS



TYPICAL SECTION A - A

NOTES:

1. Figure 4D4 provides an alternate method for protecting cables penetrating watertight decks, and the following procedure applies.
2. For stuffing tube areas other than in the weather, a method using an epoxy compound resin on the deck around the kick-pipes shall be accomplished as follows:
   a. After the kick-pipe and stuffing tubes have been installed and the cables pulled in, the 16-gauge steel coaming shall be fabricated to suit and tack-welded. Where decks have camber/shear, the coaming shall be fabricated to match the slope of the deck so that the epoxy compound will be level with the top edge of the entire coaming. This may necessitate that the coaming be made and tack welded in three separate pieces in lieu of one continuous flat bar as shown in plan view. The high side of this type of coaming shall not exceed 1½", and the corners shall be rounded.
   b. The deck around the kick-pipes within the coaming enclosure and the deck around the outside of the coaming shall be thoroughly cleaned as follows (it is not necessary to remove paint):
      (1) Blow out all dirt and loose material.
      (2) Scrub with a detergent solution.
      (3) Rinse with fresh water.
      (4) Blow out excess water.
   c. Put a small amount of Philadelphia Resin Co. adhesive resin (or equivalent) into a shallow container or on a flat plate. Add an equal amount of Philadelphia Resin Co. adhesive hardener (or equivalent) and mix thoroughly until the mixture is uniform in color (three tablespoons each of resin and hardener will provide enough adhesive for about three lineal feet of coaming).

FIGURE 4D5. Protection of cables penetrating watertight decks (alternate method) – section A-A.

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NOTES (continued):

d. Apply this adhesive "putty" around the outside of the coaming with a rounded-end ¾" wide spatula (or equivalent) and shape it to a fillet with the end of the spatula. Dip the spatula in water to provide a smooth surface.
e. Pour the Philadelphia Resin Co. (or equivalent) PRS10FR hardener (black labeled can) into the white labeled can containing PRS10FR resin and mix thoroughly for about five minutes until the mixture is a uniform light gray color and any solid material in the bottom of the can has dispersed. Do not thin the mixture. (Mixing can be accomplished by hand or with a small electric drill with a paint mixer attachment.)
f. Pour this mixture into the kick-pipe area enclosure until it begins to overflow the coaming. Wipe excess off outside of coaming. Allow to cure hard before painting.

2. The PRS10FR material is packaged in six pound units (½ gallon size) and twelve pound units (one gallon size). The table below provides a guide to unit sizes for the resin. The OV adhesive material is packaged in ½-gallon size units (consisting of one quart size resin and one quart size hardener).

Unit size	Resin size	Hardener size	Coverage
Six pound	½ gallon	One pint	9" x 12" tube area
12 pound	One gallon	One quart	12" x 18" tube area

1. The table below provides cure times for the resin. For temperatures below 60 °F, heat lamps shall be used for curing the epoxy resin. The pot life for the resin is 45 minutes, which means that the epoxy-resin compound should be used within 45 minutes after mixing. Shelf-life of the material before mixing is two years.
2.

Temperature	Cure time (hours)
85 °F	4
75 °F	8
65 °F	12

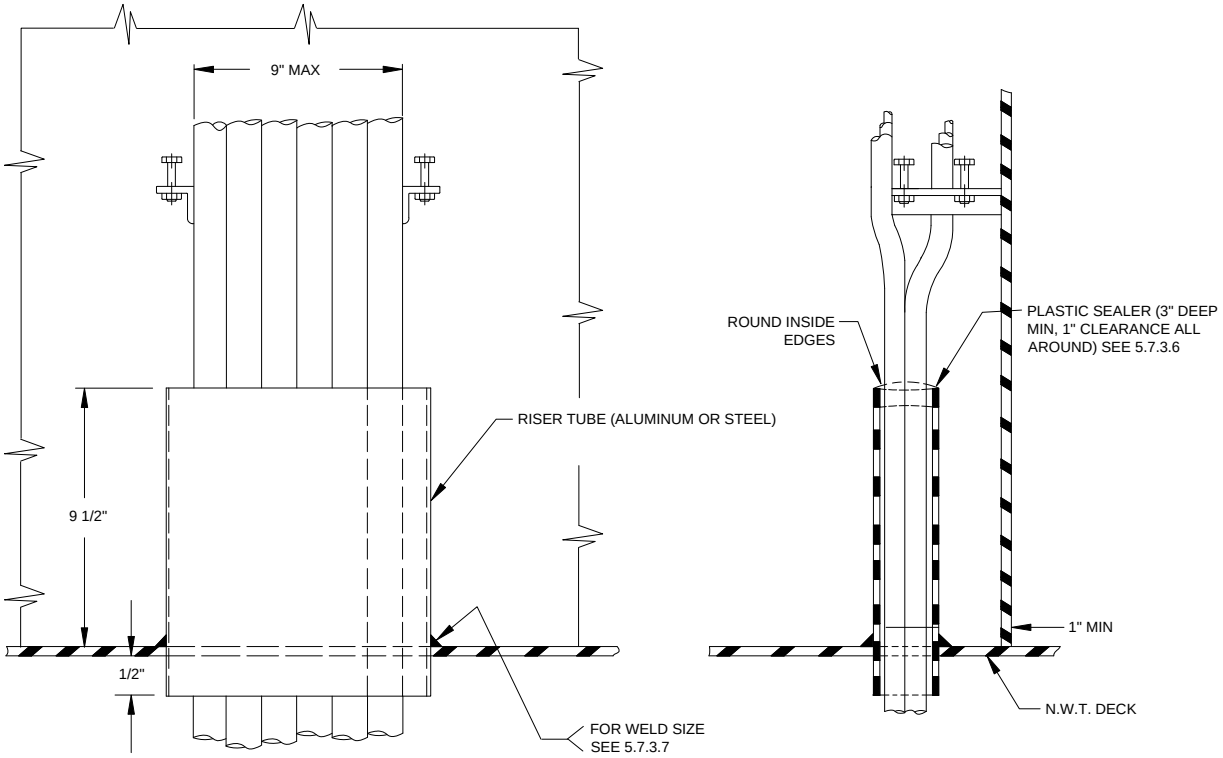
4. Local exhaust ventilation is not necessary when using these epoxy materials, however, good personnel hygiene habits should be used and contact with the skin should be avoided. After using the epoxy material, wash with borax soap or waterless skin cleaner. Solvents should be avoided as a skin cleaner, but, if necessary, alcohol should be used (denatured ethyl or isopropyl are preferred). In no case should petroleum solvents or chlorinated hydrocarbons such as gasoline, kerosene, benzene, degreasing solvents, or dry cleaning fluids be used for cleaning the skin.

FIGURE 4D5. Protection of cables penetrating watertight decks (alternate method) – section A-A - Continued.

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METHOD 4D-6-1

FOR PROTECTING THREE OR MORE CABLES
PASSING THROUGH NON-WATERIGHT DECKS



NOTE: Figure 4D6 provides information for the protection of cables penetrating a non-watertight deck.

FIGURE 4D6. Protection of cables penetrating non-watertight (NWT) decks.

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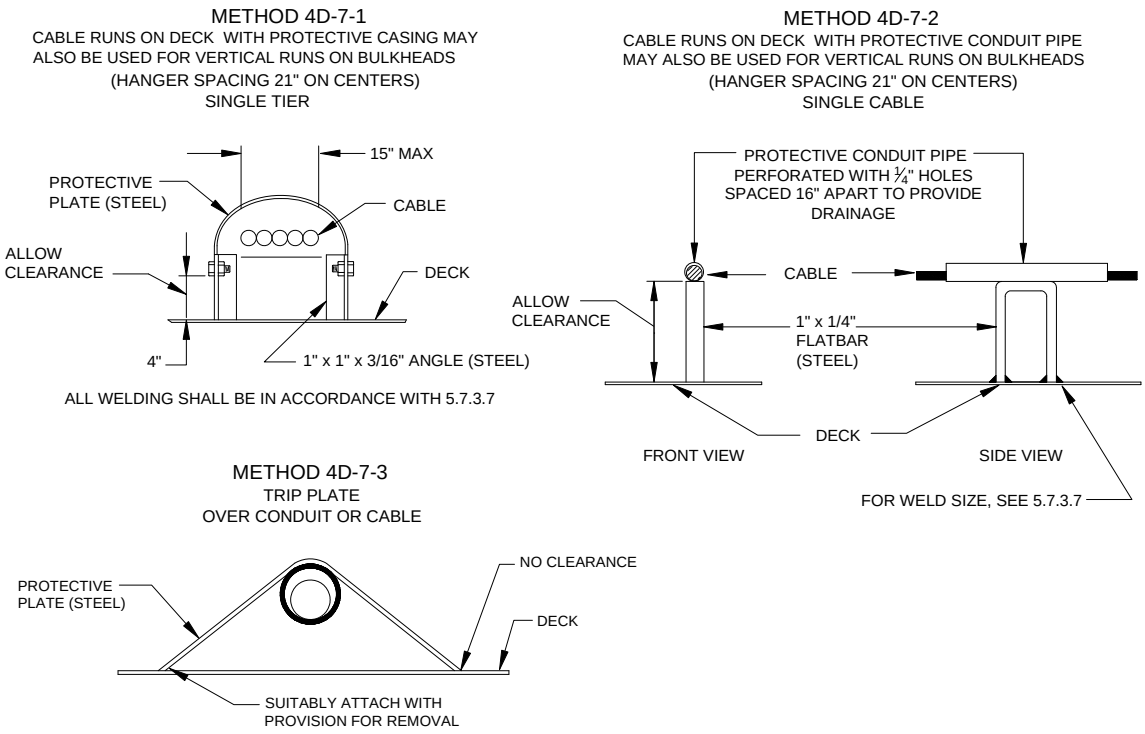


FIGURE 4D7. Protection of cables from mechanical injury.

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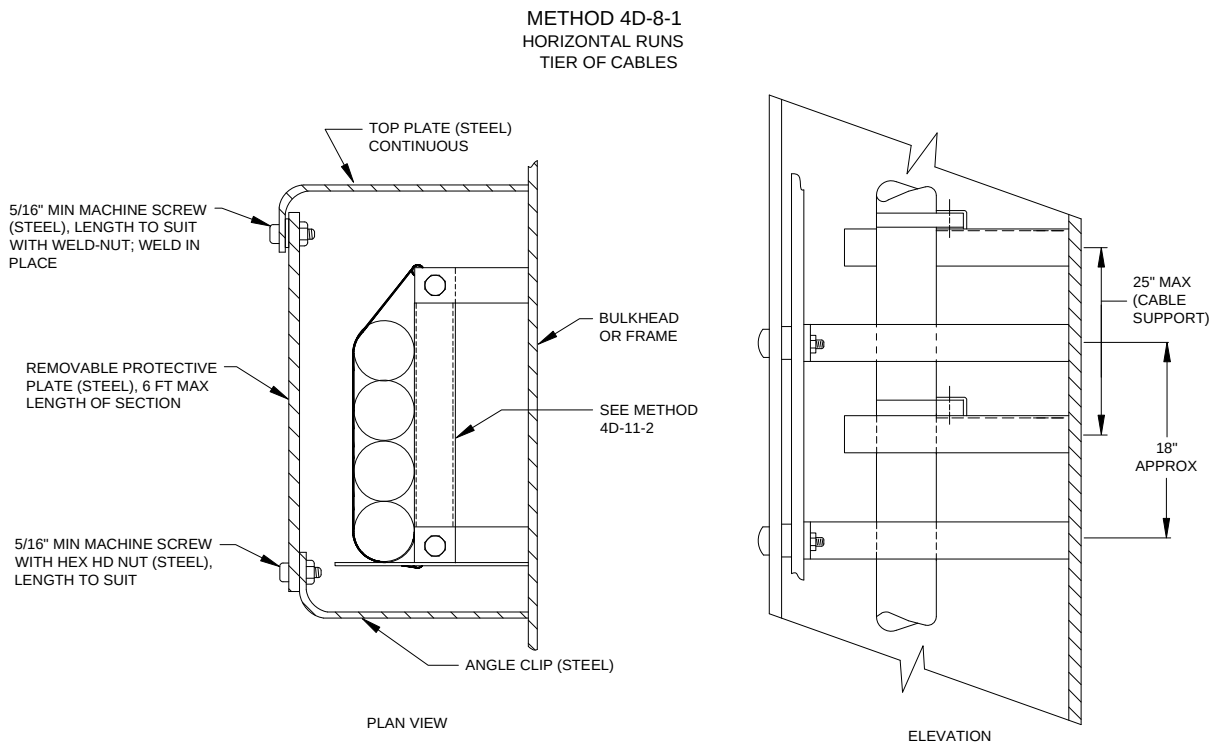


FIGURE 4D8. Protection of degaussing cables from mechanical injury – horizontal run.

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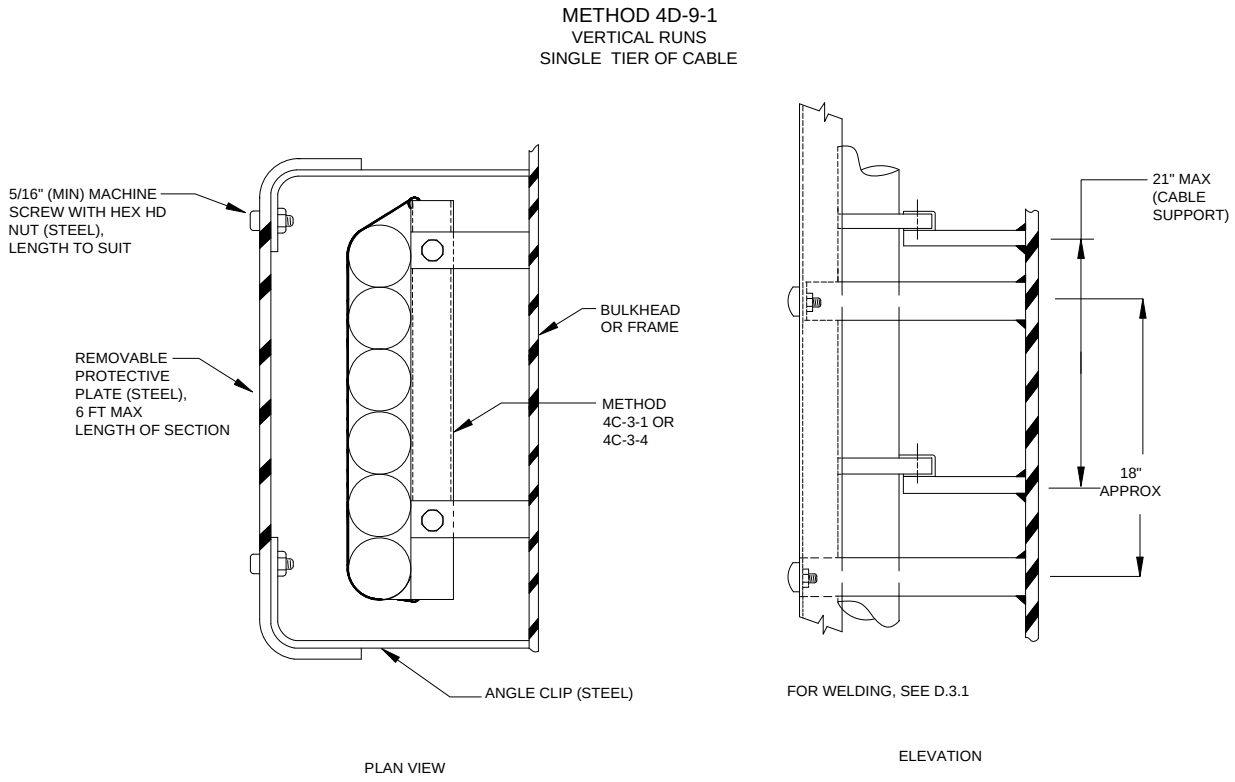
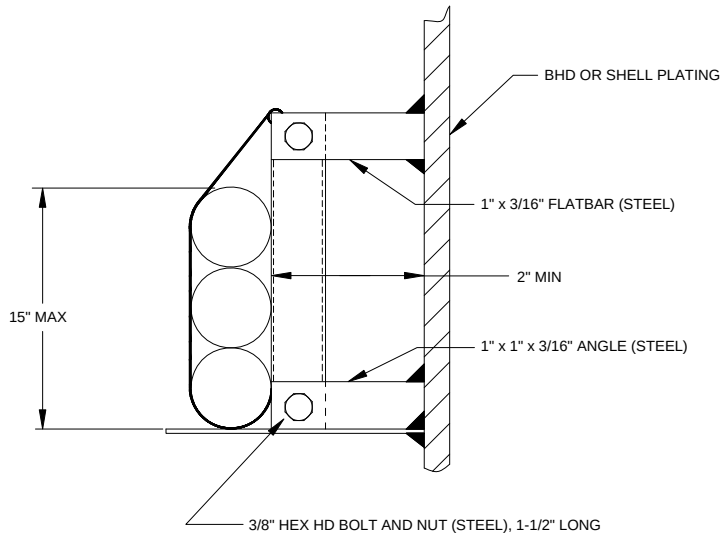


FIGURE 4D9. Protection of degaussing cables from mechanical injury – vertical run.

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METHOD 4D-10-1
CABLE HANGER DETAIL
FOR SINGLE TIER

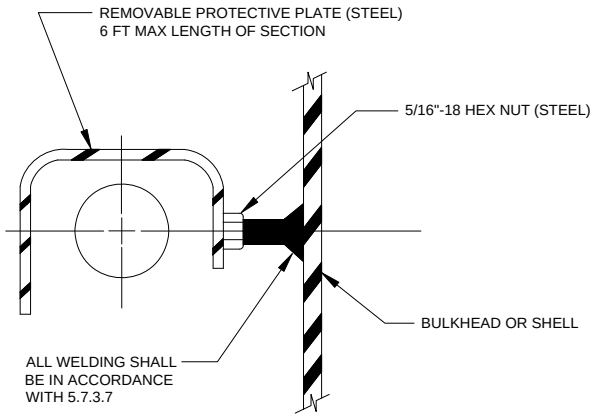


ATTACHMENT TO OUTER SHELL PLATING OF VESSEL
ALSO ATTACHMENT TO BALLISTIC BULKHEADS

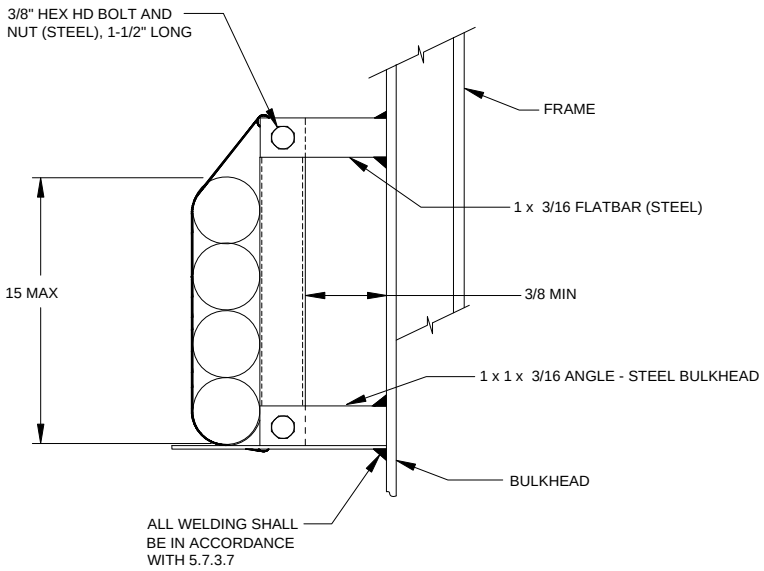
FIGURE 4D10. Protection of degaussing cables from mechanical injury.

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METHOD 4D-11-1
HORIZONTAL RUN
SINGLE CABLES



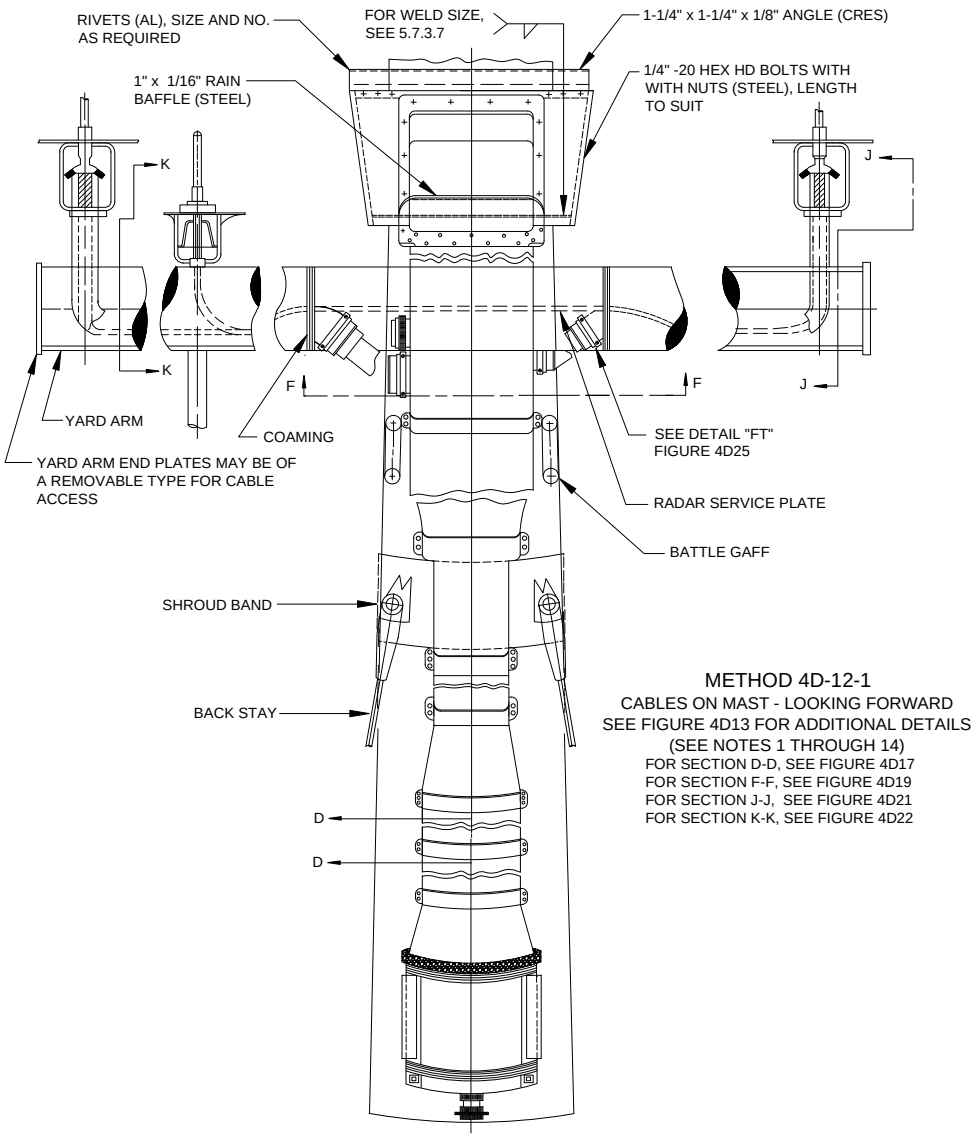
METHOD 4D-11-2
CABLE HANGER DETAIL
FOR SINGLE TIER
ATTACHMENT TO FRAME OR BULKHEAD



ALL DIMENSIONS ARE IN INCHES

FIGURE 4D11. Protection of degaussing cables from mechanical injury.

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NOTES:

1. For cables on a surface ship mast, refer to figures 4D12 through 4D26.
2. The area of the ventilation duct shall be calculated to give a minimum free flow area of 30 square inches with the closest approach of duct wall to the cable of 1".
3. The minimum distance of hanger support from the insulation to the hanger channel shall be 1½" to reduce conduction along hanger.
4. All seams, rivets, and openings in duct wall and insulating material shall be made weathertight.

FIGURE 4D12. Protection of cables on mast (surface ships) – forward view.

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NOTES (continued):

5. Provide doubler plates, as necessary, where brackets, coaming, yardarm, etc. are weakened by entrance of the ventilation duct.
6. Fraying surfaces of steel and aluminum parts shall receive one coat of primer in accordance with MIL-PRF-23236.
7. After installation, all surfaces of duct shall receive necessary coats of heat-resisting paint to match the surrounding surfaces.
8. In determining the size of duct for new construction, allowance shall be made for several additional cables to accommodate future radar and electrical requirements.
9. For vessels on which the screened speed light is located on the mainmast, the service cable shall be branched from the ventilation duct through flexible tubing.
10. Free flow areas within rain caps and air deflectors shall have 1½ to 2 times the area of the feeding pipe.
11. Flexible tubing shall be installed without twist, shall have sufficient slack to prevent the tubing from being strained upon working of mast and yardarms, and with the run pitched to avoid water pockets.
12. Flexible tubing shall be supported at 16" intervals where practicable; otherwise at such spacing and location that tubing shall take easy bends. Supports shall not be so placed that the flexing of the tubing will be restrained.
13. To reduce the number of sizes required for star supports, cable grouping may be used with a maximum clearance of 5/32", only as necessary, between star supports and corrosion resisting steel pipe.
14. Star supports shall be prevented from chafing the opposite cable by means of a spreader or steel barrier. The barrier may start longitudinally and twist 90 degrees as it progresses to a point beyond the cable outlet for inboard antenna.

FIGURE 4D12. Protection of cables on mast (surface ships) – forward view - Continued.

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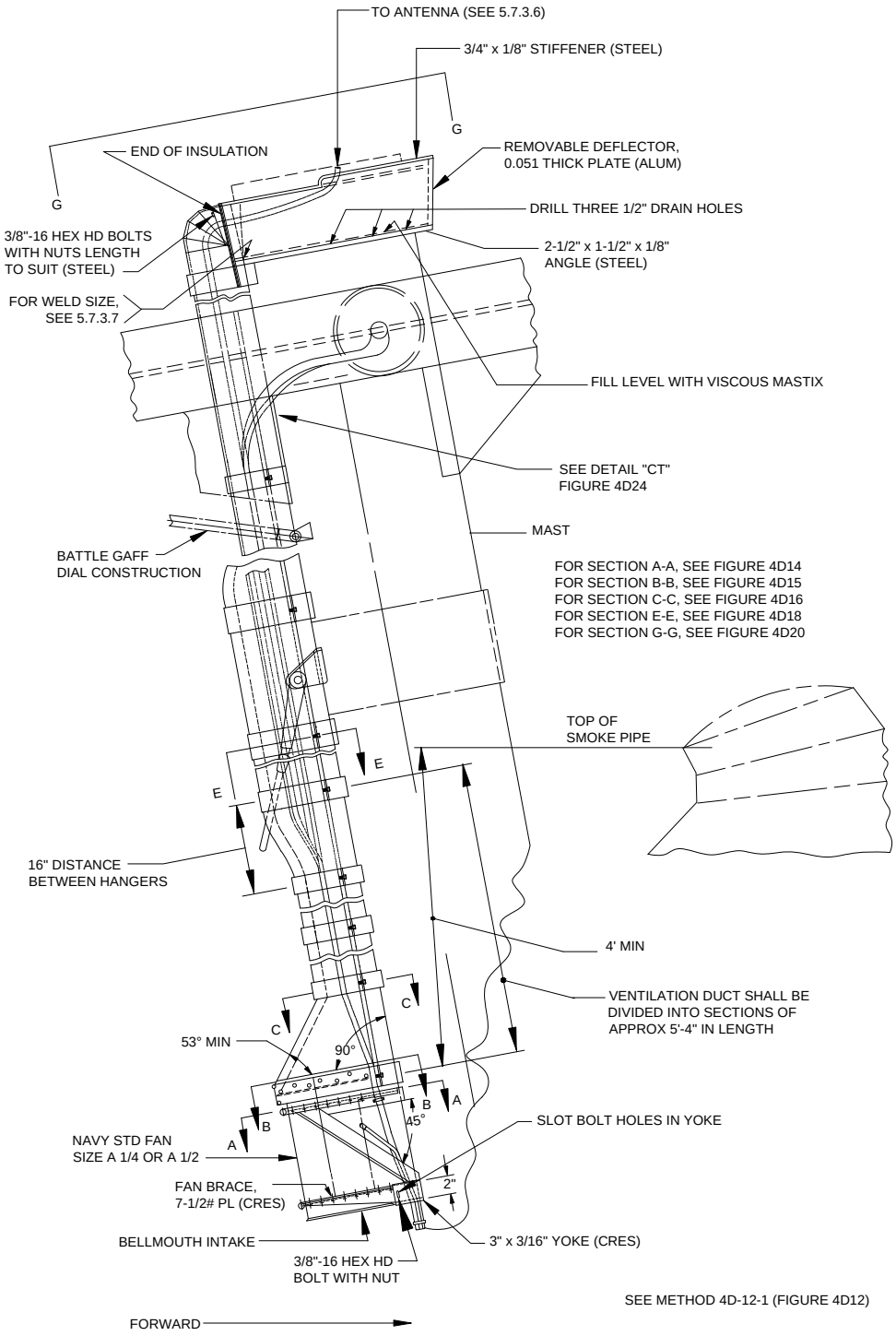


FIGURE 4D13. Protection of cables on mast (surface ships) – port view.

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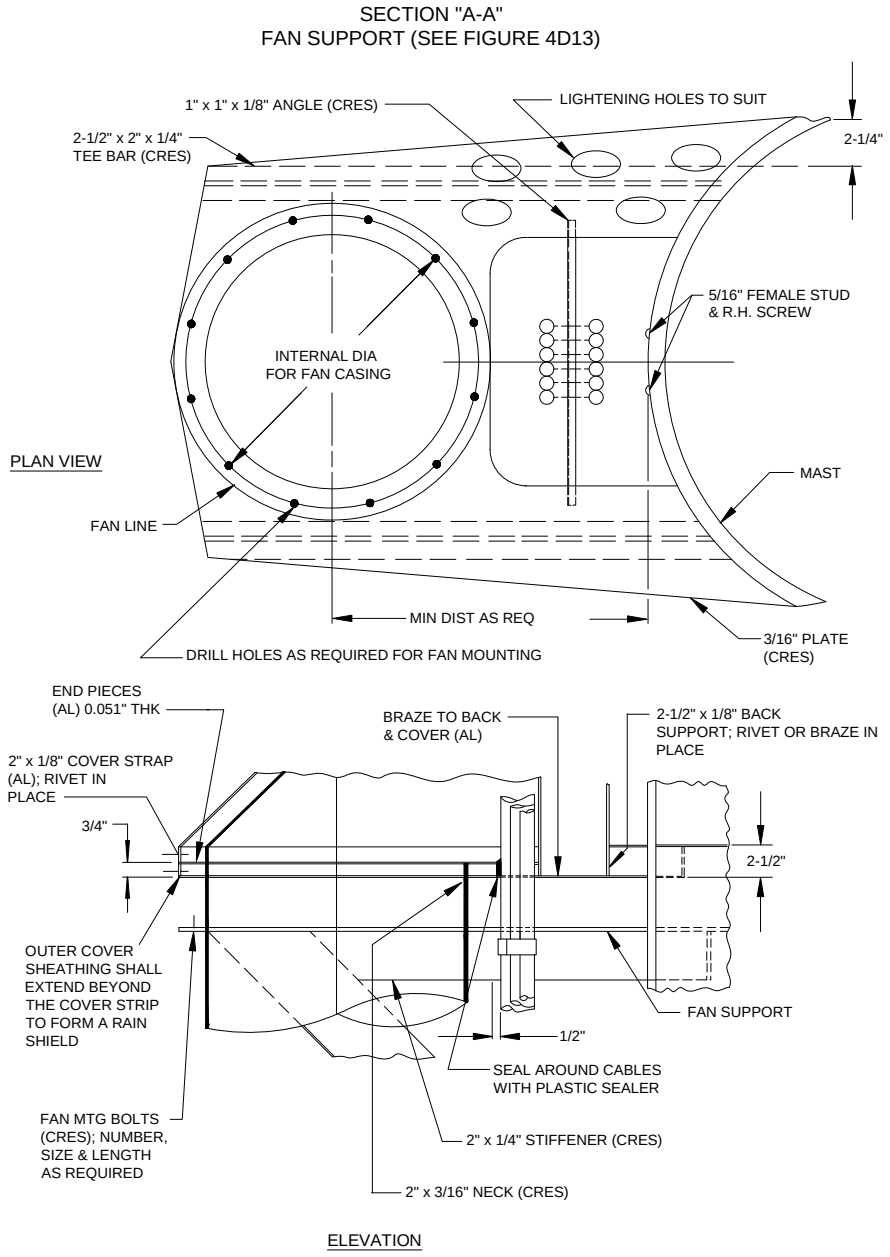


FIGURE 4D14. Protection of cables on mast (surface ships) – section "A-A".

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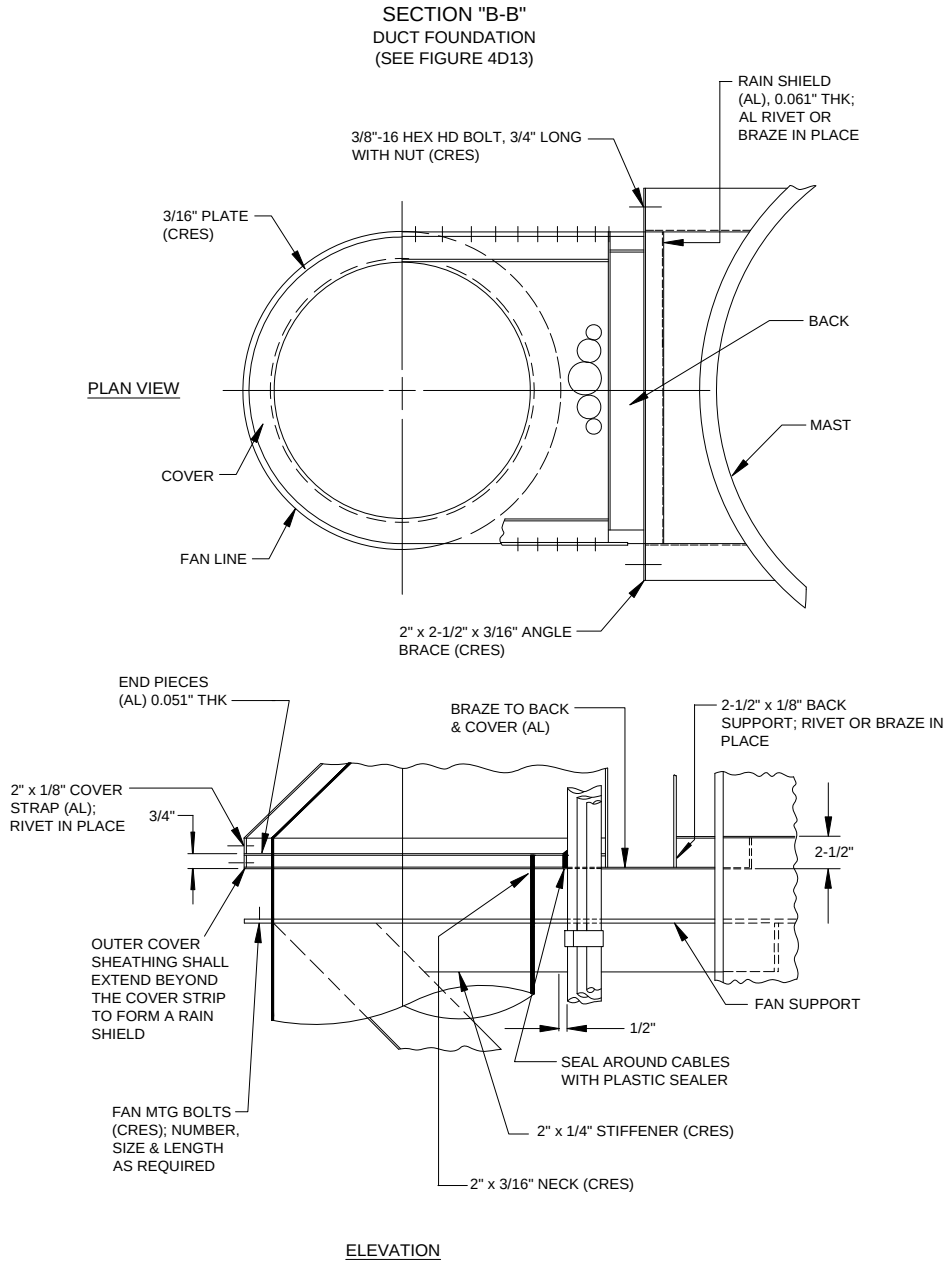
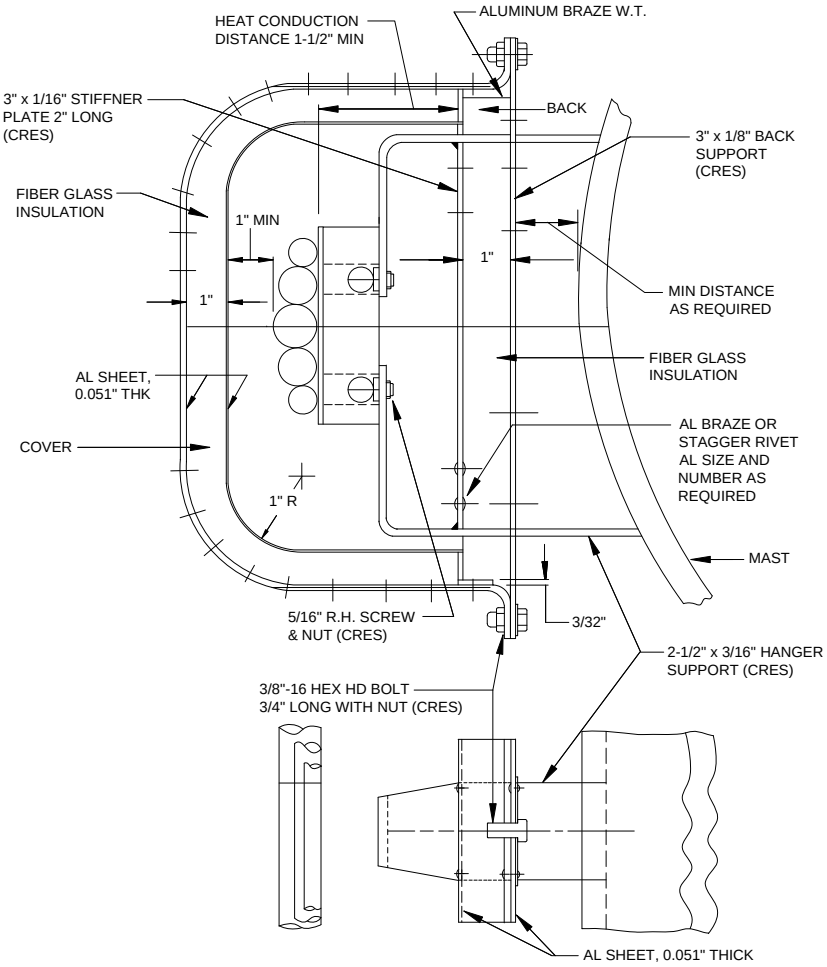


FIGURE 4D15. Protection of cables on mast (surface ships) – section "B-B".

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SECTION "C-C"
METHOD SHOWING ARRANGEMENT OF DUCT
AND A SINGLE TIER OF CABLES ON MAST WITH
ROOM FOR ADDITIONAL CABLES
(SEE FIGURE 4D13)



SIDE VIEW OF DUCT, COVER REMOVED

FIGURE 4D16. Protection of cables on mast (surface ships) – section "C-C".

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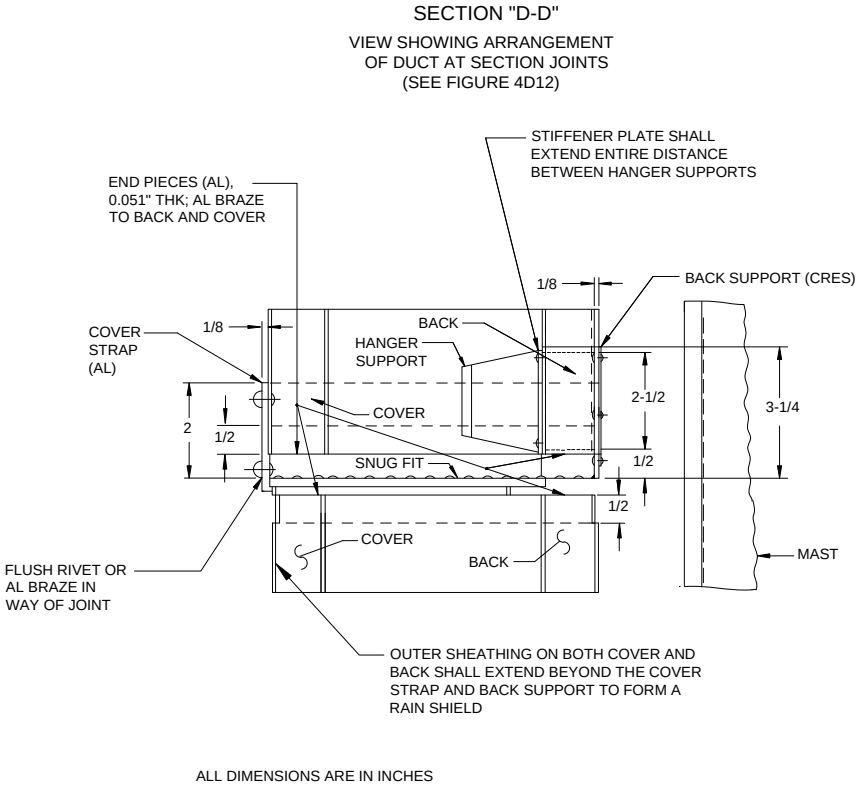


FIGURE 4D17. Protection of cables on mast (surface ships) – section "D-D".

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SECTION "E-E"

METHOD SHOWING ARRANGEMENT OF DUCT
AND CABLES TO CLEAR AN OBSTRUCTION
(SEE FIGURE 4D13)

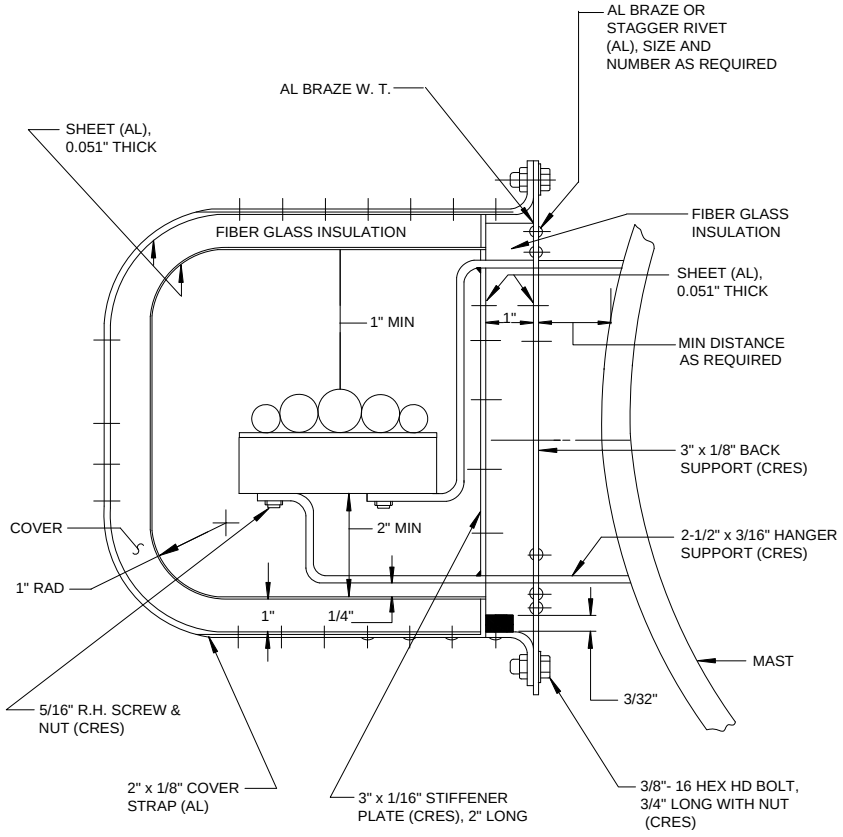
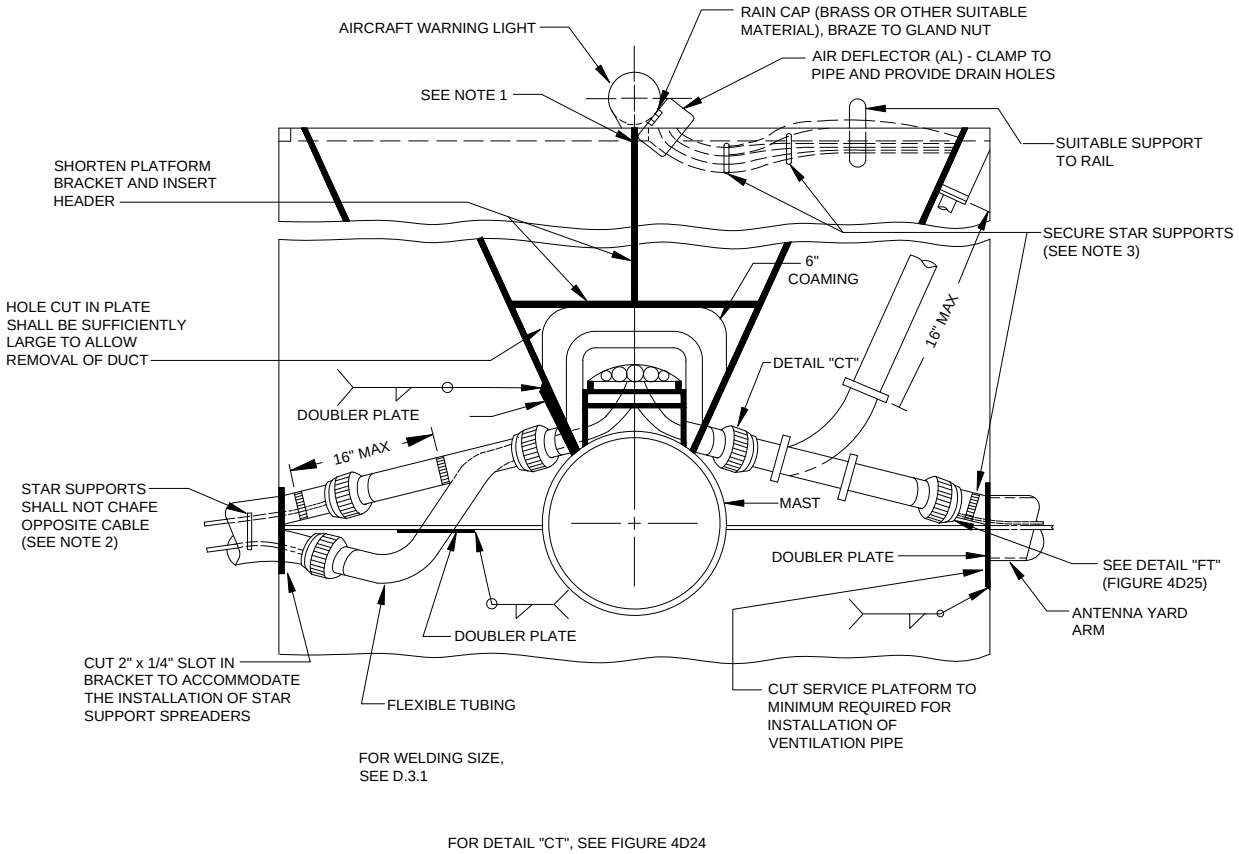


FIGURE 4D18. Protection of cables on mast (surface ships) – section “E-E”.

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SECTION "F-F"
VIEW OF RADAR SERVICE PLATFORM
(SEE FIGURE 4D12)



NOTES:

1. The area of the ventilation duct shall be calculated to give a minimum free flow area of 30 square inches with the closest approach of duct wall to the cable of 1".
2. Star supports shall be prevented from chafing the opposite cable by means of a spreader or steel barrier. The barrier may start longitudinally and twist 90° as it progresses to a point beyond the cable outlet for inboard antenna.
3. Secure star supports to cable at 16" (maximum) intervals continuing through the flexible tubing and yard arm runs and terminating 2" (minimum) from the open end of the tubing or pipe.

FIGURE 4D19. Protection of cables on mast (surface ships) – section “F-F”.

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SECTION "G-G"

VIEW OF DUCT AT RADAR PEDESTAL
(SEE FIGURE 4D13)

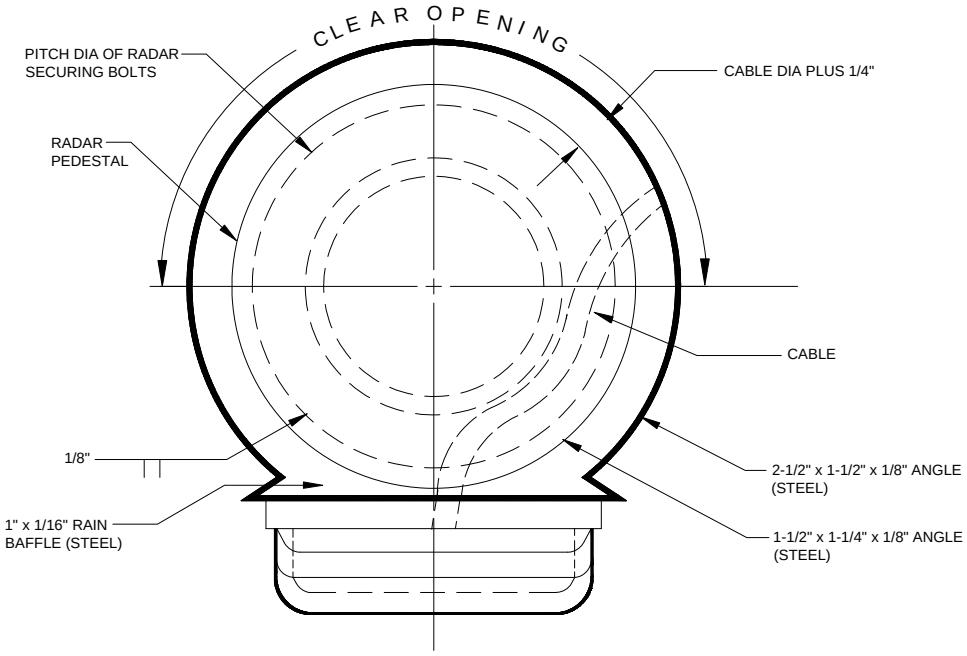
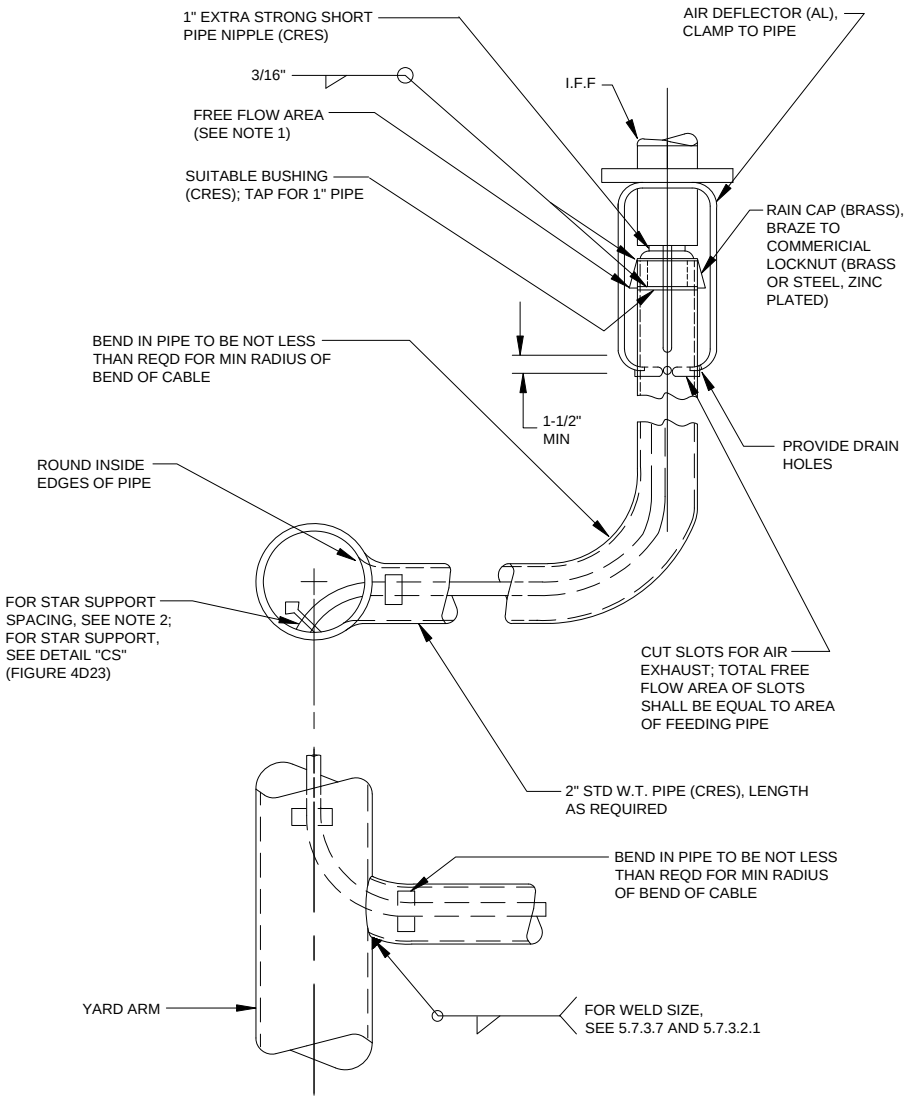


FIGURE 4D20. Protection of cables on mast (surface ships) – section "G-G".

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SECTION "J-J"
VIEW SHOWING ARRANGEMENT
OF STRUCTURAL SUPPORT
FOR I.F.F. EQUIPMENT
(SEE FIGURE 4D12)



NOTES:

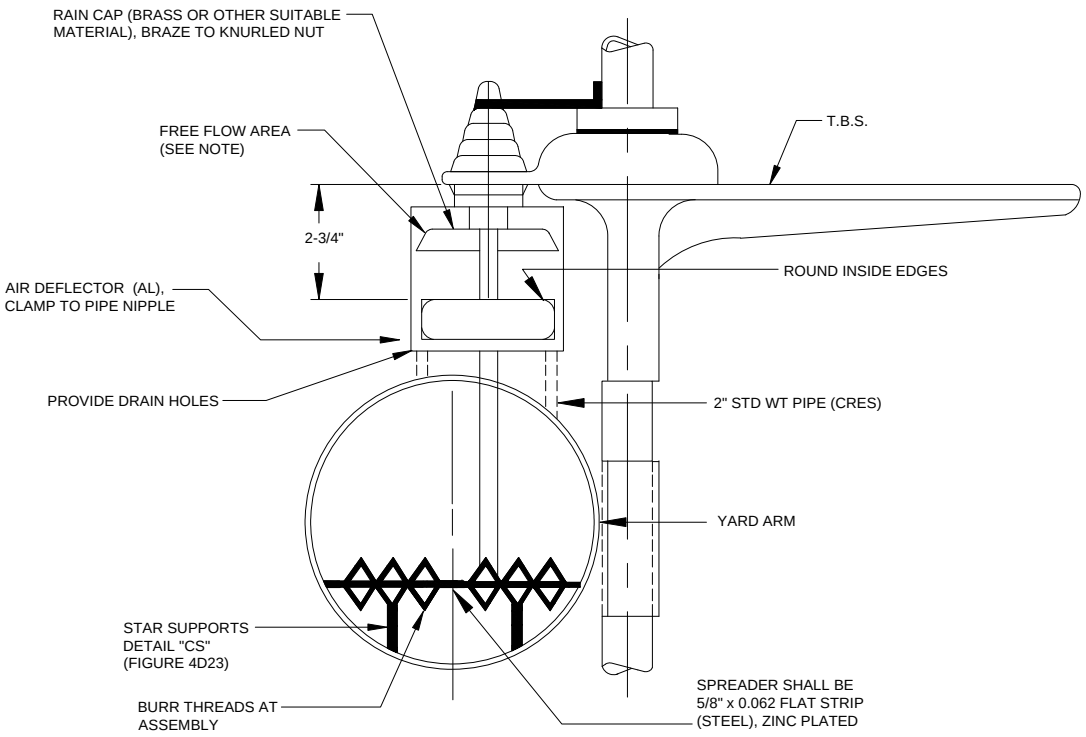
1. Free flow areas within rain caps and air deflectors shall have 1½ to 2 times the area of the feeding pipe.
2. Star supports shall be spaced a maximum of 18" on center in straight runs. Near pipe bends and in bends, they shall be provided in sufficient quantity to prevent chafing of cables and to ensure that the cable is surrounded by ventilating air.

FIGURE 4D21. Protection of cables on mast (surface ships) – section "J-J".

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SECTION "K-K"

VIEW SHOWING MOUNTING ARRANGEMENT FOR T.B.S EQUIPMENT
(SEE FIGURE 4D12)



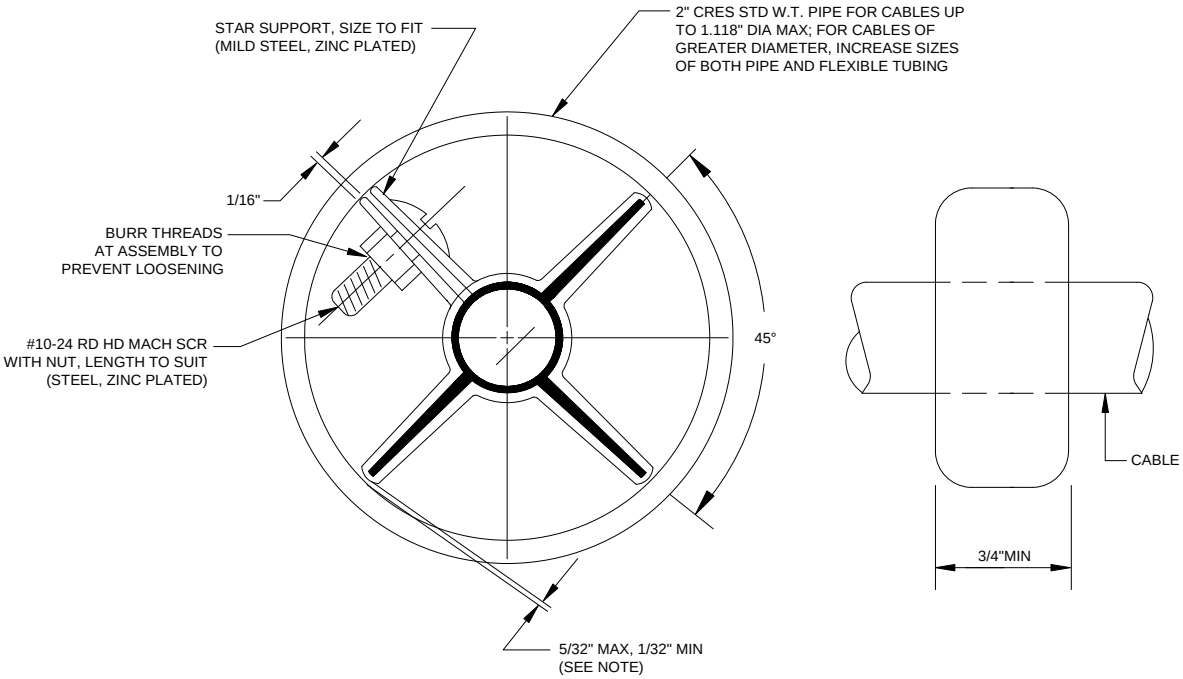
NOTE: Free flow areas within rain caps and air deflectors shall have 1½ to 2 times the area of the feeding pipe.

FIGURE 4D22. Protection of cables on mast (surface ships) – section "K-K".

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DETAIL "CS"

METHOD OF SUPPORTING CABLES IN PIPE AND FLEXIBLE TUBING
SEE FIGURES 4D21 AND 4D22



NOTE: To reduce the number of sizes required for star supports, cable grouping may be used with a maximum clearance of 5/32", only as necessary, between star supports and corrosion resisting steel pipe.

FIGURE 4D23. Protection of cables on mast (surface ships) – detail “CS”.

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DETAIL "CT"
METHOD OF SECURING FLEXIBLE TUBING TO VENTILATION DUCT
(SEE FIGURE 4D12)

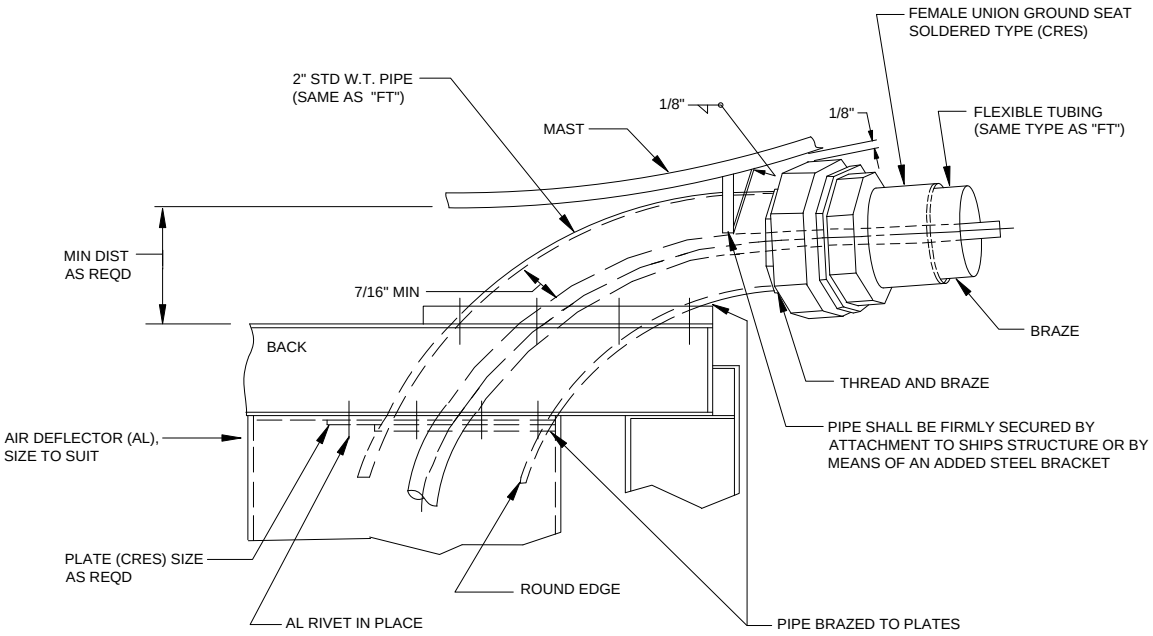


FIGURE 4D24. Protection of cables on mast (surface ships) – detail “CT”.

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DETAIL "FT"
METHOD OF SECURING FLEXIBLE TUBING TO A FLAT SURFACE
(SEE FIGURES 4D12 AND 4D19)

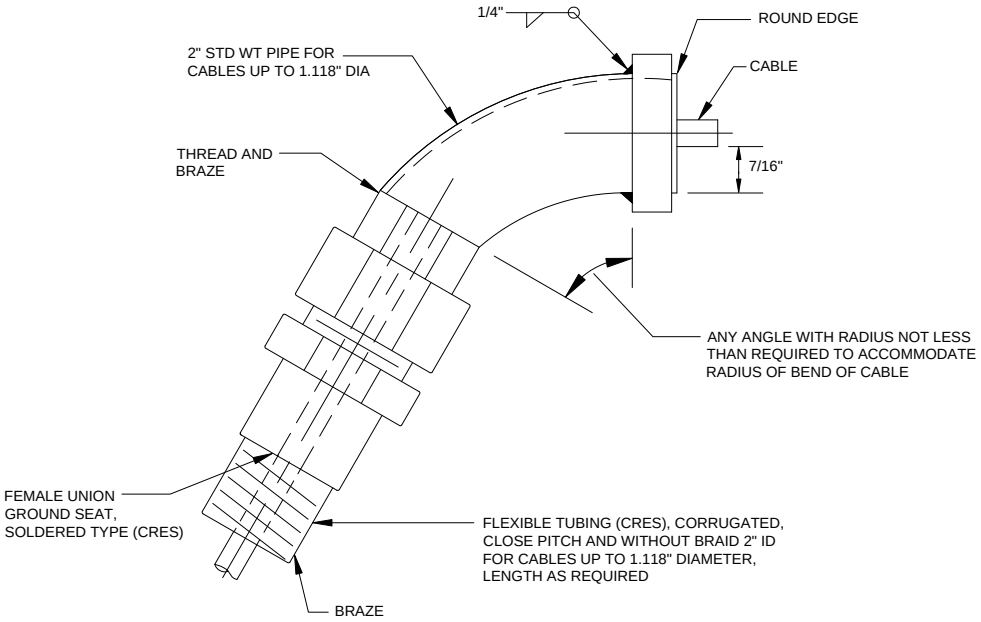


FIGURE 4D25. Protection of cables on mast (surface ships) – detail “FT”.

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METHOD 4D-26-1
ARRANGEMENT OF DUCT AND A DOUBLE TIER OF CABLES ON MAST

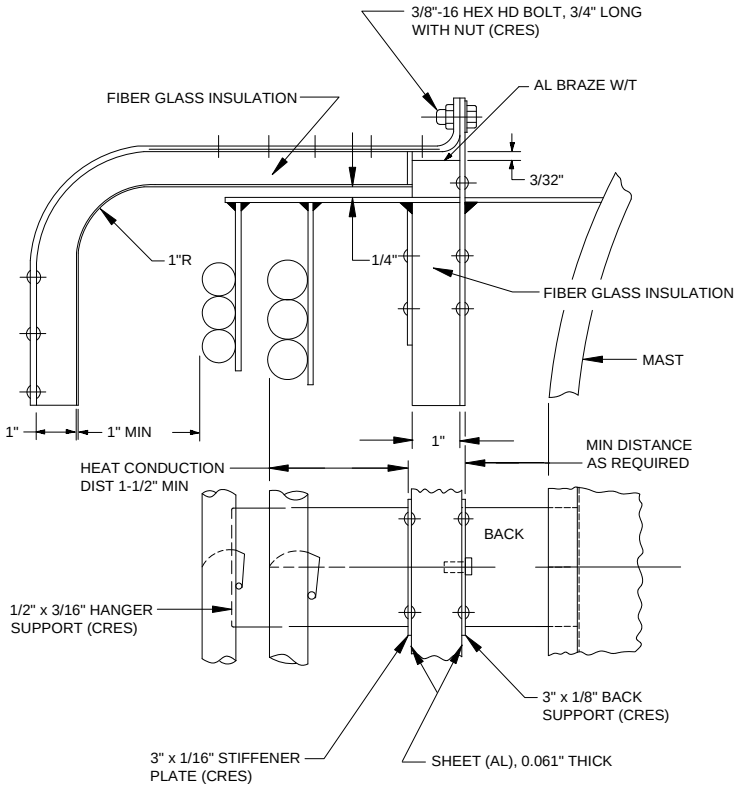


FIGURE 4D26. Arrangement of duct and double tier of cables on mast (surface ships).

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METHOD 4D-27-1
TWO TIERS OF CABLES (HANGAR SPACES)

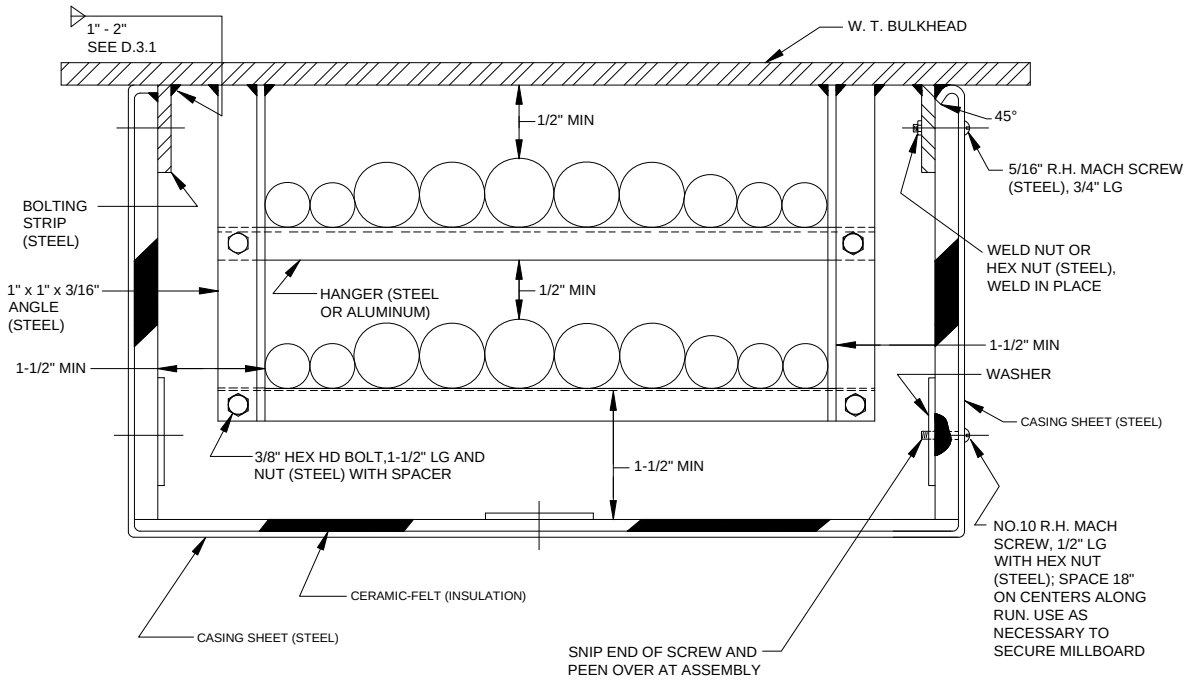


FIGURE 4D27. Protection of cables in hangar spaces (surface ships).

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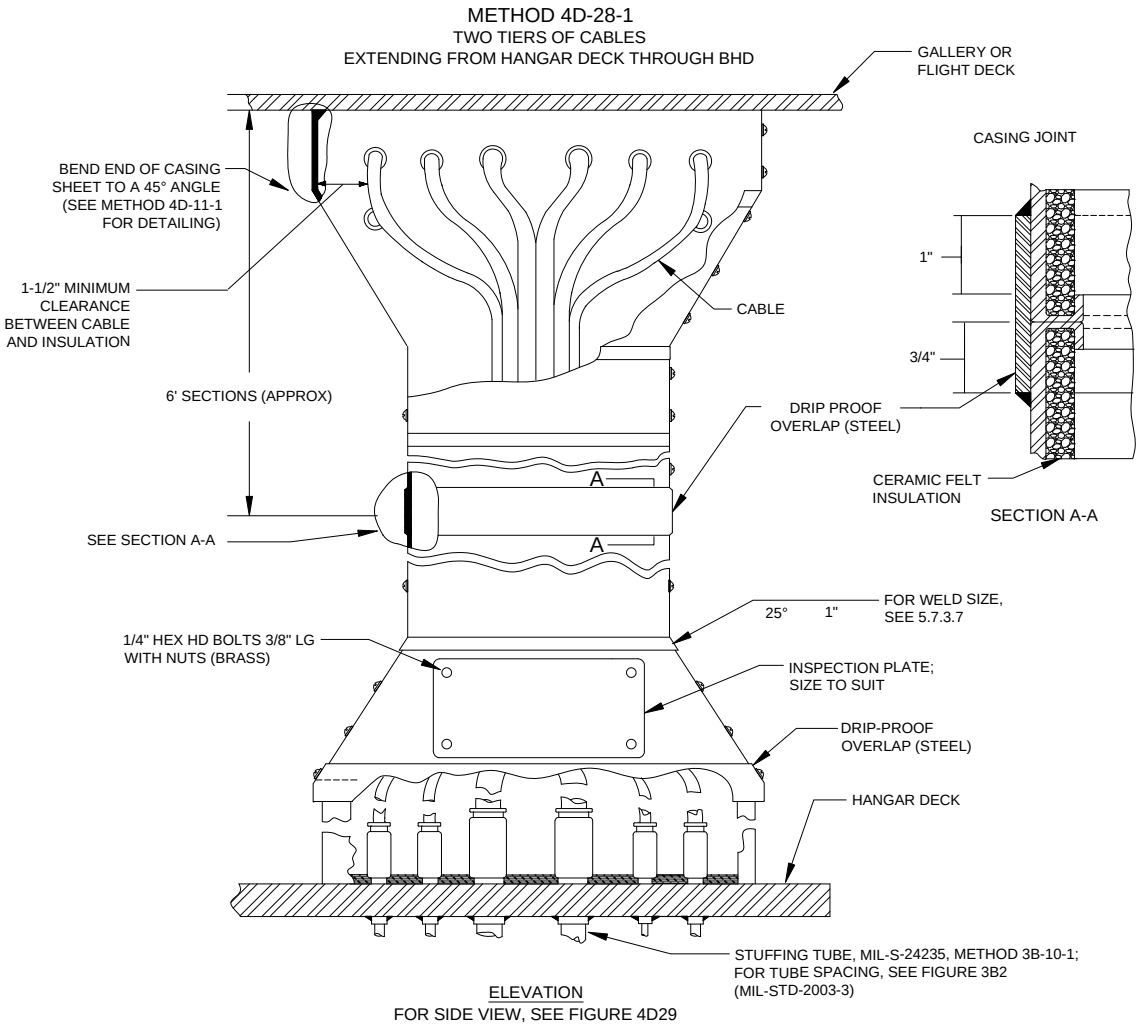


FIGURE 4D28. Protection of cables in hangar spaces (surface ships) – elevation.

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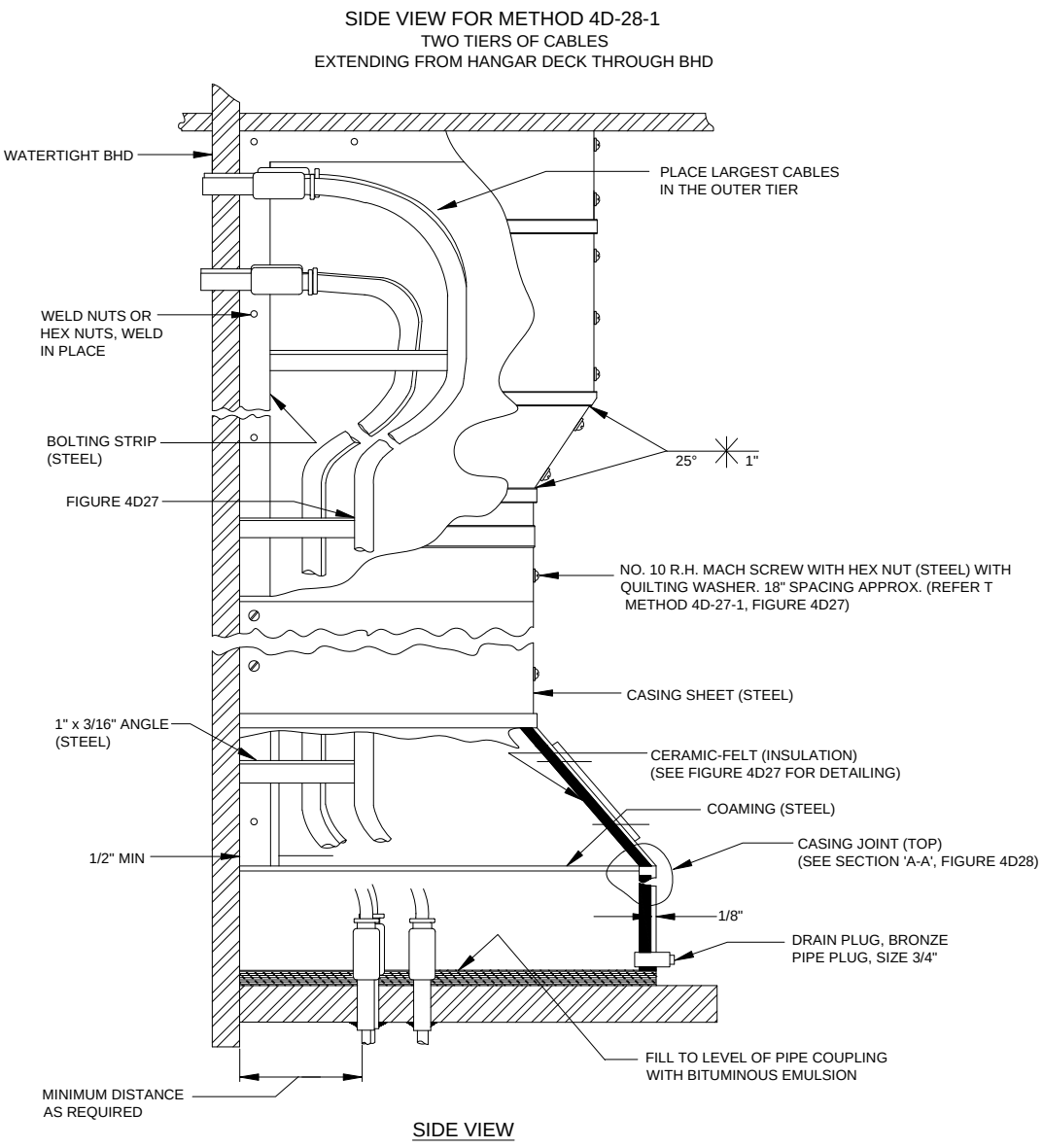


FIGURE 4D29. Protection of cables in hangar spaces (surface ships) – side view.

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Preparing activity:
Navy – SH
(Project SESS-2006-014)

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